

THE EMERGENCE OF OBJECTIVITY AND THE QUANTUM-TO-CLASSICAL TRANSITION

Thao Phuong Le

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Department of Physics and Astronomy
University College London

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I, Thao Phuong Le confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the thesis.

Signature Date

ABSTRACT

Quantum mechanics describes the smallest systems, whilst classical physics describes macroscopic scales. However, the quantum-to-classical transition is a major unsolved problem. In this thesis, we develop further understanding of the transition through the concept of objectivity. The objects we perceive in our everyday world are objective: they exist regardless of observation, and their properties are independent of observers. Quantum Darwinism is a conceptual and theoretical framework that describes the emergence of objectivity.

The thesis addresses the quantitative description of objectivity, the dynamics of emergent objectivity, and the analysis of non-objectivity in experimental and biology-inspired systems.

Firstly, we propose two new mathematical frameworks to describe objectivity, *Strong Quantum Darwinism* and *invariant spectrum broadcast structure*. Then we unite them with the pre-existing frameworks by identifying their deep connections and differences. Each framework considers varying amounts of objectivity in the system and environments and together, they create a hierarchy of objectivity.

We then analyse objectivity in a scenario where system-environment interactions are strong and non-Markovian, highlighting some practical differences between the Quantum Darwinism frameworks. Next, we describe the quantum channels that create objective states, leading us to discuss the contribution of non-Markovianity to the emergence of objectivity. Then, we introduce a witnessing scheme that simplifies the detection of non-objectivity to promote further experimental testing of Quantum Darwinism. Finally, we examine the contribution of quantum coherence and non-objectivity to the performance of a biological-inspired

quantum magnetic sensor. We find that the sensor operates in a non-objective regime, and that quantum coherence is necessary for the sensor's performance.

While the quantum-to-classical transition remains unsolved, Quantum Darwinism and its frameworks contribute to the understanding of emergent objectivity and have much potential to further explore aspects of the transition within the structure of quantum mechanics.

IMPACT STATEMENT

The research presented in the thesis contributes to our understanding of the quantum-to-classical transition and the role of system-environment correlations in the transition. The unification of the frameworks of objectivity and Quantum Darwinism gives us a greater understanding on what it means for quantum systems to be objective through the lenses of quantum information, state structure and geometry, and furthermore provides nuance on what it means to be “objective”. The new mathematical frameworks of Quantum Darwinism developed here can be immediately applied to study emergent objectivity in various system-environment scenarios.

As a step towards experimental verification with current available technological capabilities, the thesis includes research towards measuring objectivity in experimental systems. Our proposed scheme could be used to explore the quantum-to-classical transition through the lens of quantum Darwinism in larger and more realistic environment settings, which in turn will contribute to the understanding of the quantum-to-classical transition.

In the thesis, we also contribute to the understanding of the operation of quantum biological systems. Quantum biological systems operate at the quantum-classical boundary: we have analysed the important role of the environment in maintaining the quantum nature of the biological systems while simultaneously enhancing function. This work can also contribute to quantum technological advancement by exploiting environment effects to enhance their operation, as near term quantum technologies will likely operate at a quantum-classical boundary similar to quantum biological systems. In the long term, this will impact in our understanding of the natural world regarding the quantum-to-classical transition

and the nature of physical reality at the smallest scales.

The works in this thesis have also been disseminated through the publication of papers for most of the following chapters, and numerous talks and poster presentations at conferences, workshops, and university visits in the UK and beyond.

LIST OF PUBLICATIONS AND PREPRINTS

The work presented in this thesis contains results from the following papers/preprints:

- [1] Thao P. Le and Alexandra Olaya-Castro, *Objectivity (or lack thereof): Comparison between predictions of quantum Darwinism and spectrum broadcast structure*, Phys. Rev. A 98, 032103 (2018).
- [2] Thao P. Le and Alexandra Olaya-Castro, *Strong Quantum Darwinism and Strong Independence are Equivalent to Spectrum Broadcast Structure*, Phys. Rev. Lett. 122, 010403 (2019).
- [3] Thao P. Le, *Quantum discord-breaking and discord-annihilating channels*, (2019), arxiv:1907.05440v1.
- [4] Thao P. Le and Alexandra Olaya-Castro, *Witnessing non-objectivity in the framework of strong quantum Darwinism*, Quantum Sci. Technol. 5, 045012 (2020).
- Thao P. Le and Alexandra Olaya-Castro, *A non-maximally mixed initial state is necessary to detect magnetic field direction in an avian-inspired quantum magnetic sensor*, (2020, in preparation)

The following work was not including in this thesis:

- [5] Thao P. Le, Piotr Mironowicz, and Pawel Horodecki, *Blurred quantum Darwinism across quantum reference frames*, (2020), arxiv:2006.06364v1.

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Introduction

Quantum mechanics describes the smallest microscopic systems, whilst classical physics describes our everyday scales right out to the wide reaches of the universe. Quantum phenomena such as quantum entanglement and quantum coherence do not seem to appear in macroscopic objects: they are not the common experience of those outside a quantum science laboratory. This suggests the existence of a “boundary” between quantum mechanics and classical physics. Is this boundary sharp, or is the transition from quantum to classical gradual? *Where* is it? And *is* there a transition at all?

The quantum-to-classical transition is a longstanding open problem since the birth of quantum mechanics, remaining unsolved to this day, despite the intense effort and wide range of approaches. Efforts in advancing the understanding of the quantum-to-classical transition have and will continue to contribute to the foundations of quantum physics. Furthermore, greater knowledge of the transition and quantum-classical boundary equips us to better probe and develop quantum systems that exist at the boundary: for example, some biomolecular systems have a small amount of quantumness [6–9], and engineering quantum systems to retain their quantum features is important for quantum technologies.

The understanding and conceptualisation of the quantum-to-classical transition can be approached through quantum theory (*e.g.* [10–14]), with modifications of quantum theory (*e.g.* collapse models [15, 16]), and with different interpretations of quantum theory [17–22]. In this thesis, we are concerned with developing understanding of the transition retaining the framework of quantum mechanics.

Quantum coherence is a fundamental feature of quantum mechanics that arises from the quantum superposition of orthogonal states with respect to a reference basis [23]. Pragmatically, quantum coherence is described by the non-diagonal elements of a density operator of a quantum system. Decoherence is the accepted mechanism that causes the decay of quantum coherence and loss of information [11, 24, 25]. It leads to the vanishing of non-diagonal terms and subsequent measurements of the decohered state are compatible with a classical statistical mixture.

Within a closed quantum system, quantum coherence can never permanently fade away, due to the nature of closed quantum evolution. However, realistic systems are embedded in an environment, usually inaccessible or only partially accessible. Via interactions with the surrounding environment, quantum systems gradually lose coherence to the environment, *i.e.*, decohere, and become effectively classical [11, 24, 25].

However, decoherence alone cannot explain the entirety of the quantum-to-classical transition. While it is able to describe how the quantum systems become classical, decoherence does not tell us about how the *environment* behaves during the decoherence process. Quantum Darwinism is a conceptual framework that approaches this problem by explicitly describing the environment and its correlations with the system [26–28].

Beyond classicality, everyday objects are also *objective*. Their properties exist independently regardless of any hypothetical observer, and information about their properties can be widely gathered and agreed upon by multiple independent observers. The moon exists regardless of any lifeforms on Earth, and many of us can also observe the same (or extremely similar) moon. Quantum Darwinism describes the process of emergent objectivity.

As a system interacts with its surrounding environment, the system not only decoheres, but furthermore information about the system can spread into the environment. The application of decoherence in open quantum systems typically considers the environment to be a very large system, that is by definition too large to be realistically known. However, real environments are made up of multiple microscopic components: consider all the photons, all the air molecules, that surround us. Through interactions with the system in question, these environment particles can be affected and gain information about the system. Under quantum Darwinism, certain types of system-information tend to spread into multiple parts

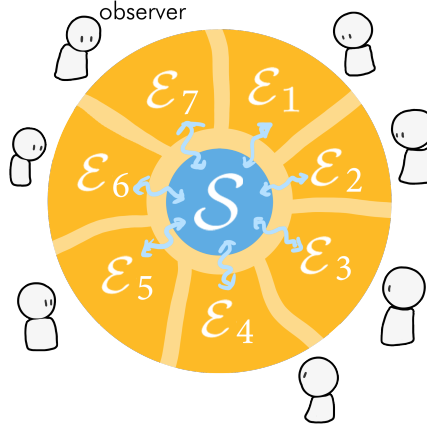


Figure 1.1: **Quantum Darwinism** recognises that the environment is made up of multiple components. For example we have $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2 \otimes \cdots \otimes \mathcal{E}_7$. Due to the system $\mathcal{S}$'s interaction with the environment, the system decoheres and information about the system can spread into the subenvironments. Different observers have access to different subenvironments, and when observers are able to determine and agree upon the state of the system, the system state is said to be objective.

of the environment at the expense of other information. *Information* is “Darwinistic”, and certain information is “fittest”, similar to biological evolution. The copies of information are a “record” to the existence of an object, regardless of observer. The multiple copies of information allows multiple observers to find out the same information about the system. Hence, the emergence of objectivity is caused by the spreading of information. We have depicted the scenario in Fig. 1.1. The quintessential example is the photon environment: photons constantly bounce off objects, carrying away information about the object’s colour, shape, position, *etc.*, and due to the abundance of photons, multiple observers are able to capture information about the original objects.

The overarching purpose of this thesis is to investigate the emergence of objectivity through the frameworks of Quantum Darwinism. The conceptual results derived in this thesis give a theoretical underpinning to the endeavour, producing a unified picture of Quantum Darwinism. This is balanced with examples and applications of the frameworks aimed at further understanding the dynamical nature of the emergence of objectivity.

1.1 Outline

While the core of this thesis is based on Quantum Darwinism, many of the following chapters intersect with other fields. Hence, in the majority of cases, the relevant pre-existing literature is given in the introduction prior to original research results in the following chapters.

Chapter 2 introduces terminology and notation for open quantum systems and has a brief review of decoherence as a precursor to Quantum Darwinism.

Chapter 3 lays out all the different theoretical frameworks of Quantum Darwinism, including a complementary framework we propose, *Strong Quantum Darwinism*. We also prove the theoretical connections between the different frameworks, and hence identify their differences. The chapter ends with the hierarchy of quantum Darwinism frameworks and a discussion of the different types of objectivity that each framework describes.

Chapter 4 numerically explores a dynamical system-environment model with non-trivial system-environment interactions. In this kind of situation, the discrepancies between the various frameworks of quantum Darwinism emerge, thus emphasising the importance in specifying the relevant form of objectivity.

Chapter 5 begins with a review on the pre-existing literature describing the emergence of objectivity and the factors that affect objectivity. We then characterise perfect objectivity-creating channels.

As first step towards experimental verification of quantum Darwinism, Chapter 6 introduces a witnessing scheme that can detect non-objectivity using a simplified version of strong quantum Darwinism. We propose a quantum photonic experiment and conduct various numerical simulations, ending with a discussion about the feasibility of the scheme.

Quantum biological systems can exist at the quantum-classical boundary: and some systems exhibit some quantum effects. In Chapter 7, we examine an avian-inspired magnetic sensor, based on the suspected mechanisms of how some birds are able to navigate the Earth's magnetic field. We pinpoint the form of quantum coherence necessary for the magnetic sensor, and find that it operates under a non-objective regime.

Chapter 8 concludes the thesis, including final outlooks on possible future research.

Decoherence: the precursor to objectivity

Quantum mechanics is widely accepted as governing the smallest things in our universe. With experimental and technological advancement, quantum mechanical effects are being observed in increasingly larger systems, having progressed from the first double-slit experiment involving tiny photons to quantum superposition experiments with large molecules [29, 30]. However, despite the advancements, we do not see quantum mechanical effects in very large macroscopic objects.

Physically, within the framework of quantum mechanics, decoherence is the dynamical mechanism that destroys microscopic quantum superpositions and leaves behind classical statistical mixtures. It relies on the observation that realistic systems interact with an environment—it is impossible for perfect isolation to occur. This environment interaction, over time, can cause the system to lose quantum coherence.

In Section 2.1, we introduce the basic formalism behind quantum dynamics and open quantum systems, which will be useful in the later chapters and also underlies the dynamical context of decoherence. In Section 2.2, we go into further detail about decoherence, including pointer states. In Section 2.3, we briefly examine the measurement problem.

For further information, in-depth reviews on decoherence include Refs. [10, 11, 31, 32].

2.1 Describing dynamics of quantum systems

For the interested reader, see Breuer and Petruccione [33] for comprehensive information regarding open quantum systems and their dynamics. In this section, we briefly state some descriptions of quantum dynamics.

The evolution of quantum systems can be described into two primary ways: one involving a continuous time evolution via differential equations, and one involving discrete transformations from initial state to final state.

Schrödinger's equation [34] governs how non-relativistic particles evolve over time,

$$i\hbar(\partial/\partial t)|\Psi\rangle = \hat{H}|\Psi\rangle, \quad (2.1)$$

where $|\Psi\rangle$ is a pure quantum state and $\hat{H}$ is the Hamiltonian that governs the dynamics.

In this thesis, we are more interested in mixed states. Consider a system $\mathcal{S}$ and an environment $\mathcal{E}$, with some combined initial state $\rho_{\mathcal{SE}}(0)$. For mixed density operators, the Liouville–von Neumann equation describes the evolution of the quantum state:

$$\frac{d}{dt}\rho_{\mathcal{SE}}(t) = -\frac{i}{\hbar}\left[\hat{H}_{total}(t), \rho_{\mathcal{SE}}(t)\right], \quad (2.2)$$

where $\hat{H}_{total}(t)$ is the total Hamiltonian of the system and environment. The formal solution of this gives rise to the unitary operator $U(t)$:

$$U(t) = \mathcal{T} \exp\left[-\frac{i}{\hbar} \int_0^t d\tau \hat{H}_{total}(\tau)\right], \quad (2.3)$$

where $\mathcal{T}$ denotes time-ordering:

$$\mathcal{T}(A(t)B(t')) = \begin{cases} A(t)B(t') & \text{for } t < t', \\ B(t')A(t) & \text{for } t' < t. \end{cases} \quad (2.4)$$

Under this closed quantum dynamics, the combined evolution of the system–environment is unitary:

$$\rho_{\mathcal{SE}}(t) = U(t)\rho_{\mathcal{SE}}(0)U^\dagger(t). \quad (2.5)$$

Very frequently though, the environment is too large to explicitly simulate or calculate—or sometimes, the state of the environment is not particularly important, and we only care about the reduced system evolution. At the same time,

realistic systems cannot be perfectly isolated from its environment—making it an *open quantum system*—meaning that although we do not know the environment, its existence can still affect the system. In these scenarios, there are a number of methods that will give us either the effective system channel, or the reduced system master equation.

If the system is initially uncorrelated with the environment, *i.e.* $\rho_{\mathcal{SE}}(0) = \rho_{\mathcal{S}}(0) \otimes \rho_{\mathcal{E}}(0)$, then the effective dynamics of the reduced system can be described with completely positive trace preserving (CPTP) maps

$$\Lambda(\rho_{\mathcal{S}}(0)) = \rho_{\mathcal{S}}(t) = \text{tr}_{\mathcal{E}}[U(t)(\rho_{\mathcal{S}}(0) \otimes \rho_{\mathcal{E}}(0))U^\dagger(t)], \quad (2.6)$$

which can be equivalently written in the operator-sum representation

$$\Lambda(\rho_{\mathcal{S}}) = \sum_k E_k \rho_{\mathcal{S}} E_k^\dagger, \quad (2.7)$$

where E_k are some Kraus operators satisfying $\sum_k E_k^\dagger E_k = \mathbb{1}$ for trace-preserving channels. If the completely positive map is merely trace-nonincreasing (*i.e.* a quantum operation), then $\sum_k E_k^\dagger E_k \leq \mathbb{1}$.

If there are initial correlations, then generalisations of CPTP maps, such as superchannels or supermaps, are able to capture this fact [35].

Beyond using discrete maps, the dynamics of open quantum systems can also be studied via master equations. Various different approximations are used to reduce the evolution of the full system-environment [Eq. (2.2)] to an equation that describes the effective evolution of the reduced system. Often, the following assumptions are made: weak system-environment interaction (also known as the Born approximation), spectrally flat environment, and the Markov approximation (which applies to environments in which the internal correlations inside the environment decay much faster than the time scale of the system evolution, and can be thought of as a “memory-less” assumption, or vanishing environmental correlation time). These assumptions/approximations lead to the Lindblad (or Lindblad-Gorini-Kossakowski-Sudarshan) master equation [36, 37],

$$\frac{d}{dt}\rho_{\mathcal{S}}(t) = -\frac{i}{\hbar}[\hat{H}, \rho_{\mathcal{S}}(t)] + \frac{1}{\hbar^2} \sum_k \Gamma_k \left(L_k \rho_{\mathcal{S}} L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho_{\mathcal{S}} \} \right), \quad (2.8)$$

where $\hat{H}$ is some Hamiltonian (not necessarily the system’s Hamiltonian $\hat{H}_{\mathcal{S}}$), $\{L_k\}_k$ are Lindblad operators, and Γ_k are the rates of the dissipation/decay.

$\{A, B\} = AB + BA$ is the anti-commutator. The Lindblad master equation describes Markovian dynamics. Systems of interest can be sufficiently isolated from the environment such that this description is suitable (especially when describing extremely large environments [33, 38]), from dissipative environment dynamics [39] to experiments involving atomic, molecular, and optical systems [40]. Furthermore, the Lindblad master equation can be derived from the repeated collision model, where multiple microscopic environments interact with the system over time, *e.g.* background gas particles [41, 42]. Specialised master equations also exist for specific scenarios [33].

However, the assumptions of Markovianity do not always hold in reality. Many experimental systems undergo non Markovian dynamics due to strong system-environment coupling [43, 44], initial system-environment correlations [45, 46] or finite-sized environments [47].

There are many different definitions for what constitutes Markovian evolution, and what constitutes non-Markovian evolution. None of these definitions are entirely equivalent, and there remains controversy on how to define Markovianity, and how to define non-Markovianity. There is a proliferation of different measures used in the literature [38, 48–50]. For example, Rivas *et al.* [51] define non-Markovianity as dynamics that are *not* CP-divisible: a quantum map Λ_t that defines dynamics over time t is CP-divisible if for all $t \geq s$, there exists completely positive trace-preserving (CPTP) maps $V_{t,s}$ such that

$$\Lambda_t = V_{t,s}\Lambda_s. \quad (2.9)$$

Time-dependent Lindblad master equations are all CP-divisible. Alternatively, Breuer *et al.* [49] define non-Markovianity as having information back-flow: a quantum map Λ_t has *no* information back-flow if

$$\frac{d}{dt}\|\Lambda_t(\rho_1 - \rho_2)\|_1 \leq 0 \quad (2.10)$$

for any pair of states ρ_1, ρ_2 , and where $\|\cdot\|_1$ is the trace-norm. CP-divisibility implies no information back-flow, but the reverse is not true [52]; Ref. [50] contains a detailed description of the hierarchy and connection of various measures of (non)-Markovianity.

To facilitate the study of non-Markovian dynamics, several theoretical and numerical tools have been developed. Projection operator techniques can be used

to give formal equations to the dynamics of the reduced system [33, 53, 54]. Non-Markovian master equations derived within a perturbative framework include the Hierarchical Equations of Motion [55], and exact solutions of system-environment dynamics can be done using the Time Evolving Density with Orthogonal Polynomial Algorithm [56, 57]. Generalised quantum maps such as superchannels [35, 58] can also be used. Another alternative are stochastic Schrodinger equations [59–61], and collisional models can also simulate non-Markovian dynamics [62], among many more methods [63].

An argument against studying non-Markovian systems *per se* is that if the system is strongly interacting with certain parts of the environment, then the “system” could just be redefined to include the parts of the environment it strongly interacts with, such that the new, larger system interacts in a Markovian manner with the remaining environment. However, studying non-Markovian systems is useful given the experimental and physical examples mentioned above, and when it is unfeasible to consider the relevant environment explicitly.

That said, in this thesis, we typically simulate the full system-environment unitarily, since we need to calculate system-environment correlations, and thus have explicit non-Markovian dynamics on the system.

2.2 Decoherence

Decoherence takes into account that quantum systems are constantly interacting with a surrounding environment [64]. This interaction carries away quantum coherence, causing the system to lose quantum properties. The particular details of the interaction singles out *which* basis the system decoheres into. These two effects are called environment-induced decoherence and environment-induced superselection respectively [10, 11, 64–72].

Environment-induced decoherence causes the system to lose quantum and classical correlations, with off-diagonal inference terms decaying, such that the system locally “collapses” into a preferred set of states. When a system interacts with its environment, transfers of information can occur. The state of the system has to be consistent with the transferred information in the environment. This can be thought as a partial measurement by the environment upon the system, and thus the environment is sometimes considered a witness of the system.

Environment-induced superselection, or *einselection*, describes how the interaction between the apparatus and the environment singles out a set of mutually commuting system observables, *i.e.* observables that are stable with respect to the interaction Hamiltonian. The preferred states of the system are then called *pointer states*. The system eventually decoheres into a mixture of pointer states, with its phase information lost.

Technically, even for decohered states, the entire system-environment still contains the original quantum coherence (as it is overall a closed system) [73]. However, the environment has *many* degrees of freedom, and recovering the coherence would require global measurements of the entire system and environment. This makes it extremely difficult to undo decoherence [74], and renders decoherence effectively irreversible [11].

A common source of decoherence that causes systems to localise in spatial position is the scattering of environment particles (such as light or air molecules) on the system: this process is known as collisional decoherence [41, 42]. Another ubiquitous force is gravity, and there has been work on how gravity can cause decoherence in position [75–77]. Experimentally, decoherence has been observed in many systems [78–81]. Note that since decoherence is a dynamical effect, whether or not a system decoheres can also depend on the (quantum) reference frame [82], as dynamics change in different frames [83]. Decoherence can also be studied in generalised probabilistic theories beyond quantum mechanics [84, 85].

Note that decoherence does not always occur. For example, if the environment measures multiple, incompatible, observables on the system, and this can lead to *quantum frustration*, which hinders decoherence altogether [67, 86, 87].

2.2.1 Pointer states

When the environment interacts with the system in some particular manner, the long-lasting state of the system becomes a so-called *pointer state*, referring to the pointer of some measurement apparatus. These pointer states form a basis. Ideally, it would depend only on the system-environment interaction, but in reality the final diagonal basis of the system also depends on its internal interactions.

If the interaction between system and environment ultimately dominates, then any observable that commutes with the interaction is a potential pointer observable. Given that there can be many possible pointer observables that

commute with the coupling interaction, Zurek suggests the *predictability sieve* (also known as stability criterion) [10, 68, 72, 88, 89], whereby different pointer observables can be ranked by some measure of “classicality” (which can include minimising entropy, or maximising purity), and the most classical pointer wins.

Let us decompose the total Hamiltonian $\hat{H}_{total} = \hat{H}_S + \hat{H}_E + \hat{H}_{SE}$ into the self (local) Hamiltonians $\hat{H}_S$ and $\hat{H}_E$ that act on the system and environment respectively, and the $\hat{H}_{SE}$ that describes the interaction between them. Depending on their relative strengths, there are three general classes of system-environment dynamics and subsequent pointer states [11, 70]:

1. The dynamics is dominated by the system-environment interactions $\hat{H}_{SE}$. The pointer states will be eigenstates of $\hat{H}_{SE}$ [68], which are typically position eigenstates [24]. Furthermore, if $\hat{H}_{total} \approx \hat{H}_{SE}$ (meaning that internal environment interactions are weak), then we are at the *quantum measurement limit* [11].
2. The dynamics is dominated by the system’s internal evolution $\hat{H}_S$. The pointer states will be the energy eigenstates of $\hat{H}_S$. This is called the *quantum limit of decoherence* [70].
3. The intermediary case, where neither $\hat{H}_{SE}$ nor $\hat{H}_S$ dominate. The pointer states will be a “compromise”, mixture, or combination between the different dynamics. For example, in quantum Brownian motion, the pointer states are localised in phase space (in both position and momentum) [11].

Because system-environment interactions tend to depend on distance or position, this type of information is most easy to proliferate, which neatly explains why spatially localised states survive whereas nonlocal superpositions decay [24]. Alternatively, if the energy states of the system is greater than that of the environment—and the system Hamiltonian dominates evolution—then the pointer states correspond to the system’s energy eigenstates [70]. For particular scenarios, the pointer states can be calculated: for example, harmonic-oscillator models in general have coherent pointer states [89, 90], and in quantum Brownian motion, the pointer states are Gaussian wave packets with particular properties [91]. In Ref. [92], scaling the strength of the system self-Hamiltonian relative to the interaction Hamiltonian leads to a sliding scale of pointer states ranging between eigenstates of the system Hamiltonian to eigenstates of the interaction

Hamiltonian. Pointer states have also been discovered in experimental systems, such as quantum dots [93, 94].

Pointer states form pointer subspaces [72], and under some conditions [25, 95], they can also coincide with *decoherence-free subspaces* [96], where the latter describes which states are immune to environment decoherence. Due to this immunity, there has been work to construct particular decoherence-free subspaces in order to encode quantum information using reservoir engineering [97]. Similarly, we could use reservoir engineering to construct particular pointer states [98].

2.3 The measurement problem

Closely related to the issue of decoherence of states is the issue of quantum measurement, which is axiomatically separate from Schrodinger’s equation of evolution. Whilst a quantum system may evolve quantum mechanically, when an observer makes a von Neumann measurement, the system suddenly becomes a concrete eigenstate of the measurement observable (and to a weaker sense in weak measurements [99, 100]). Under the quantum measurement postulate, for some state ρ and some observable A , the probability of obtaining outcome a_n associated with operator A_n is $\text{tr}[\rho A_n]$, and the post-measurement state is

$$\tilde{\rho} = \frac{A_n \rho A_n}{\text{tr}[\rho A_n]}. \quad (2.11)$$

This process is not unitary, unlike the unitary time evolution of Schrodinger’s equation. The measurement “collapses” states instantaneously, causing states to lose all other phase information. The quantum measurement problem pertains to this collapse, and how one can interpret such a collapse, if there is a collapse.

The measurement problem can be formulated as follows: consider the ideal von Neumann measurement. If the system is in some basis $\{|s_n\rangle\}_n$ and the measurement apparatus $\mathcal{A}$ has some microscopically distinguishable “pointer” positions $\{|a_n\rangle\}_n$ such that state $|a_n\rangle$ is taken if the system is in state $|s_n\rangle$, then a pre-measurement would take the combined initial system-apparatus state to:

$$\left(\sum_n c_n |s_n\rangle \right) |a_r\rangle \rightarrow \sum_n c_n |s_n\rangle |a_n\rangle, \quad (2.12)$$

where $|a_r\rangle$ is some initial state of the apparatus. Yet, now the measurement apparatus states are also in superposition—but in reality we would only measure

a single value “ n ”, not a superposition of values. Furthermore, the basis $\{|s_n\rangle\}_n$ is not unique. Which leads to the questions—why did we measure a definite outcome, rather than a superposition, and what exactly *did* we measure?

If we add in the environment to Eq. (2.12), then

$$\left(\sum_n c_n |s_n\rangle\right) |a_r\rangle |e_0\rangle \rightarrow \left(\sum_n c_n |s_n\rangle |a_n\rangle\right) |e_0\rangle \quad (2.13)$$

$$\rightarrow \sum_n c_n |s_n\rangle |a_n\rangle |e_n\rangle, \quad (2.14)$$

where $|e_n\rangle$ are the states of the environment associated with the different pointer states. Yet now the *environment* is also in superposition with the system and the apparatus, and the problem of the measurement has been delayed.

Through einselection, decoherence picks out the preferred basis of the system, $\{|s_n\rangle\}_n$. In many decoherence models [10, 25, 72], the environment states $\{|e_n\rangle\}_n$ are orthogonal, which means when we trace out the environment, we are left the decohered state

$$\sum_n |c_n|^2 |s_n\rangle\langle s_n| \otimes |a_n\rangle\langle a_n|. \quad (2.15)$$

However, the system is now in a statistical mixture of states—decoherence does not explain how the system projects into a single state. It cannot solve the quantum measurement problem [11, 25]. While decoherence can be thought of being due to the environment continuously monitoring the system, the process of decoherence is technically unitary, while the measurement collapse is non-unitary.

Rather, in order to solve the measurement problem, proposed solutions typically correspond to interpretations or modifications of quantum mechanics, *i.e.* going beyond unitary quantum evolution [15, 16, 18, 19, 101]. For the interested reader, Refs. [11, 31, 102] are reviews of some proposed solutions and progress on the measurement problem.

However, we *can* also use decoherence to help understand how information transfers from quantum system to environment. Quantum Darwinism is a framework that extends decoherence, adopting the concept of *objectivity* that is stronger than the classicality of decoherence. While quantum Darwinism does not solve the measurement problem, it does show how we can *perceive* information gained from measurements.

The frameworks of objectivity

This chapter includes content from the following publication:

Thao P. Le and Alexandra Olaya-Castro, *Strong Quantum Darwinism and Strong Independence are Equivalent to Spectrum Broadcast Structure*, Phys. Rev. Lett. 122, 010403 (2019) [2].

The original contributions in this chapter are: the construction of the two frameworks Strong Quantum Darwinism and Invariant Spectrum Broadcast Structure in Sec. 3.2, and the results of Section 3.3 and 3.4, the Corollary 2 and Theorem 3 in Sec. 3.5, and the subsequent hierarchy constructed in Sec. 3.6.

Quantum systems are constantly interacting with their surrounding environments: even in the laboratory, isolation cannot be maintained indefinitely. Decoherence theory describes how quantum systems lose their quantum coherence and decohere into a preferred pointer basis due to these external interactions [11, 24, 25].

Beyond being decoherent, macroscopic objects also have objective existences, and the information we can determine about such macroscopic objects is classical in the preferred basis. Quantum Darwinism builds upon decoherence theory, to describe how quantum systems become objective:

Definition 1. Objectivity [26, 28, 103]: A system state is *objective* if it is (1) simultaneously accessible to many observers (2) who can all determine the state independently without perturbing it and (3) all arrive at the same result.

A system state is objective when information about the state can be obtained and agreed upon by hypothetical observers. By recovering system state’s information “without perturbing the system”, we mean that the system state remains unchanged after “forgetting the results” of measurements on the system and observable environments.¹ Note, this definition of objectivity has also been called *inter-subjectivity* [104], *i.e.*, it describes a perception of objective reality relative to the collection of systems and environments. There are no *actual* measurements conducted within the framework and analysis of quantum Darwinism.

Quantum Darwinism emphasises how the environment is made up of multiple components. System-environment interactions can cause information about the system to spread into the environment. If this information about the system exists in multiple subenvironments, corresponding to multiple copies of the information, then we say that the system state is objective. We can consider the environment as “measuring” the system.

From decoherence theory, the basis in which the system decoheres into is defined by the pointer states. Information about the pointer states can be copied and proliferated into the environment, making them the “fittest” quantum states. The scenario of quantum Darwinism is illustrated in Fig. 1.1.

A paradigmatic example of the process of quantum Darwinism is the behaviour of the photon environment. As photons bounce off objects, they pick up information about the object—colour and shape and size *etc....* There are a *lot* of photons, leading to lots of copies of the same, or very similar, information. Intercepting only a small fraction of them is sufficient to gain information about the object, and many observers can independently measure their own collection of photons, and see the (approximately) same object.

Photons are in fact the ideal type of environment to consider, as they may scatter from objects and systems of interest, but tend to not interact with one another, hence maintaining the original information about the systems. In contrast, air molecules, for example, *do* interact with each other through collisions, thus delocalising the information they had initially obtained: an observer would have to measure more subenvironments to gain the original information [105].

Quantum Darwinism has been explored in many scenarios, including with

¹Formally, this is *Bohr non-disturbance* [17], which is the concept that a state is invariant after discarding/forgetting measurement results upon it. This was first introduced in the context of quantum Darwinism by Horodecki *et al.* [103]. In their paper, they take the *entire* joint system-environment state to be unchanged after discarding measurement results.

a photon environment [103, 106–108] and harmonic oscillators [109–115]. The environment can also consist of spins [27, 105, 116–121], or even internal degrees of freedom coupled to the gravitational field [76]. There are also works which consider more general finite-dimensional environments [104, 122–124] as well as infinite-dimensional environments [125, 126]. The emergence of quantum Darwinism will be further discussed in Chapter 5.

In Section 3.1, we will first introduce some notions and mathematical definitions. Section 3.2 defines all the frameworks of quantum Darwinism, including some original frameworks. Section 3.3 presents our main result connecting the major frameworks of quantum Darwinism, in particular with strong quantum Darwinism. Section 3.4 explores methods to measure strong quantum Darwinism. Section 3.5 presents the results relating the frameworks to the definition of objectivity. Finally, Section 3.6 presents the hierarchy of the different frameworks and their corresponding level of objectivity.

3.1 Notation and preliminaries

For the interested reader, see Nielsen and Chuang [127] for in-depth details surrounding quantum information topics. Here, we introduce the relevant measures of quantum information for this thesis.

We have a system $\mathcal{S}$ and numerous environments $\mathcal{E} = \{\mathcal{E}_k\}_k$. The full system-environment is described by the density operator $\rho_{\mathcal{SE}} = \rho_{\mathcal{SE}_1 \dots \mathcal{E}_N}$, if we have N total subenvironments. We will often be considering a system-*fragment* $\rho_{\mathcal{SF}} = \text{tr}_{\mathcal{E} \setminus \mathcal{F}}[\rho_{\mathcal{SE}}]$, where $\mathcal{F} \subseteq \mathcal{E}$ denotes a subset of the environment and where $\text{tr}_{\mathcal{E} \setminus \mathcal{F}}$ denotes the trace over all of the environment excluding the fragment. The Hilbert space of a quantum system X is $\mathcal{H}_X$. Quantum channels are described as completely positive trace-preserving (CPTP) maps, $\Phi : \text{BL}(\mathcal{H}^{\text{in}}) \rightarrow \text{BL}(\mathcal{H}^{\text{out}})$, where $\text{BL}(\mathcal{H})$ denote all the bounded linear operators on $\mathcal{H}$ [127], where a density state $\rho \in \text{BL}(\mathcal{H}_{\mathcal{S}})$ satisfies $\rho = \rho^\dagger$, $\rho \geq 0$ and $\text{tr}[\rho] = 1$.

The von Neumann entropy of some state ρ is

$$H(\rho) = -\text{tr} \rho \log \rho = -\sum p_i \log p_i, \quad (3.1)$$

where $\{p_i\}$ are the eigenvalues of the state ρ . The von Neumann entropy describes the amount of information encoded in the state; alternatively, the von Neumann

entropy describes the uncertainty about the state (and thus the information that we need to know to *not* be uncertain) [127].

The quantum mutual information between two systems A, B with joint state ρ_{AB} is

$$I(A : B) = H(\rho_A) + H(\rho_B) - H(\rho_{AB}) = H(A) + H(B) - H(AB), \quad (3.2)$$

where reduced states are $\rho_A = \text{tr}_B[\rho_{AB}]$ and $\rho_B = \text{tr}_A[\rho_{AB}]$. The mutual information measures how much information A and B have in common. The conditional entropy of A given B describes our uncertainty in A given that we know B :

$$H(A|B) = H(AB) - H(B) = \sum_b p_b H(\rho_{A|b}), \quad (3.3)$$

where $\{p_b\}$ are the eigenvalues of ρ_B and $\rho_{A|b}$ is the reduced state on A given that B is in the eigenstate labeled by b .

For a joint state ρ_{ABC} , the quantum mutual information between A and B , conditioned on C , is given by the conditional mutual information:

$$I(A : B|C) = H(A|C) + H(B|C) - H(AB|C). \quad (3.4)$$

The quantum discord, which describes all the genuinely quantum correlations, is defined as follows [128, 129]:

$$\mathcal{D}(A : B) = \min_{\{\Pi_{A|i}\}_i} \left[\sum_i p_i H(\rho_{B|i}) + H(\rho_A) - H(\rho_{AB}) \right], \quad (3.5)$$

where $\rho_{B|i}$ is the conditional state on B after measurement result i on A using the von Neumann measurement $\{\Pi_{A|i}\}_i$ ²:

$$\rho_{B|i} = \frac{\text{tr}_A \left[\Pi_{A|i} \rho_{AB} (\Pi_{A|i})^\dagger \right]}{p_i}, \quad \text{and} \quad p_i = \text{tr} \left[\Pi_{A|i} \rho_{AB} (\Pi_{A|i})^\dagger \right]. \quad (3.6)$$

The classical Holevo information can then be defined as the difference between the quantum mutual information and the quantum discord [130]:

$$\chi(A : B) = I(A : B) - \mathcal{D}(A : B) = H(B) - \sum_i p_i H(B|A = i). \quad (3.7)$$

Note that quantum discord, and hence the classical Holevo information, are asymmetric: here, the 'preferred' system is A , and we have implicitly allocated the first

²The measurement can be generalised to positive-operator valued measures (POVMs).

argument in $\mathcal{D}$ and χ to mark the preferred system: *i.e.*, zero discord states (from the perspective of A) are classical-quantum states with form:

$$\chi_{AB}^{(A)} = \sum_i p_i |i\rangle\langle i|_A \otimes \rho_{B|i}, \quad (3.8)$$

where $\{p_i\}$ are the eigenvalues of the state on A , $\{|i\rangle\}$ is an orthonormal basis of A and $\rho_{B|i}$ are states of B . If B was the preferred/classical system, states with zero discord from B 's perspective are quantum-classical states:

$$\chi_{AB}^{(B)} = \sum_j q_j \rho_{A|j} \otimes |j\rangle\langle j|, \quad (3.9)$$

where $\{q_j\}$ are the eigenvalues of B , $\{|j\rangle\}$ its orthonormal basis, and $\rho_{A|j}$ are some states on A . We can think of discordant states are those that cannot be determined (measured) without perturbation with local operations [131].

3.2 The mathematical frameworks of quantum Darwinism

In this section, we introduce four mathematical frameworks that quantitatively describe objectivity within the scenario of quantum Darwinism. In order of increasingly stringent conditions, they are: Zurek's quantum Darwinism [27, 28, 124], strong quantum Darwinism [2], spectrum broadcast structure [103], and lastly invariant spectrum broadcast structure. Historically, Zurek's quantum Darwinism was first described, followed by spectrum broadcast structure. Meanwhile, strong quantum Darwinism and invariant spectrum broadcast structure were described after, as these are our original contributions.

Note that in most cases, the accessible environment considered is *not* the full environment. We call the accessible environment simply the “environment” in the rest of thesis unless otherwise marked.

3.2.1 Zurek's quantum Darwinism

Definition 2. Zurek's quantum Darwinism (ZQD) [27, 28, 124]. A system-environment satisfies Zurek's quantum Darwinism when the quantum mutual information between system $\mathcal{S}$ and subenvironment $\mathcal{E}_k$ is equal to the information

contained in the system, given by the von Neumann entropy:

$$I(\mathcal{S} : \mathcal{E}_k) = H(\mathcal{S}). \quad (3.10)$$

For objectivity to emerge, Eq. (3.10) should hold for sufficiently many environments.³ Pictorially, this leads to plots such as the one in Fig. 3.1, where states with quantum Darwinism require only a small fraction $\mathcal{F}_\delta$ of the environment to recover sufficient system-information, $I(\mathcal{S} : \mathcal{F}_\delta) = (1 - \delta)H(\mathcal{S})$. In contrast, non-objective systems require half of the environment to recover the system information. With the full environment, the information rises to $2H(\mathcal{S})$, due to the overall purity of the system-full-environment state.

The number of copies of distinct system information can be estimated by the *redundancy*, where the larger the redundancy the more objective the system should be:

Definition 3. Redundancy [28]. Let f_δ be the smallest fraction of the environment for which $I(\mathcal{S} : \mathcal{F}_\delta) = (1 - \delta)H(\mathcal{S})$, where $|\mathcal{F}_\delta| = f_\delta|\mathcal{E}|$. The redundancy of the information is

$$R_\delta = \frac{1}{f_\delta}. \quad (3.11)$$

Redundancy (and therefore quantum Darwinism) would occur if $R_\delta \geq 2$, *i.e.* if at least two observers or more could view the information about the system independently. A large $R_\delta \gg 1$ is a sufficient condition for effective classicality in the sense of Zurek’s quantum Darwinism according to Ref. [65]. When an observer measures the environment, *e.g.* photons, that information about the system that they gain can be erased from environment. But due to redundancy, the information still exists in $\approx R_\delta - 1$ other fragments. By having consensus about the information among the copies suggests that the state of the system $\mathcal{S}$ exists objectively. The no-cloning theorem prohibits the copying of unknown quantum states, which appears to contradict redundancy. However, in this case, only one (technically) known quantum state is being copied—eigenstates of the pointer observable, which are orthogonal and hence do not violate the no-cloning theorem.

³This leads to the question how many environments is “sufficiently many”. Ref. [132] examined the size of the environment required for decoherence, and found that the small environments can be sufficient if they have suitable dynamics. However, the question of how many copies of information are required for a system to be “objective” is more ambiguous.

Note that redundancy as defined in Definition 3 is only unambiguous if the environment consists of symmetric and identical subsystems. However, if the coupling between system and environment subsystems varies, and different environment subsystems take on different states, then certain fragments can contain more or less information about the system and the redundancy of Definition 3 becomes inaccurate. Furthermore, there will be parts of the environment with no or very little information about the system, which would *dilute* the system information within a group containing both “informative” and “non-informative” environments. In this case, we can either define redundancy as the number of fragments that will, each independently, provide $(1 - \delta)H(\mathcal{S})$ of information. We can also take the *average*: that is,

$$\langle I(\mathcal{S} : \mathcal{F}) \rangle_{f_\delta N} = (1 - \delta)H(\mathcal{S}) \quad (3.12)$$

where the average is taken over fragments of size $f_\delta N$ (for a total of N subsystems in the environment). Zwolak and Zurek [121] prove that while there *is* a dilution of the information, the redundancy of information remains similar regardless of whether or not we include the “non-informative” environments, in models where the system evolution commutes with the interaction Hamiltonian.

3.2.2 Strong quantum Darwinism

However, Zurek’s quantum Darwinism is insufficient giving complete predictions of objectivity, because the quantum mutual information is composed of both classical *and* quantum information, and we cannot assume that the information is entirely classical in general. Furthermore, as we will explicitly explore in Chapter 4, there exists system-environments with quantum discord that satisfy Zurek’s quantum Darwinism, leading to a false objectivity—this quantum discord “hides” within the quantum mutual information in Eq. (3.10). Quantum discord can be thought of as *measurement-based disturbance* [131]: non-zero discord breaks the non-disturbance condition of objectivity (Def. 1).

We propose a new framework, strong quantum Darwinism, which avoids this problem within Zurek’s quantum Darwinism by explicitly requiring vanishing quantum discord. Our result is published in the paper *Le and Olaya-Castro* [2]. This is one of the key contributions of this thesis, and we will continue to explore strong quantum Darwinism in the remainder of the thesis:

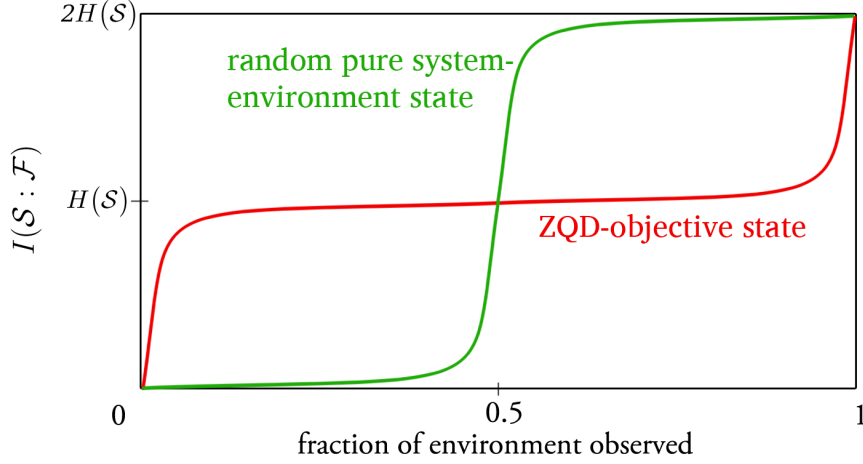


Figure 3.1: **Zurek's quantum Darwinism: depiction of a typical mutual information plot $I(\mathcal{S} : \mathcal{F})$.** Information about the system $\mathcal{S}$ stored in various fractions $\mathcal{F}$ of the environment $\mathcal{E}$. In the *ZQD-objective state* line, we have the states that emerged after decoherence and satisfy Zurek's quantum Darwinism: these require only a small fraction of the environment to obtain close to $H(\mathcal{S})$ system information. The *random pure system-environment state* line depicts the mutual information plot for random pure $\mathcal{SE}$ states, which, in general, do not have any redundancy and require a significant fraction of the environment to recover system information.

Definition 4. Strong quantum Darwinism (SQD). A system-environment satisfies strong quantum Darwinism when the quantum discord is zero, and quantum mutual information is fully classical and equal to the information contained in the system:

$$I(\mathcal{S} : \mathcal{E}_k) = I_{acc}(\mathcal{S} : \mathcal{E}_k) = \chi(\mathcal{S} : \mathcal{E}_k) = H(\mathcal{S}), \quad \mathcal{D}(\mathcal{S} : \mathcal{E}_k) = 0, \quad (3.13)$$

where $I_{acc}(\mathcal{S} : \mathcal{E}_k)$ is the classical accessible information which here is equivalent to the Holevo quantity $\chi(\mathcal{S} : \mathcal{E}_k)$.

For objectivity to emerge, Eq. (3.13) should hold for sufficiently many environments $\{\mathcal{E}_k\}_{k=1,\dots,F}$, as well as for the joint observed environment $\cup_{k=1}^F \mathcal{E}_k$. Note the observed environment is never the total environment. The full environment, (provided that the total system-environment is closed and pure), retains all quantum correlations and so will always have $I(\mathcal{S} : \mathcal{E}) = 2H(\mathcal{S})$.

The quantum discord is a measure of the quantum correlations contained between two systems, and it captures quantum correlations more general than

entanglement (*e.g.* quantum correlations in separable quantum states) [128, 129]. Discord encodes information that is only accessible via coherence quantum interactions and therefore is a quantum resource.

Thus the zero quantum discord condition ensures that there is no quantum correlations between the system and accessible environment. The quantum discord is also asymmetric: $\mathcal{D}(\mathcal{S} : \mathcal{E}_k) \neq \mathcal{D}(\mathcal{E}_k : \mathcal{S})$ [see its definition in Eq. (3.5)]. The statement $\mathcal{D}(\mathcal{S} : \mathcal{E}_k) = 0$ means that the *system* has a classical state structure, which is consistent with a decohered system state—a minimum requirement for an objective state. Meanwhile the environment has a quantum state structure, as strong quantum Darwinism is not focused on the objectivity of the environment states.

We can define an analogous concept of redundancy for strong quantum Darwinism using the classical information: suppose there are fragments $\mathcal{F}_\delta$ with size $|\mathcal{F}_\delta| = f_\delta |\mathcal{E}|$ that contain classical information that is approximately equal to the information about the system

$$I(\mathcal{S} : \mathcal{F}) \approx \chi(\mathcal{S} : \mathcal{F}) \geq (1 - \delta)H(\mathcal{S}). \quad (3.14)$$

The redundancy R_δ is the number of unique copies of that information, *i.e.*, the number of disjoint fragments $\mathcal{F}_{\delta,i}$ (where i indexes different fragments) that contain that approximate information, $R_\delta = |\mathfrak{F}_\delta|$, where $\mathfrak{F}_\delta$ describes the set of environments that contain the system information:

$$\mathfrak{F}_\delta = \{\mathcal{F}_{\delta,i} | \chi(\mathcal{S} : \mathcal{F}_{\delta,i}) \geq (1 - \delta)H(\mathcal{S}), \mathcal{F}_{\delta,i} \cap \mathcal{F}_{\delta,j} = \emptyset \forall i \neq j\}. \quad (3.15)$$

This is bounded by the minimum fraction size $f_{\delta,min}$: $R_\delta \leq 1/f_{\delta,min}$. If $\chi(\mathcal{S}^\Pi : \mathcal{F}) \geq (1 - \delta)H(\mathcal{S}^\Pi)$, then the discord is bounded by $\mathcal{D}(\mathcal{S} : \mathcal{F}) \leq \delta H(\mathcal{S})$. This is provided that the complete environment is pure, and the fragment has size $|\mathcal{F}| < |\mathcal{E}|/2$.

A rapid rise of the classical information $\chi(\mathcal{S} : \mathcal{F})$, as depicted in Fig. 3.2, implies that only a small fraction f_δ of the environment is required to have access to all the information in the system and hence suggesting that there is a large redundancy $R_\delta \approx 1/f_\delta$; this occurs for post-decohered system-environment states with strong quantum Darwinism. In contrast, Haar-random pure system-environment states⁴ will tend to have a mixture of classical and quantum correlations between

⁴A Haar-random pure state $|\psi_{\mathcal{SE}}\rangle = U|\psi_0\rangle$ can be realised using some reference state $|\psi_0\rangle$ and a random unitary matrix U drawn with a uniform probability from the set of all unitary matrices on the system and environment.

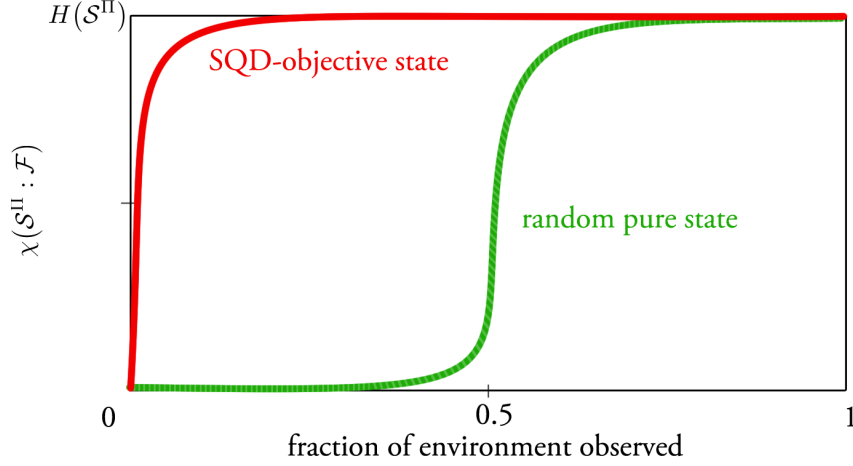


Figure 3.2: **Strong quantum Darwinism:** Classical information given by the Holevo information $\chi(\mathcal{S}^{\text{II}} : \mathcal{F})$ about the system stored in different fractions of the environment. *SQD-objective states:* if we have $\chi(\mathcal{S}^{\text{II}} : \mathcal{F}) \approx H(\mathcal{S})$ for small fractions of the environment, then the state has SQD-objectivity: one example is the reduced GHZ state. *Random pure states:* For states picked out at random, the amount of accessible classical information and quantum discord are roughly the same.

any system and fragment, and a fairly large fraction would be required to access any substantial amount of information about the system. We also give a distance measure of strong quantum Darwinism in Section 3.4.

3.2.3 Spectrum broadcast structure

Meanwhile, in the framework of spectrum broadcasting, the system state becomes objective when their *spectrum* is broadcasted into the environment. This spectrum describes the information of the system state, and by broadcasting (encoding) it into the environment, hypothetical observers can then measure the environment in order to learn about the system. State structure, rather than the entropic quantities, is used to describe objective states:

Definition 5. Spectrum broadcast structure (SBS) [103]. A system-environment has spectrum broadcast structure when the joint state can be written as

$$\rho_{\mathcal{SE}} = \sum_i p_i |i\rangle\langle i|_{\mathcal{S}} \otimes \rho_{\mathcal{E}_1|i} \otimes \cdots \otimes \rho_{\mathcal{E}_N|i}, \quad (3.16)$$

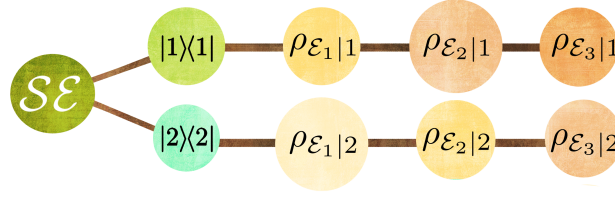


Figure 3.3: **Depiction of spectrum broadcast structure.** Conditioned on the system state $|i\rangle\langle i|$, the subenvironments $\mathcal{E}_k$ take on conditional states $\rho_{\mathcal{E}_k|i}$. These conditional states must be perfectly distinguishable: $\rho_{\mathcal{E}_k|1}\rho_{\mathcal{E}_k|2} = 0$ for all $k = 1, 2, 3$. This allows an observer with access to $\mathcal{E}_k$ to construct a general quantum measurement to perfectly determine index i .

where $\{|i\rangle\}$ is the pointer basis, p_i are probabilities, and all states $\rho_{\mathcal{E}_k|i}$ are perfectly distinguishable: $\rho_{\mathcal{E}_k|i}\rho_{\mathcal{E}_k|j} = 0 \forall i \neq j$, for each observed environment $\mathcal{E}_k$.

The spectrum broadcast structure is in fact one form of a classical-classical state, which have form $\sum_{ij} p_{ij} |i\rangle\langle i| \otimes |j\rangle\langle j|$, and it is a weaker form of quantum state broadcasting [133]. We depict a state with spectrum broadcast structure in Fig. 3.3. The states of strong quantum Darwinism and spectrum broadcast structure both have zero discord.

States with spectrum broadcast structure also satisfy strong independence, where there are no correlations between the environments conditioned on the information of the system:

Definition 6. Strong independence [103]. Subenvironments $\{\mathcal{E}_k\}_k$ have strong independence relative to the system $\mathcal{S}$ if their conditional mutual information is vanishing:

$$I(\mathcal{E}_j : \mathcal{E}_k | \mathcal{S}) = 0, \quad \forall j \neq k. \quad (3.17)$$

Note that “strong independence” is that there should be no correlations between the environments *conditioned* on the information about the system. That is, the only correlations between the environments is the common information about the system.

Strong independence is realistic for photon environments since photons rarely interact with each other. However, strong independence does not hold in general. For example, consider information spread via sound: air molecules in the environment are constantly interacting with each other, yet we are still able to

gain audio information from those air molecules. Hence, we see the first hint that spectrum broadcast structure is overly stringent in defining objective states.

The strong independence requirement is due to Horodecki *et al.* [103] explicitly using Bohr's notion of non-disturbance [17, 134], where local measurements are nondisturbing if they leave the *whole joint system-environment* state invariant after forgetting the results. That is, if $\rho_{\mathcal{SE}|z}$ is the conditional state after obtaining result z , then $\sum_z p(z) \rho_{\mathcal{SE}|z} = \rho_{\mathcal{SE}}$. In contrast, in strong quantum Darwinism, the *system* is undisturbed after forgetting conditional measurement results, but the environments may have changed.

Due to the state structure form, a distance measure can be used to determine whether a state has approximate spectrum broadcast structure. Mironowicz *et al.* [104] have developed a computable tight bound $\eta[\rho_{\mathcal{SF}}]$ on the trace distance. For a particular basis $\{|i\rangle\}$ on the system, we can write the system state as $\rho_{\mathcal{S}} = \sum_i p_i |i\rangle\langle i| + \sum_{i \neq j} p_{ij} |i\rangle\langle j|$. Then, the bound is:

$$T^{\text{SBS}}(\rho_{\mathcal{SF}}) = \frac{1}{2} \min_{\rho_{\mathcal{SF}}^{\text{SBS}}} \|\rho_{\mathcal{SF}} - \rho_{\mathcal{SF}}^{\text{SBS}}\|_1 \leq \eta[\rho_{\mathcal{SF}}] \quad (3.18)$$

$$\eta[\rho_{\mathcal{SF}}] \equiv \Gamma + \sum_{i \neq j} \sqrt{p_i p_j} \sum_{k=1}^F B(\rho_{\mathcal{E}_k|i}, \rho_{\mathcal{E}_k|j}), \quad (3.19)$$

where Γ is a decoherence factor that encodes the coherent terms $\{p_{ij}\}_{i \neq j}$, $B(\rho_i, \rho_j) = \|\sqrt{\rho_i} \sqrt{\rho_j}\|_1$ is the fidelity describing the distinguishability of the conditional environment states. Without the use of the computable bound $\eta[\rho_{\mathcal{SF}}]$, calculating the distance to the set of spectrum broadcasting states would require optimisation over both the system and all the subenvironments bases.

3.2.4 Invariant spectrum broadcast structure

Now consider the logical extreme of the frameworks: if spectrum broadcast structure holds from the perspective of all systems, then we have *invariant broadcast structure*. We have published this framework in the appendix of *Le and Olaya-Castro* [4].

Definition 7. Invariant Spectrum Broadcast Structure (ISBS). A collection of systems has invariant spectrum broadcast structure if it can be written in the following form:

$$\rho_{A_1 A_2 \dots A_N} = \sum_i p_i |i\rangle\langle i|_{A_1} \otimes \dots \otimes |i\rangle\langle i|_{A_N}, \quad (3.20)$$

where $\{|i\rangle_{A_k}\}_i$ are various orthonormal states on A_k . They are in general not the same states across each A_k .

Basically, invariant SBS is a variant of SBS where the conditional environment states are also pure.

Note that $\{|i\rangle_{A_k}\}_i$ for different subsystems A_k do not have to be the same—we have just *labelled* the orthogonal pure states the same. The state structure in Eq. (3.20) is a subset of classical-classical states [133], and have zero discord regardless of preferred system. The requirement of perfect classical correlations requires that the *relevant* dimension of all the subsystems are equal in general. Note that it is possible to have invariant spectrum broadcast structure for specific states that do not have equal dimension, so long as the state does not explore all degrees of freedom.

The states of invariant spectrum broadcast structure are highly reminiscent of incoherent states, albeit with perfect classical correlations, and no preferred basis in general. This lack of preferred/fixed basis means that ISBS objective states form a *nonconvex* set, and “classical mixing”, for example the convex combination of two objective states, is no longer objective in general.

Quantum Darwinism, strong quantum Darwinism, and spectrum broadcast structure are (primarily) focused on the objectivity of a particular preferred system as it interacts with the environment. Alternatively, we can take the subenvironments as the “protagonist”. From the perspective of a subenvironment, it can interact with the system and with other subenvironments. If all the subenvironments become objective—which is equivalent to having no conceptual preference between “system” and “environment”—then we say the systems have *collective* objectivity:

Definition 8. Collective objectivity. A collection of systems is *collectively objective* if each system has objectivity (Def. 1), *i.e.*, there is a state property that is simultaneously and independently accessible from every other system, such that measurements on the other systems are non-disturbing and measurement outcomes are the same across all systems.

In Theorem 3, we will prove the relationship between collective objectivity and invariant spectrum broadcast structure.

3.2.5 Examples of objectivity

Let us consider an illustrative example of objectivity in all frameworks—the GHZ state:

Example 1. Consider a system with initial state $a|0\rangle_{\mathcal{S}} + b|1\rangle_{\mathcal{S}}$ and an initially uncorrelated bath of two-level systems initially in the ground state $|0\rangle$. At each time step t_k , a bath subsystem $\mathcal{E}_k$ interacts with the system via a CNOT interaction: $\hat{U}_{CNOT,k}|i\rangle_{\mathcal{S}}|j\rangle_{\mathcal{E}_k} = |i\rangle_{\mathcal{S}}|j \oplus i\rangle_{\mathcal{E}_k}$ [116, 119]. After N time steps, the system-environment state is

$$|\Psi_{\mathcal{SE}}\rangle = (a|00\cdots 0\rangle_{\mathcal{SE}_1\cdots\mathcal{E}_N} + b|11\cdots 1\rangle_{\mathcal{SE}_1\cdots\mathcal{E}_N}) \otimes |0\rangle_{\text{rest of bath}}. \quad (3.21)$$

Some of the environment subsystems are unobservable or escape the observers' grasp. If only $N' < N$ environment subsystems are accessible, then the effective observable state is:

$$\rho_{\mathcal{SE}_{\text{observable}}} = |a|^2|0\cdots 0\rangle\langle 0\cdots 0|_{\mathcal{SE}_1\cdots\mathcal{E}_{N'}} + |b|^2|1\cdots 1\rangle\langle 1\cdots 1|_{\mathcal{SE}_1\cdots\mathcal{E}_{N'}}. \quad (3.22)$$

This state satisfies all frameworks of quantum Darwinism: has (invariant) spectrum broadcast structure with probabilities $p_0 = |a|^2$ and $p_1 = |b|^2$, the quantum mutual information condition is fulfilled $I(\mathcal{S} : \mathcal{E}_k) = H(\mathcal{S}) = H(|a|^2, 1 - |a|^2)$ and the state has zero discord and hence satisfies strong and Zurek's quantum Darwinism.

Example 2. A quantum-correlation breaking channel on the system combined with state broadcasting/preparation to the environment can produce SBS objectivity based on the original system information [111, 135]:

$$\Lambda_{\mathcal{S} \rightarrow \mathcal{F}}(\rho_{\mathcal{S}}(0)) = \sum_i \langle i | \rho_{\mathcal{S}}(0) | i \rangle \rho_{\mathcal{E}_1|i} \otimes \cdots \otimes \rho_{\mathcal{E}_F|i}, \quad (3.23)$$

where $\{|i\rangle\}_i$ are orthogonal and $\{\rho_{\mathcal{E}_k|i}\}_i$ are mutually orthogonal. This channel measures the system and conditionally re-prepares the subenvironments into distinguishable states.

In the following sections, we will analytically clarify the relationships between these four frameworks.

3.3 Strong quantum Darwinism and system-objectivity

With all the different mathematical frameworks of quantum Darwinism, it is not immediately clear which one to choose, nor how they relate to each other. Zurek’s quantum Darwinism and strong quantum Darwinism use entropic conditions—they focus on the *information*. In contrast, spectrum broadcast structure and invariant SBS look at geometric state structure.

These frameworks do not align and the differences are not immediately clear when we move from entropic conditions to state structure. In the next chapter, we will explore the differences between the frameworks on a particular system-environment model. But firstly, here, we show the precise mathematical relationships between the frameworks.⁵ Subsequently, we prove that *strong quantum Darwinism* gives the minimum conditions for system objectivity.

The relationship between Zurek’s quantum Darwinism and strong quantum Darwinism is clear as both conditions are posed in the same mathematical terms: simply add, or remove, the zero-quantum-discord condition. This means that strong quantum Darwinism is more stringent, and implies that states that satisfy Zurek’s quantum Darwinism can retain quantum correlations with the observed environments.

Furthermore, Zurek’s quantum Darwinism can falsely signal objectivity in situations with nonzero quantum discord and hence is insufficient to determine whether or not a state is objective [1, 103, 136], which will discuss in Chapter 4.

The relationship between strong quantum Darwinism and spectrum broadcast structure is not immediate, as the former uses entropic quantities and the latter employs state structure. Our key result of this section is the following theorem that describes the missing link between the two frameworks, which is also published in our paper [2]:

Theorem 1. *A state $\rho_{S\mathcal{F}}$ has spectrum broadcast structure if and only if it satisfies strong quantum Darwinism and has strong independence.*

The full proof is given in Appendix A.1, and in our paper Ref. [2]. The proof involves algebraic manipulations to translate entropic quantities to state structure and vice versa, but we will outline the key steps here:

⁵Historically, the work in the next Chapter 4 was done *before* the work in this chapter.

($\implies$) Starting with a general state with spectrum broadcast structure, we can directly calculate the quantum mutual information, the Holevo quantity, and the system entropy, and find that they satisfy strong quantum Darwinism. Furthermore, spectrum broadcast structure automatically satisfies strong independence.

($\impliedby$) The strong quantum Darwinism condition, $I_{acc}(\mathcal{S} : \mathcal{F}) = \chi(\mathcal{S}^\Pi : \mathcal{F})$, implies commuting conditional fragment states [137]. Combined with the condition $\chi(\mathcal{S}^\Pi : \mathcal{F}) = H(\mathcal{S})$ (which is also known as surplus decoherence [130] and is related to local broadcasting of classical correlations [133]), we can bring the system-fragment state to the following form:

$$\rho_{\mathcal{S}\mathcal{F}} = \sum_i p_i |i\rangle\langle i| \otimes \rho_{\mathcal{F}|i}, \quad (3.24)$$

where $\rho_{\mathcal{F}|i}$ are mutually distinguishable. We say the state in Eq. (3.24) has *bipartite spectrum broadcast structure*. The addition of strong independence breaks the conditional state on $\mathcal{F}$ into the tensor products over the subenvironments as given in Eq. (3.16).

If we consider the proof as up to Eq. (3.24) (*i.e.* not including the strong independence condition), we have the following corollary:

Corollary 1. *Strong quantum Darwinism is equivalent to bipartite spectrum broadcast structure.*

This corollary allows us to visualise strong quantum Darwinistic states. Unlike spectrum broadcast structure, a joint system-environment can have strong quantum Darwinism when the *environments* are correlated and entangled, for example,

$$\begin{aligned} \rho_{\mathcal{S}\mathcal{E}_1 \dots \mathcal{E}_N} = & p_1 |1\rangle\langle 1|_{\mathcal{S}} \otimes \frac{1}{2} (|1 \dots 1\rangle + |2 \dots 2\rangle)(\langle 1 \dots 1| + \langle 2 \dots 2|)_{\mathcal{E}_1 \dots \mathcal{E}_N} \\ & + p_2 |2\rangle\langle 2|_{\mathcal{S}} \otimes \frac{1}{2} (|3 \dots 3\rangle + |4 \dots 4\rangle)(\langle 3 \dots 3| + \langle 4 \dots 4|)_{\mathcal{E}_1 \dots \mathcal{E}_N}. \end{aligned} \quad (3.25)$$

In this example, the environment has a higher dimension than the system. Larger-dimensional environments are standard within Zurek's quantum Darwinism, strong quantum Darwinism and spectrum broadcast structure. If we enforce that the environments have the same dimension as the system, then we reduce to invariant spectrum broadcast structure:

Proposition 1. *Consider a joint system-fragment $\mathcal{SE}_1 \cdots \mathcal{E}_F$. If the dimensions of all subsystems are the same, i.e. $d_S = d_{\mathcal{E}_k} = d$, then the conditions of strong quantum Darwinism and spectrum broadcast structure reduces to invariant spectrum broadcast structure.*

The following proof is also published in the appendix of our paper [4].

Proof. We will prove that strong quantum Darwinism reduces to invariant spectrum broadcast structure when all the dimensions are the same, $d_S = d_{\mathcal{E}_k} = d$. Since spectrum broadcast structure is contained within strong quantum Darwinism, spectrum broadcast structure will then automatically reduce to invariant spectrum broadcast structure.

From Corollary 1, strong quantum Darwinism has bipartite spectrum broadcast structure, and furthermore the *reduced* system-single-environment states also have bipartite spectrum broadcast structure. Thus, for any single environment $k \in \{1, \dots, F\}$ that forms an objective state with the system, the reduced system-subenvironment state can be written as:

$$\rho_{S\mathcal{E}_k} = \sum_{i=1}^d p_i |i\rangle\langle i| \otimes \rho_{\mathcal{E}_k|i}, \quad \rho_{\mathcal{E}_k|i} \rho_{\mathcal{E}_k|j} = 0, \quad \forall i \neq j. \quad (3.26)$$

In a space of dimension $d_{\mathcal{E}_k}$, all sets of mutually orthogonal states contain at most $d_{\mathcal{E}_k}$ state vectors $\{|\psi_i\rangle\}_i$. Since there must be $d_S = d_{\mathcal{E}_k} = d$ different conditional states $\{\rho_{\mathcal{E}_k|i}\}_i$, this implies that they must be equivalent to one of those sets of mutually orthogonal states, which means that they are pure, i.e. $\rho_{\mathcal{E}_k|i} = |\psi_i\rangle\langle\psi_i|_{\mathcal{E}_k}$. Hence the local state on $\mathcal{SE}_k$ has invariant spectrum broadcast structure:

$$\rho_{S\mathcal{E}_k} = \sum_i p_i |i\rangle\langle i| \otimes |\psi_i\rangle\langle\psi_i|_{\mathcal{E}_k}, \quad \forall k \in \{1, \dots, F\}. \quad (3.27)$$

Now the combined state of the system-fragment also has bipartite spectrum broadcast structure:

$$\rho_{S\mathcal{F}} = \sum_i p_i |i\rangle\langle i| \otimes \rho_{\mathcal{F}|i}, \quad \rho_{\mathcal{F}|i} \rho_{\mathcal{F}|j} = 0, \quad \forall i \neq j. \quad (3.28)$$

We have that

$$\text{tr}_{\mathcal{F} \setminus \mathcal{E}_k} [\rho_{\mathcal{F}|i}] = \rho_{\mathcal{E}_k|i} = |\psi_i\rangle\langle\psi_i|_{\mathcal{E}_k}. \quad (3.29)$$

Since the reduced state is pure, $\rho_{\mathcal{E}_k|i}$ is not correlated with the other sub-environments in $\rho_{\mathcal{F}|i}$. Hence, the conditional fragment can be broken up into the tensor product

between $|\psi_i\rangle\langle\psi_i|_{\mathcal{E}_k}$ and the remainder:

$$\rho_{\mathcal{F}|i} = |\psi_i\rangle\langle\psi_i|_{\mathcal{E}_k} \otimes \rho_{\mathcal{F}\setminus\mathcal{E}_k|i}. \quad (3.30)$$

This holds for all subenvironments $\mathcal{E}_k$, $k = 1, 2, \dots, F$, in $\mathcal{F}$, hence the conditional fragment state is a product of pure states:

$$\rho_{\mathcal{F}|i} = |\psi_i\rangle\langle\psi_i|_{\mathcal{E}_1} \otimes \dots \otimes |\psi_i\rangle\langle\psi_i|_{\mathcal{E}_F}, \quad (3.31)$$

thus recovering invariant spectrum broadcast structure. $\square$

And lastly, in general, invariant spectrum broadcast structure can be recovered from spectrum broadcast structure by imposing purity on the conditional environment states.

3.4 Measuring strong quantum Darwinism

In strong quantum Darwinism, a large Holevo quantity (classical information) $\chi(\mathcal{S}^\Pi : \mathcal{F})$ is required for objectivity; whilst the discord $\mathcal{D}(\mathcal{S}^\Pi : \mathcal{F})$ is a hindrance. We can capture these relationships in the following function:

$$M^{\text{SQD}}(\rho_{\mathcal{S}\mathcal{F}}) \equiv \frac{H(\mathcal{S}) - \chi(\mathcal{S}^\Pi : \mathcal{F}) + \mathcal{D}(\mathcal{S}^\Pi : \mathcal{F})}{2H(\mathcal{S})}, \quad (3.32)$$

which takes values between $[0, 1]$. Objectivity occurs when the minimum value is obtained, $M^{\text{SQD}}(\rho_{\mathcal{S}\mathcal{F}}) = 0$ signaling perfect strong quantum Darwinism. This measure is not unique in the sense that it implies a certain partial ordering of states based on the assumption that we have equal weighting between the classical information and quantum discord. The components of Eq. (3.13) can be combined to form other measures with different orderings. The calculation of the entropic quantities of strong quantum Darwinism requires optimisation over measurements on the system.

Mironowicz *et al.* [104] defined a geometric distance bound on how close a state $\rho_{\mathcal{S}\mathcal{F}}$ is to having spectrum broadcast structure, which we had introduced back in Eq. (3.18). For clarity of comparison, we repeat it here:

$$T^{\text{SBS}}(\rho_{\mathcal{S}\mathcal{F}}) = \frac{1}{2} \min_{\Pi_{\mathcal{S}}} \|\rho_{\mathcal{S}\mathcal{F}} - \rho_{\mathcal{S}^\Pi\mathcal{F}}^{\text{SBS}}\|_1 \leq \eta[\rho_{\mathcal{S}\mathcal{F}}], \quad (3.33)$$

$$\eta[\rho_{\mathcal{S}\mathcal{F}}] \equiv \|\rho_{\mathcal{S}\mathcal{F}} - \rho_{\mathcal{S}^\Pi\mathcal{F}}\|_1 + \sum_{i \neq j} \sqrt{p_i p_j} B(\rho_{\mathcal{F}|i}, \rho_{\mathcal{F}|j}). \quad (3.34)$$

The fidelity is $B(\rho_1, \rho_2) = \|\sqrt{\rho_1}\sqrt{\rho_2}\|_1$, and $\rho_{S^\Pi\mathcal{F}} = \sum_i p_i |i\rangle\langle i| \otimes \rho_{\mathcal{F}|i}$ is the post-measurement (separable) state.

Our proposed non-objectivity measure $M^{\text{SQD}}(\rho_{S\mathcal{F}})$ and Mironowicz *et al.* [104]’s $\eta[\rho_{S\mathcal{F}}]$ are in fact related to each other. The term $\|\rho_{S\mathcal{F}} - \rho_{S^\Pi\mathcal{F}}\|_1 = \mathcal{D}_{\text{geo}}(\mathcal{S}^\Pi : \mathcal{F})$ is the *geometric* quantum discord [138], hence it is related to the entropic quantum discord: in particular, for two-qubit states, we have the explicit bound with the entropic quantum discord [139–142]:

$$\|\rho_{S\mathcal{F}} - \rho_{S^\Pi\mathcal{F}}\|_1 \geq \sqrt{2}\mathcal{D}(\mathcal{S}^\Pi : \mathcal{F}) \geq \mathcal{D}(\mathcal{S}^\Pi : \mathcal{F}). \quad (3.35)$$

Similarly, for a qubit system, the Holevo quantity is bounded as

$$\chi(\mathcal{S}^\Pi : \mathcal{F}) \geq I_{\text{acc}}(\mathcal{S} : \mathcal{F}) \geq H(\{p_1, p_2\}) - 2\sqrt{p_1 p_2} B(\rho_{\mathcal{F}|1}, \rho_{\mathcal{F}|2}), \quad (3.36)$$

[(Eq. (5) in Ref. [143]]. Hence, in the case where the system-fragment $\rho_{S\mathcal{F}}$ is a two-qubit state,

$$\eta(\rho_{S\mathcal{F}}) \geq \mathcal{D}(\mathcal{S}^\Pi : \mathcal{F}) - \chi(\mathcal{S}^\Pi : \mathcal{F}) + H(\{p_1, p_2\}) \quad (3.37)$$

$$= 2H(\mathcal{S})M^{\text{SQD}}(\rho_{S\mathcal{F}}). \quad (3.38)$$

Thus, the SBS distance upper-bound also upper-bounds the non-objectivity of strong quantum Darwinism. This is consistent with the fact that spectrum broadcast structure is more strict than strong quantum Darwinism.

Another alternative measure of strong quantum Darwinism would be the closest distance to the set of SQD-objective states

$$M^{\text{SQD}}(\rho_{S\mathcal{F}}) = \min_{\rho_{S\mathcal{F}}^{\text{SQD}}} \|\rho_{S\mathcal{F}} - \rho_{S\mathcal{F}}^{\text{SQD}}\|, \quad (3.39)$$

for any appropriate distance measure $\|\cdot\|$. Later, in Chapter 6, we will be considering subsets of SQD states and defining a distance to them as a measure of non-objectivity.

3.5 Requirements for objectivity

All the frameworks of quantum Darwinism are constructed to describe objectivity. However, in Sec. 3.3, we analysed how the frameworks are all different, *i.e.* describing different sets of objective states. That means these frameworks must all describe a different version of objectivity.

In introducing spectrum broadcast structure, Horodecki *et al.* [103] also proved the following theorem:

Theorem 2. [103]. *Assume that a system undergoes a full decoherence. Then the appearance of a spectrum broadcast structure is a sufficient condition for objectivity in the sense of Definition 1:*

$$\text{objective existence} \Leftarrow \text{spectrum broadcast structure}, \quad (3.40)$$

$$\text{objective existence} + \text{strong independence} \Rightarrow \text{spectrum broadcast structure}. \quad (3.41)$$

Note that the original theorem in Ref. [103] stated that spectrum broadcast structure is necessary *and* sufficient for objectivity. However, in their proof of Eq. (3.41), they actually applied strong independence as the main last step to acquire spectrum broadcast structure. This leads to the question: what does strong independence add? Strong independence adds nondisturbance to the environment states, making the environments partially objective too.

If we remove strong independence, then we are left with *strong quantum Darwinism*, which describes states that have *system*-objectivity but not *environment*-objectivity, which is the content of the following corollary.

Corollary 2. *Strong quantum Darwinism is sufficient and necessary for objectivity:*

$$\text{strong quantum Darwinism} \Longleftrightarrow \text{objectivity}. \quad (3.42)$$

Proof. Ref. [103] proved that (bipartite spectrum broadcast structure) $\Longleftrightarrow$ (objectivity). By Corollary 1, we obtain Eq. (3.42). $\square$

Thus, it is now clear that when we say “objectivity,” we automatically assume “system-objectivity.” Spectrum broadcast structure is *overly* strong for [system] objectivity. In contrast, Zurek’s quantum Darwinism is *insufficient* for objectivity as system-environment states can retain quantum correlations (especially quantum discord) under the ZQD conditions.

Strong quantum Darwinism, which lies in between these two frameworks, gives precisely system-(only)-objectivity. Meanwhile, invariant spectrum broadcast structure is the “most” objective, as it describes collective objectivity (from Definition 8):

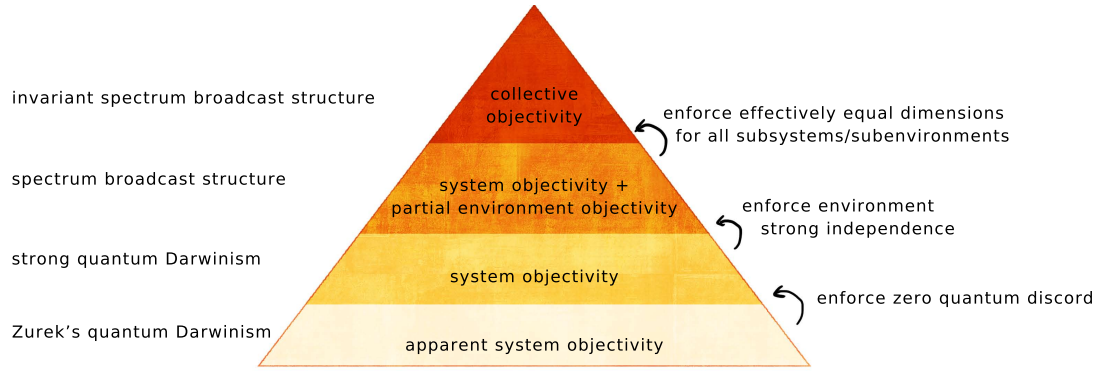


Figure 3.4: **Tiers of objectivity.** The frameworks that describe objectivity, from top to bottom, are: invariant spectrum broadcast structure (Def. 7), spectrum broadcast structure (Def. 5), strong quantum Darwinism (Def. 4), and Zurek's quantum Darwinism (Def. 2). By successive relaxations or tightening of conditions, we can move along the frameworks and the amount of objectivity contained in the system and environments.

Theorem 3. *Invariant spectrum broadcast structure is necessary and sufficient for collective objectivity:*

$$\text{invariant spectrum broadcast structure} \iff \text{collective objectivity}. \quad (3.43)$$

The full proof of Theorem 3 is given in Appendix A.2, but we outline the key steps here: the ($\implies$) direction is immediate from Corollary 2, since in ISBS, each subsystem has bipartite spectrum broadcast structure with relation to all the other subsystems and hence is objective. In the reverse ($\impliedby$) direction, Corollaries 1 and 2, along with Definition 8, give that collective objectivity implies bipartite spectrum broadcast structure relative to all subsystems. Algebraic manipulation subsequently reduces the joint state into invariant spectrum broadcast structure. Strong independence is inherent in collective objectivity and invariant spectrum broadcast structure.

In summary, different frameworks describe different versions of objectivity. We will organise this neatly in Fig. 3.4 in the following section.

3.6 Hierarchy of frameworks

We represent the compilation of the work in this chapter in Fig. 3.4, which gives a cohesive picture of the different frameworks relating to varying levels of objectivity:

- Zurek’s quantum Darwinism does not have true objectivity, instead describing an “apparent” objectivity. While environments contain information about the system, measurement in the environment will disturb the system.
- Strong quantum Darwinism describes objectivity of the system *alone*. Measurement of the environments can change the environments, but will not change the system state (up to forgetting measurement results).
- Spectrum broadcast structure describes an objective system and a partially objective environment. This partial objectivity is due to the mixed states $\rho_{\mathcal{E}_k|i}$ in Eq. (3.16). For example, an observer of $\mathcal{E}_1$ can learn the exact state of the system $|i\rangle_{\mathcal{S}}$ after a measurement of $\rho_{\mathcal{E}_1|i}$, but can only infer that the *other* observers have the conditional mixed states $\rho_{\mathcal{E}_k|i}$, $k = 2, 3 \dots$, and not the precise eigenstate.
- Finally, invariant spectrum broadcast structure describes objectivity of all subsystems, *i.e.* collective objectivity.

Further versions of objectivity could be constructed, for example partial objectivity of all systems,

$$\rho_{A_1 \dots A_N} = \sum_i p_i \rho_{A_1|i} \otimes \rho_{A_2|i} \cdots \otimes \rho_{A_N|i}, \quad \rho_{A_k|i} \rho_{A_k|j} = 0 \quad \forall i \neq j, \forall k, \quad (3.44)$$

or “trees” of objectivity where some environment subsystems have their own environment bath, or where there are sections with local objectivity. Fig. 3.4 displays the key frameworks from which more complex forms can be constructed. And thus, depending what one argues to be the “correct” level of objectivity that exists in our world, we can produce the relevant mathematical framework.

In this chapter, we have proven the major, previously missing, connection between the state-structure, geometric framework of spectrum broadcast structure and the entropic-oriented Zurek’s quantum Darwinism (and, historically, introduced strong quantum Darwinism in the process). Consequently, this gives us greater insight into the definition of objectivity itself, and how different conditions relax or strengthen objectivity within the system-and-environment. Strong quantum Darwinism captures the essence of objectivity of quantum systems by making explicit our intuition that classical information is required for the emergence of objectivity and that quantum correlations impedes objectivity and classicality.

In the next chapter, we examine a situation where the predictions of Zurek's quantum Darwinism deviate from the other frameworks and also give an example application of strong quantum Darwinism and spectrum broadcast structure.

Discrepancies between Zurek's quantum Darwinism and spectrum broadcast structure

This chapter includes content from the following publication:

Thao P. Le and Alexandra Olaya-Castro, *Objectivity (or lack thereof): Comparison between predictions of quantum Darwinism and spectrum broadcast structure*, Phys. Rev. A 98, 032103 (2018) [1].

All results in this chapter (starting in particular with our novel partial trace method in Sec. 4.2.2) are original contributions.

Historically, the development of strong quantum Darwinism was the successor to the work in this chapter.

Before the development of strong quantum Darwinism, there were two pre-existing frameworks: (Zurek's) quantum Darwinism and spectrum broadcast structure. Past literature primarily explored Zurek's quantum Darwinism in particular (for example, Refs. [27, 105–107, 109, 110, 112–114, 116–121, 130, 144–149]), and spectrum broadcast structure more rarely [76, 103, 104, 115, 123, 150], and they were treated somewhat interchangeably, depending on the situation/analysis method at hand. Before the development of strong quantum Darwinism and the associated proofs, it was not immediately clear how different Zurek's quantum Darwinism and spectrum broadcast structure were.

In this chapter, we explore how Zurek’s quantum Darwinism and spectrum broadcasting structure are nonequivalent through explicit examples, showing where Zurek’s quantum Darwinism is not suitable for making predictions about objectivity. Works such as Ref. [121] typically assume that there is zero or very little quantum discord, but in reality this assumption does not hold in general. As we argued in the previous chapter, Zurek’s quantum Darwinism can only determine *apparent* objectivity. For making predictions about objectivity of the system, checking quantum discord is needed, *i.e.*, the framework of strong quantum Darwinism (or a stricter framework) is required.

Horodecki *et al.* [103] presented the first example where the mutual information property of Zurek’s quantum Darwinism was fulfilled even though the state is entangled:

$$\rho_{SE} = pP_{(a|00)+b|11)} + (1-p)P_{(a|10)+b|01)} \quad (4.1)$$

where $P_{|\psi\rangle} = |\psi\rangle\langle\psi|$, $p \neq 1/2$, $a = \sqrt{p}$ and $b = \sqrt{1-p}$. The quantum mutual information between system and environment is $I(\mathcal{S} : \mathcal{E}) = H(\mathcal{S})$, *i.e.* Zurek’s quantum Darwinism is satisfied and thus “should” have objectivity. However, the classical information is not maximal $\mathcal{X}(\mathcal{S}^\Pi : \mathcal{E}) \neq H(\mathcal{S})$ for $p \neq 0, 1$. Strong quantum Darwinism is not satisfied, and this state does not have spectrum broadcast form. Strong quantum Darwinism and spectrum broadcast structure give the correct conclusion that the system is *not* objective.

The main argument against this example is the small size of the environment. Pleasance and Garraway [136] showed the same inconsistency in Zurek’s quantum Darwinism in a model with a two-level system interacting with multiple non-Markovian sub-environments. They found that the mutual information property Eq. (3.10) is fulfilled, but that this mutual information can comprise of mostly quantum discord. The presence of quantum discord will cause the system to be disturbed upon observers measuring their subenvironment [151].

These two examples show that Zurek’s quantum Darwinism and spectrum broadcast structure give conflicting predictions as to whether a state is objective, and furthermore, suggest that Zurek’s quantum Darwinism is incomplete given that it can falsely predict objectivity. Combined with the work in this chapter, this drove us to develop strong quantum Darwinism in the previous chapter, which resolved the prediction issues of Zurek’s quantum Darwinism.

In this chapter, we present a scenario where Zurek’s quantum Darwinism is unsuitable. We jointly investigate the predictions of Zurek’s quantum Darwinism, strong quantum Darwinism, and spectrum broadcast structure. Aside from Ref. [103], Zurek’s quantum Darwinism and spectrum broadcast structure are not explored in tandem. Our chosen model is a two-level system interacting with an N -level environment with a random matrix interaction. We would expect that any framework of quantum Darwinism should also be applicable to single environments, for example spatial continuous degrees of freedom, and continuous frequencies of light—in such a situation, suitable discretisation is required to describe effective or approximate subenvironments. In order to analyse quantum Darwinism, we propose a method to partition the N -level environment, and compare it with the partition method of Pérez [152].

We find that the predictions of Zurek’s quantum Darwinism are inconsistent with that of strong quantum Darwinism and spectrum broadcast structure when using the partitioning method of Pérez [152]. Alternatively, this shows that Zurek’s quantum Darwinism is inapplicable to the type of environment implied by Pérez’s partial trace.

This chapter is organised as follows: in Section 4.1, we introduce our system-environment model, which consists of a two-level system interacting with a single effective N -level environment via a random matrix coupling. In order to partition the N -level environment, we propose a novel partial-trace method in Section 4.2, and compare it with the partitioning method used by Pérez [152]. In Section 4.3, we analyse the predictions given by Zurek’s quantum Darwinism by examining the quantum mutual information. In Section 4.4, we analyse the predictions given by strong quantum Darwinism by calculating the classical accessible information and quantum discord. In Section 4.5, we compare that with an analysis of its state structure to determine its closeness to a spectrum broadcasting form. We conclude this chapter in Section 4.7.

4.1 Two-level system with N -level environment

Our model comprises of a two-level system interacting with an N -level environment, and the interaction between them is mediated by a random Gaussian

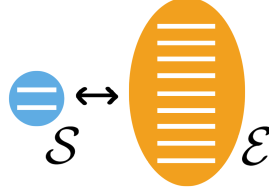


Figure 4.1: **The system-environment model** in which we investigate the predictions of the various frameworks of quantum Darwinism. The system $\mathcal{S}$ has two levels. The environment $\mathcal{E}$ has N levels equally spaced apart. The system and environment each have their own internal Hamiltonian, and their coupling is mediated by a random Gaussian matrix (c.f. Sec. 4.1).

matrix coupling. By doing so, we are assuming that the interaction between system and environment varies rapidly or that the environment is highly complex. These situations can be approximated with a random Gaussian matrix [153, 154]. Random matrix models and random matrix theory have also been used to model spectral fluctuations [155], and to explore decoherence [156–158] and quantum chaos [159].

The system Hamiltonian and environment Hamiltonians are the following, respectively ($\hbar = 1$):

$$\hat{H}_{\mathcal{S}} = \frac{\Delta E}{2} \sigma_z, \quad \hat{H}_{\mathcal{E}} = \sum_{n=0}^{N-1} \varepsilon_n |n\rangle\langle n|, \quad \varepsilon = \left(-\frac{\delta\varepsilon}{2}, -\frac{\delta\varepsilon}{2} + \frac{\delta\varepsilon}{N-1}, \dots, \frac{\delta\varepsilon}{2} \right), \quad (4.2)$$

where the environment levels $\{\varepsilon_n\}_n$ are spaced $\delta\varepsilon/(N-1)$ apart. The interaction Hamiltonian between system and environment is

$$\hat{H}_{\mathcal{SE}} = \sigma_x \otimes \lambda \frac{X}{\sqrt{8N}}, \quad (4.3)$$

where X is a Gaussian orthogonal random matrix of size N .¹ Rather than using a Gaussian unitary ensemble more typical in quantum information, we have a Gaussian orthogonal ensemble coupling following that of preceding works by other authors [152, 161], which allows us to use and compare their particular results in this study.

By changing the strength of the interactions λ relative to the system and environment energy scales ΔE and $\delta\varepsilon$ and the number of levels N in the environment,

¹That is, X is a real $N \times N$ symmetric matrix with $X_{ij} \sim \mathcal{N}(0, 1)$ and $X_{ii} \sim \sqrt{2}\mathcal{N}(0, 1)$ where $\mathcal{N}(0, 1)$ refers to the normal distribution. The probability density of X is proportional to $\exp(-\frac{1}{4} \text{tr } X^2)$. The $\sqrt{8N}$ factor is introduced fix the width of the averaged smooth density of state [160].

the system-environment correlations can be strengthened and entropy production will deviate from Markovian predictions [161]. Non-Markovian situations are of particular interest to us, as some previous work such as Refs. [112, 136] suggest that non-Markovianity can hinder the emergence of quantum Darwinism (cf. Sec. 5.3 where we go further into the relationship between non-Markovianity and the emergence of objectivity). More generally, non-Markovianity can also generate steady-state entanglement [162, 163], lead to entanglement rebirth [164], and the persistence of quantum correlations [165, 166] due to the strong system-environment interactions.

For our model, we fix the parameters to be $\Delta E = 1$, $\delta\varepsilon = \Delta E$ and $\lambda = \Delta E/5$. We take the following initial conditions: the system-environment are initially separable, $\rho_{SE}(0) = \rho_S(0) \otimes \rho_E(0)$. The system has initial state:

$$\rho_S(0) = |\Psi_S(0)\rangle\langle\Psi_S(0)|, \quad \text{with } |\Psi_S(0)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle). \quad (4.4)$$

The environment $\rho_E(0)$ is either in the pure superposition state,

$$\rho_E(0) = |\Psi_E(0)\rangle\langle\Psi_E(0)|, \quad \text{with } |\Psi_E(0)\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} |n\rangle, \quad (4.5)$$

or in a thermal state with inverse temperature β (where we fix $\beta = 10$):

$$\rho_E(0) = \frac{e^{-\beta\hat{H}_E}}{Z_E}, \quad Z_E = \text{tr} \left[e^{-\beta\hat{H}_E} \right]. \quad (4.6)$$

We will compare the evolutions due to the different environments as it has been shown that quantum Darwinism is hindered when the environment is a statistical mixture, as noise in the environment states reduces the amount of system information that can be encoded [116, 119, 120, 144, 149].

Any form of quantum Darwinism typically requires the full dynamics of $\rho_{SE}(t)$, whether to calculate the explicit correlations between the system-environment or to analyse the state structure.² We obtain the joint system by numerically evolving the state via the Liouville von Neumann equation,

$$\frac{d}{dt}\rho_{SE}(t) = -i \left[\hat{H}_S + \hat{H}_E + \hat{H}_{SE}, \rho_{SE}(t) \right], \quad (4.7)$$

²There are specific cases such as with Gaussian states [113] where the properties of Gaussian states simplify the situation, meaning that a reduced number of state quantities are required to evaluate the system-environment correlations.

which we do by employing the Python QuTip package [167, 168]. The reduced system state is then recovered via the usual trace,

$$\rho_S(t) = \text{tr}_E[\rho_{SE}(t)]. \quad (4.8)$$

Note that in our model, the system-environment coupling does not commute with the system Hamiltonian *and* their energy scales are comparable. Hence, the pointer states of the system selected by the environment are not the eigenstates of σ_x as one would “expect” from the coupling interaction in Eq. (4.3). Instead, the pointer states would be a more complicated compromise between the different internal and coupling interactions. Rather than imposing a pointer basis, we will be maximising over relevant system bases and information whenever a pointer basis is required.

In Fig. 4.2, we show how decoherence and re-coherence changes as we change N —re-coherence is typical of non-Markovian dynamics, where the environment returns back coherence to the system. For various environment with $N = 3, 10, 200$ energy levels, we plot the system von Neumann entropy $H(S) = H(\rho_S(t)) = -\text{tr} \rho_S(t) \log_2 \rho_S(t)$, the excited population $\langle 1 | \rho_S(t) | 1 \rangle$, and the absolute value of the off-diagonal component $|\langle 0 | \rho_S(t) | 1 \rangle|$ corresponding to the coherence of the system state. As the number of levels N increases, the entropy of the system shows roughly monotonic rise and decoherence is roughly monotonic (up to small-scale oscillations). Whilst for small N , the system shows cycles of gaining and losing entropy and coherence, which is consistent with non-Markovian dynamics. A thermal environment causes faster decay of the system excited population. The thermal environment also dampens the small-scale oscillations in entropy, excited state population and decoherence dynamics. This is consistent with the results of Ref. [161], where the dynamics in this model becomes increasingly Markovian-like as N is increased, *i.e.* with decreasing coherence rebirths that correspond to information back-flow. Hence, there are strong system-environment correlations at finite N . Therefore, for the remainder of the chapter, we fix $N = 10$, in order to account for non-Markovian dynamics with a reasonable number of levels in the environment that will allow us to study quantum Darwinism.

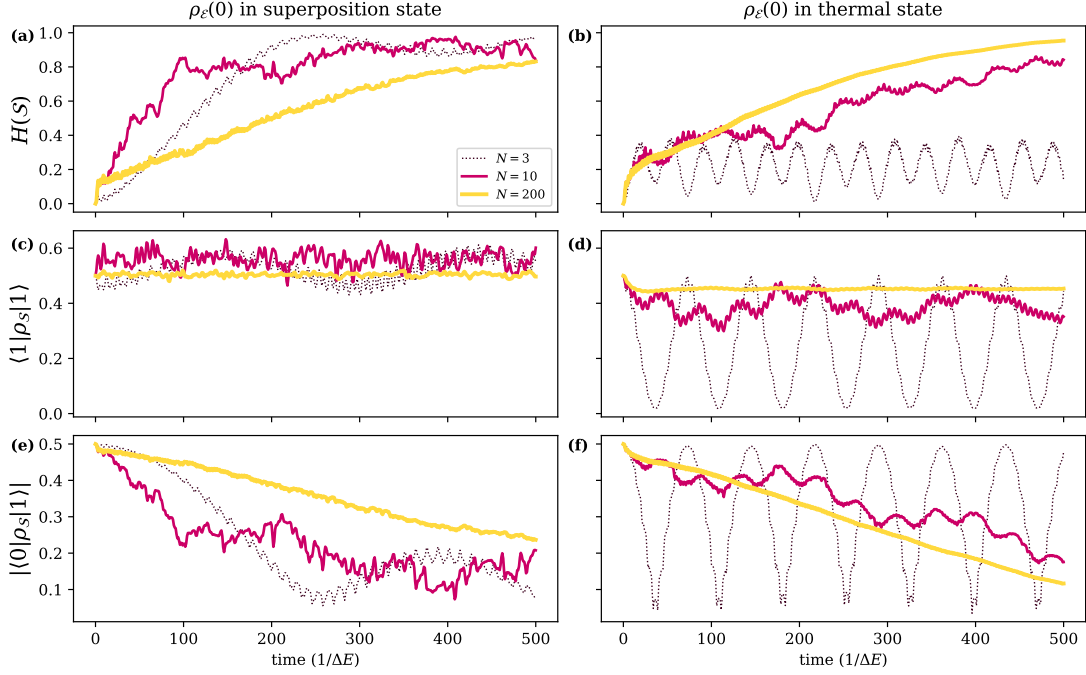


Figure 4.2: **Evolution of the system as the environment changes.** The environment begins in either the superposition state, $\rho_{\mathcal{E}}(0) = |\Psi_{\mathcal{E}}(0)\rangle\langle\Psi_{\mathcal{E}}(0)|$, with $|\Psi_{\mathcal{E}}(0)\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} |n\rangle$, or in the thermal state $\rho_{\mathcal{E}}(0) = e^{-\beta\hat{H}_{\mathcal{E}}}/Z_{\mathcal{E}}$ with inverse temperature β . The system begins in superposition state $\rho_{\mathcal{S}}(0) = |\Psi_{\mathcal{S}}(0)\rangle\langle\Psi_{\mathcal{S}}(0)|$ with $|\Psi_{\mathcal{S}}(0)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. (a), (b) Plots of the system entropy $H(\mathcal{S})$, (c), (d) the excited coefficient of the system, $\langle 1|\rho_{\mathcal{S}}|1\rangle$, and (e), (f) the absolute value of the off-diagonal coefficient $|\langle 0|\rho_{\mathcal{S}}|1\rangle|$. As N increases, the interaction between system and environment weakens and transitions from non-Markovian to Markovian. With parameters $\lambda = 0.1$, $\Delta E = 0.5$, $\delta\varepsilon = 0.5$, and $\beta = 10$.

4.2 Partitioning the environment

Our environment model consists of a single environment. Hence, we need partial trace methods to “artificially” break up the environment into suitable subcomponents for study under quantum Darwinism. In subsection 4.2.1, we reintroduce the level-partitioning and elimination method defined by Pérez [152]. In subsection 4.2.2, we propose a novel partial trace method that avoids the caveats of the first method (explained later in the text), which we call the “staircase environment” trace. Graphical depictions of how these partial traces affect the matrix structure are shown in Fig. 4.3. As we will find out in the later sections, these two methods lead to different predictions of Zurek’s quantum Darwinism.

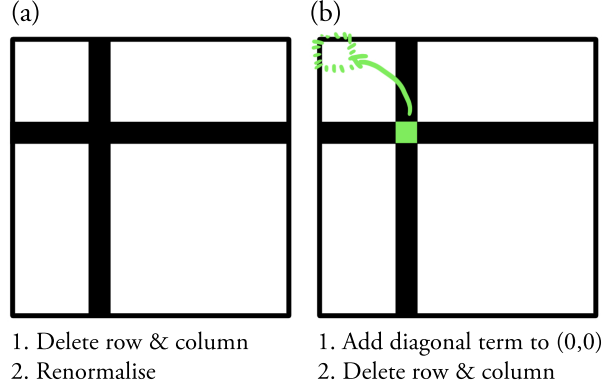


Figure 4.3: **Graphical depiction of two alternative partial traces.** (a) The level-partitioning and elimination partial trace method given by Pérez [152]. (b) The staircase environment trace we introduce, based on the assumption that the environment comprises of $N - 1$ different subsystems with increasing excited energy.

4.2.1 The Pérez trace (level-partitioning and elimination)

The Pérez [152] partial trace method presumes the situation where we can only access some fraction of levels $\mathcal{F}$, and assumes that all other levels in the environment are essentially non-existent. Therefore, for a general system-environment state,

$$\rho_{S\mathcal{E}}(t) = \sum_{i,j=0,1} \sum_{n,m=0}^{N-1} c_{ijnm}(t) |i\rangle\langle j|_{\mathcal{S}} \otimes |n\rangle\langle m|_{\mathcal{E}}, \quad (4.9)$$

the Pérez trace gives the following reduced system-fragment state:

$$\rho_{S\mathcal{F}}(t) = \frac{1}{N_{\mathcal{F}}(t)} \sum_{i,j=0,1} \sum_{n,m \in \mathcal{F}} c_{ijnm}(t) |i\rangle\langle j|_{\mathcal{S}} \otimes |n\rangle\langle m|_{\mathcal{E}}, \quad (4.10)$$

where $N_{\mathcal{F}}(t) = \sum_{i=0,1} \sum_{n \in \mathcal{F}} c_{iinn}(t)$ is a normalisation factor. Here, $\mathcal{F} \subseteq \mathcal{E} = \{0, 1, \dots, N-1\}$ is a subset of possible energy levels. If $N_{\mathcal{F}}(t) = 0$, then $\rho_{S\mathcal{F}}(t) = 0$, which corresponds to the environment being in one of the “excluded” $\mathcal{E} \setminus \mathcal{F}$ levels. A graphical depiction of this partial trace is given in Fig. 4.3(a), and can be thought of “deleting” excluded energy levels.

If the full system-environment state is pure, *i.e.*,

$$\rho_{S\mathcal{E}}(t) = |\Psi_{S\mathcal{E}}(t)\rangle\langle\Psi_{S\mathcal{E}}(t)| \quad \text{with} \quad |\Psi_{S\mathcal{E}}(t)\rangle = \sum_{i=0,1} \sum_{n=0}^{N-1} a_{in}(t) |i\rangle_{\mathcal{S}} \otimes |n\rangle_{\mathcal{E}}, \quad (4.11)$$

then the Pérez trace ensures that all system-fragment states are *also* pure:

$$\rho_{S\mathcal{F}}(t) = |\Psi_{S\mathcal{F}}(t)\rangle\langle\Psi_{S\mathcal{F}}(t)| \quad \text{with } |\Psi_{S\mathcal{F}}(t)\rangle = \frac{1}{\sqrt{N_{\mathcal{F}}(t)}} \sum_{i=0,1} \sum_{n \in \mathcal{F}} a_{in}(t) |i\rangle_S \otimes |n\rangle_{\mathcal{E}}. \quad (4.12)$$

One caveat to this method is that the apparent reduced system state is not the true system state in general. Partial traces should satisfy:

$$\rho_S = \text{tr}_{\mathcal{E}}[\rho_{S\mathcal{E}}] \stackrel{!}{=} \text{tr}_{\mathcal{F}}[\rho_{S\mathcal{F}}], \quad (4.13)$$

i.e. the system state should be recovered once any number of environments are traced out. However, the Pérez trace gives $\text{tr}_{\mathcal{F}}[\rho_{S\mathcal{F}}] \neq \rho_S = \text{tr}_{\mathcal{E}}[\rho_{S\mathcal{E}}]$ in general. As such, we also suggest a different partial trace method that does not have this problem.

4.2.2 The staircase environment trace

The staircase environment trace assumes that the environment is comprised of $N - 1$ two-level subsystems, where each subsystem has increasing energy:

$$\hat{H}_{\mathcal{E}} = \varepsilon_0 |0\rangle\langle 0|^{\otimes N-1} + \sum_{n=1}^{N-1} \varepsilon_n |1\rangle\langle 1|_{\mathcal{E}_n} \otimes \mathbb{1}_{\mathcal{E} \setminus \mathcal{E}_n}. \quad (4.14)$$

For example, $|1\rangle = |1\rangle_{\mathcal{E}_1} \otimes |0\rangle_{\mathcal{E}_2} \otimes \cdots \otimes |0\rangle_{\mathcal{E}_{N-1}}$. Under this assumption, the effective N -levels that we are working with are in fact a *subspace* of a larger 2^{N-1} environment Hilbert space. For a general system-environment state given in Eq. (4.9), tracing out some environment $\mathcal{E}_k$ corresponds to

$$\rho_{S(\mathcal{E} \setminus \mathcal{E}_k)}(t) = \text{tr}_{\mathcal{E}_k}[\rho_{S\mathcal{E}}(t)] \quad (4.15)$$

$$= \langle 0_{\mathcal{E}_k} | \rho_{S\mathcal{E}}(t) | 0_{\mathcal{E}_k} \rangle + \langle 1_{\mathcal{E}_k} | \rho_{S\mathcal{E}}(t) | 1_{\mathcal{E}_k} \rangle \quad (4.16)$$

$$\begin{aligned} &= \sum_{i,j=0,1} \sum_{\substack{n,m=0 \\ \text{s.t. } n,m \neq k}}^{N-1} c_{ijnm} |i\rangle\langle j|_S \otimes |n\rangle\langle m|_{\mathcal{E} \setminus \mathcal{E}_k} \\ &\quad + \sum_{i,j=0,1} c_{ijkk}(t) |i\rangle\langle j|_S \otimes |0\rangle\langle 0|_{\mathcal{E} \setminus \mathcal{E}_k}. \end{aligned} \quad (4.17)$$

A graphical depiction of this partial trace is given in Fig. 4.3(b). A reduced system-fragment state is therefore

$$\rho_{S\mathcal{F}}(t) = \sum_{\substack{i,j=0,1 \\ n,m \in \mathcal{F} \cup \{0\}}} c_{ijnm}(t) |i\rangle\langle j|_S \otimes |n\rangle\langle m|_{\mathcal{E} \setminus \mathcal{F}} + \sum_{\substack{i,j=0,1 \\ k \in \mathcal{E} \setminus \mathcal{F}}} c_{ijkk}(t) |i\rangle\langle j|_S \otimes |0\rangle\langle 0|_{\mathcal{E} \setminus \mathcal{F}}. \quad (4.18)$$

The advantage of this method is that the reduced system state as derived from $\rho_{S\mathcal{F}}(t)$ is equivalent to the true system state derived from $\rho_{S\mathcal{E}}(t)$, *i.e.* that $\rho_S(t) = \text{tr}_{\mathcal{E}}[\rho_{S\mathcal{E}}(t)] = \text{tr}_{\mathcal{F}}[\rho_{S\mathcal{F}}(t)]$. The caveat is that we have made the assumption about the nature of the excited energy levels for the environment sub-components.

The staircase trace could represent a set of spin particles with energy separations, or photons with varying frequency. Note that the von Neumann entropy of the reduced state $\rho_{\mathcal{F}}$ or $\rho_{S\mathcal{F}}$ in the N -dimensional subspace has the same non-zero eigenvalues as the full representation, hence working in the subspace is not detrimental.

4.3 Conflicting predictions of Zurek's quantum Darwinism

First, we study the predictions of Zurek's quantum Darwinism (ZQD) in our particular model. After evolving the system and environment as in Eq. (4.7), we use the two partial trace methods introduced above to partition a single environment into fractions. This allows us to calculate the quantum mutual information between system and fraction, $I(\mathcal{S} : \mathcal{F}) = H(\mathcal{S}) + H(\mathcal{F}) - H(\mathcal{SF})$ that is the core term in ZQD (cf. Definition 2). Recall that ZQD is said to arise when sufficiently small fragments $\mathcal{F}_\delta \subset \mathcal{E}$ of the environment contain roughly all the information about the system state, *i.e.* $I(\mathcal{S} : \mathcal{F}_\delta) \approx H(\mathcal{S})$. We find that the two different partitioning methods give *different predictions* as to whether ZQD-objectivity has emerged or not.

The results are given in Fig. 4.4, where the quantum mutual information has been plotted at times $t = 300, 400, 500(1/\Delta E)$ over different fractions f . In all cases, the mutual information has been averaged over all possible fractions of equal size, since the subenvironments are not identical. The system state is taken as the true $\rho_S = \text{tr}_{\mathcal{E}}[\rho_{S\mathcal{E}}]$ (where here the trace is the standard partial trace in quantum information). The system entropy $H(\mathcal{S})$ is calculated from this. In the later Section 4.6, we will show what happens if we take the reduced system state from the system-fragment instead, $\rho'_S = \text{tr}_{\mathcal{F}}[\rho_{S\mathcal{F}}]$ (those figures will be a little different, but qualitatively, the same conclusions can be drawn).

While the quantum mutual information value changes over different times, its relationship to the system entropy remains roughly constant at the times

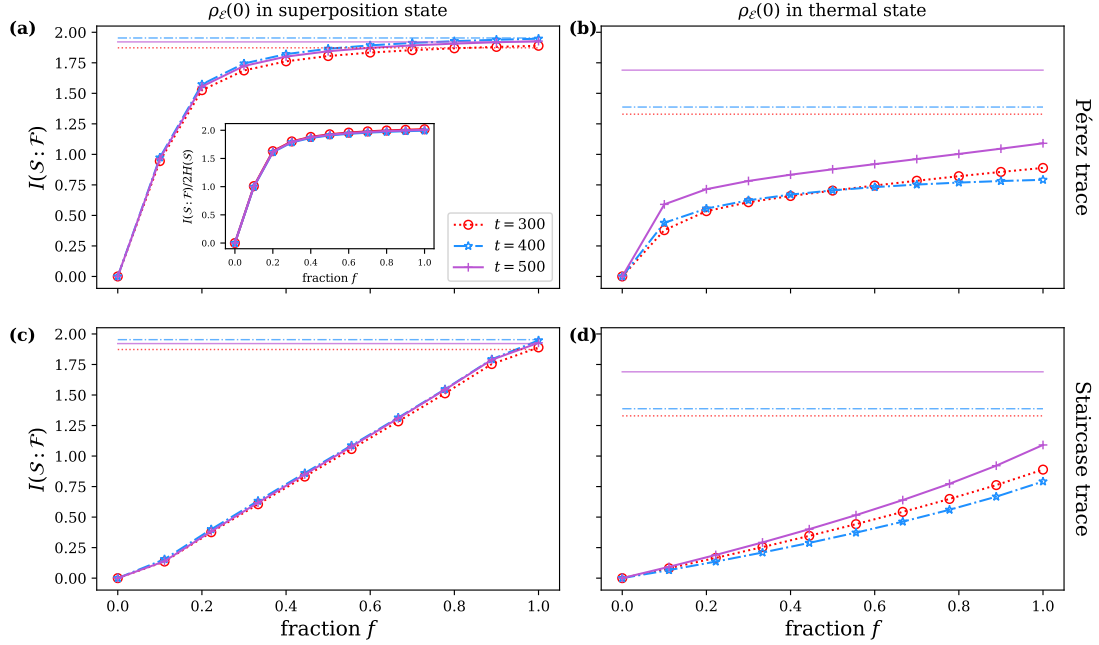


Figure 4.4: **Mutual information $I(\mathcal{S} : \mathcal{F})$ between system and environment fragments.** (a), (b) The Pérez [152] trace [Eq. (4.10)] is used to form different fractions of the environment. (c), (d) The staircase trace [Eq. (4.18)] is used. The horizontal lines correspond to twice the system entropy $2H(\mathcal{S})$ (the maximum mutual information possible if the full system-environment state is pure). The inset in (a) shows the normalised mutual information. With parameters $\Delta E = 1$, $\delta\varepsilon = \Delta E$, $\lambda = \Delta E/5$, and $\beta = 10$ for the thermal state (and $\hbar = 1$).

considered: this is illustrated in Fig. 4.4(a)[inset] which shows the normalised mutual information $I(\mathcal{S} : \mathcal{F})/2H(\mathcal{S})$. This behavior repeats for the other situations represented in Fig. 4.4, hence we have not shown them here. If the initial environment is in the thermal state [Fig. 4.4(b,d)/right side], then the maximum mutual information $I(\mathcal{S} : \mathcal{E}) \neq 2H(\mathcal{S})$ (due to the fact that the full system-thermal environment is not a pure state) but the overall shape remains qualitatively the same to the initial pure environment case.

Now consider the difference between the Pérez trace and the staircase trace plots, especially in the initial pure environment case (the left side of Fig. 4.4). The staircase trace produces mutual information plots that are more typical of Zurek's quantum Darwinism: symmetric about $f = 0.5$, corresponding to half of the environment, and reaching the full information $2H(\mathcal{S})$ with the full environment ($f = 1$) [Fig. 4.4(c)]. According to the staircase trace, there is no ZQD, and hence

no objectivity, since it takes half of the environment to achieve the ZQD condition $I(\mathcal{S} : \mathcal{F}) = H(\mathcal{S})$.

In contrast, the Pérez trace produces a quantum mutual information curve that sharply rises to the ZQD condition $I(\mathcal{S} : \mathcal{F}) = H(\mathcal{S})$, before slowly approaching its maximum value $2H(\mathcal{S})$ [Fig. 4.4(a)]. Furthermore, the curve with the Pérez trace is asymmetric—this is due to how the Pérez trace decomposes the environment in direct sums as $\mathcal{E} = \mathcal{F} \oplus (\mathcal{E} \setminus \mathcal{F})$ [152]. This asymmetry means that the plots from the Pérez trace do not match those typical of ZQD. However, given that it *does* satisfy the underlying mathematical condition, the results from the Pérez trace suggest that ZQD has emerged, and that it only requires a single fragment for it to do so. The way the plot quickly reaches $I(\mathcal{S} : \mathcal{F}) \approx 2H(\mathcal{S})$ for larger fragments $\mathcal{F}$ implies that the full quantum correlations can be accessed without requiring the majority of the environment.

In typical ZQD plots, the quantum mutual information is symmetric, and this is shown by Blume-Kohout and Zurek [27] *under the assumption that the partial trace gives the correct reduced system state*. In contrast, the Pérez trace, though physically motivated, does not give the correct system state from the reduced system-fragment state, $\text{tr}_{\mathcal{F}}[\rho_{\mathcal{S}\mathcal{F}}] \neq \rho_{\mathcal{S}} = \text{tr}_{\mathcal{E}}[\rho_{\mathcal{S}\mathcal{E}}]$, as previously mentioned in subsection 4.2.1. This is the root cause for the asymmetric mutual information plots for the Pérez trace.

The proof for the symmetric nature of typical mutual information plots, given by Blume-Kohout and Zurek [27], relies on the purity of the full system-environment state $\rho_{\mathcal{S}\mathcal{E}}$. The environment can be partitioned into two arbitrary parts $\mathcal{E}_1$ and $\mathcal{E}_2$. Since bipartitions of pure states have the same spectrum, their von Neumann entropy is the same: $H(\mathcal{E}_1) = H(\mathcal{S}\mathcal{E}_2)$, and $H(\mathcal{E}_2) = H(\mathcal{S}\mathcal{E}_1)$. Hence, the mutual information between different parts are related as $I(\mathcal{S} : \mathcal{E}_1) + I(\mathcal{S} : \mathcal{E}_2) = I(\mathcal{S} : \mathcal{E}_1\mathcal{E}_2)$, and this leads to that typical symmetric structure when the environments are identical, or when the mutual information is averaged over fragments of the same size (Fig. 3.1). Under the Pérez trace, however, the reduced system-subenvironment states $\rho_{\mathcal{S}\mathcal{E}_1}$ and $\rho_{\mathcal{S}\mathcal{E}_2}$ are pure when the full state $\rho_{\mathcal{S}\mathcal{E}_1\mathcal{E}_2}$ is pure. However, the reduced *environment* states $\rho_{\mathcal{E}_1}$ and $\rho_{\mathcal{E}_2}$ are not pure in general, hence $H(\mathcal{E}_1) \neq H(\mathcal{S}\mathcal{E}_2)$, and $H(\mathcal{E}_2) \neq H(\mathcal{S}\mathcal{E}_1)$, and hence $I(\mathcal{S} : \mathcal{E}_1) + I(\mathcal{S} : \mathcal{E}_2) \geq I(\mathcal{S} : \mathcal{E}_1\mathcal{E}_2)$. This causes the plots under the Pérez trace to be asymmetric.

In fact, this asymmetric curve and sharp rise in mutual information is a *universal feature* of the Pérez trace when the full system-environment state is

pure—we *always* have $I(\mathcal{S} : \mathcal{F}) = H(\mathcal{S})$ when $|\mathcal{F}| = 1$, and $I(\mathcal{S} : \mathcal{F}) > H(\mathcal{S})$ for fragment sizes $|\mathcal{F}| > 1$. This is because the partially reduced $\mathcal{SF}$ state is always effectively pure or zero, in which case $H(\mathcal{SF}) = 0$ always for $\mathcal{F} \neq \emptyset$. Furthermore, when $\mathcal{F}$ consists of a *single* level, then $\rho_{\mathcal{F}}$ is essentially a c-number, and due to trace preservation, is either $\rho_{\mathcal{F}} = [1]$ or $\rho_{\mathcal{F}} = [0]$, and so $H(\mathcal{F}) = 0$ for $|\mathcal{F}| = 1$.

Hence, if we use the Pérez trace, which corresponds to an environment of which we can only access certain energy levels, then Zurek’s quantum Darwinism either (1) always emerges, or (2) is not applicable. The results of the staircase trace and other results later in the chapter conflict with the first conclusion—objectivity cannot emerge in *all* scenarios simply by using the Pérez trace with environments where we can only access some levels. This leaves us with the second conclusion of non-applicability—which suggests then that Zurek’s quantum Darwinism is limited and cannot be used universally to explore the emergence of objectivity. We argue that, regardless of which conclusion (1) or (2) we take, the *overall* consequence is that Zurek’s quantum Darwinism as in Def. 2, with mathematical condition $I(\mathcal{S} : \mathcal{F}) = H(\mathcal{S})$, is inconsistent with system objectivity. This is why in Section 3.6, we say that Zurek’s quantum Darwinism at best describes an *apparent* objectivity.

4.4 Strong quantum Darwinism: quantum versus classical information

Strong quantum Darwinism assesses objectivity by considering the quantum versus classical information. Recall that the quantum mutual information of two systems can be decomposed into the sum of their accessible (Holevo) information χ and the quantum discord $\mathcal{D}$ between them (from Section 3.1):

$$I(\mathcal{S} : \mathcal{F}) = \chi(\mathcal{S} : \mathcal{F}) + \mathcal{D}(\mathcal{S} : \mathcal{F}). \quad (4.19)$$

The accessible information quantifies about the amount of classical information shared, whereas the discord describes any non-classical correlations [128, 129].

When the mutual information between system and fragment is approximately equal to the system entropy, $I(\mathcal{S} : \mathcal{F}) \approx H(\mathcal{S})$, Zurek’s quantum Darwinism posits that the fragment $\mathcal{F}$ contains (approximately) all the information about the properties of the system. Typically, past works *assumed* that states

would have a large amount of accessible information [117]. However, as we noted at the beginning of the chapter, Pleasance and Garraway [136] had found that this is not always the case—they show a scenario whereby there is apparent ZQD due to the present quantum mutual information plateau, but where the mutual information, at small fractions, is largely comprised of *quantum discord*—which would fail the test of *strong* quantum Darwinism. As such, it is imperative for us to investigate the nature of information in our particular model, especially in the case of the Pérez trace which we found is inconsistent under Zurek’s quantum Darwinism.

We calculate the accessible information by maximising over possible POVMs on the system,

$$\chi(\mathcal{S} : \mathcal{F}) = \max_{\hat{\Pi}_{\mathcal{S}}} \left\{ H \left(\sum_a p(a) \rho_{\mathcal{F}|a} \right) - \sum_a p(a) H(\rho_{\mathcal{F}|a}) \right\}, \quad (4.20)$$

where a are the measurement results of the POVM $\hat{\Pi}_{\mathcal{S}}$, $\rho_{\mathcal{F}|a}$ refers to the state of the fragment given result a and $p(a)$ is the probability of result a [130]. The discord can always be minimised using rank-one projectors [169], and so correspondingly the accessible information can always be maximised using rank-one projectors. As our system is two-level, a general observable can be written in form $\vec{r} \cdot \vec{\sigma}$, where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$, and $\vec{r}$ is a unit vector. The two associated projectors are then $\Pi_{\mathcal{S}}^{\pm} = (\mathbb{1} \pm \vec{r} \cdot \vec{\sigma})/2$ and the conditional fragment states are

$$\rho_{\mathcal{F}|\pm} = \text{tr}_{\mathcal{S}}(\Pi_{\mathcal{S}}^{\pm} \rho_{\mathcal{SF}} \Pi_{\mathcal{S}}^{\pm}) / p(\pm), \quad (4.21)$$

where $p(\pm) = \text{tr}(\rho_{\mathcal{SF}} \Pi_{\mathcal{S}}^{\pm})$. We optimise via a random search, where the unit vector $\vec{r}$ is randomly chosen across the unit-sphere.

The results are given in Fig. 4.5. We find that the amount of quantum discord is *comparable* to the amount of accessible information, regardless of the partial trace method. It is clear that relatively large environment fractions are required to obtain classical information approximately equal to the system entropy. By decomposing the mutual information into its classical and quantum components, the conclusion from both partial trace methods is clear: the system state is *not* objective, regardless of what Zurek’s quantum Darwinism would otherwise imply. Assuming that there must be classical-only information is fallacy of ZQD, and this example is yet more evidence that the explicit zero-discord-condition of strong quantum Darwinism (Definition 4) is necessary.

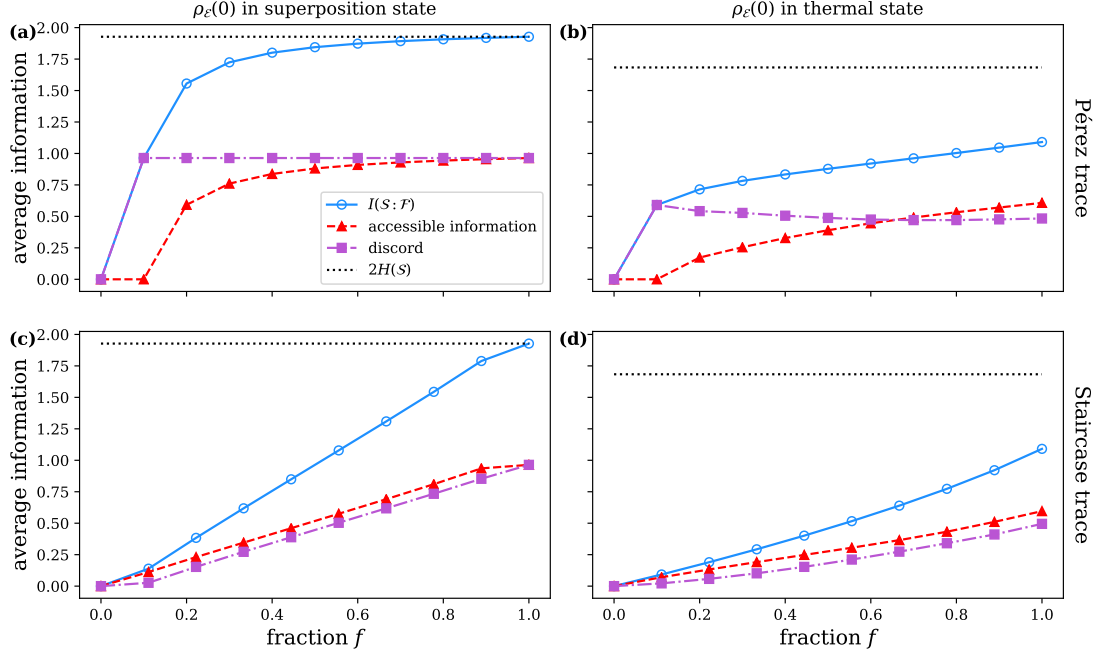


Figure 4.5: **Decomposition of mutual information into accessible information and quantum discord (at time $t = 500$).** The accessible information and quantum discord is averaged over all fractions of the same size f . The system entropy is $H(\mathcal{S})$ (corresponding to the true system state). With parameters $\Delta E = 1$, $\delta\varepsilon = \Delta E/\lambda = \Delta E/5$, and $\beta = 10$ for the thermal state (and $\hbar = 1$).

If we focused solely on Zurek’s quantum Darwinism, then the two partial traces gave different predictions as to whether or not objectivity emerged. In contrast, when we delve into the nature of the information, *à la* strong quantum Darwinism, we found that the quantum mutual information is a roughly equal mixture of accessible information and quantum discord—in both partial traces. Thus, we conclude that the system is *not* objective. Our results show that Zurek’s quantum Darwinism is insufficient to fully determine whether or not the quantum-to-classical transition has occurred, and that strong quantum Darwinism is preferable. In the following section, we consider a yet more stringent condition on objectivity: spectrum broadcast structure.

4.5 Spectrum broadcasting in the model

For ease, let us recall the form of spectrum broadcast structure (Definition 5):

$$\rho_{\mathcal{SF}} = \sum_i p_i |i\rangle\langle i|_{\mathcal{S}} \otimes \rho_{\mathcal{E}_1|i} \otimes \cdots \otimes \rho_{\mathcal{E}_{fN}|i}, \quad \rho_{\mathcal{E}_k|i} \rho_{\mathcal{E}_k|j} = 0 \quad \forall i \neq j. \quad (4.22)$$

In subsection 4.5.1, we will first construct an upper-bound of the minimum distance from the true system-fragment state $\rho_{\mathcal{SF}}$ to the nearest spectrum broadcast structure state $\rho_{\mathcal{SF}}^{\text{SBS}}$ based on the bound given in Ref. [104]. By evaluating this bound on our system-environment states, in subsection 4.5.2 we find that spectrum broadcasting has not occurred.

4.5.1 An approximate measure of spectrum broadcasting

A measure of spectrum broadcasting is the distance to the set of SBS states,

$$M^{\text{SBS}}(\rho_{\mathcal{SF}}) = \min_{\rho_{\mathcal{SF}}^{\text{SBS}}} \|\rho_{\mathcal{SF}} - \rho_{\mathcal{SF}}^{\text{SBS}}\|, \quad (4.23)$$

minimised over all states of spectrum-broadcast form, under the L_1 norm (or alternative distance measures). However, this suffers all the analogous difficulties in measuring the distance of entangled states to the set of separable states—the spectrum-broadcast states are a subset of the separable states and brute-force optimization is no easy task.

Mironowicz *et al.* [104] took the approach of constructing a computable upper-bound $\eta[\rho_{\mathcal{SF}}]$ on the distance (see Eq. (3.19)). Their upper-bound is the summation of the decoherence factors and the distinguishability of conditional environment states $\rho_{\mathcal{E}_k|i}$ under the case of the quantum measurement limit. Using the same derivation (see Appendix A.3), we rewrite that upper-bound in a form suitable for our model:

$$M^{\text{SBS}}(\rho_{\mathcal{SF}}) \leq \eta[\rho_{\mathcal{SF}}], \quad (4.24)$$

$$\eta[\rho_{\mathcal{SF}}] = \min_{\{P_i^{\mathcal{S}}\}} \left[\|\rho_{\mathcal{SF}} - \rho_{\mathcal{SF}}^{\text{sep}}(\{P_i^{\mathcal{S}}\})\|_1 + \sum_{i \neq j} \sqrt{p_i p_j} B(\rho_{\mathcal{F}|i}, \rho_{\mathcal{F}|j}) \right], \quad (4.25)$$

where $B(\rho_1, \rho_2) = \|\sqrt{\rho_1} \sqrt{\rho_2}\|_1$ is the fidelity, and where $\{P_i^{\mathcal{S}} = |i'\rangle\langle i'|\}$ is the set of rank-one *system* projectors (which need not be the standard computation

basis) that produces a separable state:

$$\rho_{\mathcal{SF}}^{\text{sep}}(\{P_i^{\mathcal{S}}\}) = \sum_i P_i^{\mathcal{S}} \otimes \mathbb{1}^{\mathcal{F}} \rho_{\mathcal{SF}} P_i^{\mathcal{S}} \otimes \mathbb{1}^{\mathcal{F}} \quad (4.26)$$

$$=: \sum_{i=0,1} p_i |i'\rangle\langle i'| \otimes \rho_{\mathcal{F}|i}, \quad (4.27)$$

where $p_i = \text{tr}[P_i^{\mathcal{S}} \rho_{\mathcal{SF}}]$ and $\rho_{\mathcal{F}|i} = \text{tr}_{\mathcal{S}}[P_i^{\mathcal{S}} \rho_{\mathcal{SF}}]/p_i$. In Eq. (4.25), the error bound is minimised over the projectors. The bound $\eta[\rho_{\mathcal{SF}}]$ is tight when $\rho_{\mathcal{SF}}$ has a spectrum broadcast structure.

4.5.2 Results

We employ the upper-bound on the distance of a state to spectrum broadcast form, $\eta[\rho_{\mathcal{SF}}]$ in Eq. (4.25), to investigate the structure of the system-fragment states. Since the system is a qubit, the projectors can be written as $P_{\pm}^{\mathcal{S}} = (\mathbb{1} \pm \hat{n} \cdot \vec{\sigma})/2$ for unit vector $\hat{n}$. Numerically, we minimise the distance given in Eq. (4.25) by sampling $\hat{n}$ over the unit sphere. We further minimise over all fractions of the same size, to produce Fig. 4.6.

We find that, in the majority of cases, there is no fraction size at which a spectrum broadcast structure is formed between system and environment fragment. This confirms our prior conclusion that objectivity has not emerged. The only “prospective” case is that of the smallest non-zero environment fraction in the Pérez trace [Fig. 4.6(a)], where the distance is vanishing. At this point however, the environment-fragment consists of a single c-number $\rho_{\mathcal{F}} = [1]$ or $\rho_{\mathcal{F}} = [0]$, and hence cannot truly satisfy spectrum broadcast structure—on a one-dimensional space, there should only be one POVM element.

The source of deviation from a spectrum broadcast structure of the system-environment state differs between the partial trace methods used. The Pérez trace finds that the system-fragment state tends to be *non-separable*, whereas the staircase trace assumptions find that for small fragments, the states are largely separable but rather *non-distinguishable*. We believe this is due to the differing assumptions of the structure of the environment implicit in the different partial traces.

By using a manageable upper bound on the distance of a state to spectrum broadcast form, we investigated the structure of the system-fragment states. Both partial traces agreed that the system-fragment states did not have spectrum broad-

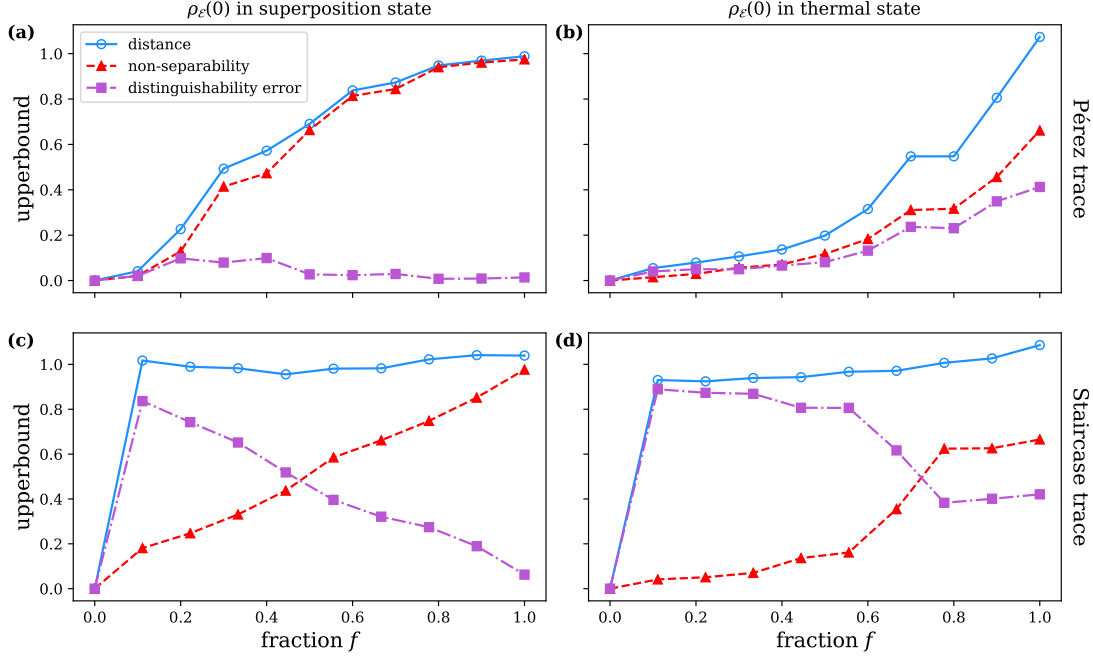


Figure 4.6: **Upper-bounds to the minimum distance of system-fragment states to the set of spectrum broadcast structure states (at time $t = 500$).** The distance $M^{\text{SBS}}(\rho_{\mathcal{SF}})$ is upper-bounded by the summation of the non-separability of the system-fragment state [first term in Eq. (4.25)] and the distinguishability error of the reduced fragment states [second term in Eq. (4.25)]. Overall, the system-fragment states do not have a spectrum broadcast structure. With parameters $\Delta E = 1$, $\delta\varepsilon = \Delta E/\lambda = \Delta E/5$, and $\beta = 10$ for the thermal state (and $\hbar = 1$).

cast structure form, strengthening our prior conclusion that objectivity has not emerged. The minimisation could, in the future, be done via more sophisticated methods than the random search employed here. Nonetheless, our work illustrates its practicality, and the combined investigation of both quantum Darwinism and spectrum broadcasting leads to stronger conclusions on the question of emergent classicality.

4.6 If the reduced state is derived from the system-fragment state

As noted previously, the true state of the system, $\rho_{\mathcal{S}} = \text{tr}_{\mathcal{E}}[\rho_{\mathcal{SE}}]$, cannot be determined from the system-fragment $\rho_{\mathcal{SF}}$ if using the Pérez partial trace. As

such, when we calculated the mutual information in the previous sections, we took $H(\mathcal{S}) = H(\rho_{\mathcal{S}})$ as being the true entropy of the system, determined by the true state of the system.

Arguably, we could have taken $H(\mathcal{S}) = H(\rho'_{\mathcal{S}})$ as being determined by the system state calculated from the reduced system-environment, $\rho'_{\mathcal{S}} = \text{tr}_{\mathcal{F}}[\rho_{\mathcal{SF}}]$. In this section, we show what happens if one had taken this route. Note that the staircase trace produces identical plots to those shown in the main text, *i.e.* the staircase trace recovers the correct system state.

The quantum mutual information plots used in Zurek's quantum Darwinism is given in Fig. 4.7, and the decomposition into accessible information and discord for strong quantum Darwinism is given in Fig. 4.8 (the plot for the distance to spectrum broadcast structure does not employ the system entropy). Qualitatively, similar conclusions can be drawn from these results as in the prior sections, *i.e.* that two different predictions are given by Zurek's quantum Darwinism for the two partial traces, with the Pérez trace predicting objectivity and the staircase trace predicting non-objectivity.

When the system-environment starts off in a pure state [Fig. 4.7(a) and Fig. 4.8(a)], the environment fragment always shares full information with the 'false' system according to the Pérez trace. Explicitly, $\rho_{\mathcal{SF}}$ are pure states under the Pérez trace, and the reduced states $\rho'_{\mathcal{S}}$ and $\rho_{\mathcal{F}}$ are obtained from $\rho_{\mathcal{SF}}$ via the typical trace, hence the known property of pure states [27] applies, *i.e.* that $I(\mathcal{S} : \mathcal{F}) = 2H(\mathcal{S})$. With reference to the discussion in Section 4.3, the fact that the plots are non-symmetric follows. In summary, the results are qualitatively the same, and hence the conclusions are unchanged.

4.7 Summary and conclusions

In this chapter, we have jointly investigated Zurek's quantum Darwinism, strong quantum Darwinism and spectrum broadcasting as frameworks to test the objectivity of a two-level system interacting with an effective N -level environment via a decohering random matrix coupling. We compared two methods to decompose the environment into fragments. First is the partial trace of Pérez [152], which partitions the levels of the environment and eliminates any levels outside the relevant fraction, but has the caveat of the system density matrix can only be truly recovered from the trace of the full environment $\rho_{\mathcal{SE}}$ but not the fraction

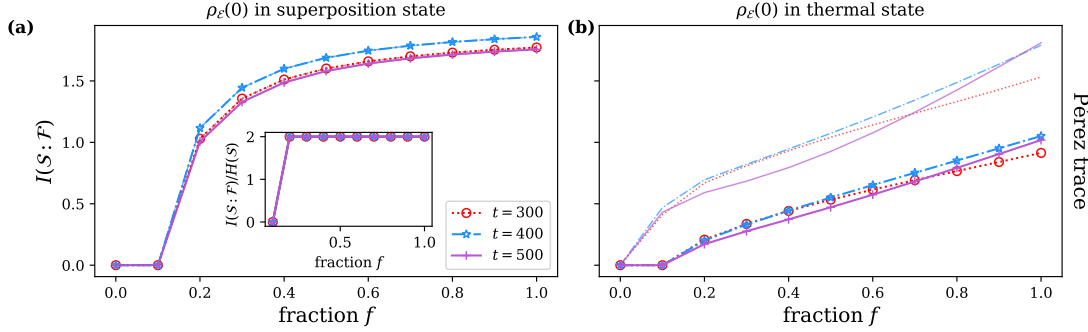


Figure 4.7: **Quantum mutual information $I(\mathcal{S} : \mathcal{F})$ between system and environment fragments using the Pérez partial trace** [Eq. (4.10)] (compare with Fig. 4.4). The Pérez [152] trace is used to form different fractions of the environment. The reduced system is calculated as $\rho'_S = \text{tr}_{\mathcal{F}}[\rho_{S\mathcal{F}}]$. The dashed lines (without markers) corresponds to twice the system entropy $2H(\mathcal{S})$ (the maximum mutual information possible if the full system-environment state is pure). The inset in (a) shows the normalised mutual information, $I(\mathcal{S} : \mathcal{F})/H(\mathcal{S})$. With parameters $\Delta E = 1$, $\delta\epsilon = \Delta E/\lambda$, $\lambda = \Delta E/5$, and $\beta = 10$ for the thermal state (and $\hbar = 1$).

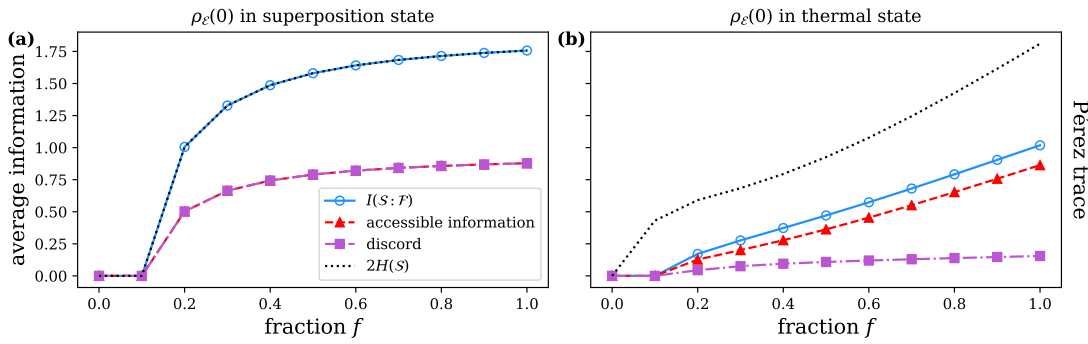


Figure 4.8: **Decomposition of mutual information $I(\mathcal{S} : \mathcal{F})$ into accessible information and quantum discord (at time $t = 500$) using the Pérez partial trace** [Eq. (4.10)] (compare with Fig. 4.5). The accessible information and quantum discord is averaged over all environment fractions of the same size f . The system entropy is $H(\mathcal{S})$, where the reduced system is calculated as $\rho'_S = \text{tr}_{\mathcal{F}}[\rho_{S\mathcal{F}}]$. With parameters $\Delta E = 1$, $\delta\epsilon = \Delta E/\lambda$, $\lambda = \Delta E/5$, and $\beta = 10$ for the thermal state (and $\hbar = 1$).

ρ_{SF} . Our proposed method, the staircase trace, does not have this problem. The staircase trace works on the presumption that each increasing energy level is the excited level of a single particular subenvironment.

Under the two different partial trace methods, we found conflicting conclusions from Zurek’s quantum Darwinism as to whether or not there is objectivity. We then analysed the classical versus quantum nature of the information, as per strong quantum Darwinism, and found that both partial traces now agreed that there was no objectivity—a considerable amount of the shared information was in fact quantum in nature. Using a numerical distance measure, we also find that the system-fragment states did not have spectrum broadcast structure, although the Pérez trace did suggest that the state was *close* to a SBS structure.

In the study of quantum Darwinism, the environment needs to be decomposable: for an environment of N subsystems, this would naively correspond to a Hilbert space of at least 2^N . In this chapter, we show that *can* feasibly to investigate monolithic environments within the quantum Darwinism, by considering environments of N subsystems with a (drastically reduced) Hilbert space of N dimensions that can be motivated by various means. The novel staircase environment trace we introduce has the advantage over the Pérez trace in that it commutes with the traditional partial trace *and* it produces traditional quantum mutual information plots as per ZQD. The staircase environment trace assumed a particular type of structured environment: the spirit of trace construction can be applied to other types of structured environments. For example, we can also construct a partial trace that assumes the environment is composed of $N - 1$ identical two-level systems, with increasing environment energy corresponding to the consecutively exciting multiple different subsystems.

By exploring the decomposition of the quantum mutual information into accessible information and quantum discord, we have been able to illustrate some discrepancies between Zurek’s quantum Darwinism and supposed state objectivity: a “quantum mutual information plateau” (cf. Fig. 3.1) is *not* a sufficient condition for state objectivity. Unlike previous work in the literature, we have also compared the results of Zurek’s quantum Darwinism with that of spectrum broadcasting, along with strong quantum Darwinism. Unlike Zurek’s quantum Darwinism, the nuanced investigation of strong quantum Darwinism and state structure do not give any false positives on the question of objectivity. Hence, future studies of quantum Darwinism should not really consider only Zurek’s

quantum Darwinism: rather, additional analysis of the discord, or alternative analysis of the spectrum broadcasting is required to correctly assess objectivity or lack there-of.

Emergence of objectivity

This chapter includes content from the following preprint:

Thao P. Le, *Quantum discord-breaking and discord-annihilating channels*, (2019), arxiv:1907.05440v1 [3].

The original contributions of this chapter can be found in Sec. 5.2.1 (results about discord-destroying channels), Sec. 5.2.2 (results about objectivity-creating channels), and Sec. 5.3.

We have defined the mathematical requirements for an objective system, *i.e.*, the frameworks of quantum Darwinism. However, *how* does objectivity emerge? How common is its emergence, given the fact that the macroscopic world we perceive appears to be classical and objective? Is the emergence of objectivity generic, or is it highly conditional? In this chapter, we examine the factors that lead to objectivity, and the conditions that hinder it.

Firstly, in Section 5.1, we review previous results in the literature. Not all quantum evolutions lead to the emergence of quantum Darwinism, and there is no complete characterisation of all the dynamics that can ever lead to quantum Darwinism. This is further complicated by the fact that *initial states* of systems and environments can affect whether or not the dynamics will produce end states with objectivity. There are a number of previous works which describe the emergence of objectivity under various assumptions including a sub-component of objectivity [122], or through rigorous averaging over certain classes of initial states and dynamics [123]. Beyond generalist approaches, quantum Darwinism has also

been studied in numerous models, from photon environments [103, 106–108] to gravitational decoherence [76]. It has also been found that certain properties hinder the emergence of quantum Darwinism such as mixed states in the environment [116, 119, 120, 144, 149], and non-Markovian dynamics [1, 112, 136]. Note that experiments so far [146, 147, 170] typically confirm objective states rather than describe a natural emergence *per se*, the review of quantum Darwinism experiments is delayed to Chapter 6.

However, none of these works have described the necessary and sufficient characteristics for quantum dynamics that will always lead to objective states. Hence, in Section 5.2, we characterise the bipartite quantum channels that always output objective states. In Section 5.3, we discuss these results and the properties of bipartite objectivity-creating channels and thus, obtain a better understanding on the factors that lead to perfect and imperfect objectivity.

5.1 Previous works on the emergence of objectivity

Recall the ideal example of emergent objectivity, Example 1, in which a series of CNOT interactions between the system and environment produces an objective state in a finite time:

$$\begin{aligned} & (a|0\rangle_{\mathcal{S}} + b|1\rangle_{\mathcal{S}})|0\rangle_{\mathcal{E}} \\ & \xrightarrow{\hat{U}_{\text{CNOT}}} (a|00\dots 0\rangle_{\mathcal{SE}_1\dots\mathcal{E}_N} + b|11\dots 1\rangle_{\mathcal{SE}_1\dots\mathcal{E}_N}) \otimes |0\rangle_{\text{rest of bath}}. \end{aligned} \quad (5.1)$$

Tracing out the rest of the bath leaves a perfect GHZ state that satisfies objectivity under all the frameworks. Many of the factors that hinder the emergence of objectivity can be related back to deviations from this ideal case. The emergence of objectivity is affected by a number of factors:

1. the initial system-environment states,
2. the system-environment dynamics,
3. the environment-environment dynamics,
4. how we define the individual environments, which each can be a collection of many more smaller sub-environments (“macrofractions”),

5. the time at which we consider the emergence of objectivity,
6. how “close” to perfect objectivity we want,
7. any other coarse-graining or averaging over the prior factors,
8. and the type of objectivity (or framework—ZQD, SQD, SBS, or ISBS) we are considering.

Initial system-environment states are almost always product, *i.e.* initially uncorrelated. Lack of initial system-environment correlations do not hinder the emergence of objectivity. In contrast, Refs. [116, 144] found that the presence of initial system-environment correlations can hinder quantum objectivity: the initial correlations disrupt the “information-spreading” of the system-environment interaction. Furthermore, Ref. [144] also considered initial states of the environment being self-correlated and Refs. [116, 118] considered different environment states that deviate from the perfect $|0\rangle$. These “imperfect” environment states contain initial information, and hence cannot encode as much system information, thus impeding objectivity.

There have been some works that found strong system-environment interactions (*i.e.* non-Markovian dynamics) and memory effects can hinder quantum Darwinism [1, 112, 136, 144, 171–174], including our own work in Chapter 4. Non-Markovianity in part hinders objectivity as it also hinders decoherence, the precursor to objectivity: as information flows back from the environment to the system, the system can re-cohere [49, 171, 175]. Galve *et al.* [112] noted that with non-Markovian dynamics, it is entirely possible that measurements by observers on the environment will change the state of the environment, which would then feed back into the system, hence changing the system. If so, some regimes with non-Markovian dynamics will be unable to sustain “measurements that do not disturb the system” and thus fail objectivity. Furthermore, strong system-environment interactions tend to produce quantum correlations between system and environment, thus disqualifying those states from objectivity under the frameworks of strong quantum Darwinism and spectrum broadcast structure, while strong environment-environment interactions produce correlations between the environments that cause those states to be non-objective according to spectrum broadcast structure [5].

However, not all non-Markovian dynamics are detrimental. Oliveira *et al.* [176] found a nuanced relationship between non-Markovianity and Zurek’s quantum Darwinism: while information back-flow/non-Markovianity reduced the amount of redundancy, the amount of ZQD redundancy was still appreciable even in the presence of non-Markovianity, and that furthermore, the average redundancy (averaged over time) was similar to the scenario with Markovian dynamics. Meanwhile, Lampo *et al.* [115] showed the emergence of states with spectrum broadcast structure under non-Markovian dynamics. We return to the relationship between non-Markovianity and objectivity in Section 5.3.

Aside from the CNOT-like interaction, one of the most straightforward dynamics to produce objective states involves pure dephasing on the system [104, 109, 177–179]. Pure dephasing immediately leads to a diagonal system structure in the preferred (pointer) basis without changing the information about the system in that basis, *i.e.*, retaining the original populations in that pointer basis. With pure dephasing dynamics, an initial pure system qubit, and uncorrelated environment, entanglement generation is generally necessary for the emergence of objectivity [178]. However, there are situations where entanglement is not necessary, if we consider discrete instants in time [179] (*i.e.* whether or not entanglement generation is required depends on the various other factors involved in evaluating objectivity).

Sometimes, the interaction between system and environment can be relatively weak, or there may be other interactions (*e.g.* environment-environment) and factors (initially impure environment states) that reduce the amount of system information that can be encoded into each separate subenvironment. In this situation, the use of *macrofractions* can enhance objectivity [5, 104, 123, 150, 180]: a macrofraction is a collection of subenvironments. Combining multiple subenvironments into a larger environment can strengthen the system-macrofraction correlations and increase the distinguishability of the macrofraction which is crucial for objectivity. Foti *et al.* [181] showed how environments that become classical in the large N limit will act as though it is a measurement on the system regardless of the original particular details of the interactions. Colafranceschi *et al.* [126] showed how quantum systems cannot share quantum correlations with $N \rightarrow \infty$ environments, extending from a result by Brandão *et al.* [122]. Meanwhile, Qi and Ranard [182] have examined the locations of the classical and quantum information in the environment after the application of a generic quantum channel,

finding that quantum information encoded in entanglement is typically localised to an $O(1)$ -sized region, which suggests that the majority of environments have only classical correlations and quantum discord with the system and hence have apparent objectivity under Zurek's quantum Darwinism. Together, these show how increasing the environment size can help lead to classicality.

Distinguishability of environment states has been linked closely with objectivity. Refs. [117, 118] find that for a spin (two-level system) in a spin environment, the redundancy is approximately

$$R_\delta \simeq N \frac{\bar{\xi}_{QCB}}{\ln 1/\delta}, \quad (5.2)$$

where

$$\bar{\xi}_{QCB} = -\ln \left\langle \text{tr} \left[\rho_{\mathcal{E}_j|\uparrow}^c \rho_{\mathcal{E}_j|\downarrow}^{1-c} \right] \right\rangle_{j \in \{1, \dots, N\}}, \quad 0 \leq c \leq 1 \text{ to be maximised}, \quad (5.3)$$

is the quantum Chernoff bound (QCB) [117]. The quantum Chernoff bound gives a measure of distinguishability of the states $\sigma_{j,k=\uparrow} = \rho_{\mathcal{E}_j|\uparrow}$ and $\sigma_{j,k=\downarrow} = \rho_{\mathcal{E}_j|\downarrow}$, which are the sub-environment states conditioned on the system's state $\uparrow$ or $\downarrow$. The greater the redundancy, the greater the expected objectivity, which in turn depends on the conditional state distinguishability. In turn, the distinguishability of these states depends on the initial states and dynamics, and so in general, there is no reason for these conditional states to be distinguishable. Ref. [183] also examined the relationship between state distinguishability and information gain in a different model. The lack of distinguishability is also of key concern when changing reference frames, causing objectivity to be subjective according to reference frame [5, 82].

The *time* at which we measure objectivity is important. In the dynamical setting (*i.e.* under continuous evolution), do we examine the objectivity of the system during the short time, mid time, or long time? In works such as Refs. [115, 116, 119], the states at the long-time/asymptotic limit are examined, showing objectivity. In contrast, the work of Riedel *et al.* [105] found objectivity (according to Zurek's quantum Darwinism) during the mid-time range, and showed how this objectivity eventually *decayed* in the long-time limit. Meanwhile, in Ref. [179], the authors described a dynamic pure dephasing model where objectivity emerges at *instants* in time, but not continuously.

Alternatively, the emergence of objectivity can be considered at an instantaneous point through the quantum channels formalism, which has input and

output states. With additional restrictions, there are a number of generic results regarding the emergence of objectivity. For example, Brandão *et al.* [122] split the definition of objectivity (Definition 1) into *objectivity of observables* and *objectivity of outcomes*:

Definition 9. Objectivity of observables [122]: Different observers that access a quantum system by probing part of its environment can only learn information extractable from a single observable of the system that, although possibly non-unique, is independent of which part of the environment is being probed.

Definition 10. Objectivity of outcomes [122]: Different observers that access different parts of the environment have (close to) full access to the information about the above observable and will agree on the outcome obtained.

The scenario is given in Fig. 5.1: there is an initial system input state, a channel from system to environments, $\Lambda : \text{BL}(\mathcal{H}_S) \rightarrow \text{BL}(\mathcal{H}_{\mathcal{E}_1} \otimes \cdots \otimes \mathcal{H}_{\mathcal{E}_N})$, with output states being the environments at a fixed time. By examining a restricted version of objectivity, Brandão *et al.* [122] proved that objectivity of *observables* emerges regardless of the quantum dynamics, assuming that the system is initially uncorrelated from the environment, and that the environment is large. That is, they prove that for a generic evolution, and a large environment, the effective map onto the environment fragments, $\Lambda_j := \text{tr}_{\setminus \mathcal{E}_j} \circ \Lambda$, is very close to a measure-and-prepare map $\mathcal{M}_j(\rho_S) := \sum_k \text{tr}[M_k \rho_S] \sigma_{j,k}$, where the “measurement” $\{M_k\}$ is the same observable for the different fractions, ρ_S is the *initial* system state and $\{\sigma_{j,k}\}$ are some output states.

Knott *et al.* [125] prove the analogous result for infinite dimensional systems under energy constraints, and improved bounds and further generalisations have been given by Colafranceschi *et al.* [126].

The alternative to fixing a instantaneous time, or a series of instantaneous times, is to average over time. Korbicz *et al.* [123] prove that spectrum broadcast structure emerges in the quantum-measurement limit with a Hamiltonian of form $A \otimes \sum_{k=1}^N B_k$ under suitable averaging over all initial states and all observables and coarse-graining over environments and over time.

Lastly, due to the differences between the frameworks of objectivity, choosing a different framework can lead to a different result, as we examined in back in Chapter 4. Furthermore, we can ask whether we want the environment to contain the *initial* system information, or to contain the current instantaneous system

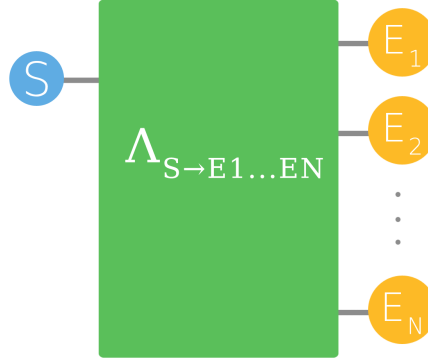


Figure 5.1: **The scenario of Brandão *et al.* [122] that leads to generic emergence of objectivity of observables.** The system $\mathcal{S}$ is initially uncorrelated from the environment, and there is a quantum channel $\Lambda_{S \rightarrow E_1 E_2 \dots E_N}$ that describes the information transmitted from the system to the various subenvironments E_1, E_2 etc. For a single environment E_j , the effective channel is $\Lambda_{S \rightarrow E_j} = \text{tr}_{\setminus E_j} \circ \Lambda_{S \rightarrow E_1 E_2 \dots E_N}$ (obtained by tracing out all the other environments.) Its difference from a measure-and-prepare channel is bounded $\left\| \Lambda_{S \rightarrow E_j} - \sum_k \text{tr}(M_k \rho_S) \sigma_k^{E_j} \right\|_{\diamond} \leq \epsilon$, where ϵ depends on a number of factors including the system and environment dimensions and the number of environments.

information—these are nonequivalent, most easily seen when the system state changes over time [176]. For example, with models of pure dephasing on the system [104, 109, 177–179], the pure dephasing of the system into the basis $\{|i\rangle\}$ typically retains the original system population in this basis and spreads this information into the environment—in these dephasing models, the original system information and the current instantaneous information is the same. Meanwhile, in Ref. [179], the objective system information does *not* lie in the pointer basis/the dephasing basis. In the results of Refs. [122, 125, 126, 182] for example, the quantum channels in their theorems are spreading the initial system information, while in works that involve dynamical evolution (and numerical evolution especially), objectivity is calculated as information between the current system-environment state.

In terms of specific scenarios and models, objectivity, under the frameworks of Zurek’s quantum Darwinism and spectrum broadcast structure, have been explored explicitly in various models, showcasing how those previous factors impact emergent objectivity: in spin-spin models [26, 27, 105, 116–121, 144, 149],

spin-bosons [115], dielectric spheres [103, 106–108], harmonic oscillators [109–114], single N -level environment [1, 152], under QED [184], under gravitational decoherence [76] and under generic interaction-only evolution [104, 123, 124]. There has been some limited experiments with open quantum dots [185–190]. Much more recently, there has also been experimental work with photonic cluster states [147], photonic quantum simulators [146] and nitrogen vacancy centers [170] (which we will discuss more in Chapter 6). Beyond quantum theory, Scandolo *et al.* [191] prove that spectrum broadcast structure also emerges in causal generalised probabilistic theories that have a classical subtheory: using the states of the classical subtheory, spectrum broadcast states can always be constructed. We can also observe how objectivity changes under change of quantum reference frame [5, 82]. Objectivity is subjective in the sense that it is dependent on quantum reference frame: only a small class of states are objective in all frames [5], and otherwise all observers need to share a large, effectively classical frame [82].

In the following section, we will characterise perfect objectivity-creating channels, and introduce yet another understanding of how non-Markovianity relates to the emergence of objectivity.

5.2 Bipartite objectivity-creating quantum channels

In general, the previous studies of the dynamics that lead to objectivity were done with approximation—with averaging, with distances to the objective state, or through numerical simulation. Of course, these are more realistic scenarios, but that also introduces a lot more complexities, such as “*when does the state become objective*”, “*how long does the state remain objective for*” and “*how close to objectivity is close enough*”.

There is yet no complete characterisation of *all* the sets of initial system-environment states and corresponding dynamics that would lead to objectivity. In this section, we approach the characterisation from a different angle: rather than finding sufficiently many conditions that will lead to mostly-objective states for some time period, we consider the quantum channels that will output perfectly objective states regardless of the initial system-environment state. In this way, imperfect objectivity-dynamics could be considered as existing some δ -distance

from the perfect channels.

States with strong quantum Darwinism have zero discord. Hence, objectivity-creating channels must be a subset of discord-destroying channels. In addition to zero discord, however, objective states also have perfect classical correlations. Hence, in Section 5.2.1, we first characterise discord-destroying channels. In Section 5.2.2 we add on the perfect-classical-correlations condition to produce bipartite objectivity-creating channels.

5.2.1 Discord-destroying quantum channels

Quantum entanglement is perhaps *the* most famous quantum correlation, and it has been and continues to be intensively studied [192]. Entanglement is the key feature in many applications, including quantum key distribution and quantum computation [193, 194]. Given its usefulness, this has lead to the study of the mechanisms that hinder and destroy entanglement, all the better to avoid such situations in quantum technologies. For example, in dynamical processes, entanglement might undergo sudden death [195] and/or a sudden birth [196]. In discrete settings, there have been investigations into the robustness of entanglement against added noise [197]. And in particular, there have been characterisations of entanglement-destroying quantum channels [198, 199] which have been thoroughly characterised in closed form, and more recently, there has also been work on entanglement-annihilating channels [200–202].

Entanglement-breaking channels have a clear form: $\Phi_{\mathbf{EB}}$ is entanglement breaking (EB) if $\mathcal{I}^A \otimes \Phi_{\mathbf{EB}}^B(\rho_{AB})$, or equally $\Phi_{\mathbf{EB}}^A \otimes \mathcal{I}^B(\rho_{AB})$, is separable for all initial states ρ_{AB} , *i.e.* the output states can always be written as $\rho_{\text{out}} = \sum_k p_k \rho_{A|k} \otimes \sigma_{B|k}$ for density matrices $\rho_{A|k}, \sigma_{B|k}$. Entanglement-breaking channels can always be written in the following form [198, 199]:

$$\Phi_{\mathbf{EB}}(X) = \sum_k \text{tr}[F_k X] R_k, \quad (5.4)$$

where $F_k \geq 0$ are positive semi-definite, $\{F_k\}$ form a positive operator valued measure (POVM) with $\sum_k F_k = \mathbb{1}$ and $\{R_k\}$ are fixed density states. In contrast, entanglement-*annihilating* channels [200] act across a multipartite system: $\Phi_{\mathbf{EA}}$ is entanglement-annihilating (EA) if $\Phi_{\mathbf{EA}}^{AB}(\rho_{ABC\dots})$ is separable across $A|B|C|\dots$ (or more complicated divisions) for all initial states $\rho_{ABC\dots}$ [200]. There is no closed-

form characterisation of entanglement-annihilating channels, although there has been further study onto their conditions [201, 202].

However, there are multiple quantum effects that are more general than entanglement, including quantum coherence and quantum discord, that are also important in their own respective applications. In fact, discord has also been shown to be useful in tasks such as quantum metrology and parameter estimation [203, 204]. As such, the preservation of quantum discord is also integral to successful quantum applications. There are many analogous studies for quantum discord, from its robustness against noise and sudden death [205], to the characterisation of discord-breaking channels [206, 207].

Quantum discord plays a key role in strong quantum Darwinism, leading to our interest. As such, understanding the mechanism of *loss* of quantum correlations is integral.

Here, we characterise bipartite discord-destroying channels, also known as *discord-annihilating* channels in analogy to entanglement-annihilating channels [200–202]. These channels destroy discord *within* the bipartite system AB they act upon, as opposed to acting locally on one system A to destroy discord between A and B . More generally, we could consider local [135, 206, 207] and nonlocal discord breaking channels, illustrated in Fig. 5.2. More extensive results can be found in our preprint Ref. [3]; here, we only present results directly relevant to the objectivity creating channels later in Section 5.2.2.

Discord-annihilating channels, also known as global discord breaking channels, are nonlocal channels that act across an extended system and break discord between the subsystems. Note that discord is *not* symmetric in general. We must choose a preferred system, say A , and subsequently define zero-discord states as classical-quantum states $\mathbf{CQ}(\mathcal{H}_{AB})$,

$$\rho_{\mathbf{CQ}} = \sum_k p_k |k\rangle\langle k|_A \otimes \rho_{B|k} \in \mathbf{CQ}(\mathcal{H}_{AB}), \quad (5.5)$$

where $\{|k\rangle\}$ is some orthonormal basis on A and $\rho_{B|k}$ are density states. Classical-quantum states can also be written as

$$\rho_{\mathbf{CQ}} = \sum_{ij} A_{ij} \otimes |i\rangle\langle j|_B, \quad (5.6)$$

where A_{ij} are mutually commuting normal operators and $\{|i\rangle\}$ is any orthonormal basis on B [208].

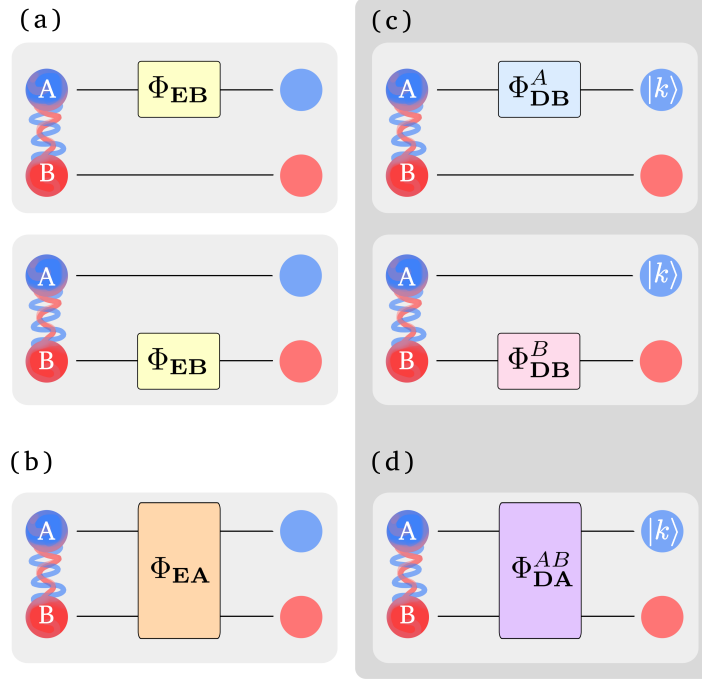


Figure 5.2: **Some quantum correlation destroying channels.** (a) Entanglement-breaking (EB) channels Φ_{EB} act locally and break entanglement with any external system [198, 199]. (b) Entanglement-annihilating (EA) channels Φ_{EA} act on a multipartite system and break entanglement among its subsystems [200–202]. (c) Discord-breaking (DB) channels act upon one of the subsystems to break discord and produce classical-quantum states. Type A channels, Φ_{DB}^A act on the first system, A , to produce the classical subsystem [135, 206, 207]. Type B channels Φ_{DB}^B act on the second system B [3]. (d) Discord-annihilating (DA) Φ_{DA}^{AB} channels, *i.e.* bipartite discord-destroying channels, act upon the bipartite system to break discord within the system (see Theorem 4).

Typically, “basis-dependent quantum discord” is used in works involving quantum discord [151, 209], however our following Lemma shows that there are more general convex restrictions of zero-discord states, which can be described more broadly as being “subspace dependent”:

Lemma 1. *All convex subsets of classical-quantum states $\mathbf{CQ}(\mathcal{H}_{AB})$ can be defined as follows: Let $\{\mathbf{V}_{A|i}\}_i$ be some orthogonal subspaces of density states on A (which do not need to cover the set of all density matrices $\mathbf{BL}(\mathcal{H}_A)$). Let $\mathfrak{J}_1$ describe the indices i ’s where $\mathbf{V}_{A|i}$ is one-dimensional, *i.e.* $\dim(\mathbf{V}_{A|i})|_{i \in \mathfrak{J}_1} = 1$, and let $\mathfrak{J}_2$ describe the indices i ’s where $\dim(\mathbf{V}_{A|i})|_{i \in \mathfrak{J}_2} > 1$. A convex CQ-subset*

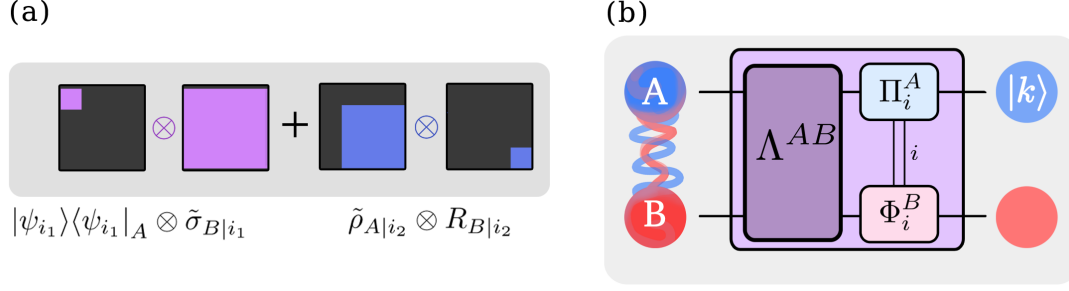


Figure 5.3: **Zero-discord states and discord-annihilating channels.** (a) Representation of a convex subset of zero-discord states, from Lemma 1 and Eq. (5.7). For different indexes i , the conditional states on the A are orthogonal and live in different subspaces. Zero-discord states are formed from the combination of: (1) pure states $|\psi_i\rangle_A$ on A and mixed states $\tilde{\sigma}_{B|i}$ on B ; and (2) mixed states $\tilde{\rho}_{A|i}$ on A and fixed states $R_{B|i}$ on B . (b) Discord-annihilating channels can be decomposed into arbitrary global dynamics Λ^{AB} , followed by projections Π_i^A on A combined with conditional point channels or identity channels Φ_i^B on B . See Theorem 4 and Eq. (5.12).

based on the orthogonal subspace division $\{\mathbf{V}_{A|i}\}_i$ contains states of the form:

$$\rho_{\mathbf{CQ}} = \sum_i t_i \tilde{\rho}_{A|i} \otimes \tilde{\sigma}_{B|i} \quad \text{where } \tilde{\rho}_{A|i} \in \mathbf{V}_{A|i} \text{ and } \tilde{\sigma}_{B|i} \text{ must be fixed for } i \in \mathfrak{I}_2, \quad (5.7)$$

where $\{t_i\}$ are some probabilities.

The full proof of Lemma 1 is given in Appendix B.1 and in our paper Ref. [3]. The proof uses the following necessary condition for zero-discord states (Ref. [210], Proposition 1):

$$[\rho_{\mathbf{CQ}}, \rho_A \otimes \mathbb{1}_B] = 0, \quad \text{where } \rho_A = \text{tr}_B[\rho_{\mathbf{CQ}}]. \quad (5.8)$$

We find that a convex mixture of two zero discord states that also satisfies Eq. (5.8) can be reduced to the form given in Lemma 1.

We can also write Eq. (5.7) as:

$$\sum_{i_1 \in \mathfrak{I}_1} t_{i_1} |\psi_{i_1}\rangle\langle\psi_{i_1}|_A \otimes \tilde{\sigma}_{B|i_1} + \sum_{i_2 \in \mathfrak{I}_2} t_{i_2} \tilde{\rho}_{A|i_2} \otimes R_{B|i_2}, \quad (5.9)$$

where $|\psi_{i_1}\rangle_A$ are the only states in the one-dimensional $\mathbf{V}_{A|i}$ for $i \in \mathfrak{I}_1$, and $R_{B|i_2}$ are the fixed density states. While $\tilde{\sigma}_{B|i_1}$ are generally free, it is also possible to

have fixed conditional states on B with fixed pure states on A , *i.e.* having terms like: $|\psi_i\rangle\langle\psi_i|_A \otimes R_{B|i}$. We depict a zero discord state in Fig. 5.3(a).

Here are some explicit examples of convex classical-quantum state sets:

Example 3. If all the orthogonal subspaces on A correspond to the fixed orthonormal basis $\{|i\rangle_A\}$ on A , then we have the following convex classical-quantum subset:

$$V'_{\text{diag}A} = \left\{ \sum_i p_i |i\rangle\langle i|_A \otimes \rho_{B|i} \left| \begin{array}{l} \sum_i p_i = 1, p_i \geq 0, \\ \rho_{B|i} \in \mathbf{V}_B \end{array} \right. \right\}, \quad (5.10)$$

where $\mathbf{V}_B$ can be any convex subset of density states on B . This is the “basis-dependent quantum discord” that is commonly used.

Example 4. If we have one single subspace $\mathbf{V}_{A|0}$ that has dimension $\dim(\mathbf{V}_{A|0}) \geq 1$, then we can have the following convex classical-quantum subset:

$$V_{\text{fixed}B} = \{ \rho_A \otimes R_B | \rho_A \in \mathbf{V}_{A|0} \}, \quad (5.11)$$

where R_B is a fixed density state.

Now, discord-destroying channels must involve at least one of two actions: projective measurement on subsystem A , or re-preparation of the state on subsystem B . In general, they contain a hybrid combination of both actions:

Theorem 4. A channel Φ_{DA}^{AB} is a discord-annihilating (discord destroying) if and only if it can be written in the following form:

$$\Phi_{DA}^{AB}(\rho_{AB}) = \sum_i \Pi_i^A \otimes \Phi_i^B [\Lambda^{AB}(\rho_{AB})], \quad (5.12)$$

where Λ^{AB} is a CPTP map. The indices i are divided into disjoint sets $\mathfrak{I}_1, \mathfrak{I}_2$: For a fixed orthonormal basis $\{|\psi_i\rangle\}$, the projector channels Π_i^A either project into a one-dimensional subspace, or a multidimensional subspace, all of which are mutually orthogonal:

$$\Pi_i^A(\cdot) = \begin{cases} |\psi_i\rangle\langle\psi_i|_A(\cdot)|\psi_i\rangle\langle\psi_i|_A, & i \in \mathfrak{I}_1 \\ P_{A|i}(\cdot)P_{A|i}, & i \in \mathfrak{I}_2, \end{cases} \quad (5.13)$$

where $P_{A|i}$ are (higher-than-rank-one) projectors into orthogonal subspaces such that $\sum_{i_1 \in \mathfrak{I}_1} |\psi_{i_1}\rangle\langle\psi_{i_1}|_A + \sum_{i_2 \in \mathfrak{I}_2} P_{A|i_2} = \mathbb{1}_A$. Meanwhile, the conditional channels on B are either point channels $\Phi_{\text{point}|i}^B$, or the identity channels $\mathcal{I}$ depending on i :

$$\Phi_i^B(\cdot) = \begin{cases} \mathcal{I}, & i \in \mathfrak{I}_1, \\ \Phi_{\text{point}|i}^B, & i \in \mathfrak{I}_2. \end{cases} \quad (5.14)$$

The proof is given in Appendix B.2 (with a more complicated proof in our paper Ref. [3]). The proof relies on Lemma 1 which characterises all the possible convex subsets of zero-discord classical-quantum states. Since quantum channels are linear, and the set of all states is convex, hence the image of any quantum channel is convex. For discord-destroying channels, this means their image must be a convex subset of zero-discord states. Using this fact, we show that any discord-destroying channel that produces such a convex subset of zero-discord states is invariant under the action of $\sum_i \Pi_i^A \otimes \Phi_i^B(\cdot)$ and hence can be written in the form in Eq. (5.12).

In Fig. 5.3(b), we depict general discord-destroying channels. Operationally, they decompose into some initial arbitrary global dynamics Λ^{AB} , followed by orthogonal-subspace-projections on A . When these subspaces have dimension greater than one, there is classical communication to B leading to a conditional point channel on B . We give two examples of discord-annihilating channels:

Example 5. When the orthogonal subspaces $\{\mathcal{V}_{A|i}\}_i$ are all one-dimensional, the channel reduces to a locally commutativity creating channel on A , with a fixed diagonal basis $\{|i\rangle\langle i|_A\}$:

$$\Phi_{\mathbf{DA}(\text{fixed})}^{AB}(\rho_{AB}) = \sum_i (|i\rangle\langle i|_A \otimes \mathbb{1}_B) [\Lambda^{AB}(\rho_{AB})] (|i\rangle\langle i|_A \otimes \mathbb{1}_B), \quad (5.15)$$

corresponding to arbitrary dynamics followed by a measurement channel on A .

Example 6. When there is just one orthogonal subspace that corresponds to the entire Hilbert space of A , the channel decomposes into a point channel on B and with arbitrary dynamics Λ^{AB} :

$$\Phi_{\mathbf{DA}(\text{prod})}^{AB} = (\mathcal{I}^A \otimes \Phi_{\text{DB}(\text{point})}^B) \circ \Lambda^{AB}, \quad (5.16)$$

where the point channel (“trash and re-prepare”) acts only on state B :

$$\Phi_{\text{DB}(\text{point})}^B(\cdot) = \rho_{B \text{ new}} \text{tr}[(\cdot)]. \quad (5.17)$$

Finally, we present the following result that will return to explicitly the following Sec. 5.3:

Proposition 2. *If Φ^{AB} is discord-destroying, then $\det \hat{\Phi}^{AB} = 0$, where $\hat{\Phi}^{AB}$ is any real representation of the channel.*

Proof. For a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and a set $A \subset \mathbb{R}^n$ that is Lebesgue measurable, then $\lambda(T(A)) = |\det T| \cdot \lambda(A)$, where $\lambda(X)$ denotes the Lebesgue measure of X [211]. So if we take $\hat{\Phi}^{AB}$ to be the real representation of a discord-destroying channel, $\text{BL}(\mathcal{H}_{AB})$ the set of all quantum states (which has nonzero measure), then $\hat{\Phi}^{AB}(\text{BL}(\mathcal{H}_{AB}))$ are a subset of zero-discord states, which have Lebesgue measure zero [210], *i.e.* $\lambda(\hat{\Phi}^{AB}(\text{BL}(\mathcal{H}_{AB}))) = 0$. This implies that $\det \hat{\Phi}^{AB} = 0$ for all discord-destroying channels. $\square$

Channels Λ with a zero determinant, $\det \Lambda = 0$, are singular. This implies that discord-destroying channels do not have a Lindblad form, $\mathcal{L} = \log \Lambda$, in the straightforward sense due to the singularity. Hence, discord-destroying channels are *non-Markovian*. Though, this is further complicated by the fact that it is sometimes possible to construct a logarithm on a suitable complex branch [212, 213] (although, we have not been able to construct a general Lindblad form for the discord-destroying channels without increasing the dimension). If we allow for imperfect channels, then slight perturbation in the channel could produce a Markovian channel [214]. Nonetheless, this non-Markovianity property for the perfect discord-annihilating channels carries over to perfect objectivity-creating channels.

5.2.2 Objectivity-creating channels

The set of objective states under strong quantum Darwinism (Def. 4) is characterised by zero discord and perfect classical correlations between system and environment. Hence, the set of objective states are a subset of zero-discord states, and objectivity-creating channels are a subclass of discord-destroying channels. Thus, to construct the convex subsets of objective states, and to construct objectivity-creating channels, we take the results for the zero-discord states in the previous section and impose upon the condition of perfect classical correlations. Then, non-objectivity destroying—*i.e.* objectivity creating—quantum channels are essentially discord-annihilating channels with broadcasting from the system to the environments.

Lemma 2. *The convex subsets of bipartite objective states have form:*

$$\rho_{S\mathcal{E}}^{\text{SQD}} = \sum_i t_i |\psi_i\rangle\langle\psi_i|_S \otimes \tilde{\sigma}_{\mathcal{E}|i}, \quad (5.18)$$

where $\{|\psi_i\rangle\}_i$ is some orthonormal basis on the system, and $\tilde{\sigma}_{\mathcal{E}|i} \in \mathcal{V}_{\mathcal{E}|i}$, where $\{\mathcal{V}_{\mathcal{E}|i}\}_i$ are orthogonal subspaces of the environment $\mathcal{E}$.

Proof. Objective states have zero-discord, hence we can consider zero-discord states from Lemma 1 (taking $A \rightarrow \mathcal{S}$ and $B \rightarrow \mathcal{E}$). The components with $i \in \mathcal{I}_2$: $\sum_{i \in \mathcal{I}_2} t_i \tilde{\rho}_{\mathcal{S}|i} \otimes \tilde{\sigma}_{\mathcal{E}|i}$, where $\dim(\mathcal{V}_{\mathcal{S}|i \in \mathcal{I}_2}) > 1$, $\tilde{\rho}_{\mathcal{S}|i} \in \mathcal{V}_{\mathcal{S}|i}$, $\tilde{\sigma}_{\mathcal{E}|i}$ fixed, are *not* objective, since $\tilde{\rho}_{\mathcal{S}|i}$ has rank two or greater, meaning that it contains two or more eigenstates, yet are correlated with the same fixed state $\tilde{\sigma}_{\mathcal{E}|i \in \mathcal{I}_2}$, the latter of which is *not* distinguishable for the different eigenstates of $\tilde{\rho}_{\mathcal{S}|i}$. This reduces us to the fixed basis $\{|\psi_i\rangle\langle\psi_i|\}$ on $\mathcal{S}$, and the subsequent immediate condition from strong quantum Darwinism is that $\tilde{\sigma}_{\mathcal{E}|i}, \tilde{\sigma}_{\mathcal{E}|j}$ for $i \neq j$ are all mutually distinguishable, *i.e.* they are orthogonal and exist effectively on disjoint, orthogonal subspaces, which we can then define as $\{\mathcal{V}_{\mathcal{E}|i}\}$. $\square$

Now, we can define non-objectivity-destroying channels, *i.e.*, objectivity-creating channels:

Proposition 3. *Bipartite objectivity-creating channels have form:*

$$\Phi_{\mathcal{S}QD}^{\mathcal{SE}}(\rho_{\mathcal{SE}}) = \sum_i \Pi_i^{\mathcal{S}} \otimes \Phi_i^{\mathcal{E}}[\Lambda^{\mathcal{SE}}(\rho_{\mathcal{SE}})], \quad (5.19)$$

where $\Lambda^{\mathcal{SE}}$ is an arbitrary CPTP map. For a fixed orthonormal basis $\{|\psi_i\rangle\}$, $\sum_i |\psi_i\rangle\langle\psi_i|_{\mathcal{S}} = \mathbb{1}_{\mathcal{S}}$, the projector channels $\Pi_i^{\mathcal{S}}$ project into a one-dimensional subspace:

$$\Pi_i^{\mathcal{S}}(\cdot) = |\psi_i\rangle\langle\psi_i|_{\mathcal{S}}(\cdot)|\psi_i\rangle\langle\psi_i|_{\mathcal{S}}. \quad (5.20)$$

Meanwhile, the conditional channels on the environment $\mathcal{E}$, $\Phi_i^{\mathcal{E}} : \text{BL}(\mathcal{H}_{\mathcal{E}}) \rightarrow \mathcal{V}_{\mathcal{E}|i}$ must map onto mutually orthogonal subspace $\mathcal{V}_{\mathcal{E}|i}$ for each i .

Proof. This is immediate from the previous Lemma 2 combined with the discord-destroying channels in Theorem 4. Firstly, we cannot have the indices $i \in \mathcal{I}_2$, as $\Pi_i^{\mathcal{S}}$ must project onto one-dimensional subspace. Secondly, instead of having identity channels on the environment $\mathcal{E}$ when $i \in \mathcal{I}_1$, those $\Phi_i^{\mathcal{E}}(\cdot)$ are required to project onto disjoint mutually orthogonal subspaces $\mathcal{V}_{\mathcal{E}|i}$ so that the conditional states on the environment satisfy bipartite spectrum broadcast structure/strong quantum Darwinism. $\square$

Bipartite objectivity-creating channels have an operational interpretation: after measurement on the system $\mathcal{S}$, there is classical communication to the

environment $\mathcal{E}$ followed by a local mapping of the environment state into a distinguishable subspace.

Example 7. Some quantum channels on the environment that would map into a subspace include: point channels $\Phi_{\text{point}}^{\mathcal{E}}(X) = \text{tr}[X]\sigma_{\mathcal{E}}$, and entanglement-breaking channels with form

$$\Phi_i^{\mathcal{E}}(X) = \sum_k \text{tr}[F_k X] \tilde{\sigma}_{\mathcal{E}|i,k}, \quad \text{where } \tilde{\sigma}_{\mathcal{E}|i,k} \in \mathcal{V}_{\mathcal{E}|i}, \sum_k F_k = \mathbb{1}_{\mathcal{E}}. \quad (5.21)$$

Note that if the $\Phi_i^{\mathcal{E}}(\cdot)$ of Eq. (5.19) were projection operators, then the full map $\Phi_{\text{SQD}}^{\mathcal{SE}}(\rho_{SE})$ would not be trace-preserving.

Example 8. We could instead have the “measurement” on the environment $\mathcal{E}$: the following is an objectivity-creating channel that has measurement on $\mathcal{E}$ with conditional point channels on the system:

$$\Phi_{\text{SQD}}^{\mathcal{SE}}(\rho_{SE}) = \sum_i \Phi_{\text{point}|i}^{\mathcal{S}} \otimes \Pi_i^{\mathcal{E}} [\Lambda^{\mathcal{SE}}(\rho_{SE})], \quad (5.22)$$

where $\Phi_{\text{point}|i}^{\mathcal{S}}(\cdot) = |\psi_i\rangle\langle\psi_i|_{\mathcal{S}} \text{tr}[\cdot]$ are point channels where output is a single pure state, and $\Pi_i^{\mathcal{E}}(\cdot) = P_{\mathcal{E}|i}(\cdot)P_{\mathcal{E}|i}$ project onto $\mathcal{V}_{\mathcal{E}|i}$ for suitable projection operators $P_{\mathcal{E}|i}$.

Technically, all forms of quantum Darwinism require multiple environments. An example of an objectivity-creating trace-preserving channel that would produce multipartite objective output states is

$$\Phi_{\text{SQD}}^{\mathcal{SE}}(\rho_{SE}) = \sum_i \Pi_i^{\mathcal{S}} \otimes \Phi_i^{\mathcal{E}_1} \otimes \Phi_i^{\mathcal{E}_2} \otimes \dots \otimes \Phi_i^{\mathcal{E}_N} [\Lambda^{\mathcal{SE}}(\rho_{SE})], \quad (5.23)$$

where each set of conditional environment channels $\{\Phi_i^{\mathcal{E}_k}\}_i$ map onto the orthogonal subspaces $\{\mathcal{V}_{\mathcal{E}_k|i}\}$.

All these channels rely on projecting the system and observed environment into an objective subspace: in Sec. 6.1, we will be describing a subspace-dependent version of strong quantum Darwinism.

5.3 The effect of (non-)Markovianity on the emergence of objectivity

Markovian evolution typically occurs in scenarios with weak system-environment interactions, a spectrally flat environment, and the Markov approximation (a “memory-less” assumption, or vanishing environmental correlation time), and such evolutions can be described with a Lindblad master equation [36, 37]. In contrast, non-Markovianity occurs in all other scenarios, such as when there are strong system-environment interactions, or if the environment can retain “memory” of past interactions and feed that information into future interactions [50].¹ Non-Markovianity is often said to hinder quantum Darwinism: this has been found in papers including Refs. [1, 112, 136, 172]. In contrast, Refs. [115, 176] studied situations where objectivity emerged despite the present non-Markovianity. Hence, non-Markovianity alone is not sufficient to hinder quantum Darwinism.

Using the objectivity-creating channels we have characterised, we further argue that non-Markovianity can be *necessary* for the emergence of objectivity. In Proposition 2 and the discussion following it, discord-destroying channels are non-Markovian, and hence (by the same proof) objectivity-creating channels are non-Markovian. Note that non-Markovianity is not sufficient, as there are numerous non-Markovian channels that are not objectivity-creating.

In particular, Proposition 2 implies that *perfect* objectivity at *finite* time for *all possible initial system-environment* states requires non-Markovianity.

Remark 1. For finite/bounded dynamics and finite time, objectivity-creating channels *cannot* have a Lindblad generator.

Proof. In general, suppose we have a time-dependent super operator $\mathcal{L}_\tau$ that is to describe the dynamics of our discord-destroying or objectivity creating channel Φ , such that at a *finite* time T , $\Phi = \exp\left[\int_0^T \mathcal{L}_\tau d\tau\right]$. Using the fact that $\det(e^A) = e^{\text{tr}[A]}$,

$$\det \Phi = \exp\left[\text{tr}\left[\int_0^T \mathcal{L}_\tau d\tau\right]\right] \stackrel{!}{=} 0, \quad (5.24)$$

where $\det \Phi = 0$ comes from Proposition 2. This is only possible if $\text{tr}\left[\int_0^T \mathcal{L}_\tau d\tau\right] = -\infty$, which is only possible if $\mathcal{L}_\tau$ is unbounded—or if $T = \infty$. Hence, for bounded

¹Note that there is no agreed upon definition for quantum Markovianity nor quantum non-Markovianity [50].

operators and finite time, there does not exist a time-dependent super operator $\mathcal{L}_\tau$ that produces the *perfect* objectivity-creating dynamics—and hence there would not exist a Lindblad master equation. $\square$

Also note that ideal projective measurements cannot be conducted with finite resources [215], and the objectivity-creating channels in fact project system-environment states into an objective space.

One *could* construct Markovian dynamics to reproduce the objectivity-creating channels—but with the caveat that perfect objectivity is only achieved at infinite time, or conversely we can only obtain imperfect objectivity at finite time.

For example, Ref. [216] shows that any quantum channel can be simulated by a Lindblad generator $\mathcal{L}$ in the *infinite time* limit, where note that here, the generator $\mathcal{L}$ acts on an enlarged space. That is, if the system-environment total dimension is d , then there is a $d + d$ Lindblad generator that takes input states in $\begin{pmatrix} 0 & 0 \\ 0 & X \end{pmatrix}$ to $\begin{pmatrix} X & 0 \\ 0 & 0 \end{pmatrix}$:

$$\frac{d\rho_{2d}}{dt} = \mathcal{L}(\rho_{2d}) = \kappa_{eff} \sum_k F_k \rho_{2d} F_k^\dagger - \frac{1}{2} \left\{ F_k^\dagger F_k, \rho_{2d} \right\}. \quad (5.25)$$

with some factor κ_{eff} and Lindblad jump operators $F_k = \begin{pmatrix} 0 & E_k \\ 0 & 0 \end{pmatrix}$, where our objectivity-creating channel is written as $\Phi(\cdot) = \sum E_k(\cdot)E_k^\dagger$ with Kraus operators $\{E_k\}_k$.

Now, if we recall Example 1, a CNOT interaction between the system and environment produces an objective state in a finite time, which is Markovian within the combined system-environment:

$$\begin{aligned} & (a|0\rangle_{\mathcal{S}} + b|1\rangle_{\mathcal{S}})|0\rangle_{\mathcal{E}} \\ & \xrightarrow{\hat{U}_{CNOT}} (a|00\dots 0\rangle_{\mathcal{S}\mathcal{E}_1\dots\mathcal{E}_N} + b|11\dots 1\rangle_{\mathcal{S}\mathcal{E}_1\dots\mathcal{E}_N}) \otimes |0\rangle_{\text{rest of bath}}. \end{aligned} \quad (5.26)$$

However, its Markovianity does not contradict our statements about necessary non-Markovianity for emergent objectivity for all initial states at finite time. This is because the initial state on the environment is $|0\rangle$. This CNOT interaction will *not* work for *all possible* initial system-environment states. We would instead need to initially re-prepare the environment states into $|0\rangle$ *before* the CNOT interaction.

However, this collapse into the point state $|0\rangle$ can be viewed as a projection (measurement) on the environment, which happens perfectly only at the $t \rightarrow \infty$ limit if the process is Markovian [217], or, for example, involves swapping the old environment states with some special $|0\rangle$ states from the unobserved environment.

On the converse, *Markovianity* is not a predictor for the emergence of objectivity either. In the following trivial example, Markovianity can hinder objectivity:

Example 9. Suppose the system-environment dynamics is given by $U_{SE}^t = U_S^t \otimes U_E^t$. If the initial system-environment state is uncorrelated, $\rho_{SE}(0) = \rho_S(0) \otimes \rho_E(0)$, then any later state is also uncorrelated,

$$\rho_{SE}(t) = (U_S^t \otimes U_E^t) \rho_{SE}(0) (U_S^t \otimes U_E^t)^\dagger = \rho_S(t) \otimes \rho_E(t). \quad (5.27)$$

There is no quantum Darwinism in general, and the local unitary dynamics satisfy quantum Markovianity. Objectivity can arise only if the system is pure, which is a trivial case of objectivity.

Thus, if we combine past results studying the effect of non-Markovianity on the emergence of objectivity [1, 112, 115, 136, 172, 176] with the results of this chapter, we come to the conclusion that using the terms “Markovianity” or “non-Markovianity” is insufficient to predict whether or not objectivity will emerge. Instead, the emergence of objectivity is dependent on more fundamental, underlying details of system-environment interactions, the various initial states of the system and environments, the time scale in which objectivity will emerge, and any other manipulations such as coarse-graining and averaging.

While we perceive the macroscopic world around us to be objective, the emergence of objectivity from quantum dynamics is not simple. If we choose some dynamics at random, objectivity is unlikely, similar to how the set of incoherent states is very small in comparison to the space of all quantum states. In this chapter, we examined the multitude of factors that contribute to the emergence of objectivity, developing the understanding that no single property is sufficient to guarantee objectivity. In particular, we showed how both Markovian and non-Markovian dynamics can contribute to objectivity. We also noted the numerous studies in the literature that show the emergence of objectivity in diverse scenarios, through analytical and numerical simulation methods. The next step then is to explore the emergence of objectivity in the experimental laboratory: under

realistic settings, what types of initial states and interactions are favoured? Are these precisely the ones that lead to objectivity? In the next chapter, we propose an experimental witness that can allow us to analyse objectivity in systems with larger environments, thus contributing a step to the experimental verification of emergent objectivity.

Witnessing non-objectivity

This chapter includes content from the following publication:

Thao P. Le and Alexandra Olaya-Castro, *Witnessing non-objectivity in the framework of strong quantum Darwinism*, Quantum Sci. Technol. 5, 045012 (2020) [4].

The original contributions are: all sections 6.1 onwards.

Quantum Darwinism has been well explored theoretically and with numerical simulations (cf. Section 5.1). However, there have been only a handful of experimental tests. Previously, there were a number of open quantum dots experiments whose transport properties were linked indirectly with quantum Darwinism [185–190], but information quantities related to quantum Darwinism (*e.g.* the quantum mutual information or the system entropy) were never extracted. More recently, advances in experimental quantum photonics and spin system control has led to three experiments specifically exploring quantum Darwinism in photonic cluster states [147], photonic quantum simulators [146] and nitrogen vacancy centers (and their surround nuclear spins) [170]. In the two photonic experiments [146, 147], full quantum state tomography of the system-environment state is used to fully characterise their (non-)objectivity. The main caveat to these experiments is that they created states with known (non-)objectivity first, and subsequently measured these known states—there was no “natural” evolution. In contrast, in the nitrogen vacancy experiment [170], the spins were able to undergo their natural evolution, which brings us closer to understanding the natural quantum-

to-classical transition in realistic matter systems. However, only the classical information (Holevo information) was determined—which is not sufficient for Zurek’s quantum Darwinism [28] nor any of the stronger frameworks such as strong quantum Darwinism or spectrum broadcast structure. Zurek’s quantum Darwinism requires the quantum mutual information, and the latter frameworks require the quantum discord. In this chapter, we want to develop a pathway to experimental exploration of strong quantum Darwinism.

Experimental tests are hindered by the requirement of full quantum state tomography to completely characterise objectivity in the system and environment (in any of the frameworks of quantum Darwinism). Quantum Darwinism is aimed at describing large and realistic environments, and this means experimental tests that can fully probe quantum Darwinism are extremely challenging. One method is to accept some allowable error and use a more efficient tomography scheme [218–221]. Alternatively, we can develop and use simplified schemes targeted towards measuring objectivity. This problem is closely linked to the difficulties in characterising quantum entanglement and other quantum correlations [222, 223]. There, one solution has been entanglement witnesses: operators and schemes which detect non-classicality much more simply, albeit at the price that a single witness alone is not capable of detecting all entangled states [224]. The most famous witness are the Bell inequalities [225]—mathematical statements about measurement statistics that are satisfied by a classical theory but can be broken under some forms of quantum entanglement [226].

Here, we introduce a non-objectivity witness in the strong quantum Darwinism framework. For this witness to work, we choose a preferred objective subspace, analogous to the preferred basis in quantum coherence. Our scheme detects non-objectivity by comparing the evolution of the system-environment state with and without objectivity-enforcing operations. To motivate experimental tests of our scheme, we also present a quantum photonic simulation based on our scheme. By comparing the number of measurements needed for state tomography versus those needed for the proposed witness, we show that, with sufficiently good components, our scheme provides a significant advantage. Thus, the witness scheme we present can further advance the experimental testing of quantum Darwinism, and the understanding the quantum-to-classical transition in terms of objectivity. In particular, there are systems that operate at the quantum-classical boundary [227, 228], and it is important to understand whether their

nonobjectivity leads to any operational advantages—and whether we can apply those discoveries to near term quantum technologies.

This chapter is organised as follows: in Section 6.1 we introduce the basis-subspace dependent version of strong quantum Darwinism. In Section 6.2 we describe the non-objective witness scheme. In Section 6.3 we propose a quantum photonic experiment and provide numerical simulation results. In Section 6.4, we consider the simplified situation when witnessing non-objectivity in invariant spectrum broadcast structure. We conclude in Section 6.5.

6.1 Subspace-dependent strong quantum Darwinism

If we have a particular state with strong quantum Darwinism, we can immediately construct a convex subset around it, like the convex subsets in Lemma 2. Here, we define subspace-dependent strong quantum Darwinism, which picks out a convex subset of objective states, and hence is closed under convex combinations. This convexity will allow us to build our witness in the next section. The restriction of the system basis and environmental subspaces is motivated by how environments and systems tend to have a limited set of bases in which we typically measure (*e.g.* position or energy). Subspace-dependent strong quantum Darwinism contains elements of basis-dependent discord [229, 230], and of basis-dependent quantum coherence [23, 230].

Recall that strong quantum Darwinism is equivalent to *bipartite* spectrum broadcast structure (Corollary 1). That is, if the state on $\mathcal{SF} = \mathcal{S}\mathcal{E}_1 \cdots \mathcal{E}_F$ has strong quantum Darwinism, then the system $\mathcal{S}$ with the full fragment $\mathcal{F}$ satisfies bipartite spectrum broadcast structure

$$\rho_{\mathcal{SF}} = \sum_i p_i |i\rangle\langle i| \otimes \rho_{\mathcal{F}|i}, \quad \rho_{\mathcal{F}|i} \rho_{\mathcal{F}|j} = 0 \quad \forall i \neq j, \quad (6.1)$$

and the system $\mathcal{S}$ with the individual environments $\mathcal{E}_k$ also satisfy bipartite spectrum broadcast structure:

$$\rho_{\mathcal{SE}_k} = \sum_i p_i |i\rangle\langle i| \otimes \rho_{\mathcal{E}_k|i}, \quad \rho_{\mathcal{E}_k|i} \rho_{\mathcal{E}_k|j} = 0 \quad \forall i \neq j, \text{ for each } k = 1, \dots, F. \quad (6.2)$$

We will be fixing the subspace of the various $\{\rho_{\mathcal{F}|i}\}_i$ to produce subspace-dependent strong quantum Darwinism as follows:

As the different conditional fragment states $\{\rho_{\mathcal{F}|i}\}_i$ are orthogonal, we can define disjoint subspaces $\{\mathcal{H}_{\mathcal{F}|i}\}_i$ in which they lie. Similarly, the different conditional sub-environment states $\{\rho_{\mathcal{E}_k|i}\}_i$ are orthogonal and we can also define the disjoint subspaces $\{\mathcal{H}_{\mathcal{E}_k|i}\}_i$. Due to the state structure above, the conditional disjoint subspaces of $\mathcal{F}$ will comprise of a tensor product of the conditional disjoint subspaces in $\mathcal{E}_k$:

$$\mathcal{H}_{\mathcal{F}|i} = \mathcal{H}_{\mathcal{E}_1|i} \otimes \mathcal{H}_{\mathcal{E}_2|i} \otimes \cdots \otimes \mathcal{H}_{\mathcal{E}_F|i}. \quad (6.3)$$

Let $\Pi_{\mathcal{E}_k|i}$ be the projector into the subspace $\mathcal{H}_{\mathcal{E}_k|i}$ such that $\Pi_{\mathcal{E}_k|i}\rho_{\mathcal{E}_k}\Pi_{\mathcal{E}_k|i} \in \mathcal{H}_{\mathcal{E}_k|i}$ for all $\rho_{\mathcal{E}_k}$. The projector onto the tensor product space $\mathcal{H}_{\mathcal{F}|i}$ is simply

$$\Pi_{\mathcal{F}|i} = \Pi_{\mathcal{E}_1|i} \otimes \cdots \otimes \Pi_{\mathcal{E}_F|i}. \quad (6.4)$$

The action of this projector $\Pi_{\mathcal{F}|i}\rho_{\mathcal{F}}\Pi_{\mathcal{F}|i}$ preserves some correlations between the environment states in general, since the projectors $\{\Pi_{\mathcal{E}_k|i}\}_i$ can have rank greater than one. This is *allowed* and expected within the framework of strong quantum Darwinism. In contrast, these correlations are not allowed in spectrum broadcast structure.

Now, in a subspace-dependent version of strong quantum Darwinism, we define the preferred basis $\{|i\rangle_S\}_i$ on the system, and we define the preferred objective subspace partitioning for the environments, which we encode in the projectors $\{\Pi_{\mathcal{F}|i}\}_i$ from Eq. (6.4). The following *objectivity operation* projects any input state into a state satisfying strong quantum Darwinistic state, in the fixed subspaces as defined:

$$\Gamma_{\mathcal{SF}}^{\text{SQD}}(\rho) = \sum_i (|i\rangle\langle i|_S \otimes \Pi_{\mathcal{F}|i} \otimes \mathbb{1}_{\mathcal{E}\setminus\mathcal{F}}) \rho (|i\rangle\langle i|_S \otimes \Pi_{\mathcal{F}|i} \otimes \mathbb{1}_{\mathcal{E}\setminus\mathcal{F}}). \quad (6.5)$$

This is comparable to the discord-breaking measurement in Ref. [231], or the decoherence operation in Ref. [232]. Unlike the objectivity creating channels in Sec. 5.2.2, the objectivity operation is not trace preserving.

From this, we can define a measure of subspace-dependent non-objectivity using the trace norm distance:

$$M^{\text{SQD}}(\rho_{\mathcal{SF}}(t)) = \left\| \rho_{\mathcal{SF}}(t) - \Gamma_{\mathcal{SF}}^{\text{SQD}}(\rho_{\mathcal{SF}}(t)) \right\|_1 \leq 1, \quad (6.6)$$

The maximum value of the measure is $\max_{\rho_{\mathcal{SF}}(t)} M^{\text{SQD}}(\rho_{\mathcal{SF}}(t)) = 1$ and occurs when the system-fragment is completely nonobjective. The typical trace-norm distance $\|\rho - \sigma\|_1$ upper-bound (of two) occurs when ρ and σ have orthogonal

support: here, however, the support is such that $\text{supp}\{\Gamma_{\mathcal{SF}}^{\text{SQD}}(\rho)\} \subseteq \text{supp}\{\rho\}$ due to the nature of the objectivity operation. Therefore orthogonality between the two terms only occurs when $\Gamma_{\mathcal{SF}}^{\text{SQD}}(\rho_{\mathcal{SF}}(t)) = 0$.

In the following section, we will employ the objectivity operation to create a witness that lower bounds the value of the measure in Eq. (6.6).

6.2 Witnessing non-objectivity

We will construct a witness that detect non-objectivity between a system and some collection of subenvironments at some specific time t . The witness lower-bounds the amount of non-objectivity, such that a non-zero witness implies non-objectivity relative to our predefined basis and objective subspaces. The witness scheme, illustrated in Fig. 6.1, works by comparing the evolutions of a disturbed versus undisturbed system-environment state. This method of comparing two evolutions is similar to previous works on witnessing quantum discord [231] and quantum coherence [232].

Note that there are other methods for alternative potential witnesses. For example, we could have two-part witness separately measuring the classical correlations, and witnessing the quantum discord [231, 233–236]. This method will not produce a single witness value for non-objectivity: as non-objectivity scales as $\mathcal{D} - \chi$ (Sec. 3.4) *i.e.* with opposing dependence on quantum and classical information whilst experimental witnesses tend to lower-bound the information.

Another possibility is to use a traditional witness operator W leading to values $\text{tr}[W\rho]$. However, we expect this to require multiple joint copies of the state at the same time, *e.g.* $\rho_{\mathcal{SE}}^{\otimes 4}$ in the discord witness of Ref. [233], which would make the experimental scheme more cumbersome given the already large dimensions of the environment.

The system-environment evolution proceeds as follows: At time 0, the system and environment starts in some initial state $\rho_{\mathcal{SE}}(0)$. The full system-environment then evolves under the action of a unitary $U(t)$ til time t , or some preparation procedure, such that the state at time t is $\rho_{\mathcal{SE}}(t)$.

$$\rho_{\mathcal{SE}}(t) = U(t)\rho_{\mathcal{SE}}(0)U^\dagger(t) =: \mathcal{U}_t[\rho_{\mathcal{SE}}(0)]. \quad (6.7)$$

Our goal is to witness non-objectivity between the system and some fragment $\mathcal{F} \subseteq \mathcal{E}$ at time t , *i.e.* to witness the non-objectivity of the state $\rho_{\mathcal{SF}}(t) = \text{tr}_{\mathcal{E}\setminus\mathcal{F}}[\rho_{\mathcal{SE}}(t)]$.

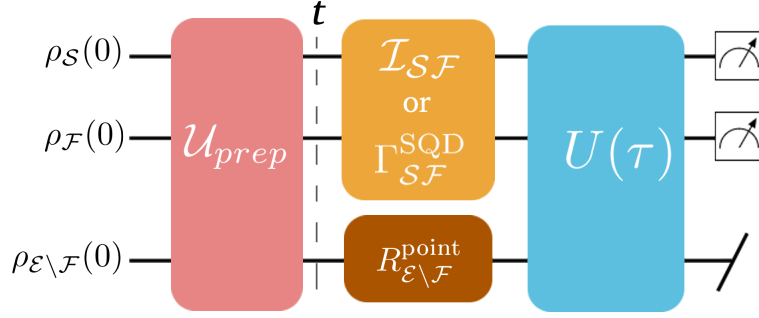


Figure 6.1: **Protocol for the non-objectivity witness:** the system-environment is prepared into some state $\rho_{SE}(t) = \mathcal{U}_{prep}(\rho_{SE}(0))$. A point channel $R_{E\setminus\mathcal{F}}^{\text{point}}$ [Eq. (6.8)] is applied on the “un-accessed” environment $E\setminus\mathcal{F}$ to ensure that the witness does not detect extraneous correlations. We can either leave the system-fragment $\mathcal{SF}$ untouched (identity channel $\mathcal{I}_{\mathcal{SF}}$) or apply the objectivity operation $\Gamma_{\mathcal{SF}}^{\text{SQD}}$ [Eq. (6.5)]. The system-environment then undergoes some unitary evolution $U(\tau)$, and is measured at the end of the protocol. The witness comes from the comparison between the final state with or without the objectivity operation.

To ensure that the witness does not pick up on extraneous correlations between the observed system-fragment and the rest of the environment, we use a point channel on the remainder environment, which discards the $\mathcal{E}_{F+1} \dots \mathcal{E}_N$ states and prepares a new (uncorrelated) state.

$$R_{E\setminus\mathcal{F}}^{\text{point}}(\rho_{SE_1 \dots E_N}(t)) = \text{tr}_{E\setminus\mathcal{F}}[\rho_{SE_1 \dots E_N}(t)] \otimes \rho_{E\setminus\mathcal{F}}^{\text{new}} = \rho_{\mathcal{SF}}(t) \otimes \rho_{E\setminus\mathcal{F}}^{\text{new}}. \quad (6.8)$$

Note that the point channel is crucial to isolate the correlations we want to test. For example, the authors of Ref. [232] profess an ambiguity in one of their witnesses as to whether it is detecting system coherences or system-environment correlations. This ambiguity would be removed with the addition of a correlation breaking channel, as we have done here.

If the point channel is the only operation that we enact at time t , then the system-environment subsequently evolves for duration τ as:

$$\rho_{SE}(t + \tau) = \mathcal{U}_\tau \circ \left(\mathbb{1}_{\mathcal{SF}} \otimes R_{E\setminus\mathcal{F}}^{\text{point}} \right) [\rho_{SE}(t)] = \mathcal{U}_\tau [\rho_{\mathcal{SF}}(t) \otimes \rho_{E\setminus\mathcal{F}}^{\text{new}}], \quad (6.9)$$

under some unitary evolution $U(\tau)$. At time $t + \tau$, we conduct some local measurement $Q_{SE} = Q_S \otimes Q_{E_1} \otimes \dots \otimes Q_{E_N}$, giving us the probability:

$$P_{\mathbb{1}_{\mathcal{SF}}} = \text{tr}[Q_{SE} \rho_{SE}(t + \tau)]. \quad (6.10)$$

One possible measurement operator $Q_{S\mathcal{E}}$ is simply the projector onto zero: $Q_{S\mathcal{E}}^{e.g.} = |0\rangle\langle 0| \otimes |0\rangle\langle 0| \otimes \cdots \otimes |0\rangle\langle 0|$.

In an alternative evolution, we could *also* apply the objectivity operation from Eq. (6.5) on the system-fragment at time t in conjunction with the point channel, and so the subsequent final state is:

$$\rho'_{S\mathcal{E}}(t + \tau) = \mathcal{U}_\tau \left[\Gamma_{S\mathcal{F}}^{\text{SQD}}(\rho_{S\mathcal{F}}(t)) \otimes \rho_{\mathcal{E}\setminus\mathcal{F}}^{\text{new}} \right], \quad (6.11)$$

leading to the alternative probability of measurement $Q_{S\mathcal{E}}$ at time $t + \tau$:

$$P_{\Gamma_{S\mathcal{F}}^{\text{SQD}}} = \text{tr}[Q_{S\mathcal{E}} \rho'_{S\mathcal{E}}(t + \tau)]. \quad (6.12)$$

The absolute difference between these probabilities (with or without the objectivity operation) gives our witness for non-objectivity:

$$W^{\text{SQD}}(Q_{S\mathcal{E}}) = \left| P_{1_{S\mathcal{F}}} - P_{\Gamma_{S\mathcal{F}}^{\text{SQD}}} \right| \quad (6.13)$$

$$= \left| \text{tr} \left[Q_{S\mathcal{E}} \mathcal{U}_\tau \left[\left\{ \rho_{S\mathcal{F}}(t) - \Gamma_{S\mathcal{F}}^{\text{SQD}}(\rho_{S\mathcal{F}}(t)) \right\} \otimes \rho_{\mathcal{E}\setminus\mathcal{F}}^{\text{new}} \right] \right] \right|. \quad (6.14)$$

The witness is a lower bound of the subspace-dependent distance measure. This can be shown as follows: if we maximised over measurement operators in the witness, then

$$W^{\text{SQD}}(Q_{S\mathcal{E}}) \leq \max_{Q_{S\mathcal{E}}} W^{\text{SQD}}(Q_{S\mathcal{E}}) \quad (6.15)$$

$$= \max_{Q'_{S\mathcal{E}}} \left| \text{tr} \left[Q'_{S\mathcal{E}} \left(\left\{ \rho_{S\mathcal{F}}(t) - \Gamma_{S\mathcal{F}}^{\text{SQD}}(\rho_{S\mathcal{F}}(t)) \right\} \otimes \rho_{\mathcal{E}\setminus\mathcal{F}}^{\text{new}} \right) \right] \right|, \quad (6.16)$$

where $Q'_{S\mathcal{E}} = U^\dagger(\tau) Q_{S\mathcal{E}} U(\tau)$ satisfies

$$\|Q'_{S\mathcal{E}}\| = \max\{|\lambda_i| : \lambda_i \text{ eigenvalue of } Q'_{S\mathcal{E}}\} \leq 1. \quad (6.17)$$

Thus:

$$\max_{Q_{S\mathcal{E}}} W^{\text{SQD}}(Q_{S\mathcal{E}}) \leq \sup_{\|B\| \leq 1} \left| \text{tr} \left[B \left(\left\{ \rho_{S\mathcal{F}}(t) - \Gamma_{S\mathcal{F}}^{\text{SQD}}(\rho_{S\mathcal{F}}(t)) \right\} \otimes \rho_{\mathcal{E}\setminus\mathcal{F}}^{\text{new}} \right) \right] \right| \quad (6.18)$$

$$= \left\| \left\{ \rho_{S\mathcal{F}}(t) - \Gamma_{S\mathcal{F}}^{\text{SQD}}(\rho_{S\mathcal{F}}(t)) \right\} \otimes \rho_{\mathcal{E}\setminus\mathcal{F}}^{\text{new}} \right\|_1 \quad (6.19)$$

$$= \left\| \rho_{S\mathcal{F}}(t) - \Gamma_{S\mathcal{F}}^{\text{SQD}}(\rho_{S\mathcal{F}}(t)) \right\|_1 \quad (6.20)$$

$$= M^{\text{SQD}}(\rho_{S\mathcal{F}}(t)), \quad (6.21)$$

where the trace norm of the measure can be written as supremum over Hermitian operators B where $\|B\| \leq 1$ ($-I \leq B \leq I$), and where $\|A \otimes B\|_1 = \text{tr} \sqrt{(A \otimes B)^\dagger (A \otimes B)} = \text{tr} [\sqrt{A^\dagger A} \otimes \sqrt{B^\dagger B}] = \|A\|_1 \cdot \|B\|_1$. This gives Eq. (6.21).

The witness depends on suitable choices of the unitary operation $\mathcal{U}_\tau$ and final measurement $Q_{\mathcal{SE}}$ to ensure that the two alternative probabilities [Eqs. (6.10), (6.12)] are different. The second unitary in particular, $U(\tau)$ can be chosen to pick up correlations that will otherwise go undetected [232] by effectively changing the basis of the final measurements, especially for nonlocal unitaries that can then implement nonlocal measurements.

Like quantum coherence and its witnesses, the witness we have presented is system-basis and environment-subspace dependent. Quantum Darwinism, in its fullest, requires optimisation over all bases. However, we wish for a scheme less intensive than full quantum state tomography. Furthermore, realistically, we are constrained in what we can measure, especially when it comes to environments: often, there are only a limited number of degrees of freedom with encoded information that we can access, and/or only a limited number of degrees of freedom that can *possibly* encode information about the system of interest. As such, there is automatically a naturally preferred basis and subspace.

6.3 Quantum photonic simulation experimental proposal

In this section, we propose an experiment to apply the witness onto a system and environment composed of photons with information encoded in the polarisation degrees of freedom. Our scheme is particularly suited to photonic qubits over spin qubits and other related systems using magnetic resonance as the objectivity operation from Eq. (6.5) relies on projective measurement. We will first present the setup in subsec. 6.3.1, some circuit elements for a hypothetical experiment in subsec. 6.3.2, then numerical simulations in subsec. 6.3.3, and followed by a comparison between the non-objectivity witness and quantum state tomography in subsec. 6.3.4.

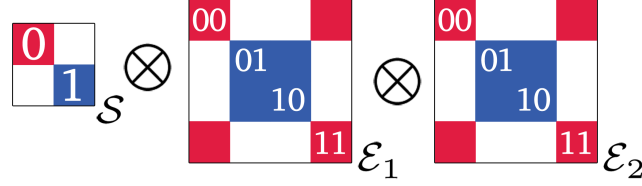


Figure 6.2: **Visualisation of our pre-determined objective subspace.** The system and environments must be in a statistical mixture of the red states and the blue states. For example, if the system is in state $|0\rangle_S$ then each environment should be in the space spanned by $\{|00\rangle, |11\rangle\}$.

6.3.1 Overall setup

The system is comprised on one photonic polarisation qubit, with fixed basis $|0\rangle_S$ and $|1\rangle_S$ (corresponding to horizontal and vertical polarisation respectively). In general, the dimension of each environment can be larger than the dimension of the system. Here, we consider each environment $\mathcal{E}_k$ as being composed of two photons, $(\mathcal{E}_k^{(1)}, \mathcal{E}_k^{(2)})$. The parity of the environment state corresponds to the two disjoint subspaces that signal strong quantum Darwinism: if the system is in state $|0\rangle_S$ then the environment should be in the space spanned by $\{|00\rangle, |11\rangle\}$; and if the system is in state $|1\rangle_S$ then the environment should be in the space spanned by $\{|01\rangle, |10\rangle\}$. The environment is composed of subenvironments $\mathcal{E}_1$ and $\mathcal{E}_2$. This allows us to probe non-objectivity for fragments $\mathcal{F} = \mathcal{E}_1$ and $\mathcal{F} = \mathcal{E}_1\mathcal{E}_2$. The objective subspace is depicted in Fig. 6.2.

Non-objectivity can be probed for fragments $\mathcal{F} = \mathcal{E}_1$ and $\mathcal{F} = \mathcal{E}_1\mathcal{E}_2$. The proposed experimental circuit consists of five overall steps (in the case of $\mathcal{F} = \mathcal{E}_1$ is shown in Fig. 6.3):

1. Preparing the states with some non-objectivity that we wish to witness,
2. Applying a point channel on $\mathcal{E} \setminus \mathcal{F}$ (which could be combined with the state preparation depending on the procedure),
3. Applying the objectivity operation on $\mathcal{SF}$, or an identity operation, depending on the run,
4. The unitary evolution,
5. And the final measurement.

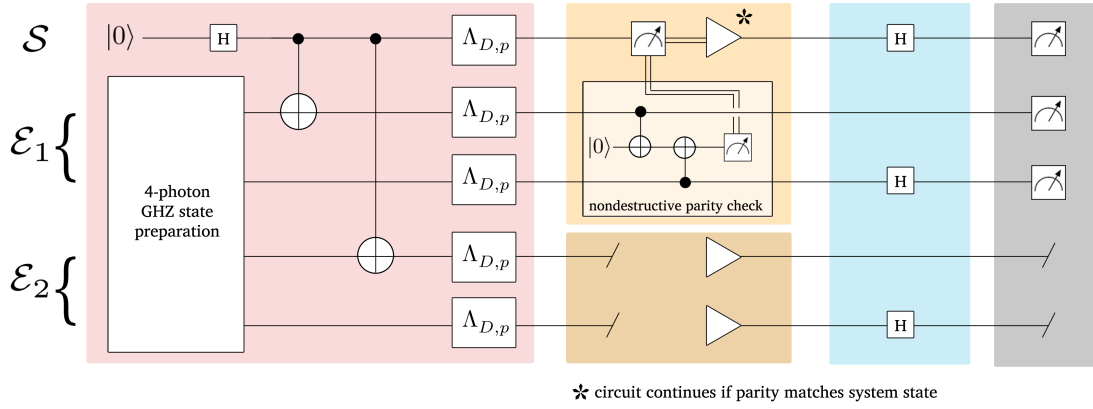


Figure 6.3: **Circuit for one particular run within the scheme to witness non-objectivity.** Each environment consists of two photon polarisation qubits. The system-environment are first prepared into the state given in Eq. (6.22), including 4-photon GHZ state preparation shown in Fig. 6.5. In this particular run, we have depolarisation channels $\Lambda_{D,p}$ [Eq. (6.24)] on all photons. The non-fragment component $\mathcal{E} \setminus \mathcal{F}$, here $\mathcal{E}_2$, undergoes a point channel, where the photons are discarded and uncorrelated photons are produced. Meanwhile, the system $\mathcal{S}$ is measured and $\mathcal{E}_1$ undergoes a nondestructive parity check involving an auxiliary photon. If the parity measurement matches the system measurement result, then the measured system state is recreated, projecting the system-environment into the objective subspace. The system-environment undergoes unitary evolution, here under Hadamard gates H (half wave plates). All measurements are in the computational basis of horizontal/vertical polarisation. If the objective operation results in a null state, then all measurement outcomes are zero.

Our initial system-environment state is

$$|\Psi(0)\rangle_{\mathcal{S}\mathcal{E}_1\mathcal{E}_2} = \frac{1}{2}|0\rangle_{\mathcal{S}}(|00,0\rangle_{\mathcal{E}_1\mathcal{E}_2} + |11,11\rangle_{\mathcal{E}_1\mathcal{E}_2}) + \frac{1}{2}|1\rangle_{\mathcal{S}}(|10,10\rangle_{\mathcal{E}_1\mathcal{E}_2} + |01,01\rangle_{\mathcal{E}_1\mathcal{E}_2}), \quad (6.22)$$

which has strong quantum Darwinism when one environment is traced out, and is entangled over the full environment. From this base initial state, we will consider two different operations that make the system-environment state more non-objective:

1. Mixing the initial state with the maximally noisy state $\mathbb{1}_{\mathcal{S}\mathcal{E}_1\mathcal{E}_2}/d_{\mathcal{S}}d_{\mathcal{E}_1}d_{\mathcal{E}_2}$, such that the state at time t is

$$\rho_{\mathcal{S}\mathcal{E}_1\mathcal{E}_2}(t) = (1-p)|\Psi(0)\rangle\langle\Psi(0)|_{\mathcal{S}\mathcal{E}_1\mathcal{E}_2} + p\frac{\mathbb{1}_{\mathcal{S}\mathcal{E}_1\mathcal{E}_2}}{d_{\mathcal{S}}d_{\mathcal{E}_1}d_{\mathcal{E}_2}}. \quad (6.23)$$

2. Local depolarisation on all photons:

$$\Lambda_{D,p}^X(\rho_X) = (1-p)\rho_X + p\frac{\mathbb{1}_X}{d_X}\rho, \quad X = \mathcal{S}, \mathcal{E}_1^{(1)}, \mathcal{E}_1^{(2)}, \mathcal{E}_2^{(1)}, \mathcal{E}_2^{(2)}, \quad (6.24)$$

that acts sequentially upon each subsystem, here $\mathcal{E}_1^{(a)}$, $a = 1, 2$ denotes the two photons in environment $\mathcal{E}_1$ *etc.*

The objective operation involves measuring the system and applying non-destructive parity measurements on each environment in the fragment. If the results match (the system is in state 0 and all the environments' parity are 0 *etc.*) then the system-fragment is objective in our pre-determined subspace. The nondestructive parity measurement on the environment means that we only need to reconstruct the system state after its measurement.

In the unitary phase of the witness, we apply Hadamard gates H_2 on alternating photons: $U = (H_2)_{\mathcal{S}} \otimes (\mathbb{1}_2 \otimes H_2)_{\mathcal{E}_1} \otimes (\mathbb{1}_2 \otimes H_2)_{\mathcal{E}_2}$.

The final measurements are all in computational basis, *i.e.* in the horizontal/vertical polarisation basis.

In the following subsection 6.3.2 we include some experimental detail on a hypothetical experiment. The numerical simulation of the circuit will be given in subsection 6.3.3.

6.3.2 Experimental components

A circuit of the procedure detailed in the previous subsection is shown in Fig. 6.3. There are a number of major components in that circuit, which we will go in to with further detail: the controlled-NOT (CNOT) operations which are required for the state preparation and the objectivity operation, the procedure to generate four-photon GHZ states required for the initial state preparation, and a direct parity measurement for the objectivity operation.

6.3.2.1 Controlled-NOT operation

The experimental procedure uses a number of nondestructive CNOT gates: in the preparation of the initial state in Eq. (6.22) and for the parity measurement for the objective operation, both seen in Fig. 6.3. This is also the main limiting factor of the witnessing scheme, and the scalability of the witnessing scheme depends

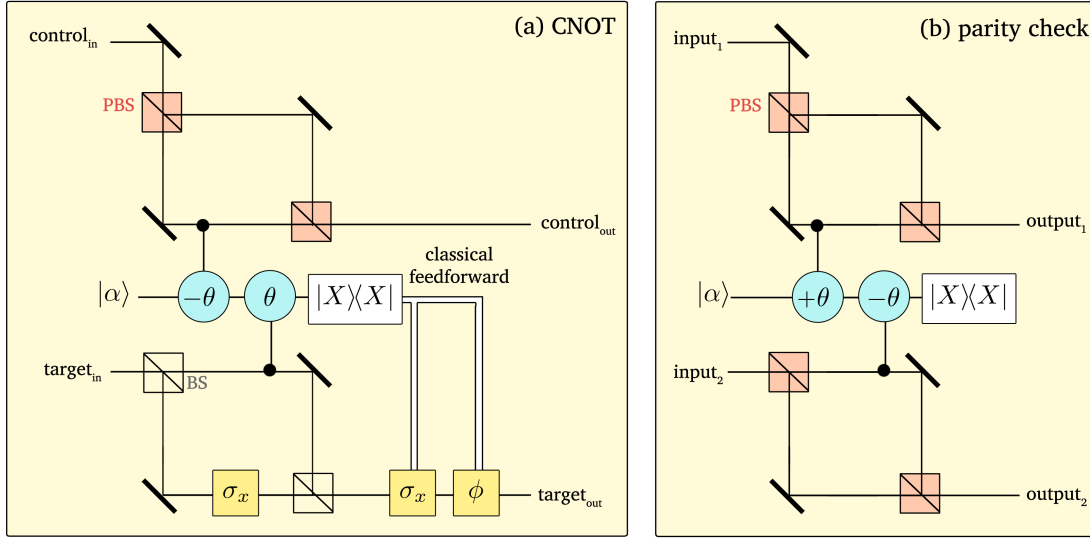


Figure 6.4: **Two circuit components with nonlinearities. (a) Controlled-NOT** for two photon polarisation qubits, based on Ref. [239]. This uses a cross-Kerr nonlinearity in the controlled-phase gates of $\pm\theta$ to boost gate success probability to $1/2$. PBS denotes polarisation beam splitter, and BS denotes beam splitter. An auxiliary coherent state $|\alpha\rangle$ is used. $|X\rangle\langle X|$ is a X homodyne measurement, and depending on its results, a further σ_x gate and phase ϕ gate may be applied to the target photon. **(b) Nondestructive parity measurement** based on Ref. [238]. Aside from polarising beam splitters (PBS), it uses cross-Kerr nonlinearities that apply a shift of $+\theta$, and $-\theta$ on the coherent state $|\alpha\rangle$ if there is photon in the corresponding control modes, followed by a X homodyne measurement $|X\rangle\langle X|$ in order to determine the parity $\{|00\rangle, |11\rangle\}$ or $\{|01\rangle, |10\rangle\}$ of the input photons.

heavily on the success probability and fidelity of the CNOT gates (see subsection 6.3.4 on this scaling).

If we only use linear optical elements, then CNOTs are probabilistic, with realistic success probabilities of the order of $1/4$ to $1/16$ [237]. In contrast, if we employ nonlinearities, then we could make deterministic or near-deterministic CNOTs [238, 239] with greater success probabilities. One explicit example is given in Fig. 6.4, which uses cross-Kerr nonlinearity to boost success probability up to $1/2$.

6.3.2.2 Initial state preparation

The system state can be created with a Hadamard H produced with a half wave plate (HWP) at $\theta = \pi/2$: $(|0\rangle + |1\rangle)/\sqrt{2} = H|0\rangle$. This is shown in the first left-hand box in Fig. 6.3. We then need to create a four-photon Greenberger-Horne-Zeilinger (GHZ) state on the environment. This procedure is depicted in Fig. 6.5, and proceeds as follows: we would first create two Bell states,

$$\left(\frac{|00\rangle + |11\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|00\rangle + |11\rangle}{\sqrt{2}}\right), \quad (6.25)$$

via spontaneous parametric down-conversion (SPDC). One photon from each pair then goes into a beam splitter with path lengths such that they arrive at the same time. Coincident detection in the outputs implies that each both photons are H -polarised or V -polarised, corresponding to projecting the four photons into the subspace spanned by $\{|0000\rangle, |1111\rangle\}$. After renormalising the state, the four-photon GHZ state is made. The probability of the four-photon GHZ state being made is $1/4$ [240].

Finally, we would apply CNOT operations, with the system as the control, onto the first photons of each of the two environments of the following system-environment state:

$$CNOT_{S:\mathcal{E}_1^{(1)}} CNOT_{S:\mathcal{E}_2^{(2)}} \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)_S \otimes \frac{1}{\sqrt{2}}(|00, 00\rangle + |11, 11\rangle)_{\mathcal{E}_1 \mathcal{E}_2} \right], \quad (6.26)$$

to produce the initial state given in Eq. (6.22).

The second component of state preparation is degrading the objectivity in the system-environment. Mixing with the maximally noisy state [Eq. (6.23)] can be done simply by creating the initial state with probability $1 - p$, and the maximally mixed state with probability p . Meanwhile, for the depolarisation channel, one possibility is create the circuit described in Ref. [241]. An alternative procedure is to mix the local state of each photon with the local unpolarised state, *i.e.* keeping a photon with $1 - p$ or replacing it with a completely unpolarised state with probability p (*e.g.* by passing it through a multimode fibre, or by simply generating a new unpolarised state). By averaging over sufficiently many runs, this would give an effective depolarisation channel.

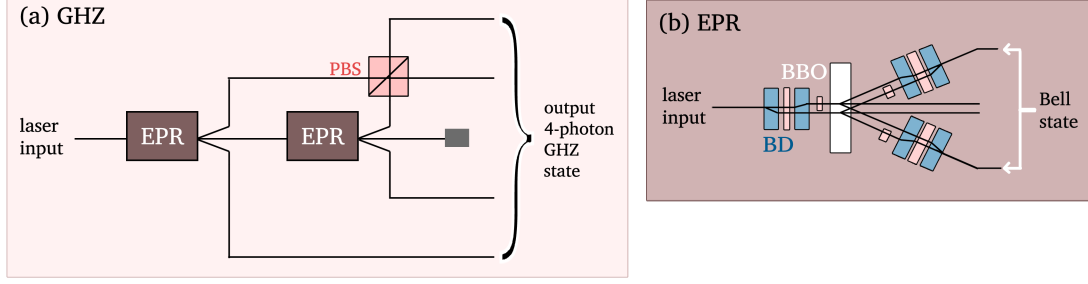


Figure 6.5: **Initial state preparation.** (a) Four-photon Greenberger-Horne-Zeilinger (GHZ) state preparation: consisting of producing two Bell states (via the EPR elements) and combining one photon from each pair at a polarising beam splitter (PBS) to produce a four-photon GHZ state. (b) The EPR element to create Bell states using spontaneous parametric down-conversion: consisting of passing laser pulses through beta-barium borate (BBO) nonlinear crystals and beam displacers (BDs), based on Ref. [146].

6.3.2.3 Objectivity operation

The objectivity operation from Eq. (6.5) can be implemented with a system polarisation measurement and nondestructive parity checks on the environment. One possible parity check scheme is shown in Fig. 6.3, which uses an auxiliary photon and two CNOT operations. Alternatively, a direct parity check scheme could be used, such as the one from [238] [see Fig. 6.4(b)]. Either methods employ nonlinear cross-Kerr nonlinearities, which is crucial for increasing the success probability of the operation.

6.3.3 Numerical simulations

In this section, we numerically simulate the circuit and scheme we proposed, shown in Figs. 6.6 and 6.8. Fig. 6.6 shows the exact values of the measure and witness (at various measurement operators detailed further below), while Fig. 6.8 considers if the state preparation involved controlled-NOT operations with fidelities of approximately 0.79. We find that we are able to often capture the original non-objectivity measure value when examining one environment in our fragment, and are able to detect non-objectivity when considering the full environment.

6.3.3.1 Measurement operators in the witness

The final measurements of the system and fragment are always in the computational basis. This reflects our motivation that only particular measurements are possible, or preferable, in a realistic system. For a particular projective rank-one measurement $\Pi_i^{\mathcal{F}}$ in the computational basis, the non-objectivity witness $W^{\text{SQD}}(\Pi_i^{\mathcal{F}})$ has a very small value in general, and this can be seen at the bottom of the plots in Figs. 6.6 and 6.8. However, we can construct a larger witness without adding any additional measurements by considering a *collection* of the measurement operators. The probabilities for each measurement operator can be determined by the nature of measurement results. Using these statistics, the ideal subset of computational measurement operators can be calculated:

$$\max_{\{Q_{\mathcal{SF}}\}} W^{\text{SQD}} = \max_{\{Q_{\mathcal{SF}}\}} \left| \sum_{\Pi_i^{\mathcal{F}} \in \{Q_{\mathcal{SF}}\}} \text{tr}[\Pi_i^{\mathcal{F}} \{\rho_{\mathcal{SE}}(t + \tau) - \rho'_{\mathcal{SE}}(t + \tau)\}] \right|. \quad (6.27)$$

As the term within the absolute magnitude, $\text{tr}[\Pi_i^{\mathcal{F}} \{\rho_{\mathcal{SE}}(t + \tau) - \rho'_{\mathcal{SE}}(t + \tau)\}]$, can be either positive or negative, this leads to:

$$\max_{\{Q_{\mathcal{SF}}\}} W^{\text{SQD}} = \max \left\{ \left| \sum_{\Pi_i^{\mathcal{F}} \text{ s.t. } \text{tr}[\dots] \geq 0} \text{tr}[\Pi_i^{\mathcal{F}} \{\rho_{\mathcal{SE}}(t + \tau) - \rho'_{\mathcal{SE}}(t + \tau)\}] \right|, \right. \quad (6.28) \\ \left. \left| \sum_{\Pi_i^{\mathcal{F}} \text{ s.t. } \text{tr}[\dots] < 0} \text{tr}[\Pi_i^{\mathcal{F}} \{\rho_{\mathcal{SE}}(t + \tau) - \rho'_{\mathcal{SE}}(t + \tau)\}] \right| \right\},$$

The value of $\max_{\{Q_{\mathcal{SF}}\}} W^{\text{SQD}}$, as can be seen in Fig. 6.6 can provide a good, sometimes *tight*, lower bound to the true value of the measure M^{SQD} . The bound is less tight when the fragment consists of the full environment, $\mathcal{F} = \mathcal{E}_1\mathcal{E}_2$: these situations correspond to greater quantum correlations between the system and fragment. This is expected, as the unitary evolution is local, and the measurements are also local, and so are unable to capture the full quantumness component of the non-objectivity. Thus, one possible improvement would be to introduce an entangling or global unitary evolution. Despite this, the witness *is* successful in detecting non-objectivity.

6.3.3.2 Simulation with operation fidelities and uncertainties with a noisy CNOT and Poisson noise

An alternative noisy CNOT is the one given with the logical table in Fig. 6.7, whose probabilities are approximations of experimental CNOT logic tables [242–

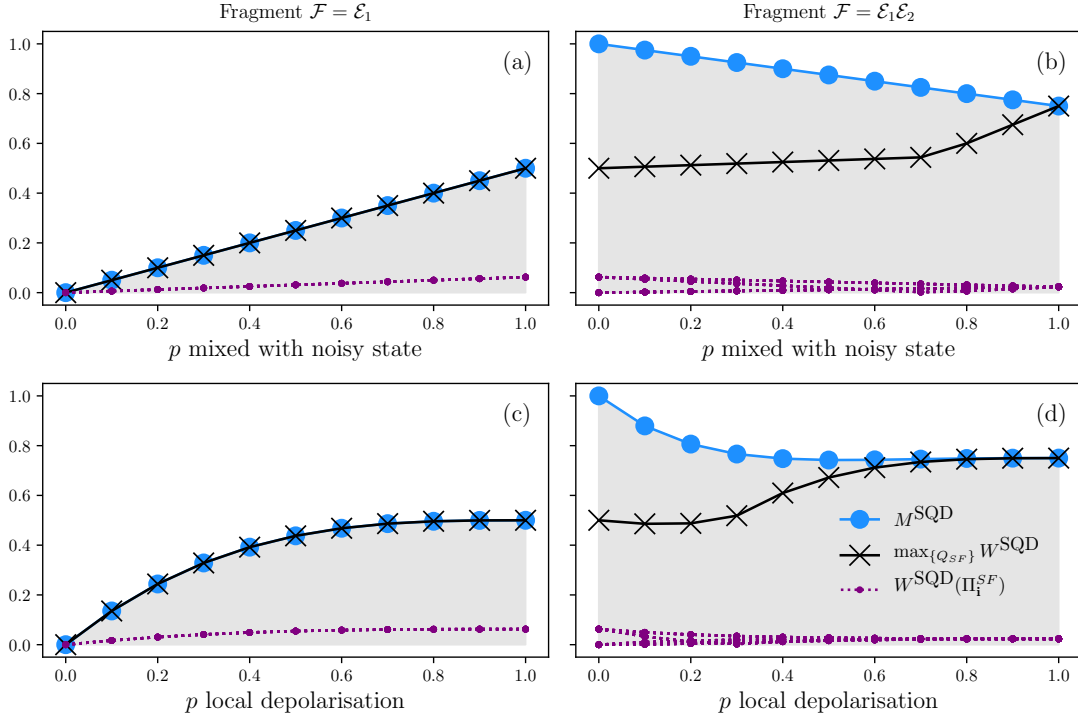


Figure 6.6: **Numerical simulation of the non-objectivity witness in the strong quantum Darwinism framework.** Fragments either consisted of the first environment $\mathcal{E}_1$ or both environments $\mathcal{E}_1\mathcal{E}_2$. M^{SQD} refers to the subspace dependent measure defined in Eq. (6.6); $\max_{\{Q_{\mathcal{SF}}\}} W^{\text{SQD}}$ refers to the maximum value of the witness when the final measurement $Q_{\mathcal{SF}}$ is the summation of a subset of computational projective measurement operators in Eq. (6.28); and $W^{\text{SQD}}(\Pi_i^{\mathcal{SF}})$ corresponds to the values of the witness with rank-one measurement $\Pi_i^{\mathcal{SF}}$ in the computational basis $\{|i\rangle\}_i$. The probability p is the additional noise (either due to mixing with the noisy state or local depolarisation) on the initial state.

244]. The average simulated gate fidelity is $\bar{F} \approx 0.79$.

Fig. 6.8 gives the results of stochastic numerical simulation. Along with the noisy CNOT gates described in Fig. 6.7, the simulation is performed thus:

1. Choose $N_\rho = 500$ random initial states ρ_i distributed close to the ideal state given in Eq. (6.22).
2. For each ρ_i , simulate $N_c = 1000$ stochastic runs of the circuit.
3. For each run, simulate measurement with total photon number picked from a Poisson distribution with mean photon number $N_p = 100$.

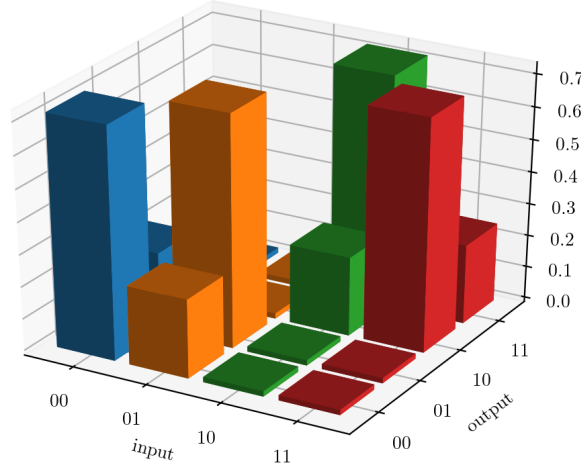


Figure 6.7: **Simulated CNOT logic table** with average gate fidelity of 0.79 used for the numerical simulation in Fig. 6.8.

4. For each ρ_i , average over all measurement outcomes and calculate the optimal witness, $\max_{\{Q_{SF}\}} W^{\text{SQD}}$.
5. Calculate the average optimal witness over all random initial states.

Thus, the simulation results in Fig. 6.8 have three primary noise sources: the initial imperfect preparation, the nonzero fidelity into the CNOT gates in the objective operation, and measurement noise.

We can see in Fig. 6.8 that the simulated experimental witness can overshoot the true measure values, especially when there is less non-objectivity [see Fig. 6.8(a), (c)]. Therefore, experimentally, the witness should pass nonzero lower bound before non-objectivity can be declared. As the CNOT gate fidelities increase, the optimal witnesses for the situation of Fig. 6.8(a), (c) would approach the true value of the measure (not shown here). Also note that the measure is never zero at $p = 0$ unlike in exact case of Fig. (6.6): due to the imperfect system-environment state preparation, the initial states are non-objective.

Without an exact experimental set up, it is unfeasible to model real and detailed error mechanisms. Nonetheless, the results of Fig. 6.8 provide an initial demonstration of a potential future experiment, showing that the scheme could realistically detect non-objectivity.

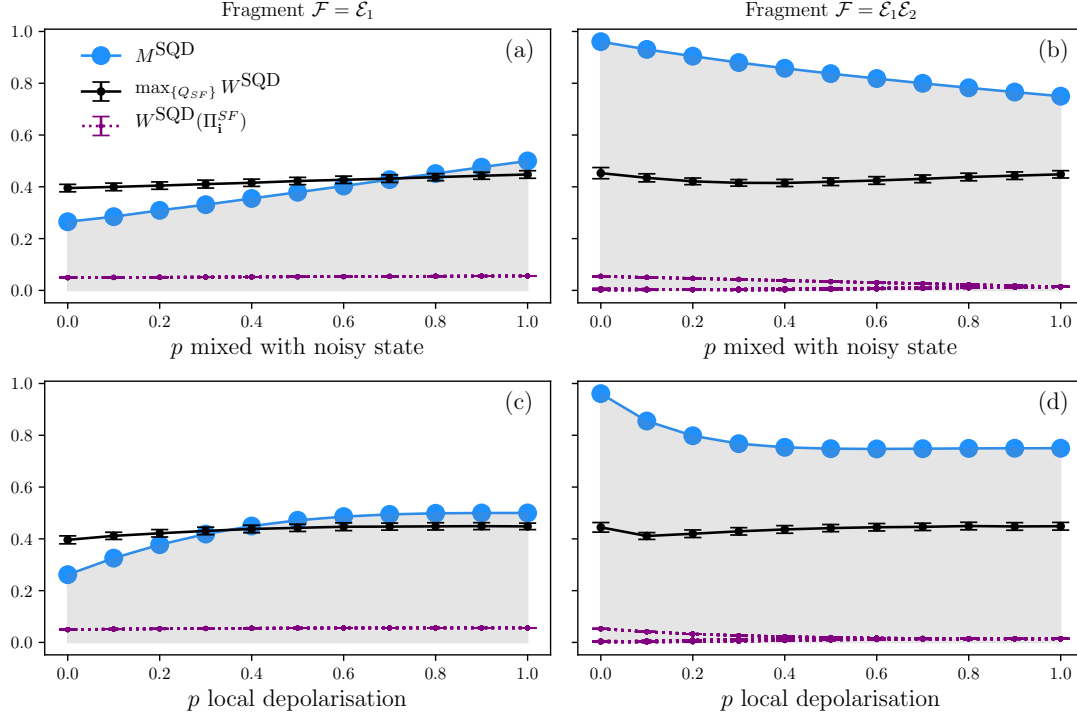


Figure 6.8: **Numerical simulation of the non-objectivity witness in strong quantum Darwinism framework, with simulated experimental results with imperfect fidelities.** Details are given in Sec. 6.3.3.2. Briefly, we have 500 initial states are randomly distributed close to the wanted initial state Eq. (6.22), 1000 copies of each random initial state, and stochastic measurement containing Poisson noise distributed with mean photon number $N_p = 100$. Monte Carlo averaging occurred over measurements and states, and the witness is averaged over all random initial states. M^{SQD} refers to the subspace-dependent measure defined in Eq. (6.6), given for the (imperfect) state at time “ t ” of Fig. 6.1 (c.f. M^{SQD} in Fig. 6.6); $\max_{\{Q_{\mathcal{SF}}\}} W^{\text{SQD}}$ refers to the maximum value of the witness when the final measurement $Q_{\mathcal{SF}}$ is the summation of a subset of computational projective measurement operators [in Eq. (6.28)]; and $W^{\text{SQD}}(\Pi_i^{\mathcal{SF}})$ corresponds to the values of the witness with rank-1 measurement $\Pi_i^{\mathcal{SF}}$ in the computational basis $\{|i\rangle\}_i$. The probability p is the additional noise, either due to mixing the initial state with the noisy state Eq. (6.23) or local depolarisation Eq. (6.24).

6.3.4 Comparison with quantum state tomography

Here we will compare quantum state tomography with our witness based on the number of trials that must be run in either scheme to produce statistics with similar confidence. We will not count the preparation component for the state at time $\rho_{SE}(t)$ as this is required regardless of technique. Rather, we want to compare the number of required measurements in the multiple different bases needed for state tomography, versus the witness which involves two different sets of runs split into (1) point channel, unitary evolution (Hadamard gates), and computational measurements in the horizontal/vertical polarisation basis and (2) point channel, objectivity operation, unitary evolution and the computational measurements.

Let us suppose there are $1 + 2M$ photons in $\mathcal{SF}$ (one system photon, and M environments with 2 photons in each environment).

For the state tomography, the photons in $\mathcal{SF}$ need to be measured in three different bases (local observables), in all the combinations (*i.e.* with polarisation analysis in horizontal/vertical, left/right, and diagonal/anti-diagonal polarisation measurements). With $1 + 2M$ photons to measure, there are 3^{1+2M} basis combination. Let C be the number of counts in one basis set required for sufficient statistics. Then quantum state tomography would require $C \cdot 3^{1+2M}$ runs in total.

Meanwhile, for the witness, we assume that the point channel and unitary evolutions are deterministic, such as the Hadamard gates we have chosen. And since we fix the final end basis, the first set of runs requires C counts (*i.e.* C copies of the state). The second set of runs requires the objectivity operation. Let p_{CNOT} be the probability of the CNOT operation succeeding. Two CNOT operations are required per environment, therefore the probability of a successful parity check on one environment has probability p_{CNOT}^2 . There are M environments, therefore the probability of all the parity checks successfully occurring is $(p_{CNOT}^2)^M$. In order for there to be a total of C successful runs, there must have been at least $C(1/p_{CNOT})^{2M}$ trials/copies of the state. Therefore, the witness scheme would require $C + C(1/p_{CNOT})^{2M}$ runs in total.

If $p_{CNOT} \gtrsim 1/3$, *i.e.* if the CNOT operation can be implemented with success probability of $1/3$ or more, then the witness scheme outperforms quantum state tomography. If we introduce in the fidelities for the CNOT operation, then we could roughly replace $p_{CNOT} \rightarrow p_{CNOT} \cdot F_{CNOT}$ where F_{CNOT} is the fidelity. If $F_{CNOT} = 0.79$, then a minimum success probability of $p_{CNOT} \geq 1/3F_{CNOT} \approx$

0.42 is required. In previous literature, for example Ref. [239], a theoretical $p_{CNOT} = 1/2$ is possible, using nonlinear elements. Although a realistic device suffers from decoherence effects [245, 246], there has been much work in analysing it and reducing it in order to increase success probabilities and fidelities [247–249]. This shows that with sufficiently good controlled-NOT operations with reasonable success probability, the witnessing scheme present here will scale better as more environments are added.

6.4 Invariant spectrum broadcast structure witnessing scheme

Invariant spectrum broadcast structure has a simple structure, describing the situation where a state is objective from the perspective of every subsystem. Recall that invariant spectrum broadcast structure (**ISBS**) leads to states of the following structure:

$$\rho_{\mathcal{S}\mathcal{E}_1\cdots\mathcal{E}_N} = \sum_i p_i |i\rangle\langle i|_{\mathcal{S}} \otimes |i\rangle\langle i|_{\mathcal{E}_1} \otimes \cdots \otimes |i\rangle\langle i|_{\mathcal{E}_N}, \quad (6.29)$$

for some local diagonal basis $\{|i\rangle_{\mathcal{E}_k}\}_i$ for the various subenvironments. Analogous to how we fixed the subspaces in strong quantum Darwinism, we fix the basis in ISBS corresponding to a preferred measurement basis. Basis-dependent invariant spectrum broadcast structure has many similarities with incoherent states. However, the set of basis-dependent ISBS states is strictly smaller than the set of incoherent states.

In this fixed basis $\{|i\rangle_X\}_i$ for all $X = \mathcal{S}, \mathcal{E}_1, \dots, \mathcal{E}_F$, we can define the objectivity operation,

$$\Gamma_{\mathcal{SF}}^{\text{ISBS}}(\rho_{\mathcal{SE}}) = \sum_i [|i\rangle\cdots i\rangle\langle i\rangle\cdots i|_{\mathcal{SF}} \otimes \mathbb{1}_{\mathcal{E}\setminus\mathcal{F}}] (\rho_{\mathcal{SE}}) [|i\rangle\cdots i\rangle\langle i\rangle\cdots i|_{\mathcal{SF}} \otimes \mathbb{1}_{\mathcal{E}\setminus\mathcal{F}}], \quad (6.30)$$

This leads to the corresponding basis-dependent measure of non-objectivity in the invariant spectrum broadcast structure framework:

$$M^{\text{ISBS}}(\rho_{\mathcal{SF}}(t)) = \|\rho_{\mathcal{SF}}(t) - \Gamma_{\mathcal{SF}}^{\text{ISBS}}(\rho_{\mathcal{SF}}(t))\|_1. \quad (6.31)$$

These states are much more simple, which allows us to construct a simple witness in an experimental scheme using photonic qubits.

6.4.1 Basis dependent witness

The overall witness scheme is identical to that of the strong quantum Darwinism witness scheme in section 6.2, but using the ISBS objectivity operation in Eq. (6.30). With that, the corresponding non-objectivity witness in this framework is:

$$W^{\text{ISBS}}(Q_{S\mathcal{E}}) = |\text{tr}[Q_{S\mathcal{E}}\mathcal{U}_\tau[\{\rho_{S\mathcal{F}}(t) - \Gamma_{S\mathcal{F}}^{\text{ISBS}}(\rho_{S\mathcal{F}}(t))\} \otimes \rho_{\mathcal{E}\setminus\mathcal{F}}^{\text{new}}]]|, \quad (6.32)$$

where $Q_{S\mathcal{E}}$ is some measurement operator and $\mathcal{U}_\tau$ is some unitary evolution.

6.4.2 Quantum photonic simulation

We take the system as one photon, and consider four environments each composed of one photon. The scheme is shown in Fig. 6.9 (compare with Fig. 6.3).

Here we consider an initial 5-photon GHZ state $(|0\rangle^{\otimes 5} + |1\rangle^{\otimes 5})/\sqrt{2}$, which can be created from the extension of the 4-photon procedure: after creating the four-photon GHZ state for the environment, one prepares the system state in $(|0\rangle + |1\rangle)/\sqrt{2}$. Then, arrange the path-lengths such that the system photon and the first environment photon arrive at a polarising beam splitter the same time. Similarly, coincident detection in the outputs implies that each both photons are H -polarised or V -polarised, and after renormalising the state, the five-photon GHZ state is made [240]. This is shown on the left-hand-side in Fig. 6.9.

In invariant spectrum broadcast structure, the objectivity operation is now very simple: correlated measurement in the horizontal/vertical polarisation basis in the system and fragment photons. If all measurements on the system and fragment photons give the same outcome, then the system-fragment state is objective, and that state can be recreated and sent down to the rest of the circuit.

Our exact numerical results are shown in Fig. 6.10. The final unitary before measurement consists of Hadamard gates on all system-environment photons. This is because such a unitary gives a better lower bound to the witness than the Hadamard arrangement used in Fig. 6.3 for strong quantum Darwinism. We see that for the mixed, reduced system-fragment states, the witness is tight with the basis-dependent measure. It is not tight for the full system-environment state, which contains global quantum correlations, but it does successfully witness non-objectivity in the state.

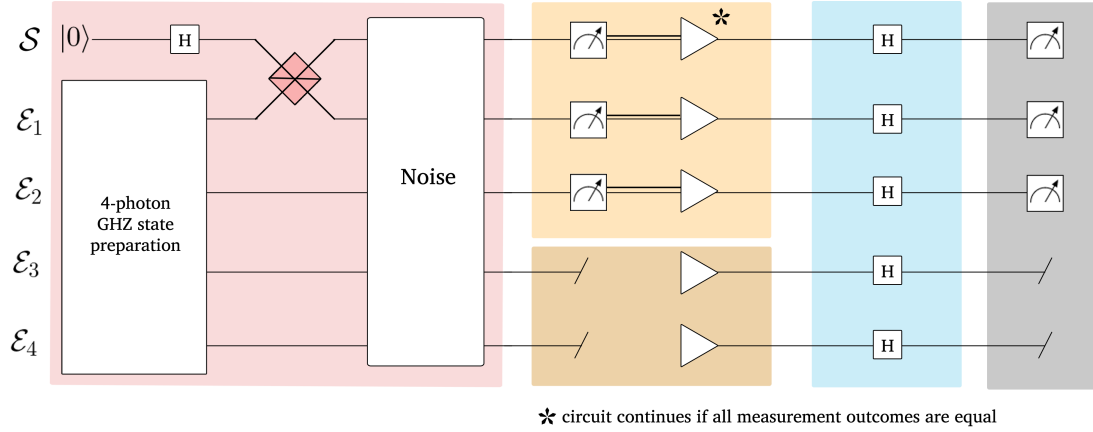


Figure 6.9: **Circuit for one particular run within the scheme to witness non-objectivity in the invariant spectrum broadcast structure framework.** The system and each environment consists of one photon polarisation qubit. The system-environment are first prepared in a five-photon GHZ state (including four-photon GHZ from Fig. 6.5). In this particular run, we have a noise channel that acts on the system-environment state [Eq. (6.23)]. The non-fragment component $\mathcal{E} \setminus \mathcal{F}$, here environments $\mathcal{E}_3$ and $\mathcal{E}_4$, undergo a point channel. Meanwhile, the system and environments $\mathcal{E}_1$ and $\mathcal{E}_2$ are measured. If all measurements results match (all horizontal polarisation, or all vertical polarisations), then that state is recreated and sent down the rest of the circuit. The system-environment undergoes unitary evolution, here under Hadamard gates H . All measurements are in the computational basis of horizontal/vertical polarisation.

This witness scheme scales extremely well due to the lack of CNOT operations, requiring only $C + C$ total runs versus the $C \cdot 3^{1+F}$ runs for F photon environments in the case of quantum state tomography (cf. with subsec. 6.3.4). Here, one could potentially afford an entangling unitary in order to witness quantum correlations in the full system-environment state.

6.5 Discussion and concluding remarks

In this chapter, we presented a subspace-dependent witness of non-objectivity based on strong quantum Darwinism. We defined an objectivity operation that projects the combined system-environment into our preferred objective subspace. The witness detects non-objectivity by comparing the evolution of the system-environment with and without the objectivity operation. We used a point channel

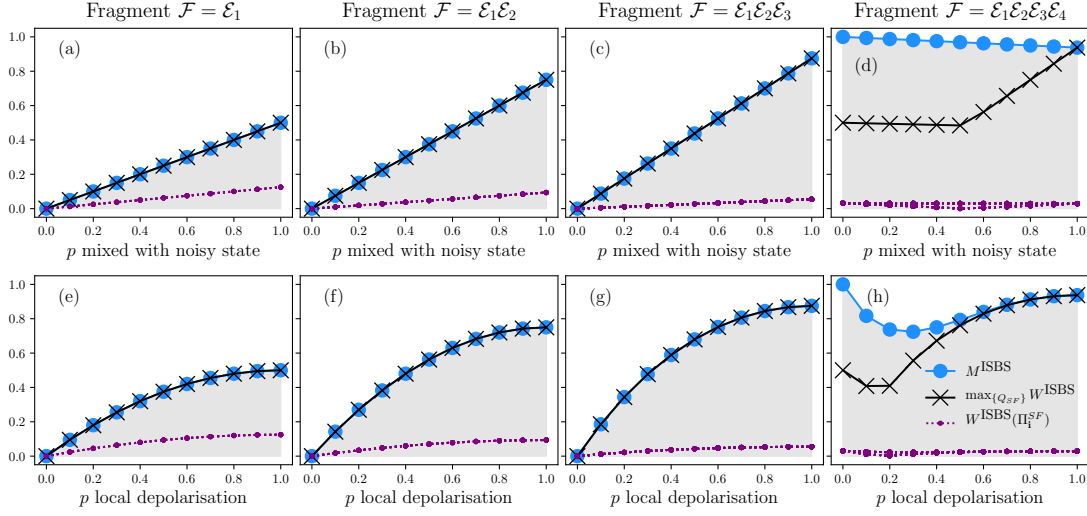


Figure 6.10: **Numerical simulation of the non-objectivity witness in the invariant spectrum broadcast structure framework.** Fragments consist of $\{\mathcal{E}_1\}$, $\{\mathcal{E}_1, \mathcal{E}_2\}$, $\{\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3\}$ and $\{\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_4\}$ from left to right. M^{ISBS} refers to the basis-dependent measure defined in Eq. (6.31); $\max_{\{Q_{\mathcal{SF}}\}} W^{\text{ISBS}}$ refers to the maximum value of the witness when the final measurement $Q_{\mathcal{SF}}$ is the summation of a subset of computational projective measurement operators [the ISBS version of Eq. (6.28)]; and $W^{\text{ISBS}}(\Pi_i^{\mathcal{SF}})$ corresponds to the values of the witness with rank-1 measurement $\Pi_i^{\mathcal{SF}}$ in the computational basis $\{|i\rangle\}_i$. The probability p is the additional noise (either due to mixing with the noisy state or local depolarisation) on the initial state.

to ensure that there is no correlations between the system-fragment and the rest of the environment that might accidentally trigger the witness. We showed that this witness scheme has the potential to scale better than quantum state tomography. We also consider the basis-dependent witness of non-objectivity based on invariant spectrum broadcast structure. In this case, controlled-NOT operations are not required for the objectivity operation, thus greatly simplifying a hypothetical experimental setup. As such, our witnessing scheme and proposal is a step towards experimental verification of quantum Darwinism with large observed environments.

Our witness scheme relies on projective measurements. Hence, photonic systems are ideal and are the preferred system in our experimental proposal. In contrast, ensemble measurements in spin and magnetic resonance systems are non-projective [250, 251]. Projective measurement results can be simulated by measuring multiple observables or with other schemes [252]. Recovering projective

measurement results would require measuring multiple relevant observables, and while it is somewhat feasible with invariant spectrum broadcast structure, it remains open to develop a scheme tailored towards witnessing strong quantum Darwinism in experimental systems with weak measurements.

Secondly, our witness scheme relies on non-invasive/non-demolition CNOT and parity check operations on photonic qubits to be realised via cross-Kerr nonlinearities. In general, such nonlinearities are difficult to implement mostly due to the extremely weak interactions between photons. However, using strong coherent pulses can reduce the required nonlinearity to orders $\theta = 10^{-5}$ [253] to $\theta = 3 \times 10^{-2}$ [238] such that strong cross-Kerr linearities are not necessary. Venkataraman *et al.* [254] have able to achieve nonlinearities up to $\theta = 10^{-2}$ using rubidium atoms confined in photonic bandgap fibres, which therefore means that our scheme's use of non-invasive operations is within experimental reach. It is also possible to achieve large phase shifts in photons without using Kerr nonlinearities [255–258].

In this chapter, we examined a witness for strong quantum Darwinism and invariant spectrum broadcast structure. What about witnessing spectrum broadcast structure, which lies between the two prior frameworks? Casting back to the discussion in subsection 6.1, our scheme would require us to chose a preferred subspace such that each conditional state $\rho_{\mathcal{E}_1|i} \otimes \cdots \otimes \rho_{\mathcal{E}_N|i} \in \mathcal{H}_{\mathcal{E}_1 \dots \mathcal{E}_N|i}$ of the broadcast state,

$$\rho_{S\mathcal{E}_1 \dots \mathcal{E}_N} = \sum_i p_i |i\rangle\langle i|_S \otimes \rho_{\mathcal{E}_1|i} \otimes \cdots \otimes \rho_{\mathcal{E}_N|i}, \quad (6.33)$$

lies in disjoint subspaces

$$\mathcal{H}_{\mathcal{E}_1 \dots \mathcal{E}_N|i} = \mathcal{H}_{\mathcal{E}_1|i} \otimes \mathcal{H}_{\mathcal{E}_2|i} \otimes \cdots \otimes \mathcal{H}_{\mathcal{E}_F|i}. \quad (6.34)$$

We can define the projectors $\Pi_{\mathcal{E}_k|i}$ into $\mathcal{H}_{\mathcal{E}_k|i}$. In strong quantum Darwinism, the environment projector onto each conditional state is simply $\Pi_{\mathcal{F}|i} = \Pi_{\mathcal{E}_1|i} \otimes \cdots \otimes \Pi_{\mathcal{E}_F|i}$, which preserves some correlations between the different environments. However, that is not allowed in spectrum broadcast structure. A spectrum broadcast structure witness in this style would require environment projectors that *also* break correlations conditional on i . Breaking all correlations requires discarding the current state and preparing a new one. Therefore, the hypothetical SBS objectivity operation is a much more intensive procedure. The local conditional

environment states $\Pi_{\mathcal{E}_k|i}\rho_{\mathcal{E}_k}\Pi_{\mathcal{E}_k|i}$ need to be known, and *re-prepared* exactly to ensure there are no extraneous correlations:

$$\Gamma_{\mathcal{SF}}^{SBS}(\rho_{\mathcal{SE}}) = \sum_i \Gamma_{S|i}(\rho_S) \otimes \Gamma_{\mathcal{E}_1|i}(\rho_{\mathcal{E}_1|i}) \otimes \cdots \otimes \Gamma_{\mathcal{E}_F|i}(\rho_{\mathcal{E}_F|i}) \otimes \text{tr}_{\mathcal{SF}}[\rho_{\mathcal{SE}|i}], \quad (6.35)$$

where $\rho_S = \text{tr}_{\mathcal{E}}[\rho_{\mathcal{SE}}]$, and the conditional environment states are reconstructed

$$\rho_{\mathcal{E}_k|i} = \frac{\text{tr}_{\mathcal{SE} \setminus \mathcal{E}_k}[|i\rangle\langle i|_S \otimes \Pi_{\mathcal{E}_k|i}\rho_{\mathcal{SE}}|i\rangle\langle i|_S \otimes \Pi_{\mathcal{E}_k|i}]}{\text{tr}_{\mathcal{SE}}[|i\rangle\langle i|_S \otimes \Pi_{\mathcal{E}_k|i}\rho_{\mathcal{SE}}|i\rangle\langle i|_S \otimes \Pi_{\mathcal{E}_k|i}]}, \quad (6.36)$$

and

$$\Gamma_{S|i}(\rho_S) = |i\rangle\langle i|_S(\rho_S)|i\rangle\langle i|_S. \quad (6.37)$$

This is very unwieldy, and therefore, spectrum broadcast structure is unsuitable for this particular method of witnessing.

Non-objectivity in a system within the framework of strong quantum Darwinism can arise from two sources: the existence of quantum correlations, or the lack of perfect classical correlations. Our scheme does not distinguish between the two, instead giving a single measure that captures non-objectivity in its own right. If the source is required, we suggest using an extra discord witness [231]. This suggests an alternative two-part witness of non-objectivity: a discord witness, followed by some kind of characterisation of the classical information. If only a binary result is required (*i.e.* whether or not there *is* non-objectivity), then the two-part protocol can terminate the moment any discord is witnessed.

In summary, by choosing a naturally preferred basis in which objectivity may arise, we have presented the first witness of non-objectivity, analogous to the witnesses of more established non-classical effects like quantum coherence and quantum entanglement. With gate operations of sufficiently high fidelity and probability of success, the witness scheme can scale better than full state tomography. The work in this chapter opens up further experimental development and testing of quantum Darwinism in larger and increasingly realistic scenarios.

Non-classicality in an avian-inspired quantum magnetic sensor

The original contributions in this chapter are: all sections [7.2.2](#) onwards.

In recent decades, there has been exploration of the quantum mechanical nature of biological processes. Fundamentally, molecular biological systems are quantum mechanical. Despite biological systems being at room temperatures with strong interactions with a polar solvent environment, the time and length scales of some biological interactions are similar to that of experimentally created persisting quantumness [[259](#), [260](#)]. The big questions then are whether quantum effects do appear and if so, whether quantum effects are a necessary component of these biological processes and whether these quantum effects *contribute* to their optimal function [[261](#)].

The greatest focus in quantum biology has been on the light-harvesting process in photosynthetic systems: there is experimental evidence, albeit still controversial, of quantum superposition during the energy transfer in the simpler light harvesting complexes of bacteria [[262](#), [263](#)] and algae [[264](#), [265](#)]. Another growing subfield is that of magnetic sensing in animals, which will be the focus of this chapter.

Behavioural studies have shown that migratory birds can orientate via the Earth's magnetic field [[266–271](#)], and there is growing traction that this

ability requires some measure of non-trivial quantum effects [272]. The two main candidates [273] are sensors based on magnetites [270, 271, 274, 275] and the radical pair mechanism [276, 277].

We are particularly interested in the radical pair mechanism (**RPM**), which involves pairs of spin-correlated radicals (molecules with unpaired electrons which have spin) that chemically recombine under additional photon energy. The chemical reaction has a sensitivity to surrounding magnetic field. If the changes in the recombination can be detected biologically, then this could be used as a means to navigate [266, 276–281].

The behaviour of a RPM magnetic sensor has been demonstrated experimentally [282–284]. The cryptochrome protein is a strong contender for an avian magnetic sensor based on the RPM [266, 272, 276, 278–281, 285–293]: it has been found in some birds [294, 295], and satisfies the constraints known for avian compasses from behavioural experiments [266–269], namely sensitivity to magnetic field inclination [296], sensitivity to oscillating magnetic fields [266, 279, 297, 298], and light-dependence [299–301]. However, it is unclear how cryptochrome can meaningfully detect *tiny* magnetic fields—behavioural studies investigating birds’ sensitivity to oscillating magnetic fields suggest that birds can detect small changes (of the order of 10 nT) in an already small Earth’s magnetic field (50 μ T) [266, 298]. One possibility is that nonclassicality can enhance performance of the birds’ internal magnetic sensor [288].

Understanding the nature and relevance of quantum effects is one of the core questions in quantum biology. Quantum effects in the radical pair magnetic (RPM) sensor have been previously studied in particular RPM models that use anisotropic hyperfine interactions [280, 281, 285–288, 302–308]. The lifetime of the radical pair mechanism is suspected to be around 1.0 μ s [272] in order to account for the observed abilities of birds to sense the magnetic field [266], and it is accepted that spin coherence times (decoherence timescales) are on the order of 1 μ s [280, 288, 309]. The typical initial state of the radical pair mechanism itself (the singlet state) is entangled. Gauger *et al.* [280] find that there can be entanglement *within* the system, although they are unclear as to whether this non-classicality contributes to the sensing. Entanglement between the system radicals lasts on the timescale of the RPM [305]. Ref. [305] hypothesised that the presence of intra-system entanglement is necessary for magnetosensitivity; in contrast, Refs. [280, 310] suggest that entanglement is a byproduct. Refs. [280, 304] found that

statistical mixtures of singlet and triplet states can also lead to magnetosensitivity, and maximally mixed initial states can also lead to magnetosensitivity provided that the singlet and triplet rates differ (*i.e.* with non-unital dynamics) [304].

How much of this entanglement directly leads to advantageous magnetic sensing? And how much of this entanglement exists simply because we have already assumed a quantum description? These past works clearly show that nonclassicality exists on relevant timescales through the evolution of the RPM magnetic sensor. In contrast, Ref. [293] considered semiclassical approximations of the radical pair magnetic sensor and found field sensitivity albeit not comparable to the full quantum treatment. This begs the question: if we take the quantum treatment, can the magnetic sensor still function without any non-classicality? More so than entanglement, what about the effect of more general quantum correlations such as quantum discord, or a more general quantum phenomenon, quantum coherence? If the magnetic sensor operates with a low amount of quantumness—*i.e.* at the quantum-classical boundary—then we can characterise its position along the quantum-to-classical transition by examining its non-objectivity.

In this chapter we consider a biological-inspired quantum magnetic sensor comprised of three spins [290–292], combined with a collisional model of its environment. We characterise the quantum effects in the magnetic sensor using basis-independent quantum coherence [311–313], thus revealing a robust relationship between quantum coherence and magnetosensing performance. In doing so, we highlight both the necessity and function of quantum coherence. Importantly, we introduced a generalised collisional environment [42, 62, 165, 166, 314–318] that, in our model, is essential for the successful operation of the sensor. This reflects how quantum biological systems are always embedded into an environment, often with strong interactions beyond Markovian approximations [261, 319]. Our results show that inclusion of the environment into analyses of the avian magnetic sensor is crucial [280, 281, 287, 302, 303, 309], and demonstrates how the combination of quantum coherence and environmental effects can enhance magnetic sensor performance. Furthermore, the use of the explicit collisional environment allows us to examine the classical and quantum correlations—and hence also non-objectivity—between the sensor and the environment and how these correlations contribute to its magnetic field sensing abilities.

This chapter is organised as follows: in Section 7.1, we introduce the

concept of basis-independent quantum coherence. In Section 7.2, we introduce the quantum magnetic sensor model and the environment model. In Section 7.3, we describe the performance measures for the magnetic sensor, and the analogous measures for the useful classical and quantum correlations over time. In Section 7.4, we present our results relating quantum coherence to the performance of the magnetic sensor. In Section 7.5 we analyse the nonobjectivity of the sensor. We conclude in Section 7.6.

7.1 Basis-independent quantum coherence

There are wealth of methods to quantify nonclassicality, from measures of entanglement [192] to negativity of the Wigner function in quantum optics [320]. Quantum coherence is the most general of all quantum phenomena and is one of the key features that distinguish quantum mechanics from classical phenomena [23, 207]. Rigorous frameworks on the resourcefulness of quantum coherence have been constructed [23, 321, 322], showing how quantum coherence enhances tasks such as quantum metrology and quantum transport on quantum networks [323].

Generally, quantum coherence is defined relative to some basis: given a preferred basis $\{|i\rangle\}_i$, incoherent states have form $\sum_i p_i |i\rangle\langle i|$ and all other states are deemed coherent. In realistic situations, there are preferred bases such as localised position or photon number. A measure of basis-dependent quantum coherence is given by the ℓ_1 norm [322]:

$$C_{\ell_1}(\rho) = \sum_{i \neq j} |\langle i | \rho | j \rangle|, \quad (7.1)$$

which picks out all “off-diagonal” terms from the density matrix of a quantum state. However, this means that basis-dependent coherence is not universal—changing into a system’s diagonal basis causes coherence to disappear.

There has been work to develop a basis-*independent* framework for quantum coherence [311–313, 324, 325]. The only basis-independent incoherent state is the maximally mixed state, $\mathbb{1}_d/d = \sum_{i=0}^{d-1} (1/d) |i\rangle\langle i|$, which is the same regardless basis chosen. We can think of basis-independent coherence as a maximisation of coherence over all possible bases. Maximal coherence is equivalent to state purity [313] and has applications in randomness [311].

The amount of basis-independent coherence can be measured with the relative entropy distance to the maximally mixed state [311]:

$$C_1(\rho) = H(\rho || \mathbb{1}_d/d) = \log_2 d - H(\rho), \quad (7.2)$$

where $H(\rho) = -\text{tr } \rho[\log_2 \rho]$ is the von Neumann entropy and $H(\rho || \sigma) = -\text{tr}[\rho \log \sigma] - H(\rho)$ is the relative entropy. Since basis-independent coherence is equivalent to maximal coherence, $C_1(\rho)$ is also called the relative entropy of purity [313]. In what follows, we use the terms basis-independent coherence and (relative entropy of) purity, and simply “coherence” interchangeably. Note that the term *coherence* is a better descriptor of the underlying quantumness rather than *purity*. If all Hamiltonians were fixed, then coherence relative to the basis of the total Hamiltonian is resourceful. However, since the external magnetic field can change direction, this leads to the fact that coherence in different bases are important during different instances. Overall, by considering basis-*independent* coherence, we can robustly quantify the resourcefulness of quantum coherence without needing to discuss some preferred basis. As we shall see, by using coherence, we will quantify the non-classical behaviour of our avian-inspired magnetic sensor.

7.2 Quantum magnetic sensor model

Typically, RPM magnetosensitivity relies on anisotropic hyperfine coupling [272, 276, 288, 326] or different g -factors [280, 327]. Meanwhile, Refs. [290–292] considered a RPM mechanism with a third “external scavenger” particle: instead of the hyperfine coupling, an electron-electron dipolar interaction creates anisotropy in the RPM chemical reaction. This third spin means that the magnetic field detection does not need to rely on anisotropic hyperfine interactions. Ref. [292] found that this three-radical model can lead to greater anisotropy in the chemical reaction, which would make it easier to detect different magnetic fields, and it is also more robust to spin relaxation [290, 291].

We consider a quantum magnetic sensor comprised of three radicals A, B, C which interact via their spin states [290–292]. Attached to each of the radicals are separate environments that interact with the system radicals in a collisional manner (this can also be considered as similar to “repeated interaction” models, albeit our environment spins each interact only once with the system). This model is advantageous as the sensor does not require hyperfine interactions with

numerous nuclear spins—up to 14—which would be numerically difficult to simulate fully [290, 292]. Note also that past papers considering environment models in the quantum magnetic sensor considered general decoherence/dephasing Lindblad operators [280]. There is no consensus on the correct environment model of the quantum magnetic sensor. Our environment model *can* model decoherence, and the use of the collision scheme means that we do not need to model a very large environment all at once.

7.2.1 Internal magnetic sensor

The internal Hamiltonian of the sensor is $\hat{H}_0 = \hat{H}_{dd} + \hat{H}_{ex}$, where $\hat{H}_{dd}$ is the electron-electron dipolar exchange (point-dipole limit), and $\hat{H}_{ex}$ is the electron exchange interaction:

$$\hat{H}_{dd} = \sum_{i < j} \hat{\mathbf{S}}_i \cdot \mathbf{D}_{ij} \cdot \hat{\mathbf{S}}_j = \sum_{i < j} d_{i,j} \left[\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j - 3 \left(\hat{\mathbf{S}}_i \cdot \vec{\mathbf{e}}_{i,j} \right) \left(\hat{\mathbf{S}}_j \cdot \vec{\mathbf{e}}_{i,j} \right) \right], \quad (7.3)$$

$$\hat{H}_{ex} = - \sum_{i < j} J_{ij} \left(\frac{1}{2} + 2 \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j \right), \quad (7.4)$$

where $\mathbf{D}_{ij}$ is a tensor, $\vec{\mathbf{r}}_{i,j}$ is the displacement between radicals i and j , with unit vector $\vec{\mathbf{e}}_{i,j} = \vec{\mathbf{r}}_{i,j} / \|\vec{\mathbf{r}}_{i,j}\|$ in the direction connecting the radicals i and j . We also have the distance-factor/scaling (with $g \approx 2$),

$$d_{i,j} = \frac{d_0}{|\vec{\mathbf{r}}_{i,j}|^3}, \quad \text{with } d_0 = \frac{\mu_0}{4\pi\hbar} g^2 \mu_B^2. \quad (7.5)$$

Note that the spin angular momentum operators are:

$$\hat{\mathbf{S}}_i = (S_i^x, S_i^y, S_i^z) = \frac{\hbar}{2} \vec{\sigma}, \quad (7.6)$$

where $\vec{\sigma} = (\sigma^x, \sigma^y, \sigma^z)$ are the Pauli matrices. The quantum magnetic sensor is affected by an external magnetic field $\vec{\mathbf{B}}_0(\theta, \phi)$:

$$\hat{H}_{\vec{\mathbf{B}}} = \gamma \vec{\mathbf{B}}_0 \cdot \sum_{i=A,B,C} \hat{\mathbf{S}}_i, \quad \gamma \equiv \frac{g\mu_B}{\hbar}, \quad (7.7)$$

$$\vec{\mathbf{B}}_0(\theta, \phi) = B_0(\cos \theta \sin \phi, \sin \theta \sin \phi, \cos \phi), \quad (7.8)$$

where $\phi \in [0, \pi]$ and $\theta \in [0, 2\pi)$. Note that $\hbar = 1$ is taken throughout this chapter, and note that the magnetic field does not affect the environment in our model.

7.2.2 Environment

We model the environment as three separate baths of two-level particles that interact with the quantum sensor radicals. The environment of the radical pair mechanism in animals has not been experimentally determined. We have chosen to use a collisional model for our environment because collisional models can simulate and represent a wide range of environment models, both Markovian and non-Markovian [42, 62, 165, 166, 314–318, 328–330], which allows our framework to easily extend to more general situations. Furthermore, the explicit environment states afforded by the collisional model also allows us to directly calculate system-environment correlations. In contrast, past works on the avian magnetic sensor typically consider either a nuclei environment [287, 302, 309] or a general Lindblad dissipation [280, 281, 287, 303].

The environment spins have no internal interactions, and we assume their initial states are all identical. The system-environment interaction proceeds as follows:

1. For a time τ_{se} period, environment particles $(\mathcal{E}_{n,A}, \mathcal{E}_{n,B}, \mathcal{E}_{n,C})$ will collide with system radicals A , B and C separately with the following controlled-NOT-like interaction

$$\hat{V}_{S\mathcal{E}}(t) = J_{se} \sum_{X=A,B,C} |1\rangle\langle 1|_X \otimes \left(\mathbb{1} - \sigma_{\mathcal{E}_{n,X}}^x \right), \quad (7.9)$$

where $t \in [nT, nT + \tau_{se})$ for $n = 0, 1, \dots$, $\hat{V}_{S\mathcal{E}}(t) = 0$ otherwise.

2. Then, for a time τ_{ee} period, the system continues to evolve but there is no system-environment interaction. This is akin to a constant time between environment collisions.

Thus, in total each collisional iteration $\#n$ takes time $T = \tau_{se} + \tau_{ee}$. This is shown in Fig. 7.1. The CNOT-like interaction is chosen for two reasons: locally on the system radicals, it induces the standard Lindblad decoherence provided the environment initial states are maximally mixed. Secondly, the CNOT-like interaction promotes the spread of system information into the environment *a la* quantum Darwinism [116]. If $J_{se}\tau_{se} = \pi/2$, then we would have the CNOT interaction between the system and environment if there were no other interactions: $U_{\text{CNOT}} = e^{-i\hat{V}_{S\mathcal{E}}\pi/2}$. In the worse case, this interaction would make the sensor system objective.

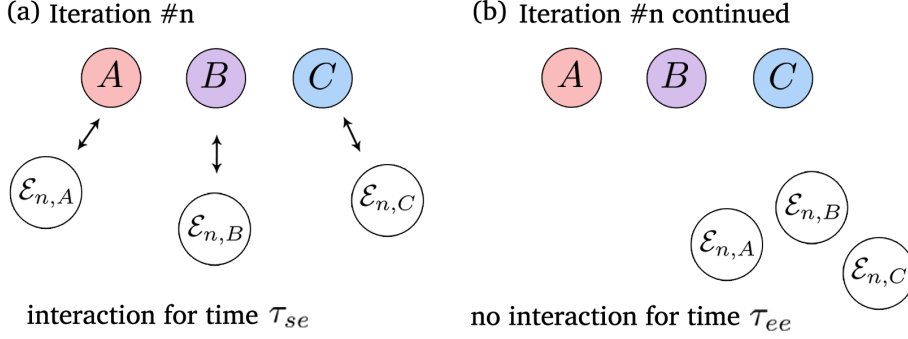


Figure 7.1: **Collisional model of the environment.** The quantum magnetic sensor consists of radical spins A , B and C . At iteration $\#n$, (a) three environment particles (also two-level) interact separately with the radicals for time τ_{se} . (b) Then for a period of τ_{ee} there is no interaction. A new set of environment particles interacts with the systems in iteration $\#n + 1$.

7.2.3 System-environment evolution

Assuming that primary radical pair A and B recombine with rate k , the evolution of the system-environment is as follows [292]:

$$\frac{d\hat{\rho}_{SE}(t)}{dt} = -i \left[\hat{H}_{total}(t), \rho_{SE}(t) \right] - k \rho_{SE}(t), \quad (7.10)$$

where $\hat{H}_{total}(t) = \hat{H}_0 + \hat{H}_{\vec{B}}[\theta, \phi] + \hat{V}_{SE}(t)$, and k is the expected lifetime of the recombination. This is disjoint for the different collisional iterations. The recursive solution to the evolution is given below.

The system-environment evolution is composed of a series of interactions between the system and fresh environment states. Let the eigendecomposition of the total Hamiltonian for times $t \in [nT, nT + \tau_{se})$ and $t \in [nT + \tau_{se}, (n+1)T)$ respectively be:

$$\begin{aligned} \hat{H}_{total}(t) &= \hat{H}_0 + \hat{H}_{\vec{B}}[\theta, \phi] + \hat{V}_{SE}(t) \\ &= \sum_i \omega_i |\phi_i\rangle\langle\phi_i|_{SE_n} \end{aligned} \quad (7.11)$$

$$\hat{H}'_{total}(t) = \hat{H}_0 + \hat{H}_{\vec{B}}[\theta, \phi] = \sum_{\alpha} \xi_{\alpha} |\alpha\rangle\langle\alpha|_S, \quad (7.12)$$

and the unital operators $\mathbb{U}_{\tau}$ and $\mathbb{W}_{\tau}$ corresponding to the unitary component of the evolution as:

$$\mathbb{U}_{\tau} \equiv \exp \left[-i\tau \hat{H}_{total} \right] = \sum_i e^{-i\omega_i \tau} |\phi_i\rangle\langle\phi_i|, \quad (7.13)$$

$$\mathbb{W}_\tau \equiv \exp \left[-i\tau \hat{H}'_{total} \right] = \sum_{\alpha} e^{-i\xi_{\alpha}\tau} |\alpha\rangle\langle\alpha|. \quad (7.14)$$

Now, for time $t \in [0, \tau_{se})$, the system-environment interacts with Hamiltonian $\hat{H}_{total} = \hat{H}_{\mathcal{SE}_0} = \hat{H}_0 + \hat{H}_{\mathbf{B}} + \hat{V}_{\mathcal{SE}_0}$ on the system environment $\rho_{\mathcal{SE}_0}$:

$$\frac{d\rho_{\mathcal{SE}_0}(t)}{dt} = -i \left[\hat{H}_{\mathcal{SE}_0}, \rho_{\mathcal{SE}_0}(t) \right] - k\rho_{\mathcal{SE}_0}(t). \quad (7.15)$$

Then, each element $|\phi_i\rangle\langle\phi_j|$ of the joint state evolves as

$$\begin{aligned} \frac{d}{dt} \langle\phi_i|\rho_{\mathcal{SE}_0}(t)|\phi_j\rangle \\ = \langle\phi_i| \left(-i \left[\hat{H}_{\mathcal{SE}_0}, \rho_{\mathcal{SE}_0}(t) \right] - k\rho_{\mathcal{SE}_0}(t) \right) |\phi_j\rangle \end{aligned} \quad (7.16)$$

$$= (-i\Delta\omega_{ij} - k) \langle\phi_i|\rho_{\mathcal{SE}_0}(t)|\phi_j\rangle, \quad (7.17)$$

where $\Delta\omega_{ij} = \omega_i - \omega_j$. Integrating over time $t \in [0, \tau_{se})$ gives

$$\langle\phi_i|\rho_{\mathcal{SE}_0}(t)|\phi_j\rangle = e^{-kt} e^{-i\omega_i t} e^{i\omega_j t} \langle\phi_i|\rho_{\mathcal{SE}_0}(0)|\phi_j\rangle. \quad (7.18)$$

Using the unitary operator $\mathbb{U}_\tau$ [Eq. (7.13)], we can write $\rho_{\mathcal{SE}_0}(t) = e^{-kt} \mathbb{U}_t \rho_{\mathcal{SE}_0}(0) \mathbb{U}_t^\dagger$.

As similar process occurs for times $[\tau_{se}, \tau_{se} + \tau_{ee})$ and so forth.

This leads to the recursive solution: starting from the initial state $\rho_{\mathcal{SE}_0}(0) = \rho_S(0) \otimes \rho_E$, for times $t \in [nT, nT + \tau_{se})$ and $t \in [nT + \tau_{se}, (n+1)T)$ respectively:

$$\rho_{\mathcal{SE}_n}(t) = e^{-k(t-nT)} \mathbb{U}_{t-nT} \rho_{\mathcal{SE}_n}(nT) \mathbb{U}_{t-nT}^\dagger, \quad (7.19)$$

$$\rho_{\mathcal{SE}_n}(t) = e^{-k(t-(nT+\tau_{se}))} \mathbb{W}_{t-(nT+\tau_{se})} \rho_{\mathcal{SE}_n}(nT + \tau_{se}) \mathbb{W}_{t-(nT+\tau_{se})}^\dagger. \quad (7.20)$$

In this chapter, we consider the radicals to lie on the z -axis, $A - B - C$ with equal spacing between $A - B$ and $B - C$, *i.e.* $A : (0, 0, -0.5)$, $B : (0, 0, 0.5)$ and $C : (0, 0, 1.5)$, in implicit units of $(d_0/d_{AB})^{\frac{1}{3}}$.

Other parameters are given in Table 7.1, with the magnetic field strength equal to the Earth's magnetic field $B_0 = 50 \mu\text{T}$, and with an expected recombination decay lifetime of $k = 1 \mu\text{s}$ [272]. All initial states (system radicals and environment particles) are product, and hence have no initial correlations.

7.3 Performance measures

In this section, we define the performance measures for the quantum magnetic sensor. We also introduce measures to capture averaged classical and quantum correlations over the operation time of the sensor.

Table 7.1: **Parameters used for the quantum magnetic sensor.** The system ABC sits in a straight line on the z axis, spaced apart as: $z_A = -0.5$ $z_B = 0.5$ and $z_C = 1.5$, in implicit units of $(d_0/d_{AB})^{1/3}$ and with $d_{AB} = 20$ angstrom ($\hbar = 1$). S-E refers to system-environment.

Parameters	Values
dipolar exchange $\hat{H}_{dd}$	N/A
electron-exchange $\hat{H}_{ex}$	isotropic $J_{ABC}/d_{AB} = 1$
decay lifetime	$k/d_{AB} = 0.0245$ ($k = 1 \mu\text{s}$)
magnetic field	$\gamma B_0/d_{AB} = 0.215$ ($B_0 = 50 \mu\text{T}$)
initial $\rho_{\mathcal{E}_n}(0)$	$\mathbb{1}/2$ unless otherwise specified
S-E interaction time	$d_{AB}\tau_{se} = 1.0$ unless otherwise specified
S-E interaction strength	$J_{se}\tau_{se} = \pi/2$
time between collisions	$d_{AB}\tau_{ee} = 1.0$

7.3.1 Singlet recombination yield and magneto-sensitivity indicators

The quantum yield of the singlet recombination product of radicals A and B is

$$\varphi_{\text{singlet}}[B_0(\theta, \phi)] = k \int_0^\infty d\tau \text{tr} \left[P_{\text{singlet}}^{(A,B)} \rho_{\mathcal{S}}(\tau) \right], \quad (7.21)$$

where $\mathcal{S} = ABC$, the evolution of $\rho_{\mathcal{S}}(\tau)$ depends on θ, ϕ , and $P_{\text{singlet}}^{(A,B)}$ is a singlet state in primary radical pair A and B :

$$P_{\text{singlet}}^{(A,B)} = \frac{1}{4}(\mathbb{1} - \vec{\sigma}_A \cdot \vec{\sigma}_B). \quad (7.22)$$

The powder-averaged (orientation-averaged) singlet yield is

$$\langle \varphi_{\text{singlet}} \rangle = \frac{1}{4\pi} \int_0^{2\pi} d\theta \int_0^\pi d\phi \sin(\phi) \varphi_{\text{singlet}}[B_0(\theta, \phi)]. \quad (7.23)$$

In general, the singlet recombination yield changes as the magnetic field direction changes: hence being able to detect these differing levels of yields would correspond to detecting different magnetic field directions. However, it is unknown exactly how birds would perceive differences in the direction of the earth's magnetic field, and thus how sensitive they are to changes in recombination yield. Hence, we will consider two different measures for the performance of the magnetic sensor: the absolute anisotropy of the yield,

$$\Delta(\varphi_{\text{singlet}}) = \max_{B_0(\theta, \phi)} \varphi_{\text{singlet}}[B_0(\theta, \phi)] - \min_{B_0(\theta, \phi)} \varphi_{\text{singlet}}[B_0(\theta, \phi)]; \quad (7.24)$$

and the relative anisotropy of the yield:

$$RA(\varphi_{\text{singlet}}) = \frac{\Delta(\varphi_{\text{singlet}})}{\langle \varphi_{\text{singlet}} \rangle}. \quad (7.25)$$

We find that the results for the relative anisotropy tend to be qualitatively similar to the absolute anisotropy.

7.3.2 Information yields

Measures of quantum coherence and correlations are typically for a static state. We are interested in the effect of the *initial* quantum coherence and how the affects subsequent evolution. However, there are zero initial correlations, so when examining quantum and classical information, it is more relevant to consider correlations averaged *over time*.

To measure the contribution of quantum and classical information over the time period where recombination occurs, we can consider *information yields*, akin to the recombination yield and the entanglement yield of Ref. [331], and coherence-like yields in prior works on the avian magnetic sensor [302]. That is, the information yield is an integration of the information $I(t)$ over time, either the classical information $I = \chi$ or the quantum discord $I = \mathcal{D}$, weighted by the integrand factor $\text{tr}[P_{\text{singlet}}^{(A,B)} \rho_S(\tau)]$ of the recombination yield:

$$I_{\text{yield}}[\theta, \phi] = \frac{k}{\varphi_{\text{singlet}}} \int_0^\infty d\tau \cdot I(\tau) \text{tr}[P_{\text{singlet}}^{(A,B)} \rho_S(\tau)]. \quad (7.26)$$

The absolute anisotropy is then defined as

$$\Delta(I_{\text{yield}}) = \max_{B_0(\theta, \phi)} I_{\text{yield}}[\theta, \phi] - \min_{B_0(\theta, \phi)} I_{\text{yield}}[\theta, \phi]. \quad (7.27)$$

7.4 Quantum coherence in the quantum magnetic sensor

Quantum coherence plays a key role in the performance of the quantum magnetic sensor. This section is organised as follows: Section 7.4.1 delves into necessity of initial basis-independent coherence. Section 7.4.2 examines the effect of adding system-environment interactions.

7.4.1 Initial basis-independent quantum coherence

In this section, we show that initial basis-independent quantum coherence—which we simplify to coherence—is necessary for a nontrivial magnetic sensor. In the next section, we show that initial coherence is likely sufficient for a nontrivial sensor for our particular model.

The Zeeman interaction between the magnetic sensor and the geomagnetic field, $\hat{H}_{\vec{B}}$, is crucial for the performance of the magnetic sensor. A necessary condition for the system state $\rho_S(0)$ to evolve nontrivially under the Zeeman interaction is that $[\rho_S(0), \hat{H}_{\vec{B}}(\theta, \phi)] \neq 0$ for at least one set of angles (θ, ϕ) . Thus, the initial system state should not take the following form:

Lemma 3. $[\rho_S(0), \hat{H}_{\vec{B}}(\theta, \phi)] = 0$ for all angles (θ, ϕ) if and only if

$$\begin{aligned} \rho_S = & \frac{\mathbb{1}_{ABC}}{8} + p_{AB}\vec{\sigma}_A \cdot \vec{\sigma}_B + p_{AC}\vec{\sigma}_A \cdot \vec{\sigma}_C + p_{BC}\vec{\sigma}_B \cdot \vec{\sigma}_C \\ & + p_{ABC} \sum_{i,j,k=x,y,z} \epsilon_{ijk} \sigma_A^i \otimes \sigma_B^j \otimes \sigma_C^k, \end{aligned} \quad (7.28)$$

where ϵ_{ijk} is the Levi-Civita symbol.

The proof is given in the Appendix C.1.1. It proceeds by writing the general system state as $\rho_S = \sum_{ijk=0,x,y,z} p_{ijk} \sigma_A^i \otimes \sigma_B^j \otimes \sigma_C^k$ and enforces the zero commutation condition for various field directions of $\hat{H}_{\vec{B}}$ until the reduced form satisfies the commutation for all angles.

Corollary 3. If $[\rho_S, \hat{H}_{\vec{B}}(\theta, \phi)] = 0$ for all angles (θ, ϕ) , then the marginal states are maximally mixed $\rho_A = \rho_B = \rho_C = \mathbb{1}/2$.

Lemma 3 suggests that initial basis-independent coherence is necessary but not sufficient. Meanwhile, Corollary 3 shows that having non-maximally mixed local states is sufficient (but not necessary) for the system to evolve under the geomagnetic field.

However, the Zeeman interaction alone is not sufficient for a magnetic sensor. Since $\hat{H}_{\vec{B}}(\theta, \phi)$ commutes with the singlet state $\hat{P}_{\text{singlet}}^{(A,B)}$ regardless of field angle, the integrand of the recombination yield from Eq. (7.21) takes the same value if the Zeeman interaction is the only interaction:

$$\text{tr}[P_{\text{singlet}}^{(A,B)} \rho_{SE}(t)] = e^{-kt} \text{tr}[P_{\text{singlet}}^{(A,B)} e^{-i\hat{H}_{\vec{B}}t} \rho_{SE}(0) e^{i\hat{H}_{\vec{B}}t}] \quad (7.29)$$

$$= e^{-kt} \text{tr}[P_{\text{singlet}}^{(A,B)} \rho_{\mathcal{SE}}(0)], \quad (7.30)$$

rendering an always constant recombination yield φ_{singlet} , regardless of field direction (θ, ϕ) .

In the more general model [Eq. (7.3) to Eq. (7.10)], initial coherence in the system-environment state remains necessary for the quantum magnetic sensor in our unital model.

Remark 2. For a nontrivial sensor (*i.e.* a nonzero performance $\Delta(\varphi_{\text{singlet}}) \neq 0$), if the environment has initial state $\rho_{\mathcal{E}_n}(0) = (\mathbb{1}_2/2)^{\otimes 3}$, then it is necessary that the initial state is not maximally mixed: $\rho_{\mathcal{SE}}(0) \neq \mathbb{1}_{d_{\mathcal{SE}}}/d_{\mathcal{SE}}$, where $d_{\mathcal{SE}} = 2^6$ is the dimensions of the combined system-environment.

Proof. To see this, we consider the converse, *i.e.*, if $\rho_{\mathcal{S}}(0) = (\mathbb{1}_2/2)^{\otimes 3}$ and where $\rho_{\mathcal{E}_n}(0) = (\mathbb{1}_2/2)^{\otimes 3}$ (as all subsystems are two-level). Then, the state evolves as:

$$\rho_{\mathcal{SE}_0}(t) = e^{-kt} \mathbb{U}_t \rho_{\mathcal{SE}_0}(0) \mathbb{U}_t^\dagger \quad (7.31)$$

$$= e^{-kt} \left(\frac{\mathbb{1}_2}{2} \right)^{\otimes 6}, \quad t \in [0, \tau_{se}) \quad (7.32)$$

$$\rho_{\mathcal{SE}_0}(t) = e^{-k(t-\tau_{se})} \mathbb{W}_{t-\tau_{se}} \rho_{\mathcal{SE}_0}(\tau_{se}) \mathbb{W}_{t-\tau_{se}}^\dagger \quad (7.33)$$

$$= e^{-k(t-\tau_{se})} e^{-k\tau_{se}} \left(\frac{\mathbb{1}_2}{2} \right)^{\otimes 6}, \quad t \in [\tau_{se}, T), \quad (7.34)$$

and so forth, where $\mathbb{U}$ and $\mathbb{W}$ are the unitary operator components of the system-environment evolution given in Sec. 7.2.3. The system state is always:

$$\rho_{\mathcal{S}}(t) = e^{-kt} \left(\frac{\mathbb{1}_2}{2} \right)^{\otimes 3}. \quad (7.35)$$

This is the same for all field angles (θ, ϕ) . Hence, the singlet yield is the same for all field angles, leading to zero performance $\Delta(\varphi_{\text{singlet}}) = 0$. $\square$

The necessity of initial coherence in the system-environment state is due to the unital nature of the evolution: if the initial state is maximally mixed, unital evolution keeps the maximally mixed state and no subsequent coherence nor state evolution (aside from the decay) is generated. This shows that the initial system-environment state should be non-maximally mixed state for nontrivial performance given a unital model.

To examine further conditions for a nontrivial magnetic sensor, we need to investigate the particular interactions and how they affect the subsequent

system evolution. The interactions beyond the geomagnetic field are crucial for symmetry breaking. In past works, this is typically done with anisotropic hyperfine interactions with nuclear spins, or with a dipolar interaction with third scavenger [290, 292]. In the next section, we do this with a system-environment interaction $\hat{V}_{SE}$.

7.4.2 The effect of a system-environment interaction in our model

Quantum biological systems are not closed: thus, their environmental interactions play a nontrivial role in their operation. In particular, we will show how environmental interactions contribute to a working magnetic sensor.

The quantum magnetic sensor requires the system to be affected differently under different field angles, and that furthermore these changes must be detectable in the recombination yield of Eq. (7.21), which recall has integrand $\text{tr}[P_{\text{singlet}}^{(A,B)}\rho_S(t)]$. As such, non-commutativity of the evolution Hamiltonians with the system state *and* with the singlet state $P_{\text{singlet}}^{(A,B)}$ is necessary.

Lemma 4. *For nontrivial performance, it is necessary that the following hold:*

$$[\rho_{SE}(0), \hat{H}_0 + \hat{V}_{SE} + \hat{H}_{\vec{B}}(\theta, \phi)] \neq 0, \quad (7.36)$$

$$[P_{\text{singlet}}^{(A,B)}, \hat{H}_0 + \hat{V}_{SE}] \neq 0, \quad (7.37)$$

$$[\hat{H}_{\vec{B}}(\theta, \phi), \hat{H}_0 + \hat{V}_{SE}] \neq 0, \quad (7.38)$$

for at least one angle (θ, ϕ) , where $\hat{H}_0$ is some internal system evolution.

Proof. Recall that the complete system-environment evolution is given in Eq. (7.10), with unitary component given by $\hat{H}_{\text{total}} = \hat{H}_0 + \hat{V}_{SE} + \hat{H}_{\vec{B}}(\theta, \phi)$.

If either $P_{\text{singlet}}^{(A,B)}$ or $\rho_{SE}(0)$ commute with the unitary evolution given by $\hat{H}_{\text{total}}$, then they become fixed points of the evolution [332], in which case the term $\text{tr}[P_{\text{singlet}}^{(A,B)}\rho_{SE}(t)] = e^{-kt} \text{tr}[P_{\text{singlet}}^{(A,B)}\rho_{SE}(0)]$ is the same for all angles [where the e^{-kt} decay factor comes from the recombination in Eq. (7.10)]. Hence non-commutativity for both are necessary, *i.e.* $[P_{\text{singlet}}^{(A,B)}, \hat{H}_{\text{total}}] \neq 0$ (Eq. (7.36) in the Lemma) and $[\rho_{SE}(0), \hat{H}_{\text{total}}] \neq 0$. For the latter commutator, since $[P_{\text{singlet}}^{(A,B)}, \hat{H}_{\vec{B}}] = 0$ commutes for all angles (due to the SU(2) symmetry in $P_{\text{singlet}}^{(A,B)}$ [292]), this reduces it the form Eq. (7.37) given in the lemma.

Finally, if the Zeeman interaction $\hat{H}_{\vec{B}}(\theta, \phi)$ commutes with all the other interactions, $\hat{H}_{\text{other}} = \hat{H}_0 + \hat{V}_{S\mathcal{E}}$ (which would mean that the other interactions have local SU(2) symmetry on the system), then $e^{-it(\hat{H}_{\vec{B}} + \hat{H}_{\text{other}})} = e^{-it\hat{H}_{\vec{B}}} e^{-it\hat{H}_{\text{other}}}$, and the effect of Zeeman interaction is unseen due to the singlet's symmetry:

$$\begin{aligned} & \text{tr}[P_{\text{singlet}}^{(A,B)} e^{-it(\hat{H}_{\vec{B}} + \hat{H}_{\text{other}})} \rho_{S\mathcal{E}}(0) e^{it(\hat{H}_{\vec{B}} + \hat{H}_{\text{other}})}] \\ &= \text{tr}\left[e^{it\hat{H}_{\vec{B}}} P_{\text{singlet}}^{(A,B)} e^{-it\hat{H}_{\vec{B}}} e^{-it\hat{H}_{\text{other}}} \rho_{S\mathcal{E}}(0) e^{it\hat{H}_{\text{other}}}\right] \end{aligned} \quad (7.39)$$

$$= \text{tr}\left[P_{\text{singlet}}^{(A,B)} e^{-it\hat{H}_{\text{other}}} \rho_{S\mathcal{E}}(0) e^{it\hat{H}_{\text{other}}}\right] \quad (7.40)$$

and subsequently the recombination yield would have no dependence on field angle. Hence, the Zeeman interaction cannot commute with $\hat{H}_0 + \hat{V}_{S\mathcal{E}}$. $\square$

For our particular model, initial basis-independent coherence is necessary and sufficient for non-commutativity of the various commutators:

Proposition 4. *For the model we consider (Sec. 7.2 with parameters in Tab. 7.1), i.e. in a model with initial environment state $\rho_{\mathcal{E}} = \mathbb{1}_{\mathcal{E}}/d_{\mathcal{E}}$ and with interactions*

$$\hat{V}_{S\mathcal{E}} = J_{se} \sum_{\alpha=A,B,C} |1\rangle\langle 1|_{\alpha} \otimes (\mathbb{1} - \sigma_{\mathcal{E}_{\alpha}}^x), \quad (7.41)$$

$$\hat{H}_{ex} = -\frac{J_{ABC}}{2} \sum_{i < j \in \{A,B,C\}} (\mathbb{1} + \vec{\sigma}_i \cdot \vec{\sigma}_j), \quad (7.42)$$

the terms of Lemma 4 holds if and only if the initial state on the system is not maximally mixed: $\rho_S \neq \mathbb{1}/d_S$.

The three components of Lemma 4 are the commutators:

1. $[\rho_S(0) \otimes \mathbb{1}_{\mathcal{E}}/d_{\mathcal{E}}, \hat{H}_0 + \hat{V}_{S\mathcal{E}} + \hat{H}_{\vec{B}}(\theta, \phi)] \neq 0$: see Proposition 5, which uses the particular model we described.
2. $[P_{\text{singlet}}^{(A,B)}, \hat{H}_0 + \hat{V}_{S\mathcal{E}}] \neq 0$: see Lemma 5 and Lemma 6 where we prove that this holds for all nontrivial $\hat{V}_{S\mathcal{E}}$.
3. $[\hat{H}_{\vec{B}}(\theta, \phi), \hat{H}_0 + \hat{V}_{S\mathcal{E}}] \neq 0$: see Lemma 7, which holds for all non-trivial system-environment interactions, provided that the external magnetic field does not interact with the environment.

Note that we consider a system configuration *without* the dipolar interaction. The dipolar interaction is *not* required for the performance of the magnetic sensor, unlike Ref. [292], as the environment is taking up the role of breaking symmetry.

Prompted by our numerical results later in the chapter (Fig. 7.2), we propose the following conjecture:

Conjecture 1. *The necessary conditions in Lemma 4 are also sufficient for nontrivial sensing.*

If Conjecture 1 is true, then combining it with Proposition 4 gives that:

Conjecture 2. *For the model we consider (Sec. 7.2 with parameters in Tab. 7.1), a sufficient condition for nontrivial sensing is $\rho_S(0) \neq \mathbb{1}_S/d_S$.*

Fig. 7.2 supports our conjecture that nonzero initial coherence is sufficient for a nontrivial sensor. In Fig. 7.2, we see that there is a general trend between greater initial coherence and large range of recombination yield φ_{singlet} which is integral to a sensitive magnetic sensor. There are four sub-trends that we identify with different colours and shapes: initial states diagonal in σ_z versus those not, and initial states with positive versus negative z coordinate (given by $\text{tr}[\sigma_z \rho_A(0)]$). The extra dependence on z is due to basis-dependence of the system-environment interaction $\hat{V}_{SE}$, while the sign dependence is due to the sign of the interaction. This can be seen clearly in the Bloch sphere depicting the initial state on $\rho_A(0)$.

In a realistic situation, we must balance both the variation in the recombination yield and the actual value of the yield, *i.e.*, while a large variation is ideal, the value of the yield itself needs to be large enough to be feasibly detectable. One measure of this balance is the objective function $\Delta[\varphi_{\text{singlet}}] \cdot \varphi_{\text{singlet}}$ *i.e.* the product of the anisotropy with the value of recombination. If this function is large, then we have both good variation and a large overall recombination values. We plot this in Fig. 7.2(d). We see that for states lying on the z -axis in particular, that there is an initial rise before a partial plateau starting from $C_{\mathbb{1}}(\rho_{ABC}(0)) \approx 1.5$, which suggests that after a certain point, increasing more initial coherence leads to diminishing returns in the performance of the sensor.

In Appendix C.3, we also numerically examine the relationship between recombination yield and quantum coherence yield. We find that they tend to be inversely linearly correlated, similar to the inverse correlation between the relative entropy of initial coherence and the recombination yield. Thus, while the increase

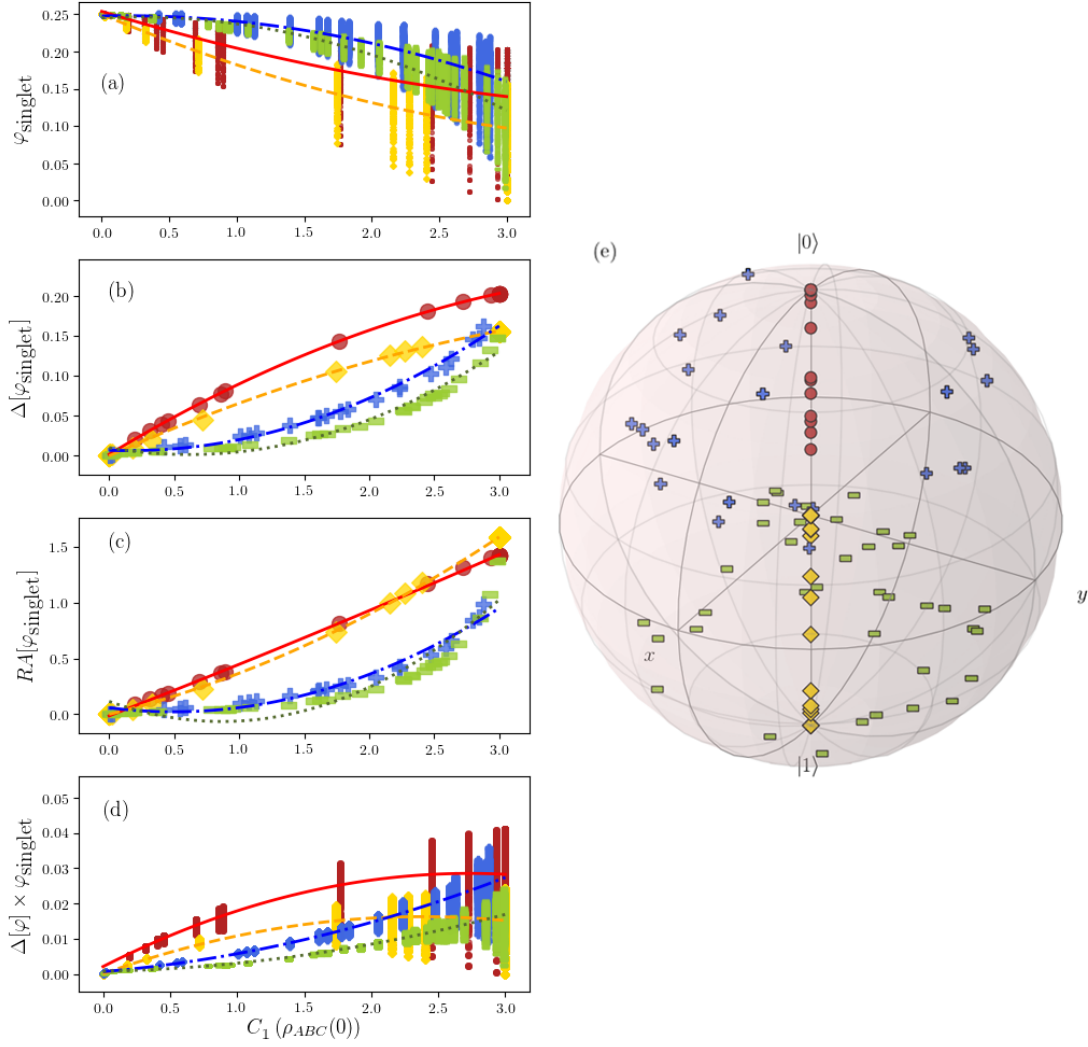


Figure 7.2: **Plots of initial system state's basis-independent quantum coherence, $C_1(\rho_{ABC}(0))$, with the (a) recombination yield φ_{singlet} , (b) absolute anisotropy of the yield, $\Delta[\varphi_{\text{singlet}}]$, (c) relative anisotropy of the yield, $RA[\varphi_{\text{singlet}}] = \Delta[\varphi_{\text{singlet}}]/\langle\varphi_{\text{singlet}}\rangle$, and (d) product of the absolute anisotropy and the yield, $\Delta[\varphi_{\text{singlet}}] \times \varphi_{\text{singlet}}$. (e) The initial state ρ_0 on A and B is randomly chosen (150 different random initial states), and displayed here on the Bloch sphere. The complete initial system state is $\rho_{ABC}(0) = \rho_0 \otimes \rho_0 \otimes \mathbb{1}/2$. Model: electron exchange interaction only, with parameters listed in Table 7.1.**

in basis-independent coherence during the process of recombination suppresses the yield amount, a certain amount of initial coherence is also needed to simply have a wide *variation* in the recombination yield. This trade-off is exemplified back in Fig. 7.2(d).

7.4.2.1 Various analytical results within our model

Here, we state the propositions and lemmas that support Proposition 4.

Proposition 5. *For the model we consider (Sec. 7.2 with parameters in Tab. 7.1), i.e. in a model with initial environment state $\rho_{\mathcal{E}} = \mathbb{1}_{\mathcal{E}}/d_{\mathcal{E}}$ and with interactions*

$$\hat{V}_{S\mathcal{E}} = J_{se} \sum_{\alpha=A,B,C} |1\rangle\langle 1|_{\alpha} \otimes (\mathbb{1} - \sigma_{\mathcal{E}_{\alpha}}^x), \quad (7.43)$$

$$\hat{H}_{ex} = -\frac{J_{ABC}}{2} \sum_{i < j \in \{A,B,C\}} (\mathbb{1} + \vec{\sigma}_i \cdot \vec{\sigma}_j), \quad (7.44)$$

the following commutator is zero $[\rho_S \otimes \mathbb{1}_{\mathcal{E}}/d_{\mathcal{E}}, \hat{H}_{ex} + \hat{V}_{S\mathcal{E}} + \hat{H}_{\vec{B}}(\theta, \phi)] = 0$ if and only if $\rho_S = \mathbb{1}_S/d_S$.

The proof is given in Appendix C.1.2. Briefly, it proceeds by writing $\rho_S = \sum_{ijk=0,x,y,z} p_{ijk} \sigma_A^i \otimes \sigma_B^j \otimes \sigma_C^k$, for $\sigma^i \in \{\sigma_0 = \mathbb{1}, \sigma^x, \sigma^y, \sigma^z\}$, and looks at the coefficient of independent terms $\sigma_A^i \otimes \sigma_B^j \otimes \sigma_C^k \otimes \sigma_{\mathcal{E}_A}^l \otimes \sigma_{\mathcal{E}_B}^m \otimes \sigma_{\mathcal{E}_C}^n$. Our particular system-environment interaction mimics decoherence on the system and imposes the $\{\mathbb{1}, \sigma_z\}$ basis on the system and calculating through with the form of electron exchange interaction $\hat{H}_{ex}$ and Zeeman interaction $\hat{H}_{\vec{B}}$ reduces the required system down to the maximally mixed state.

Lemma 5. *If $\hat{V}_{S\mathcal{E}} = \hat{V}_{A\mathcal{E}_A} + \hat{V}_{B\mathcal{E}_B} + \hat{V}_{C\mathcal{E}_C} \neq 0$ has nontrivial interaction between system $\mathcal{S}$ and environment $\mathcal{E}$ (i.e. contains two-body interaction terms), then $[P_{\text{singlet}}^{(A,B)}, \hat{V}_{S\mathcal{E}}] \neq 0$.*

The proof of Lemma 5 is given in Appendix C.1.3. Briefly, we write $\hat{V}_{A\mathcal{E}_A} = \sum_{ij=0,\dots,3} g_{ij} \sigma_A^i \otimes \sigma_{\mathcal{E}_A}^j$ (and so forth for $\hat{V}_{B\mathcal{E}_B}$). Due to how each $\hat{V}_{A\mathcal{E}_A}$, $\hat{V}_{B\mathcal{E}_B}$ and $P_{\text{singlet}}^{(A,B)}$ all act on slightly different combinations of systems and environment, their commutator is only zero when $\hat{V}_{A\mathcal{E}_A}$ acts *locally* on the system A and locally on environment $\mathcal{E}_A$ (similarly for $\hat{V}_{B\mathcal{E}_B}$), which contradicts the fact that it is an interaction between system and environment.

Lemma 6. *If $\hat{V}_{\mathcal{SE}} \neq 0$ has nontrivial interaction between $\mathcal{S}$ and $\mathcal{E}$, then $[P_{\text{singlet}}^{(A,B)}, \hat{H}_0] \neq -[P_{\text{singlet}}^{(A,B)}, \hat{V}_{\mathcal{SE}}]$ for any $\hat{H}_0$ that acts only on the system.*

Proof. From the previous Lemma 5, we see that $[P_{\text{singlet}}^{(A,B)}, \hat{V}_{\mathcal{SE}}]$ has at least one nontrivial term on the environment $\sigma_{\mathcal{E}}^j$, $j \in \{x, y, z\}$ that is nonzero. In contrast, $[P_{\text{singlet}}^{(A,B)}, \hat{H}_0] = [P_{\text{singlet}}^{(A,B)}, \hat{H}_0] \otimes \mathbb{1}_{\mathcal{E}}$ is trivial on the environment. Since $\mathbb{1}_{\mathcal{E}}$ and $\sigma_{\mathcal{E}}^j$ are linearly independent, the two commutators contain independent terms and cannot be equal. $\square$

This shows that the necessary condition $[P_{\text{singlet}}^{(A,B)}, \hat{H}_0 + \hat{V}_{\mathcal{SE}}] \neq 0$ from Lemma 4 is *always* true provided the nontrivial interaction $\hat{V}_{\mathcal{SE}}$.

Lemma 7. $[\hat{H}_{\vec{B}}(\theta, \phi), \hat{H}_0 + \hat{V}_{\mathcal{SE}}] \neq 0$ for all nontrivial $\hat{V}_{\mathcal{SE}}$.

The proof is given in Appendix C.1.4. It proceeds by contradiction and shows that if the commutator is zero, then $\hat{V}_{\mathcal{SE}}$ must be trivial (*i.e.* consisting of only local terms on system $\mathcal{S}$ and environment $\mathcal{E}$).

Previously, Cai *et al.* [302] had suggested that *entanglement* rather than quantum coherence in the initial state played the greater role, as their optimal magnetic sensitivity for incoherent states (with respect to the standard computational basis) was the same as the magnetic sensitivity for separable initial states. Meanwhile initial entangled system states performed better than both. However, their definition of magnetic field sensitivity is focused on sensitivity to changing *field strength* rather than changing field angle. Furthermore, the results of Cai *et al.* [302] also showed that incoherent and separable initial states give a nontrivial magnetic sensor, which is consistent with our results that initial basis-independent coherence is required. Our results are also consistent with Hogben *et al.* [304]’s, where initial states in statistical mixture of singlet and triplet states also lead to a magnetic sensor. Here, we have located and confirmed the source: *basis-independent* coherence. This shows that avian magnetic sensors fundamentally require coherence, whether it be in the initial state, or during the evolution itself.

7.5 Non-objectivity in the magnetic sensor

An issue with considering quantum coherence is that it is typically basis-dependent, in that changing the preferred basis can change the amount of coherence. One way to get around the basis-dependence is to consider basis-independent coherence,

which we did in the previous section. A second solution is to consider *correlations* between states. In this section, we analyse some of the quantum discord and classical correlations between the system radicals and the environment spins.

Recall that in strong quantum Darwinism (Definition 4), a system is objective when there is zero quantum discord and maximal classical correlations between itself and the observed environment:

$$\mathcal{D}(\mathcal{S} : \mathcal{E}) = 0, \quad \chi(\mathcal{S} : \mathcal{E}) = H(\rho_S). \quad (7.45)$$

A system is non-objective if it has nonzero quantum discord with the environment or non-maximal classical correlations with the environment.

In Fig. 7.3(a) and Fig. 7.4(a) we plot the normalised classical and quantum information yields [Eq. (7.26)] with the singlet recombination. The initial system-environment states are product with each other $\rho_S(0) \otimes \rho_E(0)$, hence have no initial correlations. By using the information yield, we can measure the amount of correlations generated during evolution $(\chi(t), \mathcal{D}(t))$, modulated by how much it contributed to the final recombination yield.

We consider the information between system radicals ABC and the environment spins $\mathcal{E}_{n,ABC}$. We compare two different initial environment states, $\rho_E = \mathbb{1}/2$ like in the rest of the chapter in Fig. 7.3, versus $\rho_E = |0\rangle\langle 0|$ in Fig. 7.4. This is because different environment states are known to affect subsequent quantum Darwinism [116, 119, 120, 144, 149].

In Fig. 7.3(a), the quantum discord yield is very close to zero, while the classical information is *not* maximal (maximum normalised classical information would have value = 1). Thus, the radicals ABC are non-objective whilst being close to “classical” in the sense of lacking quantum correlations with their environment. Furthermore, the greatest classical information yield corresponds to the lowest recombination yield in the bottom right of Fig. 7.3(a). This suggests that a fully objective state would be useless as a quantum magnetic sensor: if the state is fully objective, then the environment is perfectly monitoring the system, corresponding to a quantum-Zeno-like situation. Thus, non-objectivity appears to be necessary, but not sufficient, for a nontrivial quantum magnetic sensor.

In Fig. 7.3(b) we plot the absolute anisotropy for the classical and quantum correlation yields versus the absolute anisotropy of the singlet recombination. It shows that greater variation in classical information is correlated with greater sensor performance. There is a relationship between (non)-objectivity and sensor

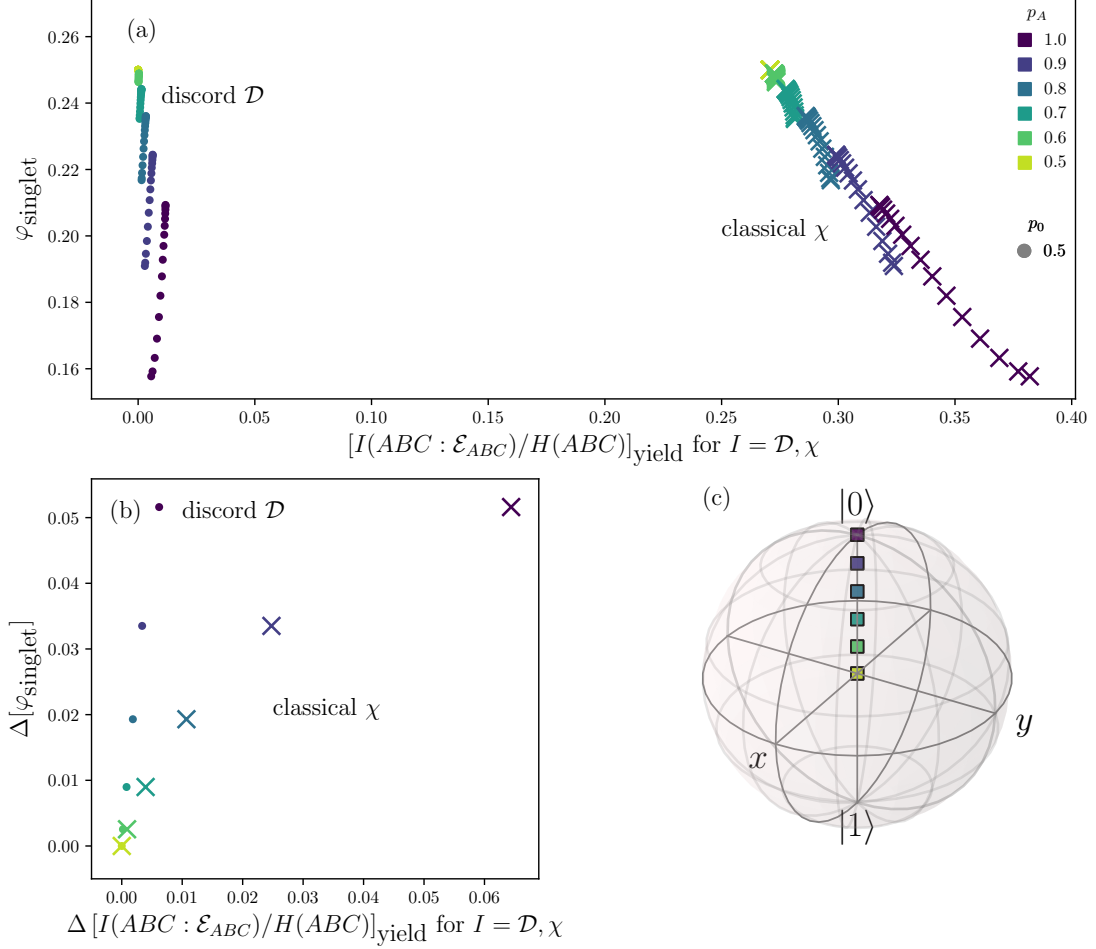


Figure 7.3: **Plots of relationship between normalised classical and quantum information versus sensor performance.** (a) Yield of normalised classical and quantum information yields between radicals ABC and their environments $\mathcal{E}_{ABC}$, $[I(ABC : \mathcal{E}_{ABC})/H(ABC)]_{\text{yield}}$, versus the singlet recombination yield φ_{singlet} , where χ denotes the classical Holevo information and $\mathcal{D}$ denotes the quantum discord. Note that the discord yield very close to zero. (b) Absolute anisotropy of the classical and quantum information yields, $\Delta[I(ABC : \mathcal{E}_{ABC})/H(ABC)]_{\text{yield}}$, versus recombination yield $\Delta[\varphi_{\text{singlet}}]$. (c) Depiction of the initial states on the Bloch sphere, *i.e.* $\rho_{ABC}(0) = \rho_A \otimes \rho_A \otimes \mathbb{1}/2$, where $\rho_A = p_A|0\rangle\langle 0| + (1 - p_A)|1\rangle\langle 1|$. Model: electron exchange interaction only, with parameters listed in Table 7.1. The initial environment states are $\rho_{\mathcal{E}} = (\mathbb{1}/2)^{\otimes 3}$.

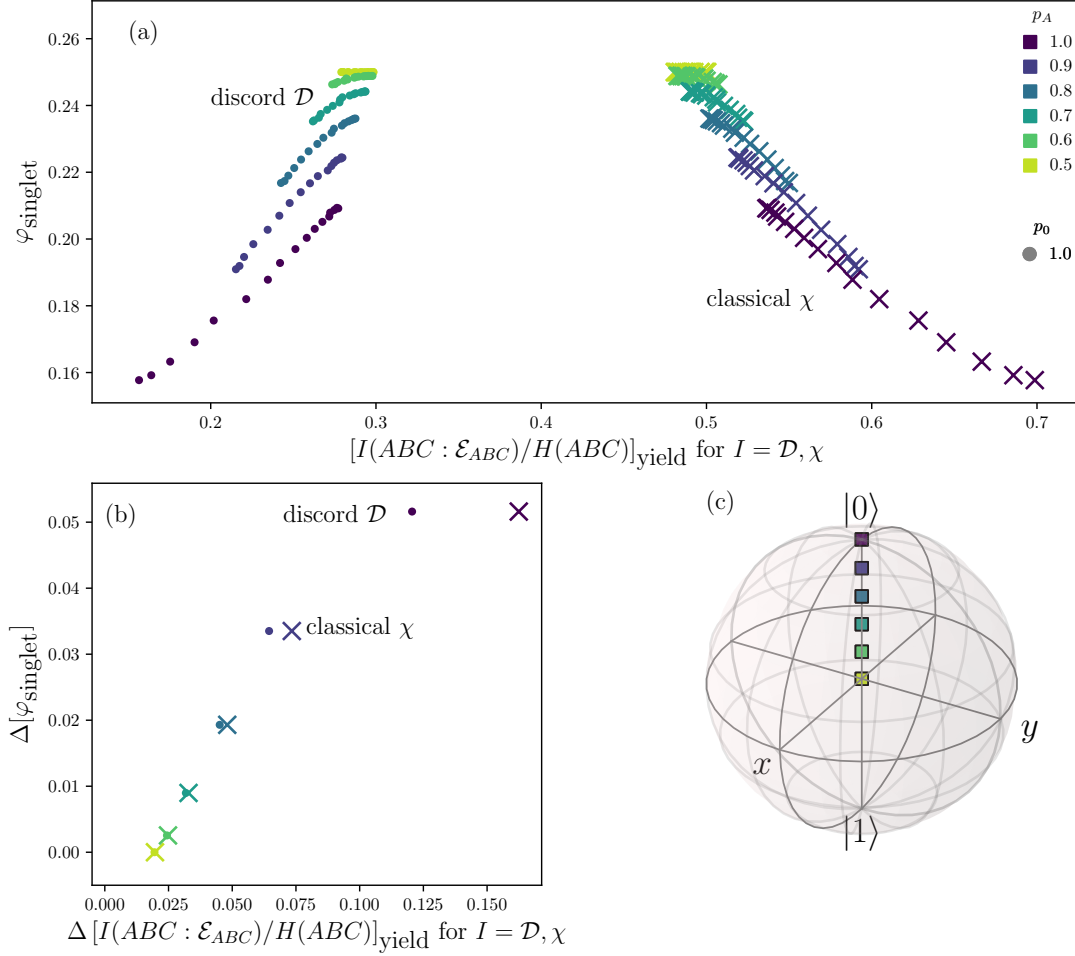


Figure 7.4: **Plots of normalised classical and quantum information yields versus the singlet recombination yield, with pure environment states.** Exactly as Fig. 7.3, except that the initial environment states are $\rho_{\mathcal{E}} = |0\rangle\langle 0|^{\otimes 3}$. (a) Yields of normalised information versus singlet recombination yield. (b) Absolute anisotropy of information versus absolute anisotropy of recombination yield. (c) Depiction of system initial states $\rho_{ABC}(0) = \rho_A \otimes \rho_A \otimes \mathbb{1}/2$, where $\rho_A = p_A|0\rangle\langle 0| + (1 - p_A)|1\rangle\langle 1|$ on the Bloch sphere.

performance, which can be explained as follows: the initial states with the greatest coherence vary more across different field angles, leading to both greater variation in the recombination yield and greater variation in the classical correlations.

This leads to an interesting point: the spins in the magnetic sensor can operate in the space while lacking quantum correlations with the external environment, and yet non-objective due to the weakness (non-maximally) of its classical correlations with the external environment. Combined with the results of the previous section, where the spins contain quantum coherence, both in the initial state and during evolution, these results pin-point the level of “quantumness” in the magnetic sensor to the nonzero quantum coherence.

In contrast, in Fig. 7.4 where the initial environment states are pure, $|0\rangle\langle 0|$, there is much more quantum discord over time. However, if we compare Fig. 7.3(b) with Fig. 7.4(b), the sensor performance given by $\Delta[\varphi_{\text{singlet}}]$ are very similar. This shows that in the model considered, quantum magnetic sensor performance relies primarily on the initial basis-independent coherence in the system radicals. This is primarily due to how the magnetic field does not directly interact with the environment state.¹

The combined results of Fig. 7.3 and Fig. 7.4 show that the quantum magnetic sensor always operates in a non-objective state with its environment, even as performance goes to zero. Hence, the presence of non-objectivity alone is not sufficient for sensor performance, though the numerical results suggest that non-objectivity may be necessary. While there is a relationship between non-objectivity and quantum sensor performance, sensor performance is much more strongly linked to initial system coherence in our model. And furthermore, non-objectivity is also linked with initial coherence. The very low quantum discord in Fig. 7.3 also confirms that quantum coherence is the key quantum effect involved in the performance of the avian-inspired quantum magnetic sensor.

In summary, the results of this section show the magnetic sensor can operate in a regime that has quantum coherence and is non-objective. The results of this section also suggest that performance is better predicted by initial system coherence rather than system-environment correlations in our model.

¹ Not shown here: when the magnetic field acts on the environment, the environment initial coherence affects the recombination yield.

7.6 Summary

In this chapter, we examined a model of a quantum magnetic sensor based on the radical pair mechanism, which studies suggest may be involved in some animals' ability to detect the direction of the Earth's magnetic field. In particular, we focused on a model with three spin radicals whose recombination depends on the direction of the external magnetic field. Unlike previous studies, we introduced an *explicit* environment, using the collisional model, that allows us to calculate correlations between the system and the environment.

There have been many studies on how quantum coherence can be a resource and enhance various quantum tasks. We investigated the relationship between coherence and one particular model of an avian-inspired magnetic sensor. We found that coherence is integral to avian-inspired magnetosensing. In our particular model, we show that initial coherence—in *any* basis—is necessary for a nontrivial sensor, and our results suggest that this is sufficient when combined with a symmetry-breaking interaction.

In the magnetic sensor, there is no (immediately) preferred basis. The Earth's magnetic field can be aligned in any direction relative to the magnetic sensor, which we assume is mobile. As the relative magnetic field direction varies, the instantaneous basis changes: hence a state that was once incoherent now appears coherent. As such, basis-*independent* coherence is relevant.

In Ref. [333], the authors examine the bases of coherence and noise in light-harvesting processes. There, they noted that quantum coherence has an effect if noise processes act on that coherence—which means that basis in which a state has coherence is *different* from the basis in which noise processes act. We see a similar situation in the magnetic sensor: as the external magnetic field can have any relative direction, it can act in any basis, hence coherence in any basis is useful.

Our results suggest that basis-independent coherence is a resource for the magnetic sensor. Realistically, even in a biological environment, the initial system radicals will not be in the maximally mixed state due to fluctuations. Furthermore, the chemical process producing radical pairs in cryptochrome produce singlet and triplet states which are non-maximally mixed states [272]. Hence, magnetic sensor is robust: the sensor continues to perform under noise, so long as it does not cause the system to become maximally mixed.

Our results also show that although the magnetic sensor is non-objective, non-objectivity itself is not a resource, but rather a product of the initial basis-independent coherence and dynamics. That said, we were also able to identify where along the quantum-classical boundary the magnetic sensor can operate—*i.e.*, it can operate with very little quantum correlations to the surrounding environment, and hence further cementing the key role of quantum coherence in the performance of the magnetic sensor.

Note that our model is *unital* (and trace-decreasing). Under this evolution, the maximally mixed state is preserved (up to a real factor). The supplemental of Hogben *et al.* [304]’s paper contains a sensor that has maximally mixed initial state $\mathbb{1}/4$ on the radicals, $\mathbb{1}/2$ on the single nuclear spin. There, the sensor is nontrivial whenever the singlet and triplet reaction rates are unequal $k_S \neq k_T$ —this leads to non-unital dynamics in the master equation, which therefore shifts the resource from the initial states with coherence to the coherence-generating dynamics. Initial non-maximally mixed states are necessary for nontrivial performance given a unital model, while non-unital dynamics can lead to nontrivial performance with initial maximally mixed states. The leading open question then is whether or not these statements hold for other models of the quantum magnetic sensor, such as the anisotropic hyperfine models [280, 302].

Purity is more easily measured than other quantum effects such as discord and entanglement: there exist schemes that do not require quantum state tomography especially for the linear purity, $\text{tr}[\rho^2]$ [252, 334–338].²

In general, we also need a symmetry breaking interaction, which can be a nuclear hyperfine interaction, or dipolar interaction with scavenger radicals, or a general system-environment interaction as we have shown here. Keens *et al.* [292] had proven that the dipolar exchange was necessary in the three-radical model—but they had no environment. We found that the dipolar exchange is not necessary for a magnetic sensor due to the environment interaction. The added environment helps break the symmetry, analogous to the symmetry breaking performed by the dipolar exchange and hyperfine interactions in previous works. Our work shows how integral it is to consider the environment when examining candidates for avian magnetosensors. While we considered a memoryless collision model, non-Markovianity could enhance performance further, and coherence in

²Not shown here: the results using the initial linear purity instead of the relative entropy of purity are qualitatively the same.

the environment itself could also act as a resource to further enhance performance [339]. The results in this chapter show that the environment of the avian magnetic sensor is an under-explored factor to the sensitivity of the sensor.

In conclusion, we have shown that the performance of the quantum magnetic sensor depends on the initial quantum coherence and purity, symmetry breaking dynamics and system-environment interactions. Our results suggest that basis-independent quantum coherence—rather than typical basis dependent coherence—could be the quantum resource for biomolecular systems operating at the quantum-classical boundary and worthwhile for future investigation.

Conclusion and outlook

The quantum-to-classical transition is a longstanding open question: it is the missing link between the microscopic world of quantum mechanics and the macroscopic world we experience. Decoherence theory describes how quantum systems lose quantum coherence, whilst quantum Darwinism describes how quantum systems become objective. In this thesis, we were concerned about the underlying details of the frameworks of quantum Darwinism, the dynamics affecting the emergence of objectivity, and some initial applications. Here, we summarise our results and present some open questions coming directly from our work.

Frameworks of objectivity: We examined the previous frameworks of quantum Darwinism: Zurek’s quantum Darwinism and spectrum broadcast structure. We presented our original frameworks, strong quantum Darwinism and invariant spectrum broadcast structure. We studied how these frameworks are unequal and constructed a hierarchy based on the stringency of their conditions and the type of objectivity they describe. Whilst Zurek’s quantum Darwinism describes an apparent objectivity, strong quantum Darwinism gives the minimal conditions for system objectivity, and spectrum broadcast structure describes system objectivity and partial environment objectivity. Meanwhile, invariant spectrum broadcast structure describes collective objectivity of all systems and environments.

Open questions: Under the wealth of frameworks describing objectivity, the two immediate questions are “which type of objectivity is best?” and “when is it necessary to differentiate between the frameworks?” For the former question,

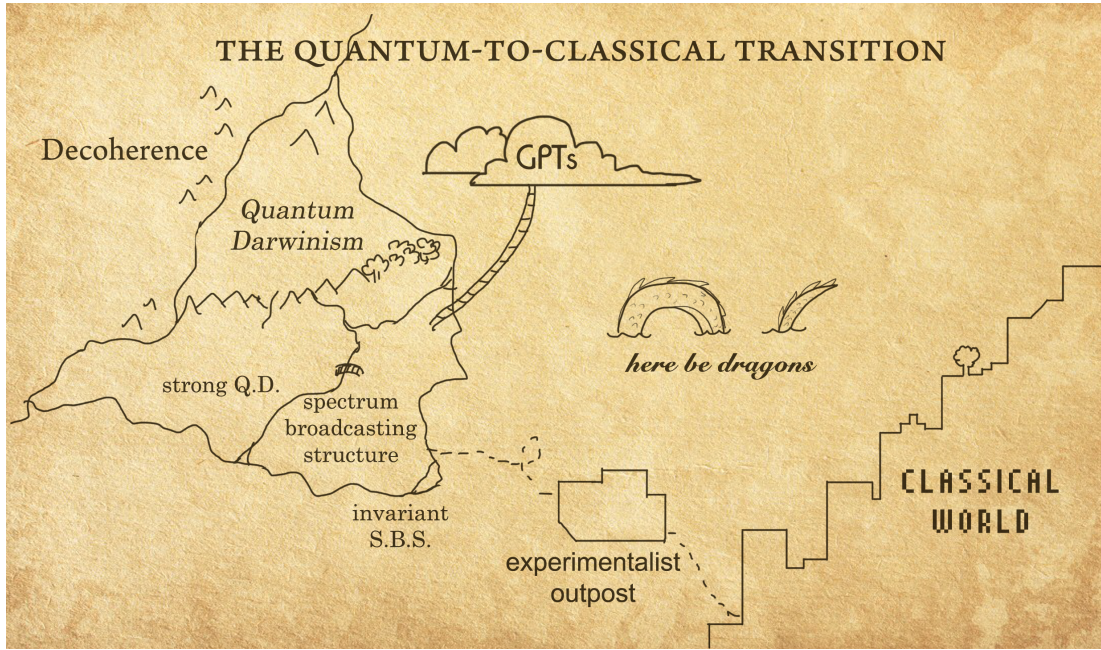


Figure 8.1: An artist’s impression of the “map” of the quantum-to-classical transition, focused on the contributions of quantum Darwinism. Decoherence and quantum Darwinism describe most of the boundary between the quantum and classical world, but not entirely. Experimental works straddle between quantum and classical, and GPTs (generalised probabilistic theories) are more general than quantum theory, hence is depicted in the “clouds” above the quantum plane.

the answer is subjective, as the different levels of objectivity each have their own set of meaningful properties. The choice depends on the application at hand and on which properties one is interested in a specific context. Also furthermore, the objectivity as described by quantum Darwinism is technically inter-subjective [104].

As for the second question, the categorisation of which frameworks suit which models and situations remains open. Historically, many models assumed independence between the environments, which meant that the subsequent objectivity was often consistent with spectrum broadcast structure, and hence also satisfied the “weaker” frameworks, Zurek’s quantum Darwinism and strong quantum Darwinism. However, we have showed this is not always true. We offer some initial thoughts:

- Invariant spectrum broadcast structure appears more suited to collections of particles (*e.g.* a spin bath, gas particles) where there is no preferred system.

- Given the strong independence condition of spectrum broadcast structure, it would likely best describe the case where the environment consists of photons that do not usually interact with each other.
- Meanwhile, strong quantum Darwinism’s framework is better for situations that tend to have non-commuting system and interaction Hamiltonians and strong system-environment interactions, as well as environment-environment interactions. We have seen this in Ref. [136] and in Chapter 4, where the predictions of strong quantum Darwinism deviates from Zurek’s quantum Darwinism due to non-negligible self Hamiltonians, a coupling Hamiltonian that does not commute with it, and strong system-environment correlations.
- Lastly, the use of Zurek’s quantum Darwinism to evaluate objective is better left to situations where one is certain no quantum correlations will develop between system and environment, and in particular when the system Hamiltonian commutes with the system-environment Hamiltonian.

An interesting theoretical extension of all of the frameworks of quantum Darwinism is a resource theory of non-objectivity. In general, the free states would be the objective states, and the free operations defined as those that transform objective states into objective states [340]. If we apply convex restrictions to the set of objective states—such as the subspace-dependent quantum Darwinism we presented—then we can apply the prescriptions known for generic convex resource theories, which have known measures and tasks in which those resourceful states are useful [340, 341].¹

Emergence of objectivity: We then focused on the emergence of objectivity. The pre-existing literature has a wealth of results for both general and specific models, exploring factors the influence the emergence of objectivity such as the dynamics, initial states, the size of the measured environment, and coarse graining. We first examined the emergence of objectivity (or lack there-of) dynamically through a system-environment model with strong interactions. We then examined the emergence from the perspective of quantum channels. We

¹Alternatively, from the viewpoint of the authors in Ref. [229], quantum discord is not a “resource”, but rather an “indicator of non-classicality”. Due to the integral component of quantum discord in strong quantum Darwinism, this would also suggest that non-objectivity may not be a *resource* in the traditional sense.

described objectivity-creating channels which always produce objective states, and we found that they corresponded to effective measurements on the system and distinguishable state preparation on the environment.

Previously, it was believed that memory of past interactions—*i.e.*, non-Markovianity—made quantum systems less objective. However, we showed that this is not always the case: all perfect-objectivity-creating channels are non-Markovian, implying that there are further nontrivial interactions between the system-observed-environment and an unobserved-environment. Perfect objectivity can only be achieved at finite times if non-Markovian dynamics are employed, whilst Markovian evolution at finite times leads to, at best, imperfect objectivity. By finding an example where Markovian evolution also hinders quantum Darwinism, we concluded that neither Markovianity nor non-Markovianity are direct predictors for the emergence of objectivity. Instead, there is evidence—in past and current literature—to suggest that whether or not objectivity emerges depends on a complex interplay between both the underlying model considered, and how we measure the environments.

Open questions: The characterisation of objectivity-creating channels is rather general and opaque, hence an immediate future work could refine the channel structures. Furthermore, the channels create *perfectly objective* quantum states, whereas in reality, objectivity at best holds up to some error. If we allow some error, then past results suggest that imperfect-objectivity-creating channels include both Markovian and non-Markovian channels.

We have not accounted for dynamic time, as quantum channels are discrete with a specific output state at a single fixed time. The description of objectivity-creating channels does not allow us to study the *rate* of emergence of objectivity. Furthermore, the objectivity-creating channels we described do not say anything about what happens to the state *after*: it is not immediately true that the state will *remain* objective, for example Ref. [105] found that after objectivity, the system can subsequently thermalise and the environment loses objective information. Hence, potential future work could include constructing the dynamical maps that replicate objectivity-creating channels in order to analyse the time-dependency in the emergence of objectivity.

Witnessing non-objectivity: In the last part of this work, we turned our focus onto realistic and experimental situations. The analysis of quantum Darwinism requires full quantum state tomography of the system and the relevant external environments, which rapidly becomes unfeasible with more realistic models with large environments. Hence we considered a simplified witnessing scheme that can detect strong quantum Darwinism and lower-bounds the amount of non-objectivity. We showed how the scheme could be implemented in a photonic circuit, and noted that the scheme provides an advantage over quantum state tomography if we have sufficiently good experimental components.

Open questions: Whilst the scheme can be adapted to spin magnetic resonance systems for simple situations, ideally a different scheme needs to be developed especially to work in spin ensemble experiments. Our approach was to compare two different evolutions of the system-environment, rather than a traditional witness observable. Considering past witnesses of quantum discord [233], a singular witness may require multiple copies of the same system-environment. Another possibility for a non-objectivity witness is entropy production, which has been used to detect of system-environment correlations [161].

Non-classicality in biomolecular-inspired systems: Finally, we looked at whether non-objectivity can also play a role in the operation of a quantum magnetic sensor, inspired by avian magnetic sensor models. We identified the key role of initial basis independent quantum coherence in the performance of the sensor. Our results suggested that non-objectivity was not a resource, in a sense that initial coherence was a more precise description of the underlying resource. However, the analysis of non-objectivity allowed us to identify the quantum-classical regime of the magnetic sensor embedded in the environment—*i.e.*, operating in the regime where the system contains very little quantum correlations, and non-maximal classical correlations, with the environment.

Open questions: In the model we examined, unital dynamics with symmetry-breaking interactions and initial non-maximally mixed states were necessary for a nontrivial magnetic sensor. Does this hold for general models based on avian magnetic sensing? Do similar results occur with *other* biomolecular systems? And

more specifically for the avian magnetic sensor, the most crucial step forward is to characterise the environment of the suspected biological magnetic sensors.

Final outlook: Quantum Darwinism does not solve the measurement problem: the environments become classically correlated with the system states, but the system states never “collapse” under the framework of quantum Darwinism. After all, quantum Darwinism is entirely unitary, unlike the non-unitary nature of measurement. Furthermore, strong quantum Darwinism and spectrum broadcast structure are partially derived from the condition that the system state remains unchanged after forgetting the environment measurement results—but this requires hypothetically measuring the environment in the first place. If the results are not forgotten, then the system-environment state faces measurement collapse. The measurement problem is focused on how a single result can be obtained, while quantum Darwinism focuses on how *correlated* results (*i.e.* objective results) can be obtained. Ultimately, quantum Darwinism, and quantum dynamics more generally, are unitary processes that are linear, while measurement is nonlinear. As such, this suggests that a resolution to the measurement problem will lie outside unitary quantum mechanics, and outside the scope of quantum Darwinism.

The overarching aim of quantum Darwinism is to work towards the understanding of the quantum-to-classical transition, which is slightly different from the measurement problem in that the transition describes a more general phenomenon. Quantum Darwinism alone does not solve the quantum-to-classical transition: but it does describes how we may perceive physical objects—that are made of quantum systems—to be objective due to the spreading of duplicate information. While the measurement problem may not be solvable within the strict confines of unitary quantum mechanics, what more can be done towards understanding the quantum-to-classical transition within the framework of quantum theory? What properties beyond objectivity do macroscopic objects have, such as some kind of “physical reality” [342]? Can the emergence of those further properties be described by quantum mechanics jointly with objectivity? Or perhaps there are only emergent properties that *behave* “classically” due to the structure the quantum states (*e.g.* objective states)? The road ahead may contain dragons (cf. Fig. 8.1) but the diversity of research and approaches done by us and countless others brings us closer to the truth.

A

Proofs for the various frameworks

A.1 Proof of Theorem 1

This proof is replicated from our paper, Thao P. Le and Alexandra Olaya-Castro, *Strong Quantum Darwinism and Strong Independence are Equivalent to Spectrum Broadcast Structure*, Phys. Rev. Lett. 122, 010403 (2019) [2].

A.1.1 Proof ($\implies$): spectrum broadcast structure implies strong quantum Darwinism and strong independence

Recall that spectrum broadcast structure states have form

$$\rho_{S\mathcal{F}}^{\text{SBS}} = \sum_i p_i |i\rangle\langle i| \otimes \rho_{\mathcal{E}_1|i} \otimes \cdots \otimes \rho_{\mathcal{E}_F|i}, \quad \rho_{\mathcal{E}_k|i} \rho_{\mathcal{E}_k|j} = 0 \quad \forall i \neq j, \quad (\text{A.1})$$

where $\{|i\rangle\}$ is the pointer basis, and p_i are probabilities. Conditioned on system state $|i\rangle$, the conditional environment states are product and hence immediately satisfy strong independence.

To recover the strong quantum Darwinism quantities [Eq. (3.13)]: Perfect distinguishability means that $\{\rho_{\mathcal{E}_k|i}\}_i$ have mutually orthogonal supports and are simultaneously diagonalisable, hence $\rho_{\mathcal{E}_k|i}, \rho_{\mathcal{E}_k|j}$ commute. Thus the products $\{\rho_{\mathcal{E}_1|i} \otimes \cdots \otimes \rho_{\mathcal{E}_F|i}\}_i$ also commute, leading to $I_{\text{acc}}(\mathcal{S} : \mathcal{F}) = \chi(\mathcal{S}^\Pi : \mathcal{F})$ [137]. Let $\{|j\rangle_{\mathcal{E}_k}\}_j$ be the eigenbasis for each subenvironment $\mathcal{E}_k$. The eigendecomposition

of each conditional state can be written as:

$$\rho_{\mathcal{E}_k|i} = \sum_j d_{\mathcal{E}_k}(j, i) |j\rangle\langle j|_{\mathcal{E}_k}, \quad (\text{A.2})$$

for suitable coefficients $d_{\mathcal{E}_k}(j, i)$: $\sum_j d_{\mathcal{E}_k}(j, i) = 1$ for all k and all i , and $d_{\mathcal{E}_k}(j, i)$ take values such that $\rho_{\mathcal{E}_k|i} \rho_{\mathcal{E}_k|j} = 0$ for all $i \neq j$. Hence,

$$\rho_{\mathcal{F}|i} = \sum_{j_1} d_1(j_1, i) |j_1\rangle\langle j_1| \otimes \cdots \otimes \sum_{j_F} d_F(j_F, i) |j_F\rangle\langle j_F| \quad (\text{A.3})$$

$$= \sum_{\mathbf{j}} d(\mathbf{j}, i) |\mathbf{j}\rangle\langle \mathbf{j}|, \quad (\text{A.4})$$

where $\mathbf{j} = (j_1, \dots, j_F)$ and we have defined $d(\mathbf{j}, i) \equiv d_1(j_1, i) \cdots d_F(j_F, i)$. With the eigenbasis $\{|\mathbf{j}\rangle\}_{\mathbf{j}}$, we can explicitly calculate the Holevo information:

$$\chi(\mathcal{S}^\Pi : \mathcal{F}) = H\left(\sum_i p_i \rho_{\mathcal{F}|i}\right) - \sum_i p_i H(\rho_{\mathcal{F}|i}), \quad (\text{A.5})$$

$$= H\left(\left\{\sum_i p_i d(\mathbf{j}, i)\right\}_{\mathbf{j}}\right) - \sum_i p_i H(\{d(\mathbf{j}, i)\}_{\mathbf{j}}) \quad (\text{A.6})$$

$$= -\sum_i p_i \sum_{\mathbf{j}} d(\mathbf{j}, i) \left[\log_2 \left(\sum_k p_k d(\mathbf{j}, k) \right) - \log_2 d(\mathbf{j}, i) \right]. \quad (\text{A.7})$$

Due to the mutually orthogonal supports, for a given j_e , $d_{\mathcal{E}_e}(j_e, i)$ is nonzero for *at most one* i . We say that $d_{\mathcal{E}_e}(j_e, i) \neq 0$ if and only if $j_e \in J_i^{\mathcal{E}_e}$, where $J_i^{\mathcal{E}_e} \cap J_{i'}^{\mathcal{E}_e} = \emptyset$ are disjoint sets. Hence, the product of those factors $d(\mathbf{j}, i) \neq 0$ if and only if $\mathbf{j} \in J_i$, where $\{J_i\}_i$ are disjoint sets $J_i \cap J_{i'} = \emptyset$. Hence,

$$\chi(\mathcal{S}^\Pi : \mathcal{F}) = -\sum_i p_i \sum_{\mathbf{j} \in J_i} d(\mathbf{j}, i) [\log_2(p_i d(\mathbf{j}, i)) - \log_2 d(\mathbf{j}, i)] \quad (\text{A.8})$$

$$= -\sum_i p_i \sum_{\mathbf{j} \in J_i} d(\mathbf{j}, i) \log_2 p_i \quad (\text{A.9})$$

$$= -\sum_i p_i \log_2 p_i = H(\mathcal{S}), \quad (\text{A.10})$$

where $\sum_{\mathbf{j} \in J_i} d(\mathbf{j}, i) = 1$ by normalisation of the state.

Now, note that

$$H(\mathcal{SF}) = H\left(\sum_i p_i |i\rangle\langle i| \sum_{\mathbf{j}} d(\mathbf{j}, i) |\mathbf{j}\rangle\langle \mathbf{j}|\right) \quad (\text{A.11})$$

$$= H\left(\{p_i d(\mathbf{j}, i)\}_{i, \mathbf{j}}\right) \quad (\text{A.12})$$

$$= - \sum_i \sum_{\mathbf{j}} p_i d(\mathbf{j}, i) \log_2(p_i d(\mathbf{j}, i)) \quad (\text{A.13})$$

$$= H(\mathcal{F}), \quad (\text{A.14})$$

hence $I(\mathcal{S} : \mathcal{F}) = H(\mathcal{S}) + H(\mathcal{F}) - H(\mathcal{SF}) = H(\mathcal{S})$. Thus, all the strong quantum Darwinism conditions have been met: since $I(\mathcal{S} : \mathcal{F}) = \chi(\mathcal{S}^\Pi : \mathcal{F}) = H(\mathcal{S})$, the quantum discord is automatically zero. Finally, given the product form of $\rho_{\mathcal{F}|i} = \rho_{\mathcal{E}_1|i} \otimes \cdots \otimes \rho_{\mathcal{E}_F|i}$, if $F > 1$, we also have SQD on multiple sub-environments $\mathcal{F} = \mathcal{E}_k$, for $k = 1, \dots, F$.

A.1.2 Proof ($\Leftarrow$) : strong quantum Darwinism with strong independence implies spectrum broadcast structure

First, we show that strong quantum Darwinism implies bipartite spectrum broadcast structure: If $\rho_{\mathcal{SF}}$ has strong quantum Darwinism, then $I_{acc}(\mathcal{S} : \mathcal{F}) = \chi(\mathcal{S}^\Pi : \mathcal{F})$: hence the ensemble $(p_i, \rho_{\mathcal{F}|i})$ of the state $\rho_{\mathcal{F}}$ has commuting $\rho_{\mathcal{F}|i}$ [137]. Let $\{|\psi_j\rangle\}_j$ be their common eigenbasis such that $\rho_{\mathcal{F}|i} = \sum_j c(j, i) |\psi_j\rangle\langle\psi_j|$ for some eigenvalues $\{c(j, i)\}_j$. Then, $H(\mathcal{S}) = \chi(\mathcal{S}^\Pi : \mathcal{F})$ implies that

$$H(\{p_i\}_i) = H\left(\sum_i p_i \sum_j c(j, i) |\psi_j\rangle\langle\psi_j|\right) - \sum_i p_i H\left(\sum_j c(j, i) |\psi_j\rangle\langle\psi_j|\right) \quad (\text{A.15})$$

$$\Rightarrow - \sum_i p_i \log_2 p_i = - \sum_i p_i \sum_j c(j, i) \left\{ \log_2 \left(\sum_k p_k c(j, k) \right) - \log_2 c(j, i) \right\}. \quad (\text{A.16})$$

This must hold for every i (conditioned on system state $|i\rangle$), so

$$\log_2 p_i = \sum_j c(j, i) \left\{ \log_2 \left(\sum_k p_k c(j, k) \right) - \log_2 c(j, i) \right\}. \quad (\text{A.17})$$

For this to hold, we require that for a given j , $c(j, k)$ is nonzero for a single k . This is the correct solution by inspection. Alternatively, we can solve it as follows: first using log laws, we can rearrange Eq. (A.17) into:

$$p_i = \prod_{j \text{ s.t. } c(j, i) \neq 0} \left[\frac{\sum_k p_k c(j, k)}{c(j, i)} \right]^{c(j, i)}. \quad (\text{A.18})$$

Since $\sum_i p_i = 1$, we can write $p_i = 1 - \sum_{k \neq i} p_k$, and for general probability distributions, we can consider $\{p_k\}_{k \neq i}$ as a set of independent variables: these equations for the environment $\mathcal{F}$ should hold regardless of particular probabilities $\{p_i\}$. Hence,

$$1 - \sum_{k \neq i} p_k = \prod_{j \text{ s.t. } c(j,i) \neq 0} \left[\frac{p_i c(j,i)}{c(j,i)} + \frac{\sum_{k \neq i} p_k c(j,k)}{c(j,i)} \right]^{c(j,i)} \quad (\text{A.19})$$

$$= \prod_{j \text{ s.t. } c(j,i) \neq 0} \left[1 - \sum_{k \neq i} p_k \left(1 - \frac{c(j,k)}{c(j,i)} \right) \right]^{c(j,i)}. \quad (\text{A.20})$$

First, if there was only one j such that $c(j,i) \neq 0$, then $c(j,i) = 1$ (since $\sum_j c(j,i) = 1$) and Eq. (A.20) reduces to

$$1 - \sum_{k \neq i} p_k = 1 - \sum_{k \neq i} p_k (1 - c(j,k)), \quad (\text{A.21})$$

and by equating the coefficients of $p_{k \neq i}$ on either side, $1 = 1 - c(j,k)$ hence, $c(j,k) = 0$ for all $k \neq i$.

Now, suppose there is two or more j 's such that $c(j,i) \neq 0$, $\{j_1, \dots, j_n\} = J_i$. Let $\sum_{k \neq i} p_k = p_{k_1} + p_{k_2} + \dots + p_{k_m}$ (*i.e.* the sum contains m terms). Using the generalised multinomial theorem, the term in the product can be written as:

$$\left[1 - \sum_{k \neq i} p_k \left(1 - \frac{c(j,k)}{c(j,i)} \right) \right]^{c(j,i)} \equiv \sum_{r_1, \dots, r_m=0}^{\infty} f(j, i, \vec{r}) p_{k_1}^{r_1} p_{k_2}^{r_2} \dots p_{k_m}^{r_m}, \quad (\text{A.22})$$

where we have defined the following for simplicity:

$$\begin{aligned} f(j, i, \vec{r}) \equiv & (-1)^{r_1 + \dots + r_m} \binom{c(j,i)}{r_1 + \dots + r_m} \binom{r_1 + \dots + r_m}{r_2 + \dots + r_m} \dots \binom{r_{m-1} + r_m}{r_m} \\ & \times \left(1 - \frac{c(j,k_1)}{c(j,i)} \right)^{r_1} \left(1 - \frac{c(j,k_2)}{c(j,i)} \right)^{r_2} \dots \left(1 - \frac{c(j,k_m)}{c(j,i)} \right)^{r_m}. \end{aligned} \quad (\text{A.23})$$

Now, Eq. (A.20) can be expanded to:

$$\begin{aligned} 1 - \sum_{k \neq i} p_k = & \sum_{\substack{\vec{r}(j_1), \vec{r}(j_2), \\ \dots, \vec{r}(j_n)=0}}^{\infty} f(j_1, i, \vec{r}(j_1)) f(j_2, i, \vec{r}(j_2)) \dots f(j_n, i, \vec{r}(j_n)) \\ & \times p_{k_1}^{r_1(j_1)+r_1(j_2)+\dots+r_1(j_n)} p_{k_2}^{r_2(j_1)+r_2(j_2)+\dots+r_2(j_n)} \dots p_{k_m}^{r_m(j_1)+r_m(j_2)+\dots+r_m(j_n)}. \end{aligned} \quad (\text{A.24})$$

Equate the coefficients of the p_k on either side of Eq. (A.24):

$$-1 = f(j_1, i, r_1 = 1) + f(j_2, i, r_1 = 1) + \cdots + f(j_n, i, r_1 = 1), \quad (\text{all other } r = 0) \quad (\text{A.25})$$

$$= -c(j_1, i) + c(j_1, k_1) - c(j_2, i) + c(j_2, k_1) - \cdots - c(j_n, i) + c(j_n, k_1) \quad (\text{A.26})$$

$$= -\sum_{j \in J_i} c(j, i) + \sum_{j \in J_i} c(j, k_1) \quad (\text{A.27})$$

$$= -1 + \sum_{j \in J_i} c(j, k_1), \quad (\text{A.28})$$

since $\sum_{j \in J_i} c(j, i) = \sum_j c(j, i) = 1$. Hence, we must have $\sum_{j \in J_i} c(j, k_1) = 0$. This is true for all the $k \neq i$, so

$$\sum_{j \text{ s.t. } c(j, i) \neq 0} c(j, k) = 0, \quad \forall k \neq i. \quad (\text{A.29})$$

Since all the terms $0 \leq c(j, i) \leq 1$, the only possible solution to Eq. (A.29) is that $c(j, k)$ is nonzero for only a single k .

Alternatively, the steps Eq. (A.16) to Eq. (A.29) can be simplified by noting that Eq. (A.16) is exactly equal to $H(I) = H(J) - H(J|I)$ for the classical probability distribution $p_{I,J}(i, j) = p_i c(j, i)$. Since we can write $H(J|I) = H(I|J) - H(I) + H(J)$, Eq. (A.16) then becomes $H(I|J) = 0$, meaning that knowing $J = j$ fixes the value of $I = i$.

Hence, the eigenstate $|\psi_k\rangle_{\mathcal{F}}$ only appears when we condition on a specific, single system state $|i\rangle$, which implies that different conditional states are distinguishable:

$$\rho_{\mathcal{F}|i} \rho_{\mathcal{F}|k} = \sum_{j,k} c(j, i) c(j, k) |\psi_j\rangle\langle\psi_j| \quad (\text{A.30})$$

$$= \sum_{j,k} c(j, i) c(j, i) \delta_{ik} |\psi_j\rangle\langle\psi_j| \quad (\text{A.31})$$

$$= 0, \quad (\text{A.32})$$

for all $i \neq k$. And hence we have bipartite spectrum broadcast structure for the post-measurement state $\rho_{S^\Pi \mathcal{F}} = \sum_i p_i |i\rangle\langle i| \otimes \rho_{\mathcal{F}|i}$.

Recall that strong quantum Darwinism also has the condition that $I(\mathcal{S} : \mathcal{F}) = \chi(\mathcal{S}^\Pi : \mathcal{F})$. This condition is called “surplus decoherence” by Zwolak and Zurek [130]—we made this a key component of strong quantum Darwinism. It implies

that the original state has the classical-quantum branching form $\sum_i p_i |i\rangle\langle i| \otimes \rho_{\mathcal{F}|i}$, *i.e.*, $\rho_{\mathcal{SF}} = \rho_{\mathcal{S}^\Pi \mathcal{F}}$ has bipartite spectrum broadcasting structure. Piani *et al.* [133] have also proven the correspondence between $I(\mathcal{S} : \mathcal{F}) = I_{acc}(\mathcal{S}^\Pi : \mathcal{F})$ and classical-classical $\mathcal{SF}$ states $\rho_{\mathcal{SF}} = \sum_{i,j} p_{ij} |i\rangle\langle i| \otimes |\psi_j\rangle\langle \psi_j|$.

A.1.2.1 Recovering multipartite spectrum broadcast structure with strong independence

To decompose $\mathcal{F}$ into distinct sub-environments, note that full strong quantum Darwinism requires sub-environments $\mathcal{E}_1, \dots, \mathcal{E}_F$ ($F > 1$) for which the strong quantum Darwinism condition holds, and therefore $\rho_{\mathcal{SE}_k} = \sum_i p_i |i\rangle\langle i| \otimes \rho_{\mathcal{E}_k|i}$, where each $\{\rho_{\mathcal{E}_k|i}\}_i$ are mutually distinguishable (by the proof above). Due to strong independence, there are no correlations between the environments conditioned on system state $|i\rangle$, hence the conditional state $\rho_{\mathcal{F}|i}$ is product. We therefore have that $\rho_{\mathcal{SF}} = \sum_i p_i |i\rangle\langle i| \otimes \rho_{\mathcal{E}_1|i} \otimes \dots \otimes \rho_{\mathcal{E}_F|i}$, *i.e.*, the multipartite spectrum broadcast structure. $\square$

A.2 Proof of Theorem 3

A.2.1 Proof ($\implies$): Invariant spectrum broadcast structure implies collective objectivity

This is immediate from Corollary 2: all systems satisfy strong quantum Darwinism (*and* spectrum broadcast structure) independently. Hence, all systems are objective, and thus by Definition 8, the joint state is collectively objective.

A.2.2 Proof ($\impliedby$): Collective objectivity implies invariant spectrum broadcast structure

From Corollary 2, strong quantum Darwinism is necessary and sufficient for objectivity. From the Definition 8 of collective objectivity, each subsystem in a collection must satisfy strong quantum Darwinism. We will prove that if we have collective objectivity, *i.e.*, strong quantum Darwinism is satisfied for all systems, then we have invariant spectrum broadcast structure:

Consider a collection of systems $\mathbf{A} = A_1 \cdots A_N$ with a joint state $\rho_{\mathbf{A}}$. If A_a is our preferred system, then we write it as $\rho_{A_a:\mathbf{A}\setminus A_a}$. If $\rho_{\mathbf{A}}$ is collectively objective,

then $\rho_{A_a:\mathbf{A}\setminus A_a}$ for all a has strong quantum Darwinism (Cor. 2), and hence has bipartite spectrum broadcast structure (Cor. 1):

$$\rho_{A_a:\mathbf{A}\setminus A_a} = \sum_{i_a} p_{i_a}^{A_a} |i_a\rangle\langle i_a|_{A_a} \otimes \mathbf{R}_i^{A\setminus A_a}, \quad \forall a = 1, \dots, N, \quad (\text{A.33})$$

where the conditional states are mutually distinguishable: $\mathbf{R}_i^{A\setminus A_a} \mathbf{R}_j^{A\setminus A_a} = 0$ for $i \neq j$. Hence, $\{\mathbf{R}_i^{A\setminus A_a}\}_i$ are mutually orthogonal, and furthermore, has support in the basis of

$$\{|i_1 \cdots i_{a-1} i_{a+1} \cdots i_N\rangle\}_{i_1 \cdots i_{a-1} i_{a+1} \cdots i_N} = \{|\mathbf{i}_{a\setminus a}\rangle\}_{\mathbf{i}_{a\setminus a}}, \quad (\text{A.34})$$

since from the state structure in Eq. (A.33), the system A_a has elements explicitly in $\{|i_a\rangle_{A_a}\}_{i_a}$. Hence we can write

$$\mathbf{R}_i^{A\setminus A_a} = \sum_{\mathbf{i}_{a\setminus a}} \langle \mathbf{i}_{a\setminus a} | \mathbf{R}_i^{A\setminus A_a} | \mathbf{i}_{a\setminus a} \rangle |\mathbf{i}_{a\setminus a}\rangle\langle \mathbf{i}_{a\setminus a}|_{\mathbf{A}\setminus A_a}. \quad (\text{A.35})$$

As the $\{\mathbf{R}_i^{A\setminus A_a}\}_i$ are mutually orthogonal/distinguishable, any particular element $|\mathbf{i}_{a\setminus a}\rangle_{\mathbf{A}\setminus A_a}$ only appears in a single $\mathbf{R}_i^{A\setminus A_a}$, that is, we can define disjoint nonzero sets of indices $\{\mathcal{A}_i^a\}_i$ where $\langle \mathbf{i}_{a\setminus a} | \mathbf{R}_i^{A\setminus A_a} | \mathbf{i}_{a\setminus a} \rangle \neq 0$ if and only if $\mathbf{i}_{a\setminus a} \in \mathcal{A}_i^a$. Hence, we can write:

$$\rho_{A_a:\mathbf{A}\setminus A_a} = \sum_{i_a} p_{i_a}^{A_a} |i_a\rangle\langle i_a|_{A_a} \otimes \sum_{\mathbf{i}_{a\setminus a} \in \mathcal{A}_i^a} \langle \mathbf{i}_{a\setminus a} | \mathbf{R}_i^{A\setminus A_a} | \mathbf{i}_{a\setminus a} \rangle |\mathbf{i}_{a\setminus a}\rangle\langle \mathbf{i}_{a\setminus a}|_{\mathbf{A}\setminus A_a}, \quad \forall a, \quad (\text{A.36})$$

$$= \sum_{i_a} \sum_{\mathbf{i}_{a\setminus a} \in \mathcal{A}_i^a} \left(p_{i_a}^{A_a} \langle \mathbf{i}_{a\setminus a} | \mathbf{R}_i^{A\setminus A_a} | \mathbf{i}_{a\setminus a} \rangle \right) |i_1 \cdots i_N\rangle\langle i_1 \cdots i_N|. \quad (\text{A.37})$$

Next, we prove that all $\mathbf{R}_{i_a}^{A\setminus A_a}$ are pure states and that there is only one effective summation. Consider $a = N$ first. This $\langle i_1, \dots, i_{N-1} | \mathbf{R}_{i_N}^{A\setminus A_N} | i_1, \dots, i_{N-1} \rangle \neq 0$ is nonzero if and only if $(i_1, \dots, i_{N-1}) \in \mathcal{A}_{i_N}^N$. This means that for a particular combination of $(i_1, \dots, i_{N-1})$, there is only one associated $|i_N\rangle$. Hence, we can write the index $i_N^* = i_N(i_1, \dots, i_{N-1})$ as a function of the other indices. (Note that we can have the same output i_N^* so long as the inputs $(i_1, \dots, i_{N-1})$ are different—so this can be a many-to-one mapping.) We can now remove the summation over i_N :

$$\rho_{\mathbf{A}} = \sum_{i_a} \sum_{\mathbf{i}_{a\setminus \{a,N\}} \in \mathcal{A}_{i_a}^a} p_{i_a}^{A_a} \langle \mathbf{i}_{a\setminus a} | \mathbf{R}_{i_a}^{A\setminus A_a} | \mathbf{i}_{a\setminus a} \rangle |i_1, \dots, i_{N-1}, i_N^*\rangle\langle i_1, \dots, i_{N-1}, i_N^*|, \quad \forall a. \quad (\text{A.38})$$

Similarly, the term $\langle i_1, \dots, i_{N-2} i_N^* | \mathbf{R}_{i_{N-1}}^{A \setminus A_{N-1}} | i_1, \dots, i_{N-2} i^* \rangle \neq 0$ if and only if $(i_1, \dots, i_{N-2}, i_N^*) \in \mathcal{A}_{i_{N-1}}^{N-1}$. This means that for a particular combination of $(i_1, \dots, i_{N-2}, i_N^*)$, there is only one associated $|i_{N-1}\rangle$. Hence, we can write the index $i_{N-1}^* = i_{N-1}(i_1, \dots, i_{N-2}, i_N^*)$ as a function of the other indices. Note that the only free indices remaining are $i_1, \dots, i_{N-2}$: there is a fixed one-to-one correspondence between i_N^* and i_{N-1}^* . By repeating this procedure, we consecutively define indices i_a in term of the others: $i_{N-2}^* = i_{N-2}(i_1, \dots, i_{N-3})$ and so forth, til we obtain that all the other indices are functions of i_1 : $i_a^* = i_a(i_1)$ for $a = 2, \dots, N$, and so

$$\rho_A = \sum_{i_1} p_{i_1}^{A_1} \langle i_2^* \dots i_N^* | \mathbf{R}_{i_1}^{A \setminus A_1} | i_2^* \dots i_N^* \rangle | i_1, i_2^*, \dots, i_N^* \rangle \langle i_1, i_2^*, \dots, i_N^* | \quad (\text{A.39})$$

$$= \sum_{i_1} p_{i_1}^{A_1} | i_1 \rangle \langle i_1 | \otimes \left(\langle i_2^* \dots i_N^* | \mathbf{R}_{i_1}^{A \setminus A_1} | i_2^* \dots i_N^* \rangle | i_2^*, \dots, i_N^* \rangle \langle i_2^*, \dots, i_N^* | \right). \quad (\text{A.40})$$

Hence, $\mathbf{R}_{i_1}^{A \setminus A_1} = | i_2^*, \dots, i_N^* \rangle \langle i_2^*, \dots, i_N^* |$ is pure, hence the joint state can be written as:

$$\rho_A = \sum_{i_1} p_{i_1}^{A_1} | i_1, i_2^*, \dots, i_N^* \rangle \langle i_1, i_2^*, \dots, i_N^* | := \sum_i p_i | i \dots i \rangle \langle i \dots i |_{A_1, \dots, A_N}, \quad (\text{A.41})$$

where we have defined $p_i := p_{i_1}^{A_1}$, and have redefined the indices labels $|i_1\rangle_{A_1} \rightarrow |i\rangle_{A_1}$, $|i_2^*\rangle_{A_2} \rightarrow |i\rangle_{A_2}$ and so forth. Since the order of the systems is not “special”, we could have equally applied the procedure to reduce the summation to any other index i_a , and found that all the states $\mathbf{R}_{i_a}^{A \setminus A_a}$ are pure. This is precisely invariant spectrum broadcast structure. $\square$

A.3 Upper-bound of the minimum distance to the set of spectrum broadcast structure states

A measure of spectrum broadcast structure (SBS) is the minimum distance to the set of SBS states. Here, we write an upper-bound to this minimum distance suitable for our particular model. The following derivation follows that of Mironowicz *et al.* [104], and can also be found in our paper, Thao P. Le and Alexandra Olaya-Castro, *Objectivity (or lack thereof): Comparison between predictions of quantum Darwinism and spectrum broadcast structure*, Phys. Rev. A 98, 032103 (2018) [1].

Let us write the general system-fragment state $\rho_{\mathcal{SF}}$ in some basis which we call $\{|i\rangle\}$ (this need not be the standard computation basis):

$$\rho_{\mathcal{SF}} = \sum_{i,j=0,1} \sum_{n,m \in \mathcal{F}} c_{ijnm} |i\rangle\langle j| \otimes |n\rangle\langle m| \equiv \rho_{\mathcal{SF}}^{\text{sep}} + \sigma_{\mathcal{SF}}. \quad (\text{A.42})$$

The separable component of the system-fragment state is

$$\rho_{\mathcal{SF}}^{\text{sep}} \equiv \sum_{i=0,1} \sum_{n,m \in \mathcal{F}} c_{iinn} |i\rangle\langle i| \otimes |n\rangle\langle m| \quad (\text{A.43})$$

$$= \sum_{i=0,1} p_i |i\rangle\langle i| \otimes \rho_{\mathcal{F}|i}, \quad (\text{A.44})$$

where

$$\rho_{\mathcal{F}|i} = (1/p_i) \sum_{n,m \in \mathcal{F}} c_{iinn} |n\rangle\langle m|, \quad \text{and} \quad p_i = \sum_{n \in \mathcal{F}} c_{iinn}. \quad (\text{A.45})$$

The non-separable remainder is

$$\sigma_{\mathcal{SF}} \equiv \sum_{i \neq j} \sum_{n,m \in \mathcal{F}} c_{ijnm} |i\rangle\langle j| \otimes |n\rangle\langle m|. \quad (\text{A.46})$$

Let $\{\Pi_i^{\mathcal{F}}\}_i$ be a complete set of projectors to measure the environment fragment (we will later optimise over the sets). A projection of Π_i will result in successfully-determined state $\Pi_i^{\mathcal{F}} \rho_{\mathcal{F}|i} \Pi_i^{\mathcal{F}}$ with probability $\text{tr}[\rho_{\mathcal{F}|i} \Pi_i^{\mathcal{F}}]$. With probability $1 - \text{tr}[\rho_{\mathcal{F}|i} \Pi_i^{\mathcal{F}}] = \text{tr}[\rho_{\mathcal{F}|i} (\mathbb{1}^{\mathcal{F}} - \Pi_i^{\mathcal{F}})]$, we will obtain an error. The error in discriminating between the different $\rho_i^{\mathcal{F}}$ states (which occur with probability p_i) is then [104]

$$\text{Err}[\{p_i, \rho_{\mathcal{F}|i}\}, \{\Pi_i^{\mathcal{F}}\}] = \sum_i p_i \text{tr}[\rho_{\mathcal{F}|i} (\mathbb{1} - \Pi_i^{\mathcal{F}})]. \quad (\text{A.47})$$

The ideal spectrum broadcast structure state for the set of measurements $\{\Pi_i^{\mathcal{F}}\}_i$ is

$$\rho_{\mathcal{SF}}^{\text{SBS}} = \sum_i q_i |i\rangle\langle i| \otimes \tilde{\rho}_{\mathcal{F}|i}, \quad (\text{A.48})$$

where $\tilde{\rho}_{\mathcal{F}|i} = \Pi_i^{\mathcal{F}} \rho_{\mathcal{F}|i} \Pi_i^{\mathcal{F}} / \text{tr}[\rho_{\mathcal{F}|i} \Pi_i^{\mathcal{F}}]$ are states that are perfectly distinguishable by $\{\Pi_i^{\mathcal{F}}\}_i$, and $q_i = p_i \text{tr}[\rho_{\mathcal{F}|i} \Pi_i^{\mathcal{F}}] / \sum_j p_j \text{tr}[\rho_{\mathcal{F}|j} \Pi_j^{\mathcal{F}}]$ are the normalised probabilities. Using this candidate form of a possible spectrum broadcast structure state,

$$M^{\text{SBS}}(\rho_{\mathcal{SF}}) \leq \|\rho_{\mathcal{SF}} - \rho_{\mathcal{SF}}^{\text{SBS}}\|_1 = \|\rho_{\mathcal{SF}}^{\text{sep}} + \sigma_{\mathcal{SF}} - \rho_{\mathcal{SF}}^{\text{SBS}}\|_1 \quad (\text{A.49})$$

$$\leq \|\rho_{\mathcal{SF}}^{\text{sep}} - \rho_{\mathcal{SF}}^{\text{SBS}}\|_1 + \|\sigma_{\mathcal{SF}}\|_1, \quad (\text{A.50})$$

following from the triangle inequality. The second term $\|\sigma_{\mathcal{SF}}\|_1 = 2T(\rho_{\mathcal{SF}}, \rho_{\mathcal{SF}}^{\text{sep}})$ can be easily calculated numerically. Focusing on the first term,

$$\|\rho_{\mathcal{SF}}^{\text{sep}} - \rho_{\mathcal{SF}}^{\text{SBS}}\|_1 \leq \sum_{i=0,1} p_i \left\| \rho_{\mathcal{F}|i} - \frac{\Pi_i \rho_{\mathcal{F}|i} \Pi_i}{\sum_j p_j \text{tr}[\rho_{\mathcal{F}|i} \Pi_j]} \right\|_1, \quad (\text{A.51})$$

following from the triangle inequality, and that $\| |i\rangle\langle i| \otimes F_i \|_1 = \text{tr}[|i\rangle\langle i|] \|F_i\|_1$. Since

$$\left\| \rho_{\mathcal{F}|i} - \frac{\Pi_i \rho_{\mathcal{F}|i} \Pi_i}{\sum_j p_j \text{tr}[\rho_{\mathcal{F}|i} \Pi_j]} \right\| = \text{tr} \left[\rho_{\mathcal{F}|i} - \frac{\Pi_i \rho_{\mathcal{F}|i} \Pi_i}{\sum_j p_j \text{tr}[\rho_{\mathcal{F}|i} \Pi_j]} \right] \quad (\text{A.52})$$

$$\leq \text{tr}[\rho_{\mathcal{F}|i}(\mathbb{1} - \Pi_i)], \quad (\text{A.53})$$

we have that $\|\rho_{\mathcal{SF}}^{\text{sep}} - \rho_{\mathcal{SF}}^{\text{SBS}}\|_1 \leq \text{Err}[\{p_i, \rho_{\mathcal{F}|i}\}, \{\Pi_i^{\mathcal{F}}\}]$, which, after minimisation, is bounded by the optimal discrimination error:

$$\min_{\rho_{\mathcal{SF}}^{\text{SBS}}} \|\rho_{\mathcal{SF}}^{\text{sep}} - \rho_{\mathcal{SF}}^{\text{SBS}}\|_1 \leq \min_{\{\Pi_i^{\mathcal{F}}\}} \text{Err}[\{p_i, \rho_{\mathcal{F}|i}\}, \{\Pi_i^{\mathcal{F}}\}] \quad (\text{A.54})$$

$$\leq \sum_{i \neq j} \sqrt{p_i p_j} B(\rho_{\mathcal{F}|i}, \rho_{\mathcal{F}|j}), \quad (\text{A.55})$$

where $B(\rho_1, \rho_2) = \|\sqrt{\rho_1} \sqrt{\rho_2}\|_1$ is the fidelity [104]. Hence,

$$M^{\text{SBS}}(\rho_{\mathcal{SF}}) \leq \sum_{i \neq j} \sqrt{p_i p_j} B(\rho_{\mathcal{F}|i}, \rho_{\mathcal{F}|j}) + \|\sigma_{\mathcal{SF}}\|_1. \quad (\text{A.56})$$

$$\Rightarrow M^{\text{SBS}}(\rho_{\mathcal{SF}}) = \min_{\sigma_{\mathcal{SF}}} \left[\|\sigma_{\mathcal{SF}}\|_1 + \sum_{i \neq j} \sqrt{p_i p_j} B(\rho_{\mathcal{F}|i}, \rho_{\mathcal{F}|j}) \right], \quad (\text{A.57})$$

where the minimisation over $\sigma_{\mathcal{SF}}$ translates to minimising over the optimal basis $\{|i\rangle\}$ to write the system in, that is, over the set of *system* projectors $\{P_i^{\mathcal{S}}\}$ that define $\rho_{\mathcal{SF}}^{\text{sep}}(\{P_i^{\mathcal{S}}\}) = \sum_i P_i^{\mathcal{S}} \rho_{\mathcal{SF}} P_i^{\mathcal{S}}$ and $\sigma_{\mathcal{SF}} = \rho_{\mathcal{SF}} - \rho_{\mathcal{SF}}^{\text{sep}}(\{P_i^{\mathcal{S}}\})$.

B

Proofs for discord-destroying channels

In this appendix, we give some proofs relating to the quantum discord-destroying channels from Chapter 5.

B.1 Proof of Lemma 1: convex subsets of classical-quantum states

This following proof can also be found in our preprint, Thao P. Le, *Quantum discord-breaking and discord-annihilating channels*, (2019), arxiv:1907.05440v1 [3].

Zero discord states must satisfy the necessary condition $[\rho_{\mathbf{CQ}}, \rho_A \otimes \mathbb{1}_B] = 0$ [210]. We want the following convex combination of two CQ states to also be zero-discord:

$$\rho_{AB} = p_1 \sum_i q_i |\psi_i\rangle\langle\psi_i| \otimes \sigma_i + p_2 \sum_j r_j |\phi_j\rangle\langle\phi_j| \otimes \varsigma_j. \quad (\text{B.1})$$

The new state must satisfy the following necessary condition for zero-discord:

$$\sum_{ij} q_i r_j [|\psi_i\rangle\langle\psi_i|, |\phi_j\rangle\langle\phi_j|] \otimes (\sigma_i - \varsigma_j) \stackrel{!}{=} 0. \quad (\text{B.2})$$

Note that the elements $\{[|\psi_i\rangle\langle\psi_i|, |\phi_j\rangle\langle\phi_j|]\}_j$ are *linearly independent* from the elements $\{[|\psi_{i'}\rangle\langle\psi_{i'}|, |\phi_j\rangle\langle\phi_j|]\}_j$ for $i \neq i'$ unless the commutator is zero.¹ Hence,

¹ We can see this by simply choosing the basis $|\psi_a\rangle = (0, \dots, 1, \dots, 0)^T$ (with 1 at position

we must have

$$\sum_j r_j [|\psi_i\rangle\langle\psi_i|, |\phi_j\rangle\langle\phi_j|] \otimes (\sigma_i - \varsigma_j) \stackrel{!}{=} 0 \quad \forall i. \quad (\text{B.4})$$

Similarly, the commutators $[|\psi_i\rangle\langle\psi_i|, |\phi_j\rangle\langle\phi_j|]$, and $[|\psi_i\rangle\langle\psi_i|, |\phi_{j'}\rangle\langle\phi_{j'}|]$, $j \neq j'$ are linearly independent unless the commutator is zero. Regardless, we reduce to the following condition:

$$[|\psi_i\rangle\langle\psi_i|, |\phi_j\rangle\langle\phi_j|] \otimes (\sigma_i - \varsigma_j) \stackrel{!}{=} 0 \quad \forall i, j, \quad (\text{B.5})$$

which is true if and only if we have at least one of: $\sigma_i = \varsigma_j$, $|\psi_i\rangle\langle\psi_i| = |\phi_j\rangle\langle\phi_j|$ or $\langle\psi_i|\phi_j\rangle = 0$. Now we will construct the state satisfying these conditions.

First, let $\mathfrak{J}_0 \ni i$ be all the indices where $|\psi_i\rangle$ which is orthogonal to *all* other $\{|\phi_j\rangle\}$. Let $\mathfrak{J}_0 \ni j$ be all the indices where $|\phi_j\rangle$ which is orthogonal to *all* other $\{|\psi_i\rangle\}$. These terms stay as they are in the resultant zero-discord state:

$$[\rho_{AB}]_{\text{pt } 0} = p_1 \sum_{i \in \mathfrak{J}_0} q_i |\psi_i\rangle\langle\psi_i| \otimes \sigma_i + p_2 \sum_{j \in \mathfrak{J}_0} r_j |\phi_j\rangle\langle\phi_j| \otimes \varsigma_j \quad (\text{B.6})$$

$$= \sum_{i \in \mathfrak{J}'_0} t_i |\psi_i\rangle\langle\psi_i| \otimes \tilde{\sigma}_{B|i}, \quad (\text{B.7})$$

where we have just relabeled $|\phi_j\rangle$ to suitable new $|\psi_{i^*}\rangle$, with $t_i \tilde{\sigma}_{B|i} = p_1 q_i \sigma_i$ for $i \in \mathfrak{J}_0$ and $t_{i^*} \tilde{\sigma}_{B|i^*} = p_2 r_{j_1(i)} \varsigma_{j_1(i)}$ for $i^* \in \mathfrak{J}_0$.

Next, take the indices i, j where $|\psi_i\rangle = |\phi_j\rangle$. Let $\mathfrak{J}_1$ be all the indices i where this occurs, with corresponding $j = j_1(i)$ where j_1 is a one-to-one function that connects the indices. This leads to the second component:

$$[\rho_{AB}]_{\text{pt } 1} = \sum_{i \in \mathfrak{J}_1} |\psi_i\rangle\langle\psi_i| \otimes (p_1 q_i \sigma_i + p_2 r_{j_1(i)} \varsigma_{j_1(i)}) \quad (\text{B.8})$$

$$= \sum_{i \in \mathfrak{J}_1} t_i |\psi_i\rangle\langle\psi_i| \otimes \tilde{\sigma}_{B|i}, \quad (\text{B.9})$$

a), leading to the following form for the commutator:

$$[|\psi_a\rangle\langle\psi_a|, |\phi_j\rangle\langle\phi_j|] = \begin{bmatrix} 0 & * & 0 & 0 & 0 \\ * & 0 & * & * & * \\ 0 & * & 0 & 0 & 0 \\ 0 & * & 0 & 0 & 0 \\ 0 & * & 0 & 0 & 0 \end{bmatrix}, \quad (\text{B.3})$$

where the nonzero elements exist along either row a or column a . Certain matrix elements only exist at a precise index $i (= a)$, hence there is linearly independence across the different indices i .

where we have defined $t_i \tilde{\sigma}_{B|i} = p_1 q_i \sigma_i + p_2 r_{j_1(i)} \varsigma_{j_1(i)}$, where $\tilde{\sigma}_{B|i}$ is some state. In fact, the form of the terms in Eq. (B.7) are the same as in Eq. (B.9) and we can merge them together.

The remaining $i \notin \mathfrak{I}_0, \mathfrak{I}_1, j \notin \mathfrak{I}_0, j_1(\mathfrak{I}_1)$ now leave terms $\{|\psi_i\rangle\}, \{|\phi_j\rangle\}$ that are neither equal with each other nor orthogonal with all others, hence we *must* have $\sigma_i = \varsigma_j = R_{B|i}$. This means that we can define subsets of indices (i, j) , $i \in \mathcal{R}_b^A$ and $j \in \mathcal{R}_b^B$ where $\sigma_i = \varsigma_j = R_{B|b}$ (and hence where $[|\psi_i\rangle\langle\psi_i|, |\phi_j\rangle\langle\phi_j|] \neq 0$) and:

$$[\rho_{AB}]_{\text{pt } 2} = \sum_b \left(\sum_{i \in \mathcal{R}_b^A} p_1 q_i |\psi_i\rangle\langle\psi_i| + \sum_{j \in \mathcal{R}_b^B} p_2 r_j |\phi_j\rangle\langle\phi_j| \right) \otimes R_{B|b} \quad (\text{B.10})$$

$$\equiv \sum_{b \in \mathfrak{I}_2} t_b \tilde{\rho}_{A|b} \otimes R_{B|b}, \quad (\text{B.11})$$

where we have defined $t_b \tilde{\rho}_{A|b} = \sum_{i \in \mathcal{R}_b^A} p_1 q_i |\psi_i\rangle\langle\psi_i| + \sum_{j \in \mathcal{R}_b^B} p_2 r_j |\phi_j\rangle\langle\phi_j|$. Furthermore, for different $b \neq b'$, the local states on A are *orthogonal*, $\tilde{\rho}_{A|b} \tilde{\rho}_{A|b'} = 0$, since if they were not orthogonal, they must have the same $R_{B|b}$. This means we can define orthogonal subspaces $\{\mathbf{V}_{A|b}\}$ on A , such that $\tilde{\rho}_{A|b} \in \mathbf{V}_{A|b}$.

This concludes all the indices, hence

$$\rho_{AB} = [\rho_{AB}]_{\text{pt } 0} + [\rho_{AB}]_{\text{pt } 1} + [\rho_{AB}]_{\text{pt } 2}. \quad (\text{B.12})$$

Each part is zero-discord, with the local state on A existing on orthogonal subspaces, and their sum is also zero discord. Without loss of generality, additional convex combinations with other zero-discord states must also match this structure. Hence, once the points in the set have been decided, the states in that set take form:

$$\sum_{i_1 \in \mathfrak{I}_1} t_{i_1} |\psi_{i_1}\rangle\langle\psi_{i_1}|_A \otimes \tilde{\sigma}_{B|i_1} + \sum_{i_2 \in \mathfrak{I}_2} t_{i_2} \tilde{\rho}_{A|i_2} \otimes R_{B|i_2}, \quad (\text{B.13})$$

where the index sets $\mathfrak{I}_1, \mathfrak{I}_2$ are disjoint, the probabilities $0 \leq t_i \leq 1$, $\sum_i t_i = 1$, $\tilde{\sigma}_i^B$ live in a convex subset of states on B and $|\psi_i\rangle\langle\psi_i| \in \mathbf{V}_{A|i}$, $\tilde{\rho}_{A|i} \in \mathbf{V}_{A|i}$ live in a convex subset of the states on A that are orthogonal subspaces: *i.e.*, $\mathbf{V}_{A|i} \cap \mathbf{V}_{A|i'} = \emptyset$ for $i \neq i'$. Simplifying gives the Lemma. $\square$

B.2 Proof of Theorem 4: characterisation of discord-destroying channels

Since $\Phi_{\mathbf{DA}}^{AB}$ is a quantum channel, it is a linear map. The set of all density states $\text{BL}(\mathcal{H}_{AB})$ is convex; hence its image, $\Phi_{\mathbf{DA}}^{AB}(\text{BL}(\mathcal{H}_{AB})) \subset \mathbf{CQ}(\mathcal{H}_{AB})$, must also be convex, and thus must be a strict subset of zero-discord states which are precisely defined by Lemma 1.

Now, by contradiction, suppose there is a more general map $\mathcal{F}^{AB}$ that produces the particular convex set with states of form

$$\mathcal{F}^{AB}(\rho_{AB}) = \rho_{\text{CQ}} = \sum_{i_1 \in \mathfrak{I}_1} t_{i_1} |\psi_{i_1}\rangle\langle\psi_{i_1}|_A \otimes \tilde{\sigma}_{B|i_1} + \sum_{i_2 \in \mathfrak{I}_2} t_{i_2} \tilde{\rho}_{A|i_2} \otimes R_{B|i_2}. \quad (\text{B.14})$$

Since the output of $\mathcal{F}^{AB}$ has the above form, it is invariant under the projection channels $\Pi_i^A(\cdot)$:

$$\sum_i \Pi_i^A(\mathcal{F}^{AB}(\rho_{AB})) = \mathcal{F}^{AB}(\rho_{AB}), \quad (\text{B.15})$$

$$\text{where } \Pi_i^A(\cdot) = \begin{cases} |\psi_i\rangle\langle\psi_i|_A(\cdot)|\psi_i\rangle\langle\psi_i|_A, & i \in \mathfrak{I}_1 \\ P_{A|i}(\cdot)P_{A|i}, & i \in \mathfrak{I}_2. \end{cases} \quad (\text{B.16})$$

If we then consider the cases for $i \in \mathfrak{I}_2$, then

$$i \in \mathfrak{I}_2 : \Pi_i^A(\mathcal{F}^{AB}(\rho_{AB})) = \Pi_i^A(\rho_{\text{CQ}}) = t_i \tilde{\rho}_{A|i} \otimes R_{B|i}. \quad (\text{B.17})$$

Since $R_{B|i}$ is fixed given i , (and is same for any input state ρ_{AB}), this is invariant under the point channel $\Phi_{\text{point}|i}^B = \text{tr}_B[\cdot]R_{B|i}$,

$$i \in \mathfrak{I}_2 : \Pi_i^A(\mathcal{F}^{AB}(\rho_{AB})) = \Phi_{\text{point}|i}^B(\Pi_i^A(\mathcal{F}^{AB}(\rho_{AB}))). \quad (\text{B.18})$$

Hence, in fact, $\mathcal{F}^{AB}$ is invariant under the combination of projection channels and point channels on $i \in \mathfrak{I}_2$:

$$\mathcal{F}^{AB}(\rho_{AB}) = \sum_i \Pi_i^A(\mathcal{F}^{AB}(\rho_{AB})) \quad (\text{B.19})$$

$$= \sum_{i \in \mathfrak{I}_1} \Pi_i^A(\mathcal{F}^{AB}(\rho_{AB})) + \sum_{i \in \mathfrak{I}_2} \Phi_{\text{point}|i}^B(\Pi_i^A(\mathcal{F}^{AB}(\rho_{AB}))), \quad (\text{B.20})$$

but this is precisely the form discord-destroying channel in the Theorem, hence the form of the discord-destroying channel in the Theorem is the more general. $\square$

C

Extended results and proofs for the quantum magnetic sensor

In this appendix, we give some extended calculations and the proofs for the results in Chapter 7, as well as extra simulations in Sec. C.2 for a different environment interaction and Sec. C.3 that examines quantum coherence yield rather than initial coherence.

C.1 Proofs of various lemmas and propositions

C.1.1 Proof of Lemma 3

Recall the statement of Lemma 3: $[\rho_S(0), \hat{H}_{\vec{B}}(\theta, \phi)] = 0$ for all angles (θ, ϕ) if and only if

$$\begin{aligned} \rho_S = & \frac{\mathbb{1}_{ABC}}{8} + p_{AB}\vec{\sigma}^A \cdot \vec{\sigma}^B + p_{AC}\vec{\sigma}^A \cdot \vec{\sigma}^C + p_{BC}\vec{\sigma}^B \cdot \vec{\sigma}^C \\ & + p_{ABC}(\sigma_x \otimes \sigma_y \otimes \sigma_z + \sigma_z \otimes \sigma_x \otimes \sigma_y + \sigma_y \otimes \sigma_z \otimes \sigma_x) \\ & - p_{ABC}(\sigma_x \otimes \sigma_z \otimes \sigma_y + \sigma_y \otimes \sigma_x \otimes \sigma_z + \sigma_z \otimes \sigma_y \otimes \sigma_x). \end{aligned} \quad (\text{C.1})$$

Proof. The state above satisfies the zero commutation relation—this can be seen by straightforward calculation.

In the reverse direction: we write the system state in the following general form $\rho_S = \sum_{ijk=0,x,y,z} p_{ijk} \sigma_i \otimes \sigma_j \otimes \sigma_k$, where $\sigma_0 = \mathbb{1}$. The commutator expands

as follows:

$$\left[\rho_S(0), \hat{H}_{\vec{B}}(\theta, \phi) \right] = \frac{\gamma}{2} \sum_{ijk=0,x,y,z} p_{ijk} \left[\sigma_i \otimes \sigma_j \otimes \sigma_k, \vec{B} \cdot \vec{\sigma}^A + \vec{B} \cdot \vec{\sigma}^B + \vec{B} \cdot \vec{\sigma}^C \right] \stackrel{!}{=} 0. \quad (\text{C.2})$$

By writing $\vec{B} \cdot \vec{\sigma} = B_x \sigma_x + B_y \sigma_y + B_z \sigma_z$, we further expand collect/separate the linearly independent terms, *i.e.* in the matrix basis $\{\mathbb{1}, \sigma_x, \sigma_y, \sigma_z\}^{\otimes 3}$,

$$\frac{1}{i\gamma} \left[\rho_S(0), \hat{H}_{\vec{B}}(\theta, \phi) \right] \quad (\text{C.3})$$

$$= \begin{pmatrix} \sigma_x(p_{y00}B_z - p_{z00}B_y) + \sigma_y(p_{z00}B_x - p_{x00}B_z) \\ + \sigma_z(p_{x00}B_y - p_{y00}B_x) \end{pmatrix} \otimes \mathbb{1} \otimes \mathbb{1} \quad (\text{C.4})$$

$$+ \mathbb{1} \otimes \begin{pmatrix} \sigma_x(p_{0y0}B_z - p_{0z0}B_y) + \sigma_y(p_{0z0}B_x - p_{0x0}B_z) \\ + \sigma_z(p_{0x0}B_y - p_{0y0}B_x) \end{pmatrix} \otimes \mathbb{1} \quad (\text{C.5})$$

$$+ \mathbb{1} \otimes \mathbb{1} \otimes \begin{pmatrix} \sigma_x(p_{00y}B_z - p_{00z}B_y) + \sigma_y(p_{00z}B_x - p_{00x}B_z) \\ + \sigma_z(p_{00x}B_y - p_{00y}B_x) \end{pmatrix} \quad (\text{C.6})$$

$$+ \mathbb{1} \otimes \left[\sum_{k=x,y,z} \begin{pmatrix} \sigma_x(p_{0yk}B_z - p_{0zk}B_y) + \sigma_y(p_{0zk}B_x - p_{0xk}B_z) \\ + \sigma_z(p_{0xk}B_y - p_{0yk}B_x) \end{pmatrix} \otimes \sigma_k \right. \\ \left. + \sum_{j=x,y,z} \sigma_j \otimes \begin{pmatrix} \sigma_x(p_{0jy}B_z - p_{0jz}B_y) + \sigma_y(p_{0jz}B_x - p_{0jx}B_z) \\ + \sigma_z(p_{0jx}B_y - p_{0jy}B_x) \end{pmatrix} \right] \quad (\text{C.7})$$

$$+ \left[\sum_{k=x,y,z} \begin{pmatrix} \sigma_x(p_{y0k}B_z - p_{z0k}B_y) + \sigma_y(p_{z0k}B_x - p_{x0k}B_z) \\ + \sigma_z(p_{x0k}B_y - p_{y0k}B_x) \end{pmatrix} \otimes \mathbb{1} \otimes \sigma_k \right. \\ \left. + \sum_{i=x,y,z} \sigma_i \otimes \mathbb{1} \otimes \begin{pmatrix} \sigma_x(p_{i0y}B_z - p_{i0z}B_y) + \sigma_y(p_{i0z}B_x - p_{i0x}B_z) \\ + \sigma_z(p_{i0x}B_y - p_{i0y}B_x) \end{pmatrix} \right] \quad (\text{C.8})$$

$$+ \left[\sum_{j=x,y,z} \begin{pmatrix} \sigma_x(p_{y j 0}B_z - p_{z j 0}B_y) + \sigma_y(p_{z j 0}B_x - p_{x j 0}B_z) \\ + \sigma_z(p_{x j 0}B_y - p_{y j 0}B_x) \end{pmatrix} \otimes \sigma_j \right. \\ \left. + \sum_{i=x,y,z} \sigma_i \otimes \begin{pmatrix} \sigma_x(p_{i y 0}B_z - p_{i z 0}B_y) + \sigma_y(p_{i z 0}B_x - p_{i x 0}B_z) \\ + \sigma_z(p_{i x 0}B_y - p_{i y 0}B_x) \end{pmatrix} \right] \otimes \mathbb{1} \quad (\text{C.9})$$

$$+ \left\{ \sum_{j,k=x,y,z} \begin{pmatrix} \sigma_x(p_{yjk}B_z - p_{zjk}B_y) + \sigma_y(p_{zjk}B_x - p_{xjk}B_z) \\ + \sigma_z(p_{xjk}B_y - p_{yjk}B_x) \end{pmatrix} \otimes \sigma_j \otimes \sigma_k \right. \\ \left. + \sum_{i,k=x,y,z} \sigma_i \otimes \begin{pmatrix} \sigma_x(p_{i y k}B_z - p_{i z k}B_y) + \sigma_y(p_{i z k}B_x - p_{i x k}B_z) \\ + \sigma_z(p_{i x k}B_y - p_{i y k}B_x) \end{pmatrix} \otimes \sigma_k \right. \\ \left. + \sum_{i,j=x,y,z} \sigma_i \otimes \sigma_j \otimes \begin{pmatrix} \sigma_x(p_{i j y}B_z - p_{i j z}B_y) + \sigma_y(p_{i j z}B_x - p_{i j x}B_z) \\ + \sigma_z(p_{i j x}B_y - p_{i j y}B_x) \end{pmatrix} \right\} \quad (\text{C.10})$$

$$\stackrel{!}{=} 0. \quad (\text{C.11})$$

Each of the lines above are linearly independent from each other. Thus, we will consider each separately.

For Eq. (C.4):

$$\begin{aligned} & (\sigma_x(p_{y00}B_z - p_{z00}B_y) + \sigma_y(p_{z00}B_x - p_{x00}B_z) + \sigma_z(p_{x00}B_y - p_{y00}B_x)) \otimes \mathbb{1} \otimes \mathbb{1} \stackrel{!}{=} 0 \\ \implies & p_{y00}B_z \stackrel{!}{=} p_{z00}B_y, \quad p_{z00}B_x \stackrel{!}{=} p_{x00}B_z, \quad p_{x00}B_y \stackrel{!}{=} p_{y00}B_x. \end{aligned} \quad (\text{C.12})$$

We wish to find the parameters p_{ijk} in which these equations hold for all allowed (B_x, B_y, B_z) magnetic fields. If, for example the magnetic field is in the z direction, with $B_z \neq 0$ and $B_y = 0$, then $p_{y00}B_z = p_{z00}B_y$ means that we must have $p_{y00} = 0$ *etc.* By considering different directions, the above holds for all field angles iff:

$$p_{x00} = p_{y00} = p_{z00} = 0. \quad (\text{C.13})$$

Similarly, Eqs. (C.5) and (C.6) give:

$$p_{0x0} = p_{0y0} = p_{0z0} = 0 \quad (\text{C.14})$$

$$p_{00x} = p_{00y} = p_{00z} = 0. \quad (\text{C.15})$$

Now consider Eq. (C.7), which expands out to:

$$0 \stackrel{!}{=} \mathbb{1} \otimes \sigma_x \otimes \sigma_x ((p_{0yx}B_z - p_{0zx}B_y + p_{0xy}B_z - p_{0xz}B_y)) \quad (\text{C.16})$$

$$+ \mathbb{1} \otimes \sigma_x \otimes \sigma_y (p_{0yy}B_z - p_{0zy}B_y + p_{0xz}B_x - p_{0xx}B_z) \quad (\text{C.17})$$

$$+ \mathbb{1} \otimes \sigma_x \otimes \sigma_z (p_{0yz}B_z - p_{0zz}B_y + p_{0xx}B_y - p_{0xy}B_x) \quad (\text{C.18})$$

$$+ \mathbb{1} \otimes \sigma_y \otimes \sigma_x (p_{0zx}B_x - p_{0xx}B_z + p_{0yy}B_z - p_{0yz}B_y) \quad (\text{C.19})$$

$$+ \mathbb{1} \otimes \sigma_y \otimes \sigma_y (p_{0zy}B_x - p_{0xy}B_z + p_{0yz}B_x - p_{0yx}B_z) \quad (\text{C.20})$$

$$+ \mathbb{1} \otimes \sigma_y \otimes \sigma_z (p_{0zz}B_x - p_{0xz}B_z + p_{0yx}B_y - p_{0yy}B_x) \quad (\text{C.21})$$

$$+ \mathbb{1} \otimes \sigma_z \otimes \sigma_x (p_{0xx}B_y - p_{0yx}B_x + p_{0zy}B_z - p_{0zz}B_y) \quad (\text{C.22})$$

$$+ \mathbb{1} \otimes \sigma_z \otimes \sigma_y (p_{0xy}B_y - p_{0yy}B_x + p_{0zz}B_x - p_{0zx}B_z) \quad (\text{C.23})$$

$$+ \mathbb{1} \otimes \sigma_z \otimes \sigma_z (p_{0xz}B_y - p_{0yz}B_x + p_{0zx}B_y - p_{0zy}B_x), \quad (\text{C.24})$$

and each of those coefficients must be zero. For the coefficient of $\mathbb{1} \otimes \sigma_x \otimes \sigma_x$,

$$p_{0yx}B_z - p_{0zx}B_y + p_{0xy}B_z - p_{0xz}B_y \stackrel{!}{=} 0 \quad (\text{C.25})$$

$$\implies B_z(p_{0yx} + p_{0xy}) = B_y(p_{0zx} + p_{0xz}), \quad (\text{C.26})$$

which holds for all magnetic field directions iff

$$p_{0yx} = -p_{0xy}, \quad p_{0zx} = -p_{0xz}, \quad (\text{C.27})$$

(i.e. by considering the value of the parameters when the field is *only* in x, y , or z direction). For the coefficient of $\mathbb{1} \otimes \sigma_x \otimes \sigma_y$,

$$p_{0yy}B_z - p_{0zy}B_y + p_{0xz}B_x - p_{0xx}B_z \stackrel{!}{=} 0 \quad (\text{C.28})$$

$$\implies (p_{0yy} - p_{0xx})B_z = p_{0zy}B_y - p_{0xz}B_x, \quad (\text{C.29})$$

$$\implies p_{0yy} = p_{0xx}, \quad p_{0xz} = 0, \quad p_{0zy} = 0. \quad (\text{C.30})$$

By considering all the linearly independent coefficients of Eq. (C.7), we obtain in summary:

$$p_{0xx} = p_{0yy} = p_{0zz} \quad (\text{C.31})$$

$$p_{0jk} = 0 \quad \text{for } j \neq k, \quad j, k \in \{x, y, z\}. \quad (\text{C.32})$$

Similarly (by considering the permutations), Eqs. (C.8) and (C.9) give us:

$$p_{x0x} = p_{y0y} = p_{z0z} \quad (\text{C.33})$$

$$p_{i0k} = 0 \quad \text{for } i \neq k, \quad i, k \in \{x, y, z\} \quad (\text{C.34})$$

$$p_{xx0} = p_{yy0} = p_{zz0} \quad (\text{C.35})$$

$$p_{ij0} = 0 \quad \text{for } i \neq j, \quad i, j \in \{x, y, z\}. \quad (\text{C.36})$$

At this point, our current reduced system state is

$$\begin{aligned} \rho_S = & p_{000} \mathbb{1} \otimes \mathbb{1} \otimes \mathbb{1} + p_{xx0} \vec{\sigma}^A \cdot \vec{\sigma}^B \otimes \mathbb{1}^C + p_{x0x} \vec{\sigma}^A \cdot \vec{\sigma}^C \otimes \mathbb{1}^B \\ & + p_{0xx} \mathbb{1}^A \otimes \vec{\sigma}^B \cdot \vec{\sigma}^C + \sum_{i,j,k=x,y,z} p_{ijk} \sigma_i \otimes \sigma_j \otimes \sigma_k. \end{aligned} \quad (\text{C.37})$$

Finally, for Eq. (C.10):

$$\begin{aligned} 0 & \stackrel{!}{=} \sum_{ijk=x,y,z} \sigma_i (p_{i\oplus 1,jk} B_{i\oplus 2} - p_{i\oplus 2,jk} B_{i\oplus 1}) \otimes \sigma_j \otimes \sigma_k \\ & + \sum_{ijk=x,y,z} \sigma_i \otimes (\sigma_j (p_{i,j\oplus 1,k} B_{j\oplus 2} - p_{i,j\oplus 2,k} B_{j\oplus 1})) \otimes \sigma_k \\ & + \sum_{ijk=x,y,z} \sigma_i \otimes \sigma_j \otimes \sigma_k (p_{i,j,k\oplus 1} B_{k\oplus 2} - p_{i,j,k\oplus 2} B_{k\oplus 1}) \\ & = \sum_{ijk=x,y,z} \sigma_i \otimes \sigma_j \otimes \sigma_k \left\{ \begin{aligned} & p_{i\oplus 1,jk} B_{i\oplus 2} - p_{i\oplus 2,jk} B_{i\oplus 1} \\ & + p_{i,j\oplus 1,k} B_{j\oplus 2} - p_{i,j\oplus 2,k} B_{j\oplus 1} \\ & + p_{i,j,k\oplus 1} B_{k\oplus 2} - p_{i,j,k\oplus 2} B_{k\oplus 1} \end{aligned} \right\} \quad (\text{C.38}) \\ \implies 0 & \stackrel{!}{=} p_{i\oplus 1,jk} B_{i\oplus 2} - p_{i\oplus 2,jk} B_{i\oplus 1} + p_{i,j\oplus 1,k} B_{j\oplus 2} - p_{i,j\oplus 2,k} B_{j\oplus 1} \end{aligned}$$

$$+ p_{i,j,k\oplus 1} B_{k\oplus 2} - p_{i,j,k\oplus 2} B_{k\oplus 1}, \quad \forall i, j, k \in \{x, y, z\}, \quad (\text{C.39})$$

where the addition is modulo 3, ordered $\{x, y, z\}$ so that, for example, $z \oplus 1 = x$.

Now consider if $i = j, k = i \oplus 1$:

$$\begin{aligned} 0 &\stackrel{!}{=} B_{i\oplus 1}(-p_{i\oplus 2,i,i\oplus 1} - p_{i,i\oplus 2,i\oplus 1}) + B_{i\oplus 2}(p_{i\oplus 1,i,i\oplus 1} + p_{i,i\oplus 1,i\oplus 1}) \\ &\quad + p_{i,i,i\oplus 2} B_i - p_{i,i,i} B_{i\oplus 2} \end{aligned} \quad (\text{C.40})$$

$$\begin{aligned} &= B_{i\oplus 1}(-p_{i\oplus 2,i,i\oplus 1} - p_{i,i\oplus 2,i\oplus 1}) + B_{i\oplus 2}(p_{i\oplus 1,i,i\oplus 1} + p_{i,i\oplus 1,i\oplus 1} - p_{i,i,i}) \\ &\quad + p_{i,i,i\oplus 2} B_i. \end{aligned} \quad (\text{C.41})$$

This must hold for all magnetic field directions, including if $B_i = B_{i\oplus 1} = 0$ or if $B_i = B_{i\oplus 2} = 0$ or if $B_{i\oplus 1} = B_{i\oplus 2} = 0$. This leads to the following conditions:

$$p_{i\oplus 1,i,i\oplus 1} + p_{i,i\oplus 1,i\oplus 1} = p_{i,i,i}, \quad p_{i\oplus 2,i,i\oplus 1} = -p_{i,i\oplus 2,i\oplus 1}, \quad p_{i,i,i\oplus 2} = 0. \quad (\text{C.42})$$

Similarly, with permutation, *i.e.* for $i = k, j = i \oplus 1$ and for $j = k, i = j \oplus 1$, we obtain the following conditions:

$$p_{i\oplus 1,i\oplus 1,i} + p_{i,i\oplus 1,i\oplus 1} = p_{i,i,i}, \quad p_{i\oplus 2,i\oplus 1,i} = -p_{i,i\oplus 1,i\oplus 2}, \quad p_{i,i\oplus 2,i} = 0, \quad (\text{C.43})$$

$$p_{j\oplus 1,j\oplus 1,j} + p_{j\oplus 1,j,j\oplus 1} = p_{j,j,j}, \quad p_{j\oplus 1,j\oplus 2,j} = -p_{j\oplus 1,j,j\oplus 2}, \quad p_{j\oplus 2,j,j} = 0. \quad (\text{C.44})$$

Since $p_{i,i,i\oplus 1} = p_{i,i\oplus 1,i} = p_{i\oplus 1,i} = p_{i,i,i\oplus 2} = p_{i,i\oplus 2,i} = p_{i\oplus 2,i,i} = 0$, we also have that $p_{i\oplus 1,i,i\oplus 1} + p_{i,i\oplus 1,i\oplus 1} = p_{i,i,i} = 0$. Hence, for, $i, j, k \in \{x, y, z\}$, the only nonzero p_{ijk} terms are when all $i \neq j \neq k$. The second conditions of Eqs. (C.42), (C.43), and (C.44) give:

$$p_{i\oplus 2,i,i\oplus 1} = -p_{i,i\oplus 2,i\oplus 1} \implies p_{i,i\oplus 1,i\oplus 2} = -p_{i\oplus 1,i,i\oplus 2} \quad (\text{C.45})$$

$$\begin{aligned} p_{i\oplus 2,i\oplus 1,i} = -p_{i,i\oplus 1,i\oplus 2} &\implies p_{i,i\oplus 1,i\oplus 2} = -p_{i\oplus 2,i\oplus 1,i} \\ &\implies p_{i\oplus 2,i,i\oplus 1} = -p_{i\oplus 1,i,i\oplus 2} \end{aligned} \quad (\text{C.46})$$

$$\begin{aligned} p_{j\oplus 1,j\oplus 2,j} = -p_{j\oplus 1,j,j\oplus 2} &\implies p_{i,i\oplus 1,i\oplus 2} = -p_{i,i\oplus 2,i\oplus 1} \\ &\implies p_{i\oplus 1,i\oplus 2,i} = -p_{i\oplus 1,i,i\oplus 2}, \end{aligned} \quad (\text{C.47})$$

so we have that:

$$p_{i,i\oplus 1,i\oplus 2} = p_{i\oplus 2,i,i\oplus 1} = p_{i\oplus 1,i\oplus 2,i} = -p_{i\oplus 1,i,i\oplus 2} = -p_{i\oplus 2,i\oplus 1,i} = -p_{i,i\oplus 2,i\oplus 1}. \quad (\text{C.48})$$

This, in short, gives us that

$$p_{xyz} = p_{yzx} = p_{zxy} = -p_{xzy} = -p_{yxz} = -p_{zyx}. \quad (\text{C.49})$$

Our reduced state is now

$$\begin{aligned} \rho_S = & p_{000} \mathbb{1} \otimes \mathbb{1} \otimes \mathbb{1} + p_{xx0} \vec{\sigma}^A \cdot \vec{\sigma}^B \otimes \mathbb{1}^C + p_{x0x} \vec{\sigma}^A \cdot \vec{\sigma}^C \otimes \mathbb{1}^B + p_{0xx} \mathbb{1}^A \otimes \vec{\sigma}^B \cdot \vec{\sigma}^C \\ & + p_{xyz} \sum_{i=x,y,z} \sigma_i \otimes \sigma_{i \oplus 1} \otimes \sigma_{i \oplus 2} - p_{xyz} \sum_{i=x,y,z} \sigma_i \otimes \sigma_{i \oplus 2} \otimes \sigma_{i \oplus 1}, \end{aligned} \quad (\text{C.50})$$

and with straightforward calculation, satisfies $[\rho_S, \hat{H}_{\vec{B}}(\theta, \phi)] = 0$ for all (θ, ϕ) . Finally, state normalisation gives $p_{000} = \mathbb{1}_8/8$. $\square$

C.1.2 Proof of Proposition 5

Recall that we want to show that $[\rho_S \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}, \hat{H}_{ex} + \hat{H}_{\vec{B}}(\theta, \phi) + \hat{V}_{S\mathcal{E}}] = 0$ if and only if $\rho_S = \mathbb{1}_S/d_S$, when the initial environment state $\rho_{\mathcal{E}} = \mathbb{1}_{\mathcal{E}}/d_{\mathcal{E}}$, and with interactions

$$\hat{V}_{S\mathcal{E}} = J_{se} \sum_{\alpha=A,B,C} |1\rangle\langle 1|^{\alpha} \otimes (\mathbb{1} - \sigma_x)^{\mathcal{E}_{\alpha}}, \quad \hat{H}_{ex} = -\frac{J_{ABC}}{2} \sum_{i < j} (\mathbb{1} + \vec{\sigma}^i \cdot \vec{\sigma}^j). \quad (\text{C.51})$$

Proof. It is immediate that $\rho_S = \mathbb{1}_S/d_S$ satisfies the zero commutator.

To prove the other direction, note that we are taking our model in particular: no $\hat{H}_{dd}$ term, and we have a particular system-environment interaction that we can write as

$$\hat{V}_{S\mathcal{E}} = \frac{J_{se}}{2} \sum_{\alpha=A,B,C} (\mathbb{1} - \sigma_z)^{\alpha} \otimes (\mathbb{1} - \sigma_x)^{\mathcal{E}_{\alpha}}. \quad (\text{C.52})$$

Now,

$$\begin{aligned} & \left[\rho_S \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}, \hat{H}_{ex} + \hat{H}_{\vec{B}}(\theta, \phi) + \hat{V}_{S\mathcal{E}} \right] \\ &= \left[\rho_S, \hat{H}_{ex} + \hat{H}_{\vec{B}}(\theta, \phi) \right] \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} + \left[\rho_S \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}, \hat{V}_{S\mathcal{E}} \right] \end{aligned} \quad (\text{C.53})$$

$$\stackrel{!}{=} 0. \quad (\text{C.54})$$

The second commutator in Eq. (C.53) contains terms of form $\sigma_{x,y,z}^{\mathcal{E}}$ that are absent from the first commutator in Eq. (C.53). Hence, the coefficients of $\sigma_{x,y,z}^{\mathcal{E}}$ in the second commutator must independently (due to linear independence) be zero for the statement to hold. Thus, let us first consider the second commutator and determine the reduced form of ρ_S needed. Let us write the general system state as $\rho_S = \sum_{ijk=0,x,y,z} p_{ijk} \sigma_i^A \otimes \sigma_j^B \otimes \sigma_k^C$.

Consider the interaction between system A and its local bath: $\hat{V}_{A\mathcal{E}_A} = \frac{J_{se}}{2}(\mathbb{1} - \sigma_z)^A \otimes (\mathbb{1} - \sigma_x)^{\mathcal{E}_A}$ (the results for $\hat{V}_{B\mathcal{E}_B}$ and $\hat{V}_{C\mathcal{E}_C}$ will be analogous):

$$\begin{aligned} & \frac{2}{J_{se}} \left[\rho_S \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}, \hat{V}_{A\mathcal{E}_A} \right] \\ &= \sum_{ijk=0,x,y,z} p_{ijk} \left[\sigma_i^A \otimes \sigma_j^B \otimes \sigma_k^C \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}, (\mathbb{1} - \sigma_z)^A \otimes (\mathbb{1} - \sigma_x)^{\mathcal{E}_A} \right] \end{aligned} \quad (\text{C.55})$$

$$= \sum_{ijk=0,x,y,z} p_{ijk} \left[\sigma_i^A \otimes \frac{\mathbb{1}^{\mathcal{E}_A}}{2}, (\mathbb{1} - \sigma_z)^A \otimes (\mathbb{1} - \sigma_x)^{\mathcal{E}_A} \right] \otimes \sigma_j^B \otimes \sigma_k^C \otimes \frac{\mathbb{1}^{\mathcal{E}_B\mathcal{E}_C}}{d_{\mathcal{E}_B\mathcal{E}_C}} \quad (\text{C.56})$$

$$= \sum_{ijk=0,x,y,z} p_{ijk} [\sigma_z^A, \sigma_i^A] \otimes \sigma_j^B \otimes \sigma_k^C \otimes \left(\frac{\mathbb{1} - \sigma_x}{2} \right)^{\mathcal{E}_A} \otimes \frac{\mathbb{1}^{\mathcal{E}_B\mathcal{E}_C}}{d_{\mathcal{E}_B\mathcal{E}_C}}. \quad (\text{C.57})$$

The (only) term containing $\sigma_x^{\mathcal{E}_A}$ is the following, and it must be zero:

$$\begin{aligned} & \sum_{ijk=0,x,y,z} p_{ijk} [\sigma_z^A, \sigma_i^A] \otimes \sigma_j^B \otimes \sigma_k^C \otimes \frac{-\sigma_x^{\mathcal{E}_A}}{2} \otimes \frac{\mathbb{1}^{\mathcal{E}_B\mathcal{E}_C}}{d_{\mathcal{E}_B\mathcal{E}_C}} \\ &= \sum_{jk=0,x,y,z} (p_{xjk} 2i\sigma_y^A - p_{yjk} 2i\sigma_x^A) \otimes \sigma_j^B \otimes \sigma_k^C \otimes \frac{-\sigma_x^{\mathcal{E}_A}}{2} \otimes \frac{\mathbb{1}^{\mathcal{E}_B\mathcal{E}_C}}{d_{\mathcal{E}_B\mathcal{E}_C}} \stackrel{!}{=} 0 \end{aligned} \quad (\text{C.58})$$

$$\Rightarrow p_{xjk} = p_{yjk} = 0, \quad \forall j, k \in \{0, x, y, z\}, \quad (\text{C.59})$$

due to linear independence of the different terms. Similarly, by examining $\hat{V}_{B\mathcal{E}_B}$ and $\hat{V}_{C\mathcal{E}_C}$, we find that $p_{ixk} = p_{iyk} = 0 \forall i, k \in \{0, x, y, z\}$ and $p_{ijx} = p_{ijy} = 0 \forall i, j \in \{0, x, y, z\}$. This means that the system state has only terms with $\mathbb{1} = \sigma_0$ and σ_z :

$$\rho_{S|z} := \sum_{ijk=0,z} p_{ijk} \sigma_i^A \otimes \sigma_j^B \otimes \sigma_k^C. \quad (\text{C.60})$$

Now in fact, this $\rho_{S|z}$ form has $[\rho_{S|z} \otimes \mathbb{1}_{\mathcal{E}}/d_{\mathcal{E}}, \hat{V}_{S\mathcal{E}}] = 0$. Thus, we now need to consider

$$\left[\rho_{S|z}, \hat{H}_{ex} + \hat{H}_{\vec{B}}(\theta, \phi) \right] \stackrel{!}{=} 0. \quad (\text{C.61})$$

Now if $\left[\rho_{S|z}, \hat{H}_{ex} + \hat{H}_{\vec{B}}(\theta, \phi) \right] \stackrel{!}{=} 0$ at all angles, then consider if $\hat{H}_{\vec{B}}(0, 0) = \frac{\gamma B_0}{2}(\sigma_z^A + \sigma_z^B + \sigma_z^C)$. In this case,

$$\left[\rho_{S|z}, \hat{H}_{\vec{B}}(0, 0) \right] = \left[\sum_{ijk=0,z} p_{ijk} \sigma_i^A \otimes \sigma_j^B \otimes \sigma_k^C, \frac{\gamma B_0}{2}(\sigma_z^A + \sigma_z^B + \sigma_z^C) \right] = 0, \quad (\text{C.62})$$

since all terms commute with each other. Hence, $\left[\rho_{S|z}, \hat{H}_{ex} + \hat{H}_{\vec{B}}(0,0)\right] \stackrel{!}{=} 0$ implies $\left[\rho_{S|z}, \hat{H}_{ex}\right] \stackrel{!}{=} 0$. Expanding,

$$- \frac{2}{J_{ABC}} \left[\rho_{S|z}, \hat{H}_{ex} \right] = \sum_{ijk=0,z} \left[p_{ijk} \sigma_i^A \otimes \sigma_j^B \otimes \sigma_k^C, 3\mathbb{1} + \vec{\sigma}^A \cdot \vec{\sigma}^B + \vec{\sigma}^A \cdot \vec{\sigma}^C + \vec{\sigma}^B \cdot \vec{\sigma}^C \right] \quad (\text{C.63})$$

$$= \sum_{ijk=0,z} p_{ijk} \left(\begin{aligned} & \left[\sigma_i^A \otimes \sigma_j^B, \vec{\sigma}^A \cdot \vec{\sigma}^B \right] \otimes \sigma_k^C \\ & + \left[\sigma_i^A \otimes \sigma_k^C, \vec{\sigma}^A \cdot \vec{\sigma}^C \right] \otimes \sigma_j^B + \sigma_i^A \otimes \left[\sigma_j^B \otimes \sigma_k^C, \vec{\sigma}^B \cdot \vec{\sigma}^C \right] \end{aligned} \right). \quad (\text{C.64})$$

Consider the first terms in Eq. (C.64):

$$\begin{aligned} & \sum_{ijk=0,z} p_{ijk} \left[\sigma_i^A \otimes \sigma_j^B, \vec{\sigma}^A \cdot \vec{\sigma}^B \right] \otimes \sigma_k^C \\ &= \sum_{ijk=0,z} p_{ijk} \left(\begin{aligned} & \left[\sigma_i^A \otimes \sigma_j^B, \sigma_x \otimes \sigma_x \right] + \left[\sigma_i^A \otimes \sigma_j^B, \sigma_y \otimes \sigma_y \right] \\ & + \cancel{\left[\sigma_i^A \otimes \sigma_j^B, \sigma_z \otimes \sigma_z \right]} \end{aligned} \right) \otimes \sigma_k^C \quad (\text{C.65}) \end{aligned}$$

$$= \sum_{k=0,z} \left(\begin{aligned} & \left[p_{0zk} \mathbb{1} \otimes \sigma_z + p_{z0k} \sigma_z \otimes \mathbb{1} + p_{zzk} \sigma_z \otimes \sigma_z, \sigma_x \otimes \sigma_x \right] \\ & + \left[p_{0zk} \mathbb{1} \otimes \sigma_z + p_{z0k} \sigma_z \otimes \mathbb{1} + p_{zzk} \sigma_z \otimes \sigma_z, \sigma_y \otimes \sigma_y \right] \end{aligned} \right) \otimes \sigma_k^C \quad (\text{C.66})$$

$$= \sum_{k=0,z} \left(\begin{aligned} & p_{0zk} \sigma_x \otimes 2i\sigma_y + p_{z0k} 2i\sigma_y \otimes \sigma_x \\ & + p_{0zk} \sigma_y \otimes (-2i\sigma_x) + p_{z0k} (-2i\sigma_x) \otimes \sigma_y \end{aligned} \right) \otimes \sigma_k^C \quad (\text{C.67})$$

$$= 2i \sum_{k=0,z} \left((p_{0zk} - p_{z0k}) \sigma_x^A \otimes \sigma_y^B + (p_{z0k} - p_{0zk}) \sigma_y^A \otimes \sigma_x^B \right) \otimes \sigma_k^C. \quad (\text{C.68})$$

Similarly,

$$\begin{aligned} & \sum_{ijk=0,z} p_{ijk} \left[\sigma_i^A \otimes \sigma_k^C, \vec{\sigma}^A \cdot \vec{\sigma}^C \right] \otimes \sigma_j^B \\ &= 2i \sum_{j=0,z} \left((p_{0jz} - p_{zj0}) \sigma_x^A \otimes \sigma_y^C + (p_{zj0} - p_{0jz}) \sigma_y^A \otimes \sigma_x^C \right) \otimes \sigma_j^B \quad (\text{C.69}) \end{aligned}$$

$$\begin{aligned} & \sum_{ijk=0,z} p_{ijk} \sigma_i^A \otimes \left[\sigma_j^B \otimes \sigma_k^C, \vec{\sigma}^B \cdot \vec{\sigma}^C \right] \\ &= 2i \sum_{i=0,z} \sigma_i^A \otimes \left((p_{i0z} - p_{iz0}) \sigma_x^B \otimes \sigma_y^C + (p_{iz0} - p_{i0z}) \sigma_y^B \otimes \sigma_x^C \right). \quad (\text{C.70}) \end{aligned}$$

Hence,

$$\begin{aligned} & - \frac{1}{J_{ABC}i} \left[\rho_{S|z}, \hat{H}_{ex} \right] \\ &= \sum_{k=0,z} \left((p_{0zk} - p_{z0k}) \sigma_x^A \otimes \sigma_y^B + (p_{z0k} - p_{0zk}) \sigma_y^A \otimes \sigma_x^B \right) \otimes \sigma_k^C \end{aligned}$$

$$\begin{aligned}
& + \sum_{j=0,z} ((p_{0jz} - p_{zj0})\sigma_x^A \otimes \sigma_y^C + (p_{zj0} - p_{0jz})\sigma_y^A \otimes \sigma_x^C) \otimes \sigma_j^B \\
& + \sum_{i=0,z} \sigma_i^A \otimes ((p_{i0z} - p_{iz0})\sigma_x^B \otimes \sigma_y^C + (p_{iz0} - p_{i0z})\sigma_y^B \otimes \sigma_x^C)
\end{aligned} \tag{C.71}$$

$$\stackrel{!}{=} 0. \tag{C.72}$$

However, note that each of the terms are linearly independent, and so the coefficients must be zero, which implies that

$$p_{0zk} = p_{z0k} \quad \text{for } k = 0, z, \tag{C.73}$$

$$p_{0jz} = p_{zj0} \quad \text{for } j = 0, z, \tag{C.74}$$

$$p_{i0z} = p_{iz0} \quad \text{for } i = 0, z. \tag{C.75}$$

Hence we can reduce the system state again:

$$\begin{aligned}
\rho'_{S|z} = & p_{000} \mathbb{1} \otimes \mathbb{1} \otimes \mathbb{1} + p_{zzz} \sigma_z \otimes \sigma_z \otimes \sigma_z \\
& + p_{z00} (\mathbb{1} \otimes \mathbb{1} \otimes \sigma_z + \mathbb{1} \otimes \sigma_z \otimes \mathbb{1} + \sigma_z \otimes \mathbb{1} \otimes \mathbb{1}) \\
& + p_{zz0} (\mathbb{1} \otimes \sigma_z \otimes \sigma_z + \sigma_z \otimes \mathbb{1} \otimes \sigma_z + \sigma_z \otimes \sigma_z \otimes \mathbb{1}).
\end{aligned} \tag{C.76}$$

Now going back to $[\rho'_{S|z}, \hat{H}_{ex} + \hat{H}_{\vec{B}}(\theta, \phi)] \stackrel{!}{=} 0$, since $[\rho'_{S|z}, \hat{H}_{ex}] = 0$, consider $[\rho'_{S|z}, \hat{H}_{\vec{B}}(\theta, \phi)]$. Consider if the magnetic field is pointed in the x direction, such that $\hat{H}_{\vec{B}}(0, \frac{\pi}{2}) = \frac{\gamma B_0}{2} (\sigma_x^A + \sigma_x^B + \sigma_x^C)$. In this case, the commutator expands out to:

$$\begin{aligned}
& \frac{1}{\gamma B_0 i} [\rho'_{S|z}, \hat{H}_{\vec{B}}(0, \frac{\pi}{2})] \\
& = \sigma_y^A \otimes (p_{z00} \mathbb{1}^B \otimes \mathbb{1}^C + p_{zz0} (\mathbb{1} \otimes \sigma_z + \sigma_z \otimes \mathbb{1}) + p_{zzz} \sigma_z \otimes \sigma_z) \\
& + \sigma_y^B \otimes (p_{z00} \mathbb{1}^A \otimes \mathbb{1}^C + p_{zz0} (\mathbb{1} \otimes \sigma_z + \sigma_z \otimes \mathbb{1}) + p_{zzz} \sigma_z \otimes \sigma_z) \\
& + \sigma_y^C \otimes (p_{z00} \mathbb{1}^A \otimes \mathbb{1}^B + p_{zz0} (\mathbb{1} \otimes \sigma_z + \sigma_z \otimes \mathbb{1}) + p_{zzz} \sigma_z \otimes \sigma_z) \stackrel{!}{=} 0
\end{aligned} \tag{C.77}$$

$$\implies p_{z00} = 0, \quad p_{zz0} = 0, \quad p_{zzz} = 0, \tag{C.78}$$

again due to the linear independence of the terms and subsequent requirements for zero coefficients. Thus, the reduced system state is $\rho_S = p_{000} \mathbb{1} \otimes \mathbb{1} \otimes \mathbb{1} = \mathbb{1}_S / d_S$ with the d_S term for normalisation. $\square$

C.1.3 Proof of Lemma 5

Recall the statement of Lemma 5: *If $\hat{V}_{S\mathcal{E}} = \hat{V}_{A\mathcal{E}_A} + \hat{V}_{B\mathcal{E}_B} + \hat{V}_{C\mathcal{E}_C} \neq 0$ has nontrivial interaction between system $\mathcal{S}$ and environment $\mathcal{E}$ (i.e. contains two-body interaction terms), then $[P_{\text{singlet}}^{(A,B)}, \hat{V}_{S\mathcal{E}}] \neq 0$.*

Let $P_{\text{singlet}}^{(A,B)} = P^{AB}$ to condense notation.

Proof. We can simplify $[P^{AB}, \hat{V}_{S\mathcal{E}}] = [P^{AB}, \hat{V}_{A\mathcal{E}_A}] + [P^{AB}, \hat{V}_{B\mathcal{E}_B}]$, as P^{AB} commutes with C .

Now, $[P^{AB}, \hat{V}_{S\mathcal{E}}]$ would be zero if either:

1. $[P^{AB}, \hat{V}_{A\mathcal{E}_A}] \stackrel{!}{=} 0$ and $[P^{AB}, \hat{V}_{B\mathcal{E}_B}] \stackrel{!}{=} 0$ are separately zero, or
2. if $[P^{AB}, \hat{V}_{A\mathcal{E}_A}] \stackrel{!}{=} -[P^{AB}, \hat{V}_{B\mathcal{E}_B}]$.

In general, we can write $\hat{V}_{A\mathcal{E}_A} = \sum_{ij=0,\dots,3} g_{ij} \sigma_i^A \otimes \sigma_j^{\mathcal{E}_A}$, for $\sigma_i \in \{\mathbb{1}, \sigma_x, \sigma_y, \sigma_z\}$. Then,

$$\begin{aligned} [P^{AB}, \hat{V}_{A\mathcal{E}_A}] &= -\frac{1}{4} \sum_{ij=0,\dots,3} g_{ij} [\sigma_x, \sigma_i^A] \otimes \sigma_x^B \otimes \sigma_j^{\mathcal{E}_A} - \frac{1}{4} \sum_{ij=0,\dots,3} g_{ij} [\sigma_y, \sigma_i^A] \otimes \sigma_y^B \otimes \sigma_j^{\mathcal{E}_A} \\ &\quad - \frac{1}{4} \sum_{ij=0,\dots,3} g_{ij} [\sigma_z, \sigma_i^A] \otimes \sigma_z^B \otimes \sigma_j^{\mathcal{E}_A}. \end{aligned} \quad (\text{C.79})$$

Due to the linear independence of the $\{\sigma_x^B, \sigma_y^B, \sigma_z^B\}$ terms, in order for $[P^{AB}, \hat{V}_{A\mathcal{E}_A}] \stackrel{!}{=} 0$ to be true, we would require

$$\sum_{ij=0,\dots,3} g_{ij} [\sigma_x, \sigma_i^A] \otimes \sigma_x^B \otimes \sigma_j^{\mathcal{E}_A} \stackrel{!}{=} 0 \quad (\text{C.80})$$

$$\sum_{ij=0,\dots,3} g_{ij} [\sigma_y, \sigma_i^A] \otimes \sigma_y^B \otimes \sigma_j^{\mathcal{E}_A} \stackrel{!}{=} 0 \quad (\text{C.81})$$

$$\sum_{ij=0,\dots,3} g_{ij} [\sigma_z, \sigma_i^A] \otimes \sigma_z^B \otimes \sigma_j^{\mathcal{E}_A} \stackrel{!}{=} 0. \quad (\text{C.82})$$

Consider Eq. (C.80):

$$\begin{aligned} \sum_{ij=0,\dots,3} g_{ij} [\sigma_x, \sigma_i^A] \otimes \sigma_x^B \otimes \sigma_j^{\mathcal{E}_A} &= \sum_{j=0,\dots,3} g_{yj} 2i \sigma_z^A \otimes \sigma_x^B \otimes \sigma_j^{\mathcal{E}_A} \\ &\quad + \sum_{j=0,\dots,3} g_{zj} (-2i) \sigma_y^A \otimes \sigma_x^B \otimes \sigma_j^{\mathcal{E}_A}. \end{aligned} \quad (\text{C.83})$$

Again, due to linearly independence of the Pauli matrices, this is zero iff $g_{yj} = g_{zj} = 0$ for all $j = 0, \dots, 3$. The only possible nonzero terms are g_{xj} and g_{0j} . Similarly, consider Eq. (C.81):

$$\begin{aligned} \sum_{ij=0,\dots,3} g_{ij} [\sigma_y, \sigma_i^A] \otimes \sigma_y^B \otimes \sigma_j^{\mathcal{E}_A} &= \sum_{j=0,\dots,3} g_{xj} (-2i) \sigma_z^A \otimes \sigma_y^B \otimes \sigma_j^{\mathcal{E}_A} \\ &+ \sum_{j=0,\dots,3} g_{zj} (2i) \sigma_x^A \otimes \sigma_y^B \otimes \sigma_j^{\mathcal{E}_A}. \end{aligned} \quad (\text{C.84})$$

which is zero iff $g_{xj} = g_{zj} = 0$ for all $j = 0, \dots, 3$. Similarly, Eq. (C.82) gives $g_{xj} = g_{yj} = 0$ for all $j = 0, \dots, 3$.

Hence, if $[P^{AB}, \hat{V}_{A\mathcal{E}_A}] \stackrel{!}{=} 0$ then we must have $\hat{V}_{A\mathcal{E}_A} = \sum_{j=0,\dots,3} g_{0j} \mathbb{1}_A \otimes \sigma_j^{\mathcal{E}_A}$. This is trivial because this is a local interaction on the environment $\mathcal{E}_A$ and contradicts the fact that $\hat{V}_{A\mathcal{E}_A}$ must be an interaction between system and environment. Similarly, $[P^{AB}, \hat{V}_{B\mathcal{E}_B}] \stackrel{!}{=} 0$ implies that $\hat{V}_{B\mathcal{E}_B} = \sum_{j=0,\dots,3} \mathbb{1}_B \otimes \sigma_j^{\mathcal{E}_B}$ is a local interaction on the environment $\mathcal{E}_B$.

Finally, in general,

$$\begin{aligned} [P^{AB}, \hat{V}_{S\mathcal{E}}] &= -\frac{1}{4} \sum_{ij=0,\dots,3} g_{ij} [\sigma_x, \sigma_i^A] \otimes \sigma_x^B \otimes \sigma_j^{\mathcal{E}_A} \otimes \mathbb{1}^{\mathcal{E}_B} \\ &- \frac{1}{4} \sum_{ij=0,\dots,3} g_{ij} [\sigma_y, \sigma_i^A] \otimes \sigma_y^B \otimes \sigma_j^{\mathcal{E}_A} \otimes \mathbb{1}^{\mathcal{E}_B} \\ &- \frac{1}{4} \sum_{ij=0,\dots,3} g_{ij} [\sigma_z, \sigma_i^A] \otimes \sigma_z^B \otimes \sigma_j^{\mathcal{E}_A} \otimes \mathbb{1}^{\mathcal{E}_B} \\ &- \frac{1}{4} \sum_{ij=0,\dots,3} f_{ij} \sigma_x^A \otimes [\sigma_x, \sigma_i^B] \otimes \mathbb{1}^{\mathcal{E}_A} \otimes \sigma_j^{\mathcal{E}_B} \\ &- \frac{1}{4} \sum_{ij=0,\dots,3} f_{ij} \sigma_y^A \otimes [\sigma_y, \sigma_i^B] \otimes \mathbb{1}^{\mathcal{E}_A} \otimes \sigma_j^{\mathcal{E}_B} \\ &- \frac{1}{4} \sum_{ij=0,\dots,3} f_{ij} \sigma_z^A \otimes [\sigma_z, \sigma_i^B] \otimes \mathbb{1}^{\mathcal{E}_A} \otimes \sigma_j^{\mathcal{E}_B}. \end{aligned} \quad (\text{C.85})$$

All the terms where $j = x, y, z$ are separately linearly independent (*i.e.* each line above is linearly independent from the others). Analogously, we find that this is zero if and only if $g_{xj} = g_{yj} = g_{zj} = 0$ for $j = x, y, z$ and $f_{xj} = f_{yj} = f_{zj} = 0$ for $j = x, y, z$. This leaves us with $\hat{V}_{A\mathcal{E}_A} = \sum_{j=0,\dots,3} g_{0j} \mathbb{1}_A \otimes \sigma_j^{\mathcal{E}_A} + \sum_{i=0,\dots,3} g_{i0} \sigma_i^A \otimes \mathbb{1}^{\mathcal{E}_A}$ and similarly for $\hat{V}_{B\mathcal{E}_B}$, *i.e.* they contain only local terms, which is a contradiction. $\square$

C.1.4 Proof of Lemma 7

Recall that we want to show that $[\hat{H}_{\vec{B}}(\theta, \phi), \hat{H}_0 + \hat{V}_{S\mathcal{E}}] \neq 0$ for all nontrivial $\hat{V}_{S\mathcal{E}}$.

Proof. Now,

$$[\hat{H}_0 + \hat{V}_{S\mathcal{E}}, \hat{H}_{\vec{B}}[\theta, \phi]] = [\hat{H}_0, \hat{H}_{\vec{B}}(\theta, \phi)] + [\hat{V}_{S\mathcal{E}}, \hat{H}_{\vec{B}}(\theta, \phi)]. \quad (\text{C.86})$$

The first term on the right hand side contains only nonzero terms on the system ABC while the $\hat{V}_{S\mathcal{E}}$ has environment terms. For this to be zero overall, it must be that any nontrivial environment terms in the second commutator are zero.

$$[\hat{V}_{S\mathcal{E}}, \hat{H}_{\vec{B}}(\theta, \phi)] = [\hat{V}_{A\mathcal{E}_A}, \hat{H}_{\vec{B}|A}(\theta, \phi)] + [\hat{V}_{B\mathcal{E}_B}, \hat{H}_{\vec{B}|B}(\theta, \phi)] + [\hat{V}_{C\mathcal{E}_C}, \hat{H}_{\vec{B}|C}(\theta, \phi)]. \quad (\text{C.87})$$

Clearly each term is independent, so let's just look at the first (and the other two follow analogously):

$$[\hat{V}_{A\mathcal{E}_A}, \hat{H}_{\vec{B}|A}(\theta, \phi)] = \left[\sum_{ij=0,\dots,3} g_{ij} \sigma_i^A \otimes \sigma_j^{\mathcal{E}_A}, \hat{H}_{\vec{B}|A}(\theta, \phi) \right] \quad (\text{C.88})$$

$$= \sum_{ij=0,\dots,3} g_{ij} [\sigma_i^A, \hat{H}_{\vec{B}|A}(\theta, \phi)] \otimes \sigma_j^{\mathcal{E}_A}. \quad (\text{C.89})$$

For the indices where $j \neq 0$ (*i.e.* for the non-identity terms on the environment E_A), we want the coefficients to be zero, *i.e.*

$$\sum_{i=0,\dots,3} g_{ij} [\sigma_i^A, \hat{H}_{\vec{B}|A}(\theta, \phi)] \otimes \sigma_j^{\mathcal{E}_A} \stackrel{!}{=} 0 \quad \forall j = x, y, z. \quad (\text{C.90})$$

If $\hat{H}_{\vec{B}|A} = \frac{\gamma B_0}{2} \sigma_x$ for example, then this would imply that

$$0 \stackrel{!}{=} \sum_{i=0,\dots,3} g_{ij} \gamma B_0 [\sigma_i^A, \sigma_x] = g_{yj} \gamma B_0 (-2i\sigma_z) + g_{zj} \gamma B_0 2i\sigma_y \quad (\text{C.91})$$

$$\implies g_{yj} = g_{zj} = 0 \quad \forall j = x, y, z. \quad (\text{C.92})$$

We can similarly show that $g_{xj} = 0$ for all $j = x, y, z$, in which case $\hat{V}_{A\mathcal{E}_A} = \mathbb{1}_A \otimes \sum_{j=0,x,y,z} g_{0j} \sigma_j^{\mathcal{E}_A} + \sum_{i=x,y,z} \sigma_i^A \otimes \mathbb{1}^{\mathcal{E}_A}$, *i.e.* containing only local terms. Similarly for $\hat{V}_{B\mathcal{E}_B}$ and $\hat{V}_{C\mathcal{E}_C}$. This is a contradiction because $\hat{V}_{S\mathcal{E}}$ must be a nontrivial interaction. Hence, provided that $\hat{V}_{S\mathcal{E}}$ is a nontrivial interaction, we always have $[\hat{H}_0 + \hat{V}_{S\mathcal{E}}, \hat{H}_{\vec{B}}[\theta, \phi]] \neq 0$. $\square$

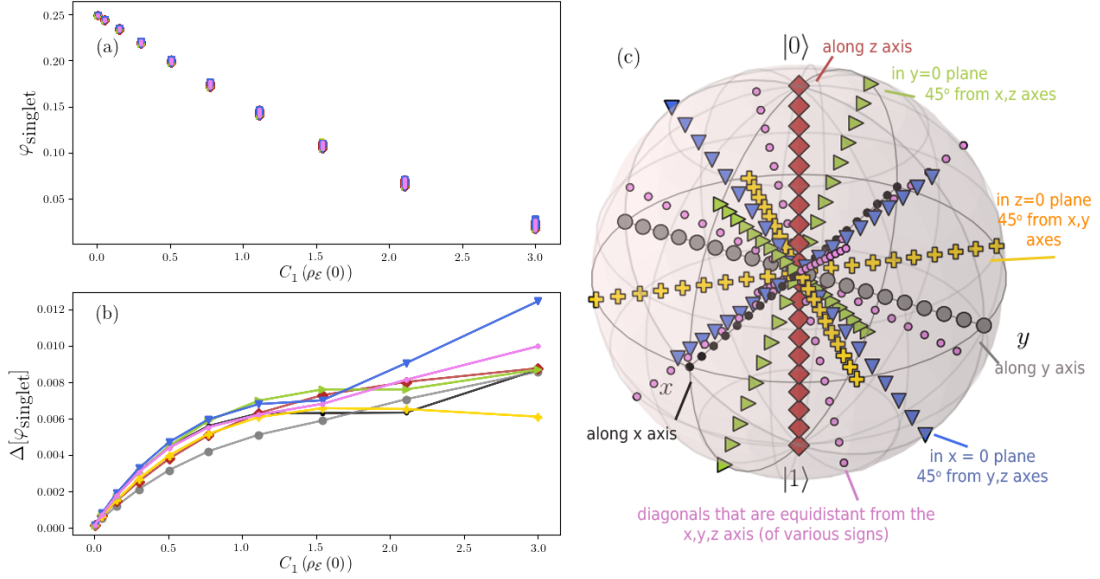


Figure C.1: **Magnetic sensor performance for a SWAP interaction** where the system is initially maximally mixed $\rho_S(0) = \mathbb{1}/d_S$ but the environment is not (see Example 10). (a) The recombination value φ_{singlet} for various initial environment states $\rho_{\mathcal{E}}(0)$ versus the initial basis-independent coherence $C_1(\rho_{\mathcal{E}}(0))$. (b) Absolute anisotropy $\Delta[\varphi_{\text{singlet}}]$. (c) Visualisation of the initial environment states on the Bloch sphere. Some points have the same colouring and shape, as their behaviour in the absolute anisotropy follow the same trend. Model details: electron exchange interaction only, with all other parameters not mentioned explicitly here listed in Table 7.1 .

C.2 Another system-environment interaction: the SWAP

What about more general system-environment interactions and more general environment initial states? If so, the condition of $\rho_S(0) \neq \mathbb{1}_S/d_S$ is not always necessary. In the main chapter 7, we mentioned that the ultimate generation of quantum coherence is key. In the following example, we can do this by transferring environment coherence into the system.

Example 10. The initial state $\rho_S(0) = \mathbb{1}/d_S$ on the system can lead to nontrivial sensing if (for example) $\hat{V}_{S\mathcal{E}}$ is a SWAP interaction between system and environment, and the initial environment states are not maximally mixed. The SWAP interaction will introduce the environment coherence into the system. Consider

the isotropic interaction

$$\hat{V}_{X\mathcal{E}_X} = J_{se} \sum_{i=x,y,z} \sigma_i^X \otimes \sigma_i^{\mathcal{E}_X}, \quad \text{for } X = A, B, C. \quad (\text{C.93})$$

Then $\exp[-it_{se}\hat{V}_{A\mathcal{E}_A}]$ performs a perfect SWAP if $J_{se}t_{se} = \pi/4$ [314]. We can see in Fig. C.1 that the coherence in the environment leads to a small change in the recombination value and hence a nontrivial magnetic sensor.

Fig. C.1(b) also shows that there is an initial sharp rise in the sensor perform versus initial environment coherence, before a partial plateau emerges around $C_1(\rho_{\mathcal{E}}(0)) \approx 1.5$. As the coherence increases, the dependence on the basis increases, such that different pure states give different $\Delta[\varphi_{\text{singlet}}]$. This dependence on the basis that is suitable for further investigation. Nevertheless these results suggest that small environment coherences are most effective, $C_1(\rho_{\mathcal{E}}(0)) \leq 1.5$, where there is less difference between different states with the same coherence.

This example shows that *environment* coherence can contribute to the performance of the magnetic sensor. Here, the environment coherence introduces effective non-unital dynamics on the system, which then creates anisotropy in the system state, which then allows the magnetic field to affect it.

C.3 Quantum coherence yield results

In section 7.4, we compared how the *initial* quantum coherence affected the subsequent magnetic sensor performance. However, it is also interesting to examine the coherence *yield*, *i.e.* how much coherence is relevant during the recombination process itself.

Due to the trace-decreasing nature of $\rho_{S\mathcal{E}}(t)$ in our model (*i.e.* $\rho_{S\mathcal{E}}(t)$ becomes unnormalised as time passes), we will use the modified version as the correct distance to the trace-decreasing maximally mixed state for times $t > 0$ as our measure of basis-independent coherence:

$$C_1^*(\rho) = H\left(\rho \left\| \frac{\mathbb{1}_d}{d} \text{tr } \rho\right.\right) \quad (\text{C.94})$$

$$= (\log_2 d - \log_2 \text{tr } \rho) \text{tr } \rho - H(\rho). \quad (\text{C.95})$$

The coherence yield is then defined as

$$[C_1^*[\theta, \phi]]_{\text{yield}} = \frac{k}{\varphi_{\text{singlet}}} \int_0^\infty d\tau \cdot I(\tau) \text{tr}[P_{\text{singlet}}^{(A,B)} C_1^*(\rho_S(\tau))]. \quad (\text{C.96})$$

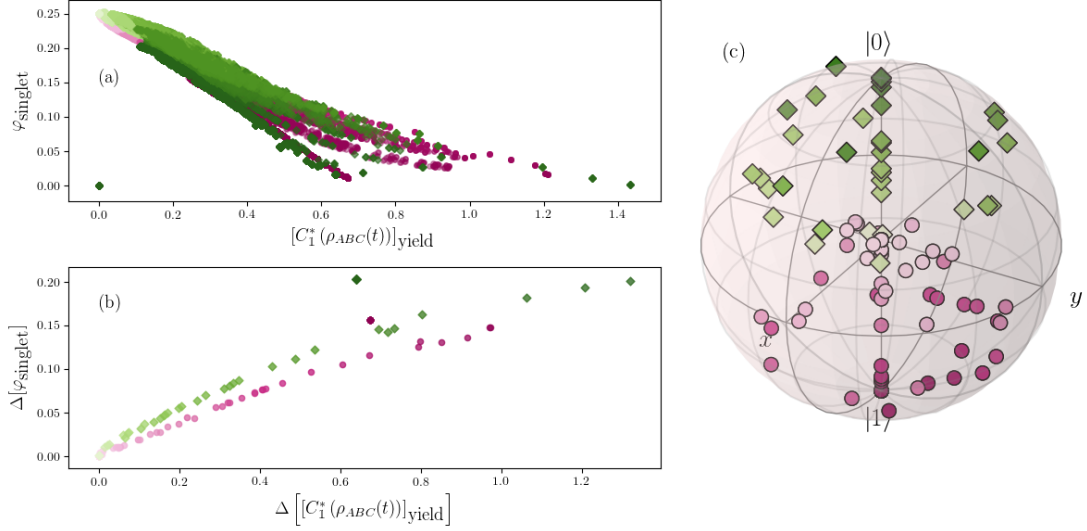


Figure C.2: **Relationship between the basis-independent quantum coherence yield vs sensor performance.** (a) Yield of basis-independent quantum coherence $[C_1^*(\rho_{ABC})]_{\text{yield}}$ vs singlet recombination yield φ_{singlet} . (b) Absolute anisotropy of the basis-independent coherence $\Delta[C_1^*(\rho_{ABC})]_{\text{yield}}$ vs absolute anisotropy of the singlet recombination yield $\Delta[\varphi_{\text{singlet}}]$. (c) Bloch sphere depicting the initial state ρ_A , where $\rho_{ABC}(0) = \rho_A \otimes \rho_A \otimes \mathbb{1}/2$. Model: electron exchange interaction only, with parameters listed in Table 7.1.

The absolute anisotropy is then defined as

$$\Delta([C_1^*]_{\text{yield}}) = \max_{B_0(\theta, \phi)} [C_1^*[\theta, \phi]]_{\text{yield}} - \min_{B_0(\theta, \phi)} [C_1^*[\theta, \phi]]_{\text{yield}}. \quad (\text{C.97})$$

Fig. C.2(a) shows the basis-independent coherence yield versus the singlet recombination yield. There is a clear association between the two: greater coherence yield leads to a *lower* recombination. However, the range of these yields is important for the sensor performance. We see also in Fig. C.2(b) that in which there is a strong positive linear relationship between the anisotropies of coherence and recombination yields. The sub-trends can be well separated by considering the sign of the z coordinate on the Bloch sphere, *i.e.* $\text{tr}[\sigma_z \rho_A(0)]$, of the initial state.

These results show that basis-independent coherence, both initial and during, suppresses the raw value of the recombination yield. Combined with the results in the main chapter, we see that larger initial basis-independent coherence leads to greater variation in both recombination yield and coherence yield over time.

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